

ASVAB Math Workbook

1,000 Practice Problems with Step-by-Step Solutions

Arithmetic Reasoning & Mathematics Knowledge

Key concepts and worked examples for every topic

A diagnostic test with a personal study plan

750 drill problems in 27 chapters

4 full-length math practice tests

Every answer checked by computer algebra

Ivan Sikuten

ASVAB Math Workbook: 1,000 Practice Problems with Step-by-Step Solutions

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Test formats, question counts, time limits, and qualifying scores are set by the Department of Defense and the individual services and can change. Always confirm current details with your recruiter or at officialasvab.com. The author makes no guarantee of any particular test score.

The practice problems in this book were generated and their answers verified with the SymPy computer algebra system; every answer was computed exactly and then confirmed by a second, independent calculation.

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How to Use This Book

The math on the ASVAB is mostly middle-school and early high-school math—percents, fractions, equations, areas, word problems—asked quickly and without a calculator. What raises scores is *practice*: seeing each kind of question often enough that you recognize it instantly and know the first step.

This workbook gives you that practice: **1,000 problems**, each with a complete step-by-step solution.

The plan

1. **Take the Diagnostic Test** (page 5). Thirty questions, about 35 minutes, no calculator. The answer key tells you which chapter each question comes from.
2. **Study the chapters you need.** Start with the topics you missed on the diagnostic. Each chapter has three parts:
 - *Key Concepts*—the rules, formulas, a worked example, a no-calculator tip, and the traps test writers love.
 - *Practice Set*—warm-up problems first, then test-level problems, then challenge problems.
 - *Answers & Explanations*—a quick answer key plus a full solution for every problem.
3. **Take the four Practice Tests** under real time limits once you have worked through the chapters. Each has an Arithmetic Reasoning section (30 questions, 36 minutes) and a Mathematics Knowledge section (25 questions, 24 minutes), the format of the paper ASVAB.
4. **Review every miss.** The answer keys of the tests list the chapter for every question, so you always know where to go back.

Reading the solutions

Every solution is broken into numbered steps that you can follow with pencil and paper. Many solutions also include:

- **TIP**—a shortcut, an estimate, or a way to check your answer.
- **WHY NOT**—what went wrong if you picked one of the tempting wrong answers. The wrong choices in this book are not random numbers; each comes from a real, common mistake (dividing by the wrong number, forgetting to convert minutes to hours, adding percents that should be multiplied). If your answer matches one of them, the explanation tells you exactly which mistake you made.

Good habits

- **No calculator.** The ASVAB does not allow one, so don't practice with one. Every problem here is designed to be done by hand.
- **Write your work down.** Scratch paper is provided at the test; use it here too.
- **Time yourself.** Each practice set shows a target time. Don't worry about speed at first—accuracy comes first, speed follows.
- **Redo your misses a week later** and log them in the Progress Tracker at the back of the book.

The ASVAB Math Subtests

The Armed Services Vocational Aptitude Battery (ASVAB) is the entrance test for the U.S. Army, Navy, Air Force, Marine Corps, Space Force, and Coast Guard. It consists of several subtests. Two of them are math:

Arithmetic Reasoning (AR)
Word problems. Each question describes a real-life situation—a purchase, a trip, a paycheck, a recipe, a work schedule—and you have to decide which operations to use. Typical topics: percents and discounts, ratios, rates and distance, averages, interest, work, measurement, and multi-step reasoning.
Mathematics Knowledge (MK)
Direct math questions: number properties, fractions, exponents and roots, algebra (expressions, equations, inequalities, factoring), and geometry (angles, triangles, area, circles, volume, coordinates). Knowing the vocabulary and formulas is half the battle.

Format and timing

The ASVAB is given in two main formats. On the **computer-adaptive** version (CAT-ASVAB), the difficulty of each question depends on how you answered the previous ones, and you cannot go back to change an answer. On the **paper-and-pencil** version, everyone answers the same questions and you can move around within a subtest. Typical question counts and time limits:¹

	Arithmetic Reasoning	Mathematics Knowledge
CAT-ASVAB (computer)	15 questions, 55 minutes	15 questions, 31 minutes
Paper and pencil	30 questions, 36 minutes	25 questions, 24 minutes

Every question is multiple choice with four answer choices. **Calculators are not allowed**; scratch paper is provided. Formulas are *not* given, so memorize the ones on the Formula Sheet at the back of this book.

Why math matters so much

Your Armed Forces Qualification Test (AFQT) score—the score that decides whether you can enlist at all—is computed from four subtests: Arithmetic Reasoning, Mathematics Knowledge, Word Knowledge, and Paragraph Comprehension. The two math subtests make up roughly **half** of the AFQT. The AFQT is reported as a percentile from 1 to 99, and each service sets its own minimum.

Arithmetic Reasoning and Mathematics Knowledge also feed many of the *line scores* (composite scores) that qualify you for specific jobs, such as technical, electronics, and administrative specialties. A few extra correct math answers can open up a wider choice of careers and enlistment bonuses.

¹These figures can change; confirm them with your recruiter or at officialasvab.com.

Test-Day Strategies

1. Read the last sentence first

In a word problem, the last sentence tells you what to find. Read it first, then read the whole problem looking for the numbers you need. Many wrong answer choices are the right answer to a *different* question (the discount instead of the sale price, the number who passed instead of the number who failed).

2. Estimate before you calculate

A rough answer tells you which choices are impossible. If a \$48 item is 25% off, the sale price must be a bit more than \$35—so \$12 and \$60 are out before you do any exact arithmetic.

3. Work backward from the choices

When a question asks “which number...” or gives an equation, you can test the answer choices. Start with a middle value; if it is too big, the answer is smaller, and vice versa. For “which value satisfies” questions this is often the fastest method.

4. Pick numbers

If the problem uses variables or percents of an unknown amount, choose an easy number (100 for percents, a common multiple for fractions) and see which answer choice gives the same result.

5. Watch the units

Minutes vs. hours, inches vs. feet, cents vs. dollars, radius vs. diameter. Convert everything to the units the question asks for *before* you calculate.

6. Manage your time

On the paper test you have a little over a minute per question. If a question takes more than two minutes, make your best guess, mark it, and move on. Never leave a question blank: a blank is always wrong, a guess might be right, and on the computer test unanswered questions at the end of a subtest lower your score.

7. Check that your answer makes sense

A sale price cannot be higher than the original price. A probability cannot be more than 1. An average must lie between the smallest and largest values. A quick reasonableness check catches many careless errors.

No-Calculator Tip Learn by heart: the squares from 1^2 to 15^2 , the fraction–decimal–percent equivalents ($\frac{1}{8} = 0.125 = 12.5\%$, $\frac{1}{3} \approx 33.3\%$, ...), and the formulas on the Formula Sheet. Instant recall of these saves minutes on test day.

Mental Math Toolkit

Because the ASVAB bans calculators, a handful of shortcuts pay off on almost every page of this book. Practice them until they are automatic.

Trick	Example
Multiply by 5	Multiply by 10, then halve: $48 \times 5 = 480 \div 2 = 240$.
Multiply by 25	Multiply by 100, then divide by 4: $36 \times 25 = 3,600 \div 4 = 900$.
Multiply by 9 or 11	$47 \times 9 = 470 - 47 = 423$; $34 \times 11 = 340 + 34 = 374$.
Break a number apart	$7 \times 98 = 7 \times 100 - 7 \times 2 = 700 - 14 = 686$.
Square a number ending in 5	35^2 : $3 \times 4 = 12$, then write 25: 1,225.
Divide by 5	Double, then divide by 10: $340 \div 5 = 680 \div 10 = 68$.
Divide by a decimal	Move both decimal points: $4.8 \div 0.06 = 480 \div 6 = 80$.
Build percents from 10%	15% of 60: 10% = 6, 5% = 3, total 9.
Swap the percent	8% of 50 = 50% of 8 = 4.
Compare fractions	Cross-multiply: $\frac{5}{8}$ vs. $\frac{3}{5}$: $5 \times 5 = 25 > 8 \times 3 = 24$, so $\frac{5}{8}$ is larger.
Estimate first	$398 \times 51 \approx 400 \times 50 = 20,000$ rules out answers near 2,000 or 200,000.

Diagnostic Test

This test samples every chapter of the book. Take it before you begin studying, without a calculator, and try to finish in about 33 minutes. Don't guess wildly—if you have no idea how to start a problem, leave it blank so that the result shows what you really need to study.

Score: ____ / 30

Target time: 33 minutes

- Multiply: 63×24
A 378 B 1,212 C 1,512 D 12,852
- Which list shows the numbers 8, -9, 12, -3, -7 in order from least to greatest?
A -9, -7, -3, 8, 12 B -3, -7, 8, -9, 12
C 12, 8, -3, -7, -9 D -3, -7, -9, 8, 12
- A training class has 42 soldiers from Alpha Company and 28 from Bravo Company. The instructor wants to form teams so that every team has the same number of Alpha soldiers and the same number of Bravo soldiers, with no one left out. What is the greatest number of teams that can be formed?
A 6 B 7 C 8 D 14
- Which of the fractions $\frac{5}{12}$, $\frac{7}{15}$, $\frac{11}{12}$, and $\frac{3}{4}$ is the largest?
A $\frac{11}{12}$ B $\frac{5}{12}$ C $\frac{7}{15}$ D $\frac{3}{4}$
- Nora buys 3 bottles of water at \$1.70 each and 2 rolls of tape at \$3.60 each. What is the total cost?
A \$5.30 B \$8.70 C \$12.20 D \$12.30
- Which of the following is equal to 0.78?
A 0.78% B 7.8% C 58% D 78%
- The label on a bag of lawn fertilizer says to use 5 pounds for every 1,000 square feet of lawn. How many pounds are needed for 2,500 square feet?
A 2 B 7.5 C 12.5 D 12,500
- Which of the following is equal to 6.18×10^5 ?
A 30.9 B 61,800
C 618,000 D 6,180,000
- What is $\sqrt{6} \cdot \sqrt{12}$ in simplest form?
A $5\sqrt{2}$ B $6\sqrt{2}$ C $36\sqrt{2}$ D 72
- At a movie theater, adult tickets cost \$12 and child tickets cost \$8. Which expression gives the total cost, in dollars, of x adult tickets and y child tickets?
A $8x + 12y$ B $20(x + y)$
C $20xy$ D $12x + 8y$
- Solve $-15x + 5y = 15$ for y .
A $y = 3x + 3$ B $y = 15x + 3$
C $y = -3x + 3$ D $y = 15x + 15$
- Lance Corporal Jensen needs a total score of at least 250 points on a fitness test. Jensen has already earned 180 points on the first events. Which inequality shows the points p that Jensen still needs from the last event?
A $180 + p > 250$ B $180p \geq 250$
C $180 + p \geq 250$ D $180 + p < 250$
- Solve the system for x :
$$y = -4x + 9$$
$$3x + y = 7$$

A -16 B -2 C 1 D 2
- What is $(-3x^4)^2$?
A $-9x^8$ B $9x^6$ C $9x^8$ D $-3x^8$

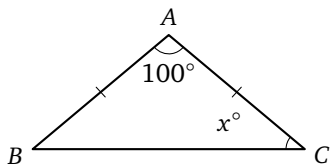
15. Which of the following is a solution of $x^2 - 4x - 32 = 0$?

A -32 B -8 C 4 D 8

16. Find the sum of the interior angles of a polygon with 8 sides.

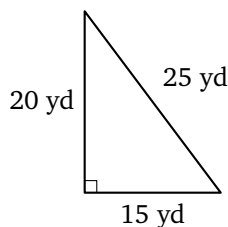
A 360° B $1,080^\circ$ C $1,260^\circ$ D $1,440^\circ$

17. In isosceles triangle ABC , the vertex angle $\angle A$ is 100° . What is the value of x , the measure of a base angle?



A 20 B 40 C 80 D 100

18. The right triangle shown has sides of 20 yards, 15 yards, and 25 yards. What is its area?



A 150 yd^2 B 250 yd^2
C 300 yd^2 D $3,750 \text{ yd}^2$

19. If the radius of a circle is cut in half, the area of the new circle is what fraction of the area of the original circle?

A $\frac{1}{8}$ B 1 C $\frac{1}{2}$ D $\frac{1}{4}$

20. The volume of a rectangular prism is 220 cubic feet. Its length is 11 feet and its width is 2 feet. What is its height?

A 8 ft B 10 ft C 20 ft D 22 ft

21. How far apart are the points $(3, 4)$ and $(-6, -8)$?

A 3 B 14 C 15 D 21

22. Omar has \$150 for a new fishing setup. He buys a rod and reel for \$70 and wants to spend the rest on lures that cost \$4.50 each. How many lures can he buy?

A 17 B 18 C 33 D 48

23. A military surplus store buys used rucksacks for \$39 each and marks up the cost by 60%. What is the selling price of each rucksack?

A \$23.40 B \$62.40 C \$82.40 D \$99.00

24. Private Walker puts a \$6,500 enlistment bonus into a savings account at the base credit union. The account pays 5% simple interest per year. How much interest will she earn in 3 months?

A \$51.25 B \$71.25 C \$81.25 D \$975.00

25. Two cars leave a town at the same time and travel in opposite directions. One goes 45 miles per hour and the other goes 55 miles per hour. How far apart are they after 2 hours?

A 20 miles B 200 miles
C 220 miles D 260 miles

26. A crew of 5 roofers can put a new roof on a warehouse in 7 days. How many *more* roofers are needed to finish the job in 5 days?

A 1 B 2 C 7 D 35

27. A bathtub faucet can fill the tub in 6 minutes. With the plug pulled, the drain empties a full tub in 18 minutes. If the faucet is turned on with the plug pulled, how long will it take to fill the tub?

A 6 minutes B 9 minutes
C 12 minutes D 24 minutes

28. A bag holds 9 green, 6 yellow, and 2 white marbles. One marble is picked at random, its color is noted, and it is put back. Then a second marble is picked. What is the probability that both are yellow?

A $\frac{36}{289}$ B $\frac{6}{17}$ C $\frac{12}{17}$ D $\frac{18}{17}$

29. Range training starts at 0645 and ends at 1410. How long does the training last?

A 7 hours 25 minutes B 8 hours 5 minutes
C 8 hours 25 minutes D 8 hours 35 minutes

30. Diego's pancake recipe uses 5 cups of milk per batch, and 7 batches are needed for a fundraiser breakfast. How many half-gallon cartons of milk must be bought? (1 half gallon = 8 cups)

A 3 B 4 C 5 D 8

Answers and Your Study Plan

Check your answers against the table. Each row shows the chapter the question comes from. **Circle the chapter number of every question you missed or left blank**—those chapters are your study plan. Work through them first, then the rest of the book in order.

#	Ans.	Ch.	Topic	#	Ans.	Ch.	Topic
				16	B	16	Lines & Angles
1	C	1	Whole Numbers & Order of Operations	17	B	17	Triangles & the Pythagorean Theorem
2	A	2	Negative Numbers & Absolute Value	18	A	18	Perimeter & Area
3	D	3	Factors, Multiples, Primes, GCF & LCM	19	D	19	Circles
4	A	4	Fractions & Mixed Numbers	20	B	20	Volume & Surface Area
5	D	5	Decimals	21	C	21	Coordinate Geometry
6	D	6	Percents	22	A	22	Money: Shopping, Discounts, Tax & Wages
7	C	7	Ratios & Proportions	23	B	23	Interest, Commission & Percent Change
8	C	8	Exponents & Scientific Notation	24	C	23	Interest, Commission & Percent Change
9	B	9	Square Roots & Radicals	25	B	24	Rate, Distance & Time
10	D	10	Algebraic Expressions	26	B	25	Work Rates & Mixtures
11	A	11	Linear Equations	27	B	25	Work Rates & Mixtures
12	C	12	Inequalities	28	A	26	Averages, Statistics & Probability
13	D	13	Systems of Equations	29	A	27	Measurement, Units & Multi-Step Problems
14	C	14	Polynomials & Exponent Rules	30	C	27	Measurement, Units & Multi-Step Problems
15	D	15	Factoring & Quadratic Equations				

How did you do?

26–30 correct	Strong. Skim the Key Concepts, do the challenge problems, then the practice tests.
18–25 correct	Solid base. Study your circled chapters fully, then the rest.
10–17 correct	Work through every chapter in order; take your time with the warm-ups.
0–9 correct	Start at Chapter 1. Every chapter builds on the ones before it.

Explanations

1. **C** 1,512

- Split 24 into $20 + 4$ and multiply 63 by each part.
- $63 \times 4 = 252$ and $63 \times 20 = 1,260$.
- Add the partial products: $252 + 1,260 = 1,512$.

WHY NOT (A) forgets the placeholder zero, multiplying by 2 instead of 20. (B) multiplies only tens by tens and ones by ones. (D) puts two placeholder zeros in the second row instead of one.

2. **A** $-9, -7, -3, 8, 12$

- Negative numbers come first. Among them, the one farthest from 0 is the least: $-9 < -7 < -3$.
- Then the positive numbers: $8 < 12$.
- So the order is $-9, -7, -3, 8, 12$.

WHY NOT (B) orders the numbers by their distance from 0, ignoring the signs. (C) lists the numbers from greatest to least. (D) puts the negative numbers in the wrong order; for negatives, the larger absolute value is the smaller number.

3. **D** 14

1 Each item count must split evenly into the groups, so the number of groups must divide both numbers. The greatest such number is the GCF.

2 Write each number as a product of primes: $42 = 2 \times 3 \times 7$, $28 = 2^2 \times 7$.

3 Take each prime they *all* share, with its lowest power: $2 \times 7 = 14$.

WHY NOT (B) is a common factor, but not the greatest one.

 4. **A** $\frac{11}{12}$

1 Rewrite each fraction with the common denominator 60: $\frac{5}{12} = \frac{25}{60}$, $\frac{7}{15} = \frac{28}{60}$, $\frac{11}{12} = \frac{55}{60}$, $\frac{3}{4} = \frac{45}{60}$.

2 The largest numerator is 55, so $\frac{11}{12}$ is the largest.

WHY NOT (B) equals $\frac{25}{60}$, which is less than $\frac{55}{60}$. (C) equals $\frac{28}{60}$, which is less than $\frac{55}{60}$. (D) equals $\frac{45}{60}$, which is less than $\frac{55}{60}$.

 5. **D** \$12.30

1 First item: $3 \times 1.70 = 5.10$.

2 Second item: $2 \times 3.60 = 7.20$.

3 Total: $5.10 + 7.20 = 12.30$, so \$12.30.

WHY NOT (A) adds one of each price and ignores the quantities. (B) forgets to multiply the second price by 2.

 6. **D** 78%

1 To change a decimal to a percent, multiply by 100 (move the decimal point two places to the right).

2 $0.78 \times 100 = 78$, so $0.78 = 78\%$.

WHY NOT (A) forgets to move the decimal point at all. (B) moves the decimal point only one place.

 7. **C** 12.5

1 Set up a proportion with the same order on both sides (pounds over square feet): $\frac{5}{1,000} = \frac{x}{2,500}$.

2 Cross-multiply: $1,000x = 5 \times 2,500 = 12,500$.

3 Divide by 1,000: $x = 12,500 \div 1,000 = 12.5$ pounds.

TIP Unit rate: 1 pound is enough for 200 square feet, so $2,500 \div 200 = 12.5$ pounds.

WHY NOT (A) sets up the proportion upside down. (B) finds only the extra amount needed and forgets the original 5 pounds. (D) multiplies by 2,500 but forgets to divide by 1,000.

 8. **C** 618,000

1 Multiplying by 10^5 moves the decimal point 5 places to the right.

2 6.18 becomes 618,000 (fill the empty places with zeros).

WHY NOT (A) multiplies by the exponent instead of by a power of 10. (B) moves the decimal point one place too few. (D) moves the decimal point one place too many.

 9. **B** $6\sqrt{2}$

1 Multiply the numbers under the radicals: $\sqrt{6} \cdot \sqrt{12} = \sqrt{6 \times 12} = \sqrt{72}$.

2 Simplify: $\sqrt{72} = \sqrt{36 \cdot 2} = 6\sqrt{2}$.

WHY NOT (C) pulls out 36 instead of its square root, 6. (D) drops the radical sign.

 10. **D** $12x + 8y$

1 Each adult ticket costs \$12, so x of them cost $12x$ dollars.

2 Each child ticket costs \$8, so y of them cost $8y$ dollars.

3 Add the two costs: $12x + 8y$.

WHY NOT (A) matches each price with the wrong item. (B) charges both prices for every item. (C) combines unlike terms.

 11. **A** $y = 3x + 3$

1 Move the x -term to the right side: add $15x$ to both sides. $5y = 15x + 15$.

2 Divide both sides by 5: $y = \frac{15x+15}{5}$.

3 Divide *each* term on top by 5: $y = 3x + 3$.

WHY NOT (B) divides only 15 by 5, not the x -term. (C) does not change the sign of $-15x$ when moving it to the other side. (D) forgets to divide by 5.

 12. **C** $180 + p \geq 250$

1 "At least 250" means the quantity is greater than or equal to 250: it includes 250 itself, so the symbol is $\geq$.

2 The quantity is $180 + p$, so the inequality is $180 + p \geq 250$.

WHY NOT (A) leaves out the boundary number, which is allowed. (B) multiplies instead of adding the points. (D) points the sign the wrong way and gets the boundary wrong.

13. **D** 2

1 The first equation tells you what y equals. Replace y in the second equation with $(-4x + 9)$: $3x + (-4x + 9) = 7$.

2 Distribute: $3x - 4x + 9 = 7$, so $-x + 9 = 7$.

3 Solve: $-x = -2$, so $x = 2$.

WHY NOT (A) makes a sign error with the 9. (B) makes a sign error. (C) gives the value of y , not x .

14. **C** $9x^8$

1 Raise *every* factor inside the parentheses to the power 2.

2 Coefficient: $(-3)^2 = 9$. Variables: multiply exponents, $(x^4)^2 = x^8$.

3 So the result is $9x^8$.

WHY NOT (A) keeps the negative sign, but a negative number squared is positive. (B) adds the exponents instead of multiplying. (D) forgets to raise the coefficient -3 to the power.

15. **D** 8

1 Factor: two numbers that multiply to -32 and add to -4 are -8 and 4 , so $(x - 8)(x + 4) = 0$.

2 The solutions are $x = 8$ and $x = -4$. Only 8 is among the choices.

3 Check: $(8)^2 - 4(8) - 32 = 64 - 32 - 32 = 0$. ✓

WHY NOT (A) is the constant term, not a solution. (B) has the wrong sign: it is the number in a factor, not a solution. (C) is the opposite of the other solution.

16. **B** $1,080^\circ$

1 The polygon has $n = 8$ sides. Use: sum of interior angles $= (n - 2) \times 180^\circ$.

2 $(8 - 2) \times 180^\circ = 6 \times 180^\circ = 1,080^\circ$.

TIP Why it works: diagonals from one corner cut the polygon into $8 - 2 = 6$ triangles, and each triangle holds 180° .

WHY NOT (A) gives the sum of the exterior angles, which is always 360° . (C) uses $n - 1$ instead of $n - 2$. (D) multiplies by the number of sides instead of $n - 2$.

17. **B** 40

1 In an isosceles triangle the two base angles (opposite the equal sides) are equal. Call each one x .

2 The angles add up to 180: $x + x + 100 = 180$, so $2x = 180 - 100 = 80$.

3 Divide by 2: $x = 80 \div 2 = 40$.

WHY NOT (C) forgets to split what is left between the two equal base angles. (D) assumes the base angle equals the vertex angle.

18. **A** 150 yd^2

1 In a right triangle the two legs (the sides that form the right angle) are the base and the height. The longest side, 25 yards, is the hypotenuse and is not used.

2 Area of a triangle $= \frac{1}{2} \times \text{base} \times \text{height}$.

3 $\frac{1}{2} \times 15 \times 20 = \frac{1}{2} \times 300 = 150$, so the area is 150 square yards.

WHY NOT (B) uses the hypotenuse as the height. (C) forgets to multiply by $\frac{1}{2}$. (D) multiplies all three sides.

19. **D** $\frac{1}{4}$

1 The new radius is $\frac{1}{2}r$.

2 New area: $\pi \left(\frac{1}{2}r\right)^2 = \frac{1}{4}\pi r^2$.

3 So the new area is $\frac{1}{4}$ of the original.

WHY NOT (A) cubes the factor; that is how volume changes. (C) assumes the area shrinks by the same factor as the radius.

20. **B** 10 ft

1 Use $V = l \times w \times h$: $220 = 11 \times 2 \times h$.

2 Area of the base: $11 \times 2 = 22$.

3 Divide: $h = 220 \div 22 = 10$ feet.

WHY NOT (C) divides by only one of the two given dimensions. (D) gives the area of the base.

21. **C** 15

1 Horizontal change: $|-6 - 3| = 9$. Vertical change: $|-8 - 4| = 12$.

2 These are the legs of a right triangle, and the distance is the hypotenuse: $d = \sqrt{9^2 + 12^2} = \sqrt{81 + 144} = \sqrt{225} = 15$.

TIP The legs 9 and 12 belong to the Pythagorean triple 9-12-15.

WHY NOT (A) subtracts the two distances. (D) adds the horizontal and vertical distances instead of using the Pythagorean theorem.

22. **A** 17

1 First take out the fixed cost: $\$150 - \$70 = \$80$ is left to spend.

2 Divide by the price of one: $\$80 \div \4.50 is 17 with some money left over, because $17 \times \$4.50 = \76.50 .

3 One more would cost \$81, which is more than \$80. Round *down*: 17 lures.

TIP Left over: $\$80 - \$76.50 = \$3.50$, not enough for one more.

WHY NOT (B) rounds up, but 18 lures would cost \$81. (C) forgets to subtract the \$70 first. (D) adds the \$70 instead of subtracting it.

23. **B** \$62.40

- 1 Markup: $0.6 \times \$39 = \23.40 .
- 2 Selling price: $\$39 + \$23.40 = \$62.40$.

TIP In one step: $1.6 \times \$39 = \62.40 .

WHY NOT (A) is the markup, not the selling price. (D) adds the percent as if it were dollars.

24. **C** \$81.25

- 1 Time must be in years: $3 \text{ months} = \frac{3}{12} = \frac{1}{4} = 0.25$ year.
- 2 One year's interest: $\$6,500 \times 0.05 = \325 .
- 3 For $\frac{1}{4}$ of a year: $\$325 \times \frac{1}{4} = \81.25 .

WHY NOT (D) uses 3 years instead of 3 months.

25. **B** 200 miles

- 1 Moving in opposite directions, the gap grows at the sum of the speeds: $45 + 55 = 100$ miles per hour.
- 2 After 2 hours: $100 \times 2 = 200$ miles.

WHY NOT (A) subtracts the speeds, but the two are moving apart in opposite directions.

26. **B** 2

- 1 Total work: $5 \times 7 = 35$ worker-days.
- 2 To finish in 5 days: $35 \div 5 = 7$ roofers are needed.
- 3 There are already 5 roofers, so $7 - 5 = 2$ more are needed.

WHY NOT (C) is the total number of workers needed, not how many more. (D) gives the total number of worker-days, not the number of workers.

27. **B** 9 minutes

- 1 Rates: the faucet fills $\frac{1}{6}$ of the tub per minute, and the drain empties $\frac{1}{18}$ of it per minute.

- 2 The drain works against the fill, so *subtract*: $\frac{1}{6} - \frac{1}{18} = \frac{3}{18} - \frac{1}{18} = \frac{2}{18} = \frac{1}{9}$ of the tub per minute.

- 3 Time = 1 tub $\div$ rate = $1 \div \frac{1}{9} = 9$ minutes.

TIP Shortcut for fill and drain: $\frac{6 \times 18}{18 - 6} = \frac{108}{12} = 9$.

WHY NOT (A) ignores the drain. (C) subtracts the two times. (D) adds the two times.

28. **A** $\frac{36}{289}$

- 1 There are $9 + 6 + 2 = 17$ marbles. Because the first one is put back, the total is 17 for both picks.

- 2 Multiply: $\frac{6}{17} \times \frac{6}{17} = \frac{36}{289}$.

WHY NOT (B) stops after the first pick. (C) adds the probabilities instead of multiplying them.

29. **A** 7 hours 25 minutes

- 1 10 minutes minus 45 minutes won't go, so borrow 1 hour: 1410 becomes 13 hours 70 minutes.

- 2 Subtract: $13 - 6 = 7$ hours and $70 - 45 = 25$ minutes.

- 3 The time is 7 hours 25 minutes.

WHY NOT (B) subtracts $1410 - 0645$ like ordinary numbers (borrows 100 minutes instead of 60). (C) borrows the hour but forgets to take it away from the hours. (D) subtracts the smaller number of minutes from the larger instead of borrowing.

30. **C** 5

- 1 Total milk: $7 \times 5 = 35$ cups.

- 2 Divide: $35 \div 8$ is 4 with a remainder of 3.

- 3 4 is not enough, so round up to 5.

WHY NOT (A) uses 16 cups per carton (that is a full gallon). (B) rounds down, which is not enough.

PART I

Number Skills

Whole numbers, signed numbers, factors, fractions, decimals, percents, ratios, exponents, and roots—the toolkit behind every other question.

- 1 Whole Numbers & Order of Operations
- 2 Negative Numbers & Absolute Value
- 3 Factors, Multiples, Primes, GCF & LCM
- 4 Fractions & Mixed Numbers
- 5 Decimals
- 6 Percents
- 7 Ratios & Proportions
- 8 Exponents & Scientific Notation
- 9 Square Roots & Radicals

Whole Numbers & Order of Operations

Tested on: Mathematics Knowledge • Arithmetic Reasoning

Key Concepts

Every other chapter in this book rests on whole-number arithmetic. With no calculator on the ASVAB, you need three things: a feel for place value, one firm rule for the order of operations, and a habit of reading word problems for what they *really* ask.

Place value and rounding

In 4,725,316 the digits name, from the left: millions (4), hundred-thousands (7), ten-thousands (2), thousands (5), hundreds (3), tens (1), ones (6). The *value* of the 7 is 700,000.

To round, look at the digit just to the *right* of the rounding place: 5 or more → round up; 4 or less → keep the digit. Then change every digit after the rounding place to 0. 38,462 → 38,000 (nearest thousand) and 38,500 (nearest hundred).

Order of operations (PEMDAS)

1. Parentheses (innermost first)
2. Exponents ($4^2 = 4 \times 4 = 16$)
3. Multiplication and Division, a tie: work *left to right*
4. Addition and Subtraction, a tie: work *left to right*

Example: $20 - 12 \div 4 \times 2 = 20 - 3 \times 2 = 20 - 6 = 14$.

Properties that make mental math easy

Commutative (order) $a + b = b + a, \quad a \times b = b \times a$

Associative (grouping) $(a + b) + c = a + (b + c)$

Distributive $a \times (b + c) = a \times b + a \times c$

Identity $a + 0 = a, \quad a \times 1 = a$

The distributive property turns hard products into easy ones: $7 \times 98 = 7 \times 100 - 7 \times 2 = 700 - 14 = 686$.

WORKED EXAMPLE

A convoy must move 250 soldiers. Each truck carries 24 soldiers. How many trucks are needed?

Solution. $250 \div 24 = 10$ remainder 10. Ten trucks carry 240 soldiers, and the other 10 still need a ride, so you need 11 trucks. When a question asks how many containers, vehicles, or trips are *needed*, round *up*.

No-Calculator Tip Estimate by rounding each number to its first digit, then count zeros: $398 \times 51 \approx 400 \times 50$; $4 \times 5 = 20$, and the three zeros give 20,000. Estimating first also catches answers that are off by a factor of 10.

Common Traps

- Adding before multiplying: $6 + 4 \times 3 = 18$, not 30.
- Multiplying before an earlier division: $24 \div 4 \times 2 = 12$, not 3.
- Rounding *down* the number of buses, boxes, or trips needed.
- Giving the remainder when the question asks for the quotient (or the reverse).

Practice Set 1

Score: ____ / 25

Target time: 27 minutes

Problems 1–8 are warm-ups, 9–18 are test level, and 19–25 are challenge problems.

- Simplify: $31 - 6 \times 2$
A 13 B 15 C 19 D 50
- Which digit is in the ten-thousands place in 54,315?
A 3 B 4 C 5 D 7
- Simplify: $8 \times 4 + 2 \times 6$
A 29 B 44 C 204 D 288
- Round 8,184 to the nearest hundred.
A 8,000 B 8,100 C 8,180 D 8,200
- The equation $45 \times 1 = 45$ is an example of which property?
A Associative property
B Identity property
C Commutative property
D Distributive property
- What is 728×8 ?
A 4,824 B 5,664 C 5,814 D 5,824
- Kevin buys 4 cases of soda at \$6 each and 6 bags of ice at \$3 each. What is the total cost?
A \$9 B \$27 C \$42 D \$90
- A group of 6 friends splits a restaurant bill of \$138 equally. How much does each friend pay?
A \$2 B \$21 C \$23 D \$132
- Simplify: $7 + 3^2 \times 3$
A 25 B 34 C 48 D 49
- Round 426,382 to the nearest thousand.
A 426,000 B 426,400 C 427,000 D 430,000
- What is the value of $(2 + 5) \times 3 - 3$?
A 0 B 14 C 18 D 24
- Find the product: 11×199
A 1,389 B 2,189 C 2,199 D 2,211
- Without a calculator, which value is closest to $1,781 \div 91$?
A 20 B 30 C 200 D 2,000
- What is $810 \div 3$?
A 27 B 270 C 280 D 2,700
- Multiply: 84×54
A 756 B 4,536 C 4,636 D 42,336
- A convoy must move 181 soldiers. Each truck can carry 18 soldiers. How many trucks are needed?
A 1 B 10 C 11 D 12
- A bakery packs 224 cookies into boxes of 24. How many full boxes can be filled?
A 5 B 6 C 8 D 9
- Tyrone buys 4 sleeping bags at \$47 each and 8 flashlights at \$19 each and pays with 4 \$100 bills. How much change should Tyrone get back?
A \$60 B \$193 C \$334 D \$340
- What is the value of $28 - 3 \times (9 + 3) \div 9 + 11$?
A 13 B 15 C 35 D 45

20. A library is packing 444 books. Each box holds 20 books. Each box costs \$4. What is the total cost of the boxes needed?
A \$88.00 B \$88.80 C \$92.00 D \$96.00
21. What is the value of $14 + 3 \times (9 - 6)^2 - 2$?
A 3 B 24 C 30 D 39
22. 74 troops must be flown to a landing zone, and each helicopter makes one trip. Each helicopter carries 12 troops. Each helicopter trip costs \$1,400 in fuel. What is the total cost of the helicopters needed?
A \$7,000 B \$8,400 C \$9,800 D \$11,200
23. Simplify: $53 - 6 \times (2 + 7) \div 2 + 6$
A 20 B 22 C 27 D 32
24. Priya buys a used motorcycle for \$2,560 and 2 helmets for \$235 each. She pays \$550 up front and splits the rest into 4 equal monthly payments. How much is each monthly payment?
A \$620.00 B \$757.50
C \$895.00 D \$2,480.00
25. 5 friends rent a cabin for 2 nights at \$220 per night. They split the total cost equally. How much does each friend pay?
A \$44 B \$68 C \$83 D \$88

Answers & Explanations 1

1 C	6 D	11 C	16 C	21 D
2 C	7 C	12 B	17 D	22 C
3 B	8 C	13 A	18 A	23 D
4 D	9 B	14 B	19 C	24 A
5 B	10 A	15 B	20 C	25 D

1. **C** 19

1 Multiply before you add or subtract: $6 \times 2 = 12$, which leaves $31 - 12$.

2 Subtract: $31 - 12 = 19$.

WHY NOT (D) adds or subtracts before multiplying or dividing.

2. **C** 5

1 Read the digits from the right, naming each place: 5 (ones), 1 (tens), 3 (hundreds), 4 (thousands), 5 (ten-thousands).

2 So the ten-thousands digit is 5.

WHY NOT (A) is the digit in the hundreds place. (B) is the digit in the thousands place.

3. **B** 44

1 Multiply before you add or subtract: $8 \times 4 = 32$, which leaves $32 + 2 \times 6$.

2 Next, multiply: $2 \times 6 = 12$, which leaves $32 + 12$.

3 Add: $32 + 12 = 44$.

WHY NOT (C) works straight from left to right instead of multiplying and dividing first. (D) adds or subtracts before multiplying or dividing.

4. **D** 8,200

1 The hundreds digit is 1. Look at the digit just to its right (the tens digit): 8.

2 Because 8 is 5 or more, round up: the hundreds digit 1 becomes 2.

3 Change every digit after the hundreds place to zero: 8,200.

WHY NOT (A) rounds to the nearest thousand instead. (B) rounds down even though the next digit, 8, is 5 or more. (C) rounds to the nearest ten instead.

5. **B** Identity property

1 Compare the two sides of the equation and ask what changed.

2 Adding 0 or multiplying by 1 leaves the number unchanged. That is the identity property.

WHY NOT (A) is about moving parentheses among numbers that are all added or all multiplied, which does not happen here. (C) is about changing the order of numbers, which does not happen here. (D) is about multiplying a number by each part of a sum or difference, which does not happen here.

6. **D** 5,824

1 Break 728 into place values: $700 + 20 + 8$.

2 Multiply each part by 8: $700 \times 8 = 5,600$, $20 \times 8 = 160$, $8 \times 8 = 64$.

3 Add the parts: $5,600 + 160 + 64 = 5,824$.

WHY NOT (B) forgets to add the carried digits.

7. **C** \$42

1 Find the cost of each kind of item: $4 \times 6 = 24$ and $6 \times 3 = 18$.

2 Add: $24 + 18 = 42$ dollars.

WHY NOT (A) adds one of each price and ignores the quantities. (B) forgets to multiply the second price by 6. (D) multiplies the total number of items by the sum of the prices.

8. **C** \$23

1 Sharing equally means dividing the total by the number of shares.

2 $138 \div 6 = 23$. Check: $6 \times 23 = 138$.

WHY NOT (A) stops dividing too soon and drops the last digit of the quotient. (D) subtracts instead of dividing.

9. **B** 34

1 Exponent next: $3^2 = 3 \times 3 = 9$, which leaves $7 + 9 \times 3$.

2 Multiply before you add or subtract: $9 \times 3 = 27$, which leaves $7 + 27$.

3 Add: $7 + 27 = 34$.

WHY NOT (A) doubles instead of squaring. (C) adds or subtracts before multiplying or dividing.

10. **A** 426,000

- 1 The thousands digit is 6. Look at the digit just to its right (the hundreds digit): 3.
- 2 Because 3 is less than 5, keep the thousands digit 6 as it is.
- 3 Change every digit after the thousands place to zero: 426,000.

WHY NOT (B) rounds to the nearest hundred instead. (C) rounds up even though the next digit, 3, is less than 5. (D) rounds to the nearest ten thousand instead.

11. **C** 18

- 1 Parentheses first: $2 + 5 = 7$, which leaves $7 \times 3 - 3$.
- 2 Multiply before you add or subtract: $7 \times 3 = 21$, which leaves $21 - 3$.
- 3 Subtract: $21 - 3 = 18$.

WHY NOT (A) adds or subtracts before multiplying or dividing. (B) ignores the parentheses.

12. **B** 2,189

- 1 Write 199 as $200 - 1$, an easy number minus a little.
- 2 Distribute the 11: $11 \times 200 - 11 \times 1 = 2,200 - 11$.
- 3 Subtract: $2,200 - 11 = 2,189$.

WHY NOT (C) subtracts 1 instead of 11×1 . (D) adds 11×1 instead of subtracting it.

13. **A** 20

- 1 Round to compatible numbers: 1,781 is about 1,800, and 91 is about 90.
- 2 Divide: $1,800 \div 90 = 20$ (think $90 \times 20 = 1,800$).

WHY NOT (C) adds an extra zero. (D) adds two extra zeros.

14. **B** 270

- 1 Start with 8: $8 \div 3 = 2$, remainder 2.
- 2 Bring down the 1 to make 21: $21 \div 3 = 7$.
- 3 Bring down the 0: 3 does not go into 0, so write 0 in the quotient.
- 4 The quotient is 270. Check: $3 \times 270 = 810$.

WHY NOT (A) skips the zero in the quotient. (D) adds an extra zero.

15. **B** 4,536

- 1 Split 54 into $50 + 4$ and multiply 84 by each part.
- 2 $84 \times 4 = 336$ and $84 \times 50 = 4,200$.
- 3 Add the partial products: $336 + 4,200 = 4,536$.

WHY NOT (A) forgets the placeholder zero, multiplying by 5 instead of 50. (D) puts two placeholder zeros in the second row instead of one.

16. **C** 11

- 1 Divide: $181 \div 18 = 10$ remainder 1.
- 2 10 full trucks hold $10 \times 18 = 180$ soldiers, so 1 soldier is still left.
- 3 They need one more truck: $10 + 1 = 11$ trucks.

WHY NOT (A) is the number of soldiers left over after filling 10 trucks. (B) rounds down, which leaves 1 soldier with no truck.

17. **D** 9

- 1 Divide: $224 \div 24 = 9$ remainder 8.
- 2 Only 9 boxes are complete; the 8 extra cookies are not enough for another one.

WHY NOT (C) is the number of cookies left over.

18. **A** \$60

- 1 Find the cost of each kind of item: $4 \times 47 = 188$ and $8 \times 19 = 152$.
- 2 Total cost: $188 + 152 = 340$ dollars.
- 3 Change: $400 - 340 = 60$ dollars.

WHY NOT (B) forgets to multiply the second price by 8. (C) subtracts one of each price and ignores the quantities. (D) is the total cost, not the change.

19. **C** 35

- 1 Parentheses first: $9 + 3 = 12$, which leaves $28 - 3 \times 12 \div 9 + 11$.
- 2 Multiply (multiplication and division go left to right): $3 \times 12 = 36$, which leaves $28 - 36 \div 9 + 11$.
- 3 Divide before you add or subtract: $36 \div 9 = 4$, which leaves $28 - 4 + 11$.
- 4 Subtract (addition and subtraction go left to right): $28 - 4 = 24$, which leaves $24 + 11$.
- 5 Add: $24 + 11 = 35$.

WHY NOT (A) adds before subtracting; addition and subtraction go left to right. (B) adds or subtracts before multiplying or dividing.

20. **C** \$92.00

- 1 Divide: $444 \div 20 = 22$ remainder 4.
- 2 The 4 leftover books need one more box, so 23 boxes are needed.
- 3 Cost: $23 \times 4 = 92$ dollars.

WHY NOT (A) rounds the number of boxes down. (B) pays for a fraction of a box.

21. **D** 39

- 1 Parentheses first: $9 - 6 = 3$, which leaves $14 + 3 \times (3)^2 - 2$.
- 2 Exponent next: $3^2 = 3 \times 3 = 9$, which leaves $14 + 3 \times 9 - 2$.
- 3 Multiply before you add or subtract: $3 \times 9 = 27$, which leaves $14 + 27 - 2$.
- 4 Add (addition and subtraction go left to right): $14 + 27 = 41$, which leaves $41 - 2$.
- 5 Subtract: $41 - 2 = 39$.

WHY NOT (A) ignores the parentheses. (C) doubles instead of squaring.

22. **C** \$9,800

- 1 Divide: $74 \div 12 = 6$ remainder 2.
- 2 The 2 leftover troops need one more helicopter, so 7 helicopters are needed.
- 3 Cost: $7 \times 1,400 = 9,800$ dollars.

WHY NOT (B) rounds the number of helicopters down.

23. **D** 32

- 1 Parentheses first: $2 + 7 = 9$, which leaves $53 - 6 \times 9 \div 2 + 6$.
- 2 Multiply (multiplication and division go left to right): $6 \times 9 = 54$, which leaves $53 - 54 \div 2 + 6$.
- 3 Divide before you add or subtract: $54 \div 2 = 27$, which leaves $53 - 27 + 6$.
- 4 Subtract (addition and subtraction go left to right): $53 - 27 = 26$, which leaves $26 + 6$.
- 5 Add: $26 + 6 = 32$.

WHY NOT (A) adds before subtracting; addition and subtraction go left to right.

24. **A** \$620.00

- 1 Total cost: $2,560 + 2 \times 235 = 2,560 + 470 = 3,030$ dollars.
- 2 Subtract the up-front payment: $3,030 - 550 = 2,480$ dollars.
- 3 Split into 4 equal payments: $2,480 \div 4 = 620$ dollars.

WHY NOT (B) forgets to subtract the up-front payment. (C) adds the up-front payment instead of subtracting it. (D) is the amount still owed, not each monthly payment.

25. **D** \$88

- 1 Total cost: $2 \times 220 = 440$ dollars.
- 2 Split it 5 ways: $440 \div 5 = 88$ dollars each.

WHY NOT (A) forgets to multiply by the number of nights.

Negative Numbers & Absolute Value

Tested on: Mathematics Knowledge • Arithmetic Reasoning

Key Concepts

Negative numbers show up whenever something goes below a starting point: temperatures below zero, depths below sea level, an overdrawn bank account. A handful of sign rules covers every problem in this chapter.

The number line and absolute value

Numbers increase to the right. Every negative number is less than 0, and the farther left, the smaller: $-9 < -4 < 0 < 3$.

The *absolute value* $|x|$ is the distance from 0, so it is never negative: $|-7| = 7$ and $|7| = 7$. The distance between two numbers is the absolute value of their difference: from -3 to 5 is $|5 - (-3)| = 8$.

Adding and subtracting

- **Same signs:** add the absolute values, keep the sign: $-6 + (-9) = -15$.
- **Different signs:** subtract the absolute values, keep the sign of the number with the larger absolute value: $-11 + 4 = -7$.
- **Subtracting** means adding the opposite: $5 - 8 = 5 + (-8) = -3$ and $-7 - (-12) = -7 + 12 = 5$.

Multiplying and dividing

Same signs give a positive answer; different signs give a negative answer: $(-6)(-7) = 42$, $-48 \div 6 = -8$. With several factors, count the negatives: an even number of them gives a positive product, an odd number a negative one. Watch exponents: $(-3)^2 = (-3)(-3) = 9$, but $-3^2 = -(3 \times 3) = -9$.

WORKED EXAMPLE

At 5 a.m. the temperature was -8°F . By noon it was 13°F . How many degrees did the temperature rise?
Solution. Rise = new – old = $13 - (-8) = 13 + 8 = 21$ degrees. On a number line: 8 degrees up to 0, then 13 more.

No-Calculator Tip For a change or a distance, think “how far apart on the number line.” If the two numbers are on opposite sides of 0, *add* their absolute values (-8 to 13 : $8 + 13 = 21$); if they are on the same side, *subtract* them.

Common Traps

- Reading $-7 - (-12)$ as $-7 - 12$. Two negatives in a row make a plus.
- Thinking -12 is greater than -5 because $12 > 5$.

- Squaring before the sign: $-4^2 = -16$, while $(-4)^2 = 16$.
- Keeping a negative sign inside absolute value bars: $|3 - 9| = 6$, not -6 .

Practice Set 2

Score: ____ / 25

Target time: 26 minutes

Problems 1–8 are warm-ups, 9–18 are test level, and 19–25 are challenge problems.

- Evaluate: $-14 - 5$
A -19 B -14 C -9 D 19
- What is $(-2) \times (-7)$?
A -14 B 14 C 15 D 16
- What is $-9 + 8$?
A -17 B -3 C -2 D -1
- What is $-88 \div (-11)$?
A -99 B -8 C 8 D 12
- Evaluate: $|14| - |-13|$
A -2 B -1 C 0 D 1
- What is the distance between -3 and 1 on a number line?
A -4 B 4 C 5 D 7
- Which of the numbers -14 , -17 , -19 , -15 is the greatest?
A -19 B -15 C -14 D -17
- At sunset the temperature at a cold-weather training site in Alaska was 12°F . By midnight it had dropped 23 degrees. What was the temperature at midnight?
A -35°F B -14°F C -11°F D 35°F
- What is $-16 + 12 - (-10)$?
A 6 B 8 C 9 D 38
- Rosa is scuba diving 10 feet below the surface, a position of -10 feet. She swims down 20 more feet and then rises 15 feet. What is her new position?
A -45 ft B -30 ft C -25 ft D -15 ft
- Evaluate: $-16 - (-11) + 9$
A -4 B 4 C 5 D 36
- A Navy helicopter hovers at an altitude of 300 feet. Directly below it, a submarine is at -100 feet (below sea level). What is the vertical distance between the helicopter and the submarine?
A 200 ft B 400 ft C 450 ft D 500 ft
- Evaluate: $(6 \div (-2)) \times (-4)$
A -12 B 6 C 10 D 12
- Evaluate: $|4 - 12| - |6|$
A -14 B -2 C 2 D 14
- Which list shows the numbers 7, 2, -15 , -5 , -13 in order from least to greatest?
A 7, 2, -5 , -13 , -15 B -15 , -13 , -5 , 2, 7
C 2, -5 , 7, -13 , -15 D -5 , -13 , -15 , 2, 7
- What number is halfway between -14 and 12 on a number line?
A -6 B -5 C -1 D 13
- At 6 a.m. the temperature at a ski resort was -13°F . At noon it was 7°F . By how many degrees did the temperature rise?
A -20 degrees B -6 degrees
C 7 degrees D 20 degrees
- Sergeant Jensen has \$110 in a checking account. The sergeant pays \$25 for a gym membership and \$128 for a utility bill. Because the balance went below \$0, the bank also charges a \$25 overdraft fee. What is the account balance now?
A $-\$288$ B $-\$78$ C $-\$68$ D $-\$43$
- If $x = -5$ and $y = -13$, what is $-x^2 - y$?
A -12 B 3 C 12 D 38

20. What is the value of $-30 \div 6 - (-5) \times 6$?
A -35 B -25 C 0 D 25
21. Simplify: $-3^2 + (-6)^2$
A 27 B 29 C 31 D 45
22. What is the value of $-2 \times (3 - 10) + (-20) \div 4$?
A -19 B -6 C 7 D 9
23. If $x = -5$ and $y = 7$, what is the value of $|x + y| - |x| - |y|$?
A -13 B -10 C 0 D 10
24. A submarine is at $-1,050$ feet (below sea level). It rises 40 feet per minute for 3 minutes and then dives 450 feet. What is its new position?
A -1,620 ft B -1,460 ft
C -1,430 ft D -1,380 ft
25. Victor has \$70 in a checking account. He pays \$66 for a utility bill and \$57 for a concert ticket. Because the balance went below \$0, the bank also charges a \$34 overdraft fee. How much must Victor deposit to bring the balance back up to \$25?
A \$112 B \$117 C \$122 D \$146

Answers & Explanations 2

1 A	6 B	11 B	16 C	21 A
2 B	7 C	12 B	17 D	22 D
3 D	8 C	13 D	18 C	23 B
4 C	9 A	14 C	19 A	24 D
5 D	10 D	15 B	20 D	25 A

1. **A** -19

1 Rewrite each subtraction as adding the opposite:
 $-14 - 5 = -14 + (-5)$.

2 Same signs: add the absolute values ($14 + 5 = 19$) and keep the sign: $-14 + (-5) = -19$.

WHY NOT (C) treats $-14 - 5$ as $-(14 - 5)$. (D) has the right size but the wrong sign.

2. **B** 14

1 Multiply the absolute values: $2 \times 7 = 14$.

2 The signs are the same, so the answer is positive:
 $(-2) \times (-7) = 14$.

WHY NOT (A) uses the wrong sign: same signs give a positive answer.

3. **D** -1

1 Different signs: subtract the absolute values ($9 - 8 = 1$) and keep the sign of -9, which has the larger absolute value: $-9 + 8 = -1$.

WHY NOT (A) adds 9 and 8 instead of finding their difference.

4. **C** 8

1 Divide the absolute values: $88 \div 11 = 8$.

2 The signs are the same, so the answer is positive:
 $-88 \div (-11) = 8$.

WHY NOT (B) uses the wrong sign: same signs give a positive answer.

5. **D** 1

1 Find each absolute value: $|14| = 14$ and $|-13| = 13$.

2 Subtract: $14 - 13 = 1$.

WHY NOT (B) has the right size but the wrong sign.

6. **B** 4

1 Distance is the larger number minus the smaller number (or the absolute value of the difference).

2 $1 - (-3) = 1 + 3 = 4$, or count 3 units to 0 and 1 more.

WHY NOT (A) subtracts in the wrong order; a distance is never negative.

7. **C** -14

1 All four numbers are negative. On a number line, the greatest one is the one farthest to the right, which is the one closest to 0.

2 -14 is closest to 0, so it is the greatest: $-19 < -17 < -15 < -14$.

WHY NOT (A) has the largest absolute value, but it is the farthest left of 0. (B) is farther left of 0 than -14. (D) is farther left of 0 than -14.

8. **C** -11°F

1 A drop means subtract: $12 - 23$.

2 $12 - 23 = 12 + (-23)$. Different signs: subtract the absolute values ($23 - 12 = 11$) and keep the sign of -23, which has the larger absolute value: $12 + (-23) = -11$. The temperature was -11°F .

WHY NOT (D) adds the drop instead of subtracting it.

9. **A** 6

1 Rewrite each subtraction as adding the opposite:
 $-16 + 12 - (-10) = -16 + 12 + 10$.

2 Different signs: subtract the absolute values ($16 - 12 = 4$) and keep the sign of -16, which has the larger absolute value: $-16 + 12 = -4$.

3 Different signs: subtract the absolute values ($10 - 4 = 6$) and keep the sign of 10, which has the larger absolute value: $-4 + 10 = 6$.

WHY NOT (D) ignores the signs and adds all the numbers.

10. **D** -15 ft

1 Going down is negative and going up is positive:
 $-10 - 20 + 15$.

2 $-10 - 20 = -30$, then $-30 + 15 = -15$.

3 The new position is -15 feet, that is, 15 feet below the surface.

WHY NOT (A) treats the rise as going farther down.

11. **B** 4

- 1 Rewrite each subtraction as adding the opposite:
 $-16 - (-11) + 9 = -16 + 11 + 9$.
- 2 Different signs: subtract the absolute values ($16 - 11 = 5$) and keep the sign of -16 , which has the larger absolute value: $-16 + 11 = -5$.
- 3 Different signs: subtract the absolute values ($9 - 5 = 4$) and keep the sign of 9 , which has the larger absolute value: $-5 + 9 = 4$.

WHY NOT (A) has the right size but the wrong sign. (D) ignores the signs and adds all the numbers.

12. **B** 400 ft

- 1 Distance = higher position – lower position: $300 - (-100)$.
- 2 Subtracting a negative means adding: $300 + 100 = 400$ feet.

TIP Sea level is 0: 300 feet above it plus 100 feet below it.

WHY NOT (A) subtracts the depth instead of adding it.

13. **D** 12

- 1 Parentheses first: $6 \div (-2) = -3$ (different signs, negative).
- 2 Then multiply: $(-3) \times (-4) = 12$ (same signs, positive).

WHY NOT (A) gets the sign wrong in one of the steps.

14. **C** 2

- 1 Work inside the bars first: $4 - 12 = -8$, so $|4 - 12| = |-8| = 8$.
- 2 Also $|6| = 6$.
- 3 Subtract: $8 - 6 = 2$.

WHY NOT (A) treats $|4 - 12|$ as -8 instead of 8. (B) has the right size but the wrong sign. (D) treats $-|6|$ as $+6$.

15. **B** $-15, -13, -5, 2, 7$

- 1 Negative numbers come first. Among them, the one farthest from 0 is the least: $-15 < -13 < -5$.
- 2 Then the positive numbers: $2 < 7$.
- 3 So the order is $-15, -13, -5, 2, 7$.

WHY NOT (A) lists the numbers from greatest to least. (C) orders the numbers by their distance from 0, ignoring the signs. (D) puts the negative numbers in the wrong order; for negatives, the larger absolute value is the smaller number.

16. **C** -1

- 1 The halfway point is the average of the two numbers: $(-14 + 12) \div 2$.
- 2 $-14 + 12 = -2$, and $-2 \div 2 = -1$.
- 3 Check: -1 is 13 units from each number.

WHY NOT (D) is half the distance between the numbers, not the point halfway between them.

17. **D** 20 degrees

- 1 Change = final – starting: $7 - (-13)$.
- 2 Subtracting a negative means adding: $7 + 13 = 20$ degrees.

TIP On a number line: 13 degrees up to 0, then 7 more.

WHY NOT (A) subtracts in the wrong order. (B) subtracts the numbers as if both were positive. (C) measures only from 0 up to the final temperature.

18. **C** $-\$68$

- 1 Start with \$110 and subtract both payments: $110 - 25 - 128 = -43$.
- 2 Subtract the fee: $-43 - 25 = -68$.
- 3 The balance is $-\$68$: the account is overdrawn by \$68.

WHY NOT (A) treats the starting balance as money owed instead of money in the account. (D) forgets the overdraft fee.

19. **A** -12

- 1 Substitute, keeping negatives in parentheses: $-(-5)^2 - (-13)$.
- 2 Square first: $(-5)^2 = 25$, so $-(-5)^2 = -25$.
- 3 Subtracting -13 means adding 13: $-25 + 13 = -12$.

WHY NOT (B) multiplies by 2 instead of squaring. (C) makes both signs wrong. (D) treats $-x^2$ as $(-x)^2$, which is positive.

20. **D** 25

- 1 Divide first: $-30 \div 6 = -5$ (different signs, negative).
- 2 Multiply: $(-5) \times 6 = -30$ (different signs, negative).
- 3 Subtract: $-5 - (-30) = 25$ (subtracting a negative means adding 30).

WHY NOT (A) gets the sign of $(-5) \times 6$ wrong. (B) has the right size but the wrong sign. (C) works from left to right instead of multiplying before subtracting.

21. **A** 27

1 In -3^2 the exponent applies only to 3: $-3^2 = -(3 \times 3) = -9$.

2 In $(-6)^2$ the parentheses include the sign: $(-6)(-6) = 36$.

3 Add: $-9 + 36 = 27$.

WHY NOT (D) treats -3^2 as $(-3)^2$.

22. **D** 9

1 Parentheses first: $3 - 10 = -7$.

2 Multiply: $-2 \times (-7) = 14$ (same signs, positive).

3 Divide: $(-20) \div 4 = -5$ (different signs, negative).

4 Add: $14 + (-5) = 9$.

WHY NOT (A) gets the sign of $-2 \times (-7)$ wrong. (B) adds before dividing.

23. **B** -10

1 Substitute: $|-5 + 7| - |-5| - |7|$.

2 Inside the first bars: $-5 + 7 = 2$, and $|2| = 2$. Also $|-5| = 5$ and $|7| = 7$.

3 Subtract from left to right: $2 - 5 - 7 = -10$.

WHY NOT (C) keeps the signs instead of taking absolute values. (D) has the right size but the wrong sign.

24. **D** -1,380 ft

1 The rise: $40 \times 3 = 120$ feet up.

2 After rising: $-1,050 + 120 = -930$ feet.

3 After diving: $-930 - 450 = -1,380$ feet.

WHY NOT (A) treats the rise as going down. (B) forgets to multiply the rate by the number of minutes.

25. **A** \$112

1 Balance after the payments and the fee: $70 - 66 - 57 - 34 = -87$.

2 Deposit needed = target - balance: $25 - (-87) = 25 + 87 = 112$.

3 Victor must deposit \$112.

WHY NOT (D) subtracts the fee twice.

Factors, Multiples, Primes, GCF & LCM

Tested on: Mathematics Knowledge • Arithmetic Reasoning

Key Concepts

Factors and multiples are the building blocks of fractions (Chapter 4) and of many “when will it happen again?” word problems. Keep the vocabulary straight and the rest is careful listing.

Vocabulary

- A **factor** of n divides n evenly: the factors of 12 are 1, 2, 3, 4, 6, 12.
- A **multiple** of n is n times a whole number: 12, 24, 36, ...
- A **prime** number has exactly two factors, 1 and itself: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, ... The number 1 is *not* prime, and 2 is the only even prime.
- The **reciprocal** of $\frac{a}{b}$ is $\frac{b}{a}$; a number times its reciprocal is 1.

Prime factorization, GCF and LCM

Split a number into primes with a factor tree: $360 = 2^3 \times 3^2 \times 5$. For $36 = 2^2 \times 3^2$ and $48 = 2^4 \times 3$:

- **GCF** (greatest common factor): shared primes, *lowest* powers: $2^2 \times 3 = 12$.
- **LCM** (least common multiple): every prime, *highest* powers: $2^4 \times 3^2 = 144$.

Divisibility rules

by 2: last digit even	by 5: last digit 0 or 5
by 3: digit sum divisible by 3	by 6: divisible by 2 <i>and</i> 3
by 4: last two digits divisible by 4	by 9: digit sum divisible by 9

WORKED EXAMPLE

One bus leaves the depot every 12 minutes and another every 18 minutes. Both leave at 8:00 a.m. When do they next leave together?

Solution. You need the LCM of 12 and 18. Multiples of 18: 18, 36, ...; 36 is also a multiple of 12. They leave together again after 36 minutes, at 8:36 a.m.

No-Calculator Tip Word-problem signal: “greatest number of identical kits,” “largest equal groups,” or “longest equal pieces” means **GCF**. “Next time together,” “least number of packs so the counts match” means **LCM**.

Common Traps

- Mixing up GCF and LCM; the GCF is never larger than the smaller number, and the LCM is never smaller than the larger number.

- Multiplying two numbers and calling it the LCM ($12 \times 18 = 216$, but the LCM is 36).
- Calling 1 prime, or calling numbers like $51 = 3 \times 17$ and $91 = 7 \times 13$ prime.

Practice Set 3

Score: ____/25

Target time: 27 minutes

Problems 1–8 are warm-ups, 9–18 are test level, and 19–25 are challenge problems.

- Which of the numbers 1, 33, 49, and 59 is prime?
A 1 B 33 C 49 D 59
- What is the reciprocal of 16?
A 16 B $\frac{1}{16}$ C $-\frac{1}{16}$ D -16
- What is the GCF of 35 and 15?
A 3 B 4 C 5 D 105
- What is the LCM of 4 and 16?
A 16 B 18 C 32 D 64
- What is the greatest common factor of 28 and 35?
A 7 B 8 C 28 D 140
- What is the LCM of 28 and 6?
A 84 B 114 C 124 D 168
- Which of the following is a factor of 27?
A 9 B 11 C 26 D 54
- Which one of 9, 21, 49, or 53 is a prime number?
A 9 B 21 C 49 D 53
- Which shows 400 written as a product of prime numbers?
A 2×5 B $2^4 \times 5^2$
C $2^3 \times 5 \times 10$ D $2^3 \times 5^2$
- Which one of 144, 310, 609, or 840 is divisible by 3 and by 5?
A 144 B 310 C 609 D 840
- Two buses leave a station at the same time. One bus leaves every 6 minutes and the other every 9 minutes. After how many minutes will they next leave at the same time?
A 3 minutes B 18 minutes
C 36 minutes D 54 minutes
- Petty Officer Murphy has 72 protein bars and 27 energy drinks. Every snack pack must be identical, with nothing left over. What is the greatest number of snack packs that can be made?
A 3 B 5 C 9 D 216
- Two warning lights blink together. One blinks every 12 seconds and the other every 18 seconds. After how many seconds will they next blink at the same time?
A 36 seconds B 41 seconds
C 72 seconds D 216 seconds
- A volunteer group has 27 pencils and 45 erasers. Every school-supply bag must be identical, with nothing left over. What is the greatest number of school-supply bags that can be made?
A 3 B 6 C 8 D 9
- Which of the numbers 112, 405, 542, and 630 is divisible by both 2 and 9?
A 112 B 405 C 542 D 630
- What is the prime factorization of 420?
A $2^3 \times 3 \times 5 \times 7$ B $2^2 \times 3 \times 5 \times 7$
C $2 \times 3 \times 7 \times 10$ D $2 \times 3 \times 5 \times 7$
- What is the greatest common factor of 12 and 8?
A 2 B 4 C 8 D 24

18. What is the least common multiple of 33 and 15?
A 165 B 205 C 330 D 495
19. How many different whole numbers divide 153 evenly?
A 3 B 4 C 6 D 8
20. Derek is buying pens and notepads. Pens come in packs of 12, and notepads come in packs of 9. He wants exactly the same number of pens and notepads. What is the least number of packs of pens he can buy?
A 3 B 4 C 9 D 36
21. A church group has 60 pairs of socks and 75 bars of soap. Every comfort kit must be identical, with nothing left over. If the greatest possible number of comfort kits is made, how many pairs of socks go in each one?
A 2 B 3 C 4 D 15
22. Ethan is buying hot dogs and buns. Hot dogs come in packs of 12, and buns come in packs of 20. He wants exactly the same number of hot dogs and buns. What is the least number of packs of hot dogs he can buy?
A 3 B 5 C 20 D 60
23. A training class has 45 soldiers from Alpha Company and 36 from Bravo Company. The instructor wants to form teams so that every team has the same number of Alpha soldiers and the same number of Bravo soldiers, with no one left out. If the greatest possible number of teams is formed, how many Alpha soldiers are on each team?
A 4 B 5 C 9 D 15
24. A gardener wants to plant 153 tomato plants in equal rows with none left over. How many different row lengths are possible, counting 1 per row and all 153 in one row?
A 3 B 4 C 5 D 6
25. What is the least common multiple of 7, 10, 4, and 8?
A 40 B 56 C 140 D 280

Answers & Explanations 3

1 D	6 A	11 B	16 B	21 C
2 B	7 A	12 C	17 B	22 B
3 C	8 D	13 A	18 A	23 B
4 A	9 B	14 D	19 C	24 D
5 A	10 D	15 D	20 A	25 D

1. **D** 59

- 1 A prime number has exactly two factors: 1 and itself.
- 2 Each of the other choices has another factor: $33 = 3 \times 11$; $49 = 7 \times 7$. And 1 is not prime.
- 3 59 is not divisible by 2, 3, 5, 7, and $11 \times 11 = 121$ is already more than 59, so 59 is prime.

WHY NOT (A) is not prime; a prime has exactly two factors, and 1 has only one. (B) is not prime: $33 = 3 \times 11$. (C) is not prime: $49 = 7 \times 7$.

2. **B** $\frac{1}{16}$

- 1 The reciprocal flips the fraction: the numerator and denominator trade places.
- 2 Write 16 as $\frac{16}{1}$; flipped, it becomes $\frac{1}{16}$. Check: $16 \times \frac{1}{16} = 1$.

WHY NOT (A) is the number itself, not its reciprocal. (C) is the opposite of the reciprocal; a reciprocal keeps the sign. (D) is the opposite, not the reciprocal.

3. **C** 5

- 1 List the factors of 15: 1, 3, 5, 15.
- 2 List the factors of 35: 1, 5, 7, 35.
- 3 The greatest number in both lists is 5.

WHY NOT (D) is the least common multiple, not the greatest common factor.

4. **A** 16

- 1 16 is itself a multiple of 4: $4 \times 4 = 16$.
- 2 So the least common multiple is the larger number, 16.

WHY NOT (C) is a common multiple, but not the least one. (D) multiplies the numbers; that is a common multiple, but not the least one.

5. **A** 7

- 1 List the factors of 28: 1, 2, 4, 7, 14, 28.
- 2 List the factors of 35: 1, 5, 7, 35.
- 3 The greatest number in both lists is 7.

WHY NOT (C) is the smallest number, but it is not a factor of 35. (D) is the least common multiple, not the greatest common factor.

6. **A** 84

- 1 List multiples of the larger number, 28: 28, 56, 84.
- 2 The first one that is also a multiple of 6 is 84 ($6 \times 14 = 84$).

WHY NOT (D) multiplies the numbers; that is a common multiple, but not the least one.

7. **A** 9

- 1 A factor of 27 divides it with no remainder.
- 2 $27 \div 9 = 3$, so 9 is a factor. 54 is bigger than 27, so it can only be a multiple.

WHY NOT (B) does not divide 27 evenly. (C) does not divide 27 evenly. (D) is a multiple of 27, not a factor.

8. **D** 53

- 1 A prime number has exactly two factors: 1 and itself.
- 2 Each of the other choices has another factor: $9 = 3 \times 3$; $21 = 3 \times 7$; $49 = 7 \times 7$.
- 3 53 is not divisible by 2, 3, 5, 7, and $11 \times 11 = 121$ is already more than 53, so 53 is prime.

WHY NOT (A) is not prime: $9 = 3 \times 3$. (B) is not prime: $21 = 3 \times 7$. (C) is not prime: $49 = 7 \times 7$.

9. **B** $2^4 \times 5^2$

- 1 Divide by 2 as many times as you can: $400 \rightarrow 200 \rightarrow 100 \rightarrow 50 \rightarrow 25$ (4 factors of 2).
- 2 Divide by 5 as many times as you can: $25 \rightarrow 5 \rightarrow 1$ (2 factors of 5).
- 3 So $400 = 2^4 \times 5^2$.

WHY NOT (A) lists each prime only once; that product is only 10. (C) uses 10, which is not prime. (D) multiplies out to 200, not 400.

10. **D** 840

- 1 Divisible by 3: its digit sum is divisible by 3. Divisible by 5: its last digit is 0 or 5.
- 2 840 passes both tests (digit sum 12; ends in 0): $840 \div 15 = 56$.
- 3 Each of the other choices fails one of the tests.

WHY NOT (A) is divisible by 3, but not by 5: it does not end in 0 or 5. (B) is divisible by 5, but not by 3: its digit sum, 4, is not divisible by 3. (C) is divisible by 3, but not by 5: it does not end in 0 or 5.

11. **B** 18 minutes

- 1 The events line up again after a common multiple of 6 and 9; the first time is the least common multiple.
- 2 Multiples of 9: 9, 18. The first one that is also a multiple of 6 is 18.

WHY NOT (A) is the greatest common factor, not the least common multiple. (C) is the second time they line up, not the first. (D) multiplies the two numbers; that is a common multiple, but not the first one.

12. **C** 9

- 1 Each item count must split evenly into the groups, so the number of groups must divide both numbers. The greatest such number is the GCF.
- 2 Write each number as a product of primes: $72 = 2^3 \times 3^2$, $27 = 3^3$.
- 3 Take each prime they *all* share, with its lowest power: $3^2 = 9$.

WHY NOT (A) is a common factor, but not the greatest one. (D) is the least common multiple, not the greatest common factor.

13. **A** 36 seconds

- 1 The events line up again after a common multiple of 12 and 18; the first time is the least common multiple.
- 2 Multiples of 18: 18, 36. The first one that is also a multiple of 12 is 36.

WHY NOT (C) is the second time they line up, not the first. (D) multiplies the two numbers; that is a common multiple, but not the first one.

14. **D** 9

- 1 Each item count must split evenly into the groups, so the number of groups must divide both numbers. The greatest such number is the GCF.
- 2 Write each number as a product of primes: $27 = 3^3$, $45 = 3^2 \times 5$.
- 3 Take each prime they *all* share, with its lowest power: $3^2 = 9$.

WHY NOT (A) is a common factor, but not the greatest one.

15. **D** 630

- 1 Divisible by 2: its last digit is even. Divisible by 9: its digit sum is divisible by 9.
- 2 630 passes both tests (even; digit sum 9): $630 \div 18 = 35$.
- 3 Each of the other choices fails one of the tests.

WHY NOT (A) is divisible by 2, but not by 9: its digit sum, 4, is not divisible by 9. (B) is divisible by 9, but not by 2: it is odd. (C) is divisible by 2, but not by 9: its digit sum, 11, is not divisible by 9.

16. **B** $2^2 \times 3 \times 5 \times 7$

- 1 Divide by 2 as many times as you can: $420 \rightarrow 210 \rightarrow 105$ (2 factors of 2).
- 2 Divide by 3 once: $105 \rightarrow 35$ (1 factor of 3).
- 3 Divide by 5 once: $35 \rightarrow 7$ (1 factor of 5), and the last factor is 7.
- 4 So $420 = 2^2 \times 3 \times 5 \times 7$.

WHY NOT (A) multiplies out to 840, not 420. (C) uses 10, which is not prime. (D) multiplies out to 210, not 420.

17. **B** 4

- 1 Write each number as a product of primes: $8 = 2^3$, $12 = 2^2 \times 3$.
- 2 Take each prime they *all* share, with its lowest power: $2^2 = 4$.

WHY NOT (A) is a common factor, but not the greatest one. (C) is the smallest number, but it is not a factor of 12. (D) is the least common multiple, not the greatest common factor.

18. **A** 165

- 1 Write each number as a product of primes: $15 = 3 \times 5$, $33 = 3 \times 11$.
- 2 Take every prime that appears, with its highest power: $3 \times 5 \times 11 = 165$.

WHY NOT (C) is a common multiple, but not the least one. (D) multiplies the numbers; that is a common multiple, but not the least one.

19. **C** 6

- 1 List the factor pairs, starting from 1: 1×153 , 3×51 , 9×17 .
- 2 The factors are 1, 3, 9, 17, 51, 153.
- 3 Count them: 6 factors.

TIP Shortcut: $153 = 3^2 \times 17$; add 1 to each exponent and multiply: $(2 + 1) \times (1 + 1) = 6$.

WHY NOT (A) counts only the prime factors. (B) leaves out 1 and 153.

20. **A** 3

- 1 The number of each item must be a common multiple of 12 and 9; the least one is the LCM.
- 2 Multiples of 12: 12, 24, 36. The first that is also a multiple of 9 is 36, so he needs 36 of each.
- 3 Packs of pens: $36 \div 12 = 3$.

WHY NOT (B) is the number of packs of notepads. (C) buys one pack of pens for each item in a pack of notepads; that works, but it is not the least. (D) is the number of pens, not the number of packs.

21. **C** 4

- 1 First find the greatest number of comfort kits: the GCF.
- 2 Write each number as a product of primes: $60 = 2^2 \times 3 \times 5$, $75 = 3 \times 5^2$.
- 3 Take each prime they *all* share, with its lowest power: $3 \times 5 = 15$.
- 4 There are 15 comfort kits, so each gets $60 \div 15 = 4$ pairs of socks.

WHY NOT (D) is the number of comfort kits, not the number of pairs of socks in each.

22. **B** 5

- 1 The number of each item must be a common multiple of 12 and 20; the least one is the LCM.
- 2 Multiples of 20: 20, 40, 60. The first that is also a multiple of 12 is 60, so he needs 60 of each.
- 3 Packs of hot dogs: $60 \div 12 = 5$.

WHY NOT (A) is the number of packs of buns. (C) buys one pack of hot dogs for each item in a pack of buns; that works, but it is not the least. (D) is the number of hot dogs, not the number of packs.

23. **B** 5

- 1 First find the greatest number of teams: the GCF.
- 2 Write each number as a product of primes: $45 = 3^2 \times 5$, $36 = 2^2 \times 3^2$.
- 3 Take each prime they *all* share, with its lowest power: $3^2 = 9$.
- 4 There are 9 teams, so each gets $45 \div 9 = 5$ Alpha soldiers.

WHY NOT (A) is the number of Bravo soldiers in each. (C) is the number of teams, not the number of Alpha soldiers in each. (D) splits into only 3 groups, a common factor that is not the greatest.

24. **D** 6

- 1 Each possible row length is a factor of 153, so count the factors of 153.
- 2 List the factor pairs, starting from 1: 1×153 , 3×51 , 9×17 .
- 3 The factors are 1, 3, 9, 17, 51, 153.
- 4 Count them: 6 factors.

TIP Shortcut: $153 = 3^2 \times 17$; add 1 to each exponent and multiply: $(2 + 1) \times (1 + 1) = 6$.

WHY NOT (A) counts only the prime factors. (B) leaves out 1 and 153.

25. **D** 280

- 1 Write each number as a product of primes: $4 = 2^2$, 7 (prime), $8 = 2^3$, $10 = 2 \times 5$.
- 2 Take every prime that appears, with its highest power: $2^3 \times 5 \times 7 = 280$.
- 3 Check: 280 divided by each of the numbers gives a whole number.

WHY NOT (A) is not divisible by 7. (B) is not divisible by 10. (C) is not divisible by 8.

Fractions & Mixed Numbers

Tested on: Mathematics Knowledge • Arithmetic Reasoning

Key Concepts

Fractions are everywhere on the ASVAB: in recipes, distances, shares of a budget, and parts of a group. Every fraction skill comes down to a few moves done carefully, one step at a time.

Simplifying and comparing

Divide the top and bottom by their greatest common factor: $\frac{18}{24} = \frac{18 \div 6}{24 \div 6} = \frac{3}{4}$. Multiplying the top and bottom by the same number gives an *equivalent* fraction: $\frac{3}{4} = \frac{15}{20}$. To compare, rewrite over a common denominator: $\frac{2}{3} = \frac{8}{12}$ and $\frac{3}{4} = \frac{9}{12}$, so $\frac{3}{4}$ is larger.

The four operations

- **Add/subtract:** common denominator first, then combine the numerators: $\frac{1}{4} + \frac{2}{3} = \frac{3}{12} + \frac{8}{12} = \frac{11}{12}$.
- **Multiply:** top times top, bottom times bottom (cancel first): $\frac{4}{9} \times \frac{3}{8} = \frac{1}{3} \times \frac{1}{2} = \frac{1}{6}$.
- **Divide:** keep the first, flip the *second*, multiply: $\frac{2}{3} \div \frac{4}{5} = \frac{2}{3} \times \frac{5}{4} = \frac{5}{6}$.

Mixed numbers

$3\frac{2}{5} = \frac{3 \times 5 + 2}{5} = \frac{17}{5}$. To add or subtract, work with the whole numbers and the fractions; when the top fraction is too small, *regroup*: $5\frac{1}{4} = 4\frac{5}{4}$. To multiply or divide, change to improper fractions first.

WORKED EXAMPLE

Maria spent $\frac{1}{3}$ of her \$1,200 paycheck on rent and then $\frac{1}{4}$ of what was left on groceries. How much money does she have now?

Solution. Rent: $\frac{1}{3} \times 1,200 = 400$, leaving 800. Groceries: $\frac{1}{4}$ of 800 is 200, leaving $800 - 200 = \$600$.

No-Calculator Tip To find a fraction of a number, divide by the bottom, then multiply by the top: $\frac{3}{8}$ of 32 is $32 \div 8 = 4$, then $4 \times 3 = 12$.

Common Traps

- Adding tops and bottoms: $\frac{1}{2} + \frac{1}{3}$ is $\frac{5}{6}$, not $\frac{2}{5}$.
- Flipping the wrong fraction when dividing (flip the second one only).
- Taking the second fraction of the *original* amount instead of what was left.
- Forgetting to take 1 from the whole number when you regroup.

Practice Set 4

Score: ____/25

Target time: 26 minutes

Problems 1–8 are warm-ups, 9–18 are test level, and 19–25 are challenge problems.

1. Simplify $\frac{18}{99}$ to lowest terms.
A $\frac{6}{33}$ B $\frac{11}{2}$ C $\frac{2}{11}$ D $\frac{2}{99}$
2. Add: $\frac{4}{5} + \frac{14}{15}$
A $\frac{9}{10}$ B $1\frac{11}{15}$ C $1\frac{4}{5}$ D $1\frac{14}{15}$
3. A platoon has 20 soldiers. $\frac{1}{5}$ of them are on guard duty tonight. How many soldiers are on guard duty?
A 4 B 5 C 16 D 100
4. What is $\frac{5}{6} - \frac{1}{2}$?
A $\frac{1}{3}$ B $\frac{2}{3}$ C 1 D $1\frac{1}{3}$
5. A parking lot holds 48 vehicles. $\frac{1}{3}$ of them are pickup trucks. How many vehicles are pickup trucks?
A 15 B 16 C 32 D 144
6. What is $\frac{10}{30}$ in simplest form?
A $\frac{2}{6}$ B $\frac{2}{3}$ C $\frac{1}{3}$ D $\frac{1}{30}$
7. If $\frac{5}{8} = \frac{x}{40}$, what is the value of x ?
A 5 B 10 C 24 D 25
8. What is $\frac{4}{5} \times 35$?
A 7 B 28 C $43\frac{3}{4}$ D 140
9. Which is the smallest: $\frac{1}{6}$, $\frac{3}{4}$, $\frac{1}{5}$, or $\frac{3}{5}$?
A $\frac{1}{6}$ B $\frac{1}{5}$ C $\frac{3}{4}$ D $\frac{3}{5}$
10. Divide: $15 \div \frac{3}{4}$
A $\frac{1}{20}$ B $11\frac{1}{4}$ C 20 D 28
11. Add: $3\frac{1}{2} + 7\frac{11}{12}$
A $10\frac{5}{12}$ B $11\frac{5}{12}$ C $12\frac{5}{12}$ D $27\frac{17}{24}$
12. Private Jensen's platoon is on a 15-mile ruck march. So far the platoon has covered $9\frac{1}{2}$ miles. How many miles are left?
A $2\frac{3}{4}$ miles B $4\frac{1}{2}$ miles
C $5\frac{1}{2}$ miles D $6\frac{1}{2}$ miles
13. What is $4\frac{1}{2} + 7\frac{4}{5}$?
A $11\frac{3}{10}$ B $12\frac{3}{10}$ C $13\frac{3}{10}$ D $35\frac{1}{10}$
14. A recipe calls for $3\frac{1}{2}$ cups of rice. Leila is making 3 batches. How many cups of rice does she need?
A $6\frac{1}{2}$ cups B $9\frac{1}{2}$ cups
C $10\frac{1}{2}$ cups D $11\frac{1}{2}$ cups
15. Divide: $\frac{1}{6} \div \frac{2}{3}$
A $\frac{1}{4}$ B 1 C 4 D 9
16. Which of the fractions $\frac{9}{10}$, $\frac{1}{6}$, $\frac{2}{3}$, and $\frac{13}{15}$ is the smallest?
A $\frac{13}{15}$ B $\frac{9}{10}$ C $\frac{2}{3}$ D $\frac{1}{6}$
17. What is $\frac{7}{10} + \frac{1}{3}$?
A $\frac{4}{15}$ B $\frac{11}{30}$ C $\frac{8}{13}$ D $1\frac{1}{30}$
18. What is $\frac{1}{6} \times 12$?
A 2 B 10 C 12 D 72
19. Subtract: $8\frac{1}{12} - 1\frac{3}{4}$
A $6\frac{1}{3}$ B $6\frac{2}{3}$ C $7\frac{1}{3}$ D $7\frac{2}{3}$
20. Multiply: $2\frac{1}{2} \times 2\frac{1}{2}$
A $2\frac{1}{4}$ B $4\frac{1}{4}$ C $6\frac{1}{4}$ D $12\frac{1}{2}$
21. Corporal Lopez received a \$4,300 enlistment bonus, used $\frac{1}{5}$ of it to pay off a car loan, and then put $\frac{2}{5}$ of the remaining money into savings. How much money is left?
A \$1,720 B \$2,064 C \$3,440 D \$3,956

22. What is $4\frac{2}{3} \div 1\frac{2}{3}$?
 A $2\frac{4}{5}$ B $3\frac{2}{5}$ C 5 D $7\frac{7}{9}$
23. A truck started a trip with 90 gallons of fuel. It used $\frac{5}{6}$ of the fuel on the first day and $\frac{1}{5}$ of what was left on the second day. How many gallons of fuel are left?
 A 12 gallons B 13 gallons
 C 15 gallons D 75 gallons
24. What is $9\frac{1}{2} - 5\frac{4}{5}$?
 A $2\frac{7}{10}$ B $3\frac{3}{10}$ C $3\frac{7}{10}$ D $4\frac{3}{10}$
25. Airman Okafor has 16 recruits. $\frac{1}{4}$ of them passed the swim test on the first try, and $\frac{1}{2}$ of those also passed the rope climb. How many recruits passed both?
 A 2 B 4 C 8 D 12

Answers & Explanations 4

1 C	6 C	11 B	16 D	21 B
2 B	7 D	12 C	17 D	22 A
3 A	8 B	13 B	18 A	23 A
4 A	9 A	14 C	19 A	24 C
5 B	10 C	15 A	20 C	25 A

1. **C** $\frac{2}{11}$

1 Find the greatest common factor of 18 and 99: it is 9.

2 Divide the top and bottom by 9: $\frac{18}{99} = \frac{18 \div 9}{99 \div 9} = \frac{2}{11}$.

TIP You can also divide in smaller steps (for example by 3 first); just keep going until the top and bottom share no factor.

WHY NOT (A) is equal to $\frac{18}{99}$, but not in lowest terms: 6 and 33 still share a factor of 3. (B) flips the fraction. (D) divides only the top by 9.

2. **B** $1\frac{11}{15}$

1 15 is a multiple of 5, so use 15 as the common denominator.

2 Rewrite $\frac{4}{5}$ with denominator 15: $\frac{4}{5} = \frac{12}{15}$.

3 Add the numerators: $\frac{12}{15} + \frac{14}{15} = \frac{26}{15} = 1\frac{11}{15}$.

WHY NOT (A) adds the numerators and adds the denominators.

3. **A** 4

1 Find $\frac{1}{5}$ of 20: divide by the bottom, $20 \div 5 = 4$.

2 Multiply by the top: $4 \times 1 = 4$.

WHY NOT (C) is the other part of the group. (D) divides by the fraction instead of multiplying.

4. **A** $\frac{1}{3}$

1 6 is a multiple of 2, so use 6 as the common denominator.

2 Rewrite $\frac{1}{2}$ with denominator 6: $\frac{1}{2} = \frac{3}{6}$.

3 Subtract the numerators: $\frac{5}{6} - \frac{3}{6} = \frac{2}{6} = \frac{1}{3}$.

WHY NOT (B) uses the common denominator without rewriting the numerators. (C) subtracts the numerators and subtracts the denominators. (D) adds instead of subtracting.

5. **B** 16

1 Find $\frac{1}{3}$ of 48: divide by the bottom, $48 \div 3 = 16$.

2 Multiply by the top: $16 \times 1 = 16$.

WHY NOT (C) is the other part of the group. (D) divides by the fraction instead of multiplying.

6. **C** $\frac{1}{3}$

1 Find the greatest common factor of 10 and 30: it is 10.

2 Divide the top and bottom by 10: $\frac{10}{30} = \frac{10 \div 10}{30 \div 10} = \frac{1}{3}$.

TIP You can also divide in smaller steps (for example by 2 first); just keep going until the top and bottom share no factor.

WHY NOT (A) is equal to $\frac{10}{30}$, but not in lowest terms: 2 and 6 still share a factor of 2. (D) divides only the top by 10.

7. **D** 25

1 The bottom changed from 8 to 40: $40 \div 8 = 5$, so it was multiplied by 5.

2 Multiply the top by the same number: $5 \times 5 = 25$.

WHY NOT (A) stops after finding $40 \div 8 = 5$.

8. **B** 28

1 Divide by the bottom: $35 \div 5 = 7$.

2 Multiply by the top: $7 \times 4 = 28$.

WHY NOT (A) finds only $\frac{1}{5}$ of 35. (C) divides by the fraction instead of multiplying. (D) multiplies by 4 but forgets to divide by 5.

9. **A** $\frac{1}{6}$

1 Rewrite each fraction with the common denominator 60: $\frac{1}{6} = \frac{10}{60}$, $\frac{3}{4} = \frac{45}{60}$, $\frac{1}{5} = \frac{12}{60}$, $\frac{3}{5} = \frac{36}{60}$.

2 The smallest numerator is 10, so $\frac{1}{6}$ is the smallest.

WHY NOT (B) equals $\frac{12}{60}$, which is more than $\frac{10}{60}$. (C) equals $\frac{45}{60}$, which is more than $\frac{10}{60}$. (D) equals $\frac{36}{60}$, which is more than $\frac{10}{60}$.

10. **C** 20

1 Dividing by $\frac{3}{4}$ means multiplying by its reciprocal, $\frac{4}{3}$.

2 $15 \times \frac{4}{3} = \frac{60}{3} = 20$.

WHY NOT (A) divides in the wrong order. (B) multiplies instead of dividing.

11. **B** $11\frac{5}{12}$

- 1 Add the whole numbers: $3 + 7 = 10$.
- 2 Add the fractions with the common denominator 12: $\frac{6}{12} + \frac{11}{12} = \frac{17}{12} = 1\frac{5}{12}$.
- 3 Combine: $10 + 1\frac{5}{12} = 11\frac{5}{12}$.

WHY NOT (A) drops the extra whole number from $\frac{17}{12}$.
(D) multiplies instead of adding.

12. **C** $5\frac{1}{2}$ miles

- 1 Subtract: $15 - 9\frac{1}{2}$.
- 2 Regroup 15 as $14\frac{2}{2}$.
- 3 $14\frac{2}{2} - 9\frac{1}{2} = 5\frac{1}{2}$ miles.

WHY NOT (D) subtracts the whole numbers and keeps the fraction.

13. **B** $12\frac{3}{10}$

- 1 Add the whole numbers: $4 + 7 = 11$.
- 2 Add the fractions with the common denominator 10: $\frac{5}{10} + \frac{8}{10} = \frac{13}{10} = 1\frac{3}{10}$.
- 3 Combine: $11 + 1\frac{3}{10} = 12\frac{3}{10}$.

WHY NOT (A) drops the extra whole number from $\frac{13}{10}$.
(D) multiplies instead of adding.

14. **C** $10\frac{1}{2}$ cups

- 1 Multiply the whole number and the fraction by 3: $3 \times 3 = 9$ and $\frac{1}{2} \times 3 = \frac{3}{2} = 1\frac{1}{2}$.
- 2 Add: $9 + 1\frac{1}{2} = 10\frac{1}{2}$.

WHY NOT (A) adds 3 instead of multiplying by 3. (B) multiplies only the whole number.

15. **A** $\frac{1}{4}$

- 1 Keep the first fraction, change $\div$ to $\times$, and flip the second fraction: $\frac{1}{6} \times \frac{3}{2}$.
- 2 Cancel before multiplying: 3 and 6 share 3, so the problem becomes $\frac{1}{2} \times \frac{1}{2}$.
- 3 Multiply top times top and bottom times bottom: $\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$.

WHY NOT (C) divides in the wrong order (flips the answer). (D) flips both fractions.

16. **D** $\frac{1}{6}$

- 1 Rewrite each fraction with the common denominator 30: $\frac{9}{10} = \frac{27}{30}$, $\frac{1}{6} = \frac{5}{30}$, $\frac{2}{3} = \frac{20}{30}$, $\frac{13}{15} = \frac{26}{30}$.
- 2 The smallest numerator is 5, so $\frac{1}{6}$ is the smallest.

WHY NOT (A) equals $\frac{26}{30}$, which is more than $\frac{5}{30}$. (B) equals $\frac{27}{30}$, which is more than $\frac{5}{30}$. (C) equals $\frac{20}{30}$, which is more than $\frac{5}{30}$.

17. **D** $1\frac{1}{30}$

- 1 The least common denominator of 10 and 3 is 30.
- 2 Rewrite: $\frac{7}{10} = \frac{21}{30}$ and $\frac{1}{3} = \frac{10}{30}$.
- 3 Add the numerators: $\frac{21}{30} + \frac{10}{30} = \frac{31}{30} = 1\frac{1}{30}$.

WHY NOT (A) uses the common denominator without rewriting the numerators. (B) subtracts instead of adding. (C) adds the numerators and adds the denominators.

18. **A** 2

- 1 Divide by the bottom: $12 \div 6 = 2$.
- 2 Multiply by the top: $2 \times 1 = 2$.

WHY NOT (B) is the part that is left, not the part asked for. (C) multiplies by 1 but forgets to divide by 6. (D) divides by the fraction instead of multiplying.

19. **A** $6\frac{1}{3}$

- 1 Rewrite the fractions with the common denominator 12: $8\frac{1}{12} - 1\frac{9}{12}$.
- 2 Since $\frac{1}{12}$ is less than $\frac{9}{12}$, regroup: take 1 from 8, so $8\frac{1}{12} = 7\frac{13}{12}$.

3 Subtract: whole numbers $7 - 1 = 6$, fractions $\frac{13}{12} - \frac{9}{12} = \frac{4}{12} = \frac{1}{3}$. The answer is $6\frac{1}{3}$.

WHY NOT (B) takes 1 from the whole number but then subtracts the fractions in the wrong order. (C) regroups but forgets to take 1 from the whole number. (D) subtracts the smaller fraction from the larger instead of regrouping.

20. **C** $6\frac{1}{4}$

- 1 Change each mixed number to an improper fraction: $2\frac{1}{2} = \frac{2 \times 2 + 1}{2} = \frac{5}{2}$ and $2\frac{1}{2} = \frac{5}{2}$.
- 2 Multiply top times top and bottom times bottom: $\frac{5}{2} \times \frac{5}{2} = \frac{25}{4} = 6\frac{1}{4}$.

WHY NOT (A) adds the whole number to the numerator instead of multiplying it by the denominator. (B) multiplies the whole numbers and the fractions separately.

21. **B** \$2,064

- 1 The car loan: $\frac{1}{5} \times 4,300 = 860$, which leaves $4,300 - 860 = 3,440$ dollars.
- 2 Savings: $\frac{2}{5}$ of what was left is $\frac{2}{5} \times 3,440 = 1,376$.
- 3 What remains: $3,440 - 1,376 = 2,064$ dollars.

TIP Shortcut: keep $\frac{4}{5}$ and then $\frac{3}{5}$ of that: $\frac{4}{5} \times \frac{3}{5} \times 4,300 = 2,064$.

WHY NOT (A) takes $\frac{2}{5}$ of the original amount instead of what was left. (C) stops after the car loan. (D) multiplies the two fractions and subtracts that from the original amount.

22. **A** $2\frac{4}{5}$

1 Change each mixed number to an improper fraction: $4\frac{2}{3} = \frac{4 \times 3 + 2}{3} = \frac{14}{3}$ and $1\frac{2}{3} = \frac{5}{3}$.

2 Dividing means multiplying by the reciprocal: $\frac{14}{3} \div \frac{5}{3} = \frac{14}{3} \times \frac{3}{5}$.

3 Cancel before multiplying: 3 and 3 share 3, so the problem becomes $14 \times \frac{1}{5}$.

4 Multiply top times top and bottom times bottom: $14 \times \frac{1}{5} = \frac{14}{5} = 2\frac{4}{5}$.

WHY NOT (C) divides the whole numbers and the fractions separately. (D) multiplies instead of dividing.

23. **A** 12 gallons

1 The first day: $\frac{5}{6} \times 90 = 75$, which leaves $90 - 75 = 15$ gallons.

2 The second day: $\frac{1}{5}$ of what was left is $\frac{1}{5} \times 15 = 3$.

3 What remains: $15 - 3 = 12$ gallons.

TIP Shortcut: keep $\frac{1}{6}$ and then $\frac{4}{5}$ of that: $\frac{1}{6} \times \frac{4}{5} \times 90 = 12$.

WHY NOT (C) stops after the first day. (D) multiplies the two fractions and subtracts that from the original amount.

24. **C** $3\frac{7}{10}$

1 Rewrite the fractions with the common denominator 10: $9\frac{5}{10} - 5\frac{8}{10}$.

2 Since $\frac{5}{10}$ is less than $\frac{8}{10}$, regroup: take 1 from 9, so $9\frac{5}{10} = 8\frac{15}{10}$.

3 Subtract: whole numbers $8 - 5 = 3$, fractions $\frac{15}{10} - \frac{8}{10} = \frac{7}{10}$. The answer is $3\frac{7}{10}$.

WHY NOT (B) takes 1 from the whole number but then subtracts the fractions in the wrong order. (D) subtracts the smaller fraction from the larger instead of regrouping.

25. **A** 2

1 First group: $\frac{1}{4} \times 16 = 4$.

2 Then take $\frac{1}{2}$ of that group: $\frac{1}{2} \times 4 = 2$.

WHY NOT (B) is the size of the first group only. (C) takes $\frac{1}{2}$ of the whole group instead of the smaller group. (D) adds the two fractions.

Decimals

Tested on: Mathematics Knowledge • Arithmetic Reasoning

Key Concepts

Decimals are fractions with denominators of 10, 100, 1,000, and so on. Money is the everyday example, and most decimal word problems on the ASVAB are about prices, pay, and fuel. The whole chapter rests on keeping track of the decimal point.

Place value

In 3.472: 4 tenths, 7 hundredths, 2 thousandths. Compare decimals place by place from the left: $0.5 > 0.45 > 0.405 > 0.054$ (look at the tenths first). Extra zeros on the right change nothing: $0.5 = 0.50$. To **round**, look at the next digit to the right: 5 or more rounds up. $7.486 \approx 7.49$ (nearest hundredth) ≈ 7.5 (nearest tenth).

The four operations

- **Add/subtract:** line up the decimal points; fill empty places with zeros: $15.00 - 3.84 = 11.16$.
- **Multiply:** multiply as whole numbers, then count the decimal places in both factors: 0.6×0.04 : $6 \times 4 = 24$, three places, 0.024.
- **Divide by a decimal:** move the point in *both* numbers until the divisor is whole: $7.2 \div 0.08 = 720 \div 8 = 90$.

Fractions and decimals

Fraction $\rightarrow$ decimal: divide the top by the bottom ($\frac{3}{8} = 3 \div 8 = 0.375$). Decimal $\rightarrow$ fraction: read the place value and simplify ($0.45 = \frac{45}{100} = \frac{9}{20}$). Worth memorizing: $\frac{1}{2} = 0.5$, $\frac{1}{4} = 0.25$, $\frac{3}{4} = 0.75$, $\frac{1}{5} = 0.2$, $\frac{1}{8} = 0.125$.

WORKED EXAMPLE

Gas costs \$3.49 per gallon. How much do 12 gallons cost?

Solution. $3.49 \times 12 = 3.49 \times 10 + 3.49 \times 2 = 34.90 + 6.98 = \41.88 .

No-Calculator Tip Estimate first to place the decimal point: 3.49×12 is about $3.5 \times 12 = 42$, so the answer must be about \$42, not \$4.19 or \$418.80.

Common Traps

- Lining up the last digits instead of the decimal points.
- Thinking a longer decimal is larger: $0.405 < 0.5$.
- Moving the decimal point in only one number when dividing.

- Forgetting to borrow across zeros: $20.00 - 7.36 = 12.64$.

Practice Set 5

Score: ____ / 25

Target time: 27 minutes

Problems 1–8 are warm-ups, 9–18 are test level, and 19–25 are challenge problems.

- What is $29.36 - 8.901$?
A 20.459 B 20.461 C 21.661 D 38.261
- What is 32.527 rounded to the nearest hundredth?
A 32.52 B 32.53 C 32.54 D 32.55
- Subtract: $12.9 - 2.34$
A 10.52 B 10.56 C 10.64 D 15.24
- Round 20.281 to the nearest hundredth.
A 20.26 B 20.27 C 20.28 D 20.29
- What is 0.7×0.5 ?
A 0.35 B 1.2 C 3.5 D 35
- What is $70.2 \div 9$?
A 7.8 B 8.4 C 78 D 631.8
- Notebooks cost \$2.03 each. How much do 7 notebooks cost?
A \$9.03 B \$14.03 C \$14.21 D \$142.10
- Diego buys a bag of chips for \$2.16 and a pack of gum for \$1.99. He pays with a \$20 bill. How much change should he get back?
A \$4.15 B \$14.85 C \$15.85 D \$17.84
- Divide: $0.855 \div 0.09$
A 9.5 B 10.5 C 19 D 95
- Convert 0.71 to a fraction in simplest form.
A $\frac{71}{1000}$ B $\frac{7}{10}$ C $\frac{71}{100}$ D $\frac{71}{10}$
- Which of the numbers 0.21, 0.0102, 0.012, and 0.201 is the least?
A 0.201 B 0.0102 C 0.21 D 0.012
- What is $\frac{39}{50}$ written as a decimal?
A 0.078 B 0.395 C 0.76 D 0.78
- Which is the least: 0.305, 0.5, 0.35, or 0.035?
A 0.5 B 0.305 C 0.35 D 0.035
- What is $21.5 \div 0.5$?
A 43 B 48 C 86 D 430
- Subtract: $50 - 48.61$
A 1.39 B 2.39 C 18.61 D 98.61
- Multiply: 1.4×3.5
A 0.49 B 4.9 C 49 D 490
- Sofia buys 3 folders at \$0.75 each and 2 granola bars at \$1.10 each. What is the total cost?
A \$1.85 B \$3.35 C \$4.45 D \$5.50
- A package of 16 AA batteries costs \$14.56. What is the price per battery?
A \$0.81 B \$0.91 C \$1.01 D \$9.10
- Store A sells a bag of 2 pounds of rice for \$1.86. Store B sells a bag of 5 pounds of rice for \$4.55. How much less is the price per pound at the cheaper store?
A \$0.02 B \$0.91 C \$0.93 D \$2.69
- Omar works on a construction crew and earns \$12.00 per hour for the first 40 hours each week. Every hour over 40 is overtime and pays 1.5 times the regular rate. Last week he worked 43 hours. How much did he earn?
A \$480.00 B \$516.00 C \$520.50 D \$534.00
- Store A sells a pack of 6 cans of soda for \$3.72. Store B sells a pack of 8 cans of soda for \$5.76. How much less is the price per can at the cheaper store?
A \$0.09 B \$0.10 C \$0.62 D \$2.04

22. Aisha works at a hardware store and earns \$11.00 per hour for the first 40 hours each week. Every hour over 40 is overtime and pays 1.5 times the regular rate. Last week she worked 45 hours. How much did she earn?
A \$440.00 B \$495.00 C \$502.50 D \$522.50
23. A motorcycle traveled 131.2 miles on 3.2 gallons of fuel. How many miles per gallon did it get?
A 4.1 miles per gallon B 39 miles per gallon
C 40 miles per gallon D 41 miles per gallon
24. Brandon buys 3.5 pounds of cheese at \$3.50 per pound and 3 pounds of ground beef at \$5.70 per pound. How much does he spend?
A \$29.35 B \$29.45 C \$139.60 D \$183.25
25. Tasha buys 2 notebooks at \$2.70 each and 2 bags of chips at \$1.47 each. She pays with a \$20 bill. How much change should she get back?
A \$11.66 B \$14.60 C \$15.83 D \$28.34

Answers & Explanations 5

1 A	6 A	11 B	16 B	21 B
2 B	7 C	12 D	17 C	22 D
3 B	8 C	13 D	18 B	23 D
4 C	9 A	14 A	19 A	24 A
5 A	10 C	15 A	20 D	25 A

1. **A** 20.459

1 Line up the decimal points and fill in zeros so both numbers have 3 decimal places: $29.360 - 8.901$.

2 Subtract column by column, keeping the point in place: $29.360 - 8.901 = 20.459$.

WHY NOT (C) subtracts the smaller digit from the larger in each column instead of borrowing. (D) adds instead of subtracting.

2. **B** 32.53

1 The hundredths digit is 2. The digit to its right (the thousandths digit) is 7.

2 Since 7 is 5 or more, round up and drop the digits after it: 32.53.

WHY NOT (A) drops the extra digits without rounding up (the next digit, 7, is 5 or more).

3. **B** 10.56

1 Line up the decimal points and fill in zeros so both numbers have 2 decimal places: $12.90 - 2.34$.

2 Subtract column by column, keeping the point in place: $12.90 - 2.34 = 10.56$.

WHY NOT (C) subtracts the smaller digit from the larger in each column instead of borrowing. (D) adds instead of subtracting.

4. **C** 20.28

1 The hundredths digit is 8. The digit to its right (the thousandths digit) is 1.

2 Since 1 is less than 5, keep the hundredths digit and drop the digits after it: 20.28.

WHY NOT (D) rounds up even though the next digit, 1, is less than 5.

5. **A** 0.35

1 Ignore the decimal points and multiply: $7 \times 5 = 35$.

2 Count the decimal places: 1 in 0.7 and 1 in 0.5, 2 in all.

3 Put 2 decimal places in the product: 0.35.

WHY NOT (B) adds instead of multiplying. (C) puts too few decimal places in the answer. (D) counts the decimal places in only one of the numbers.

6. **A** 7.8

1 Divide as with whole numbers and put the point in the answer directly above the point in 70.2.

2 $70.2 \div 9 = 7.8$. Check: $9 \times 7.8 = 70.2$.

WHY NOT (C) drops the decimal point. (D) multiplies instead of dividing.

7. **C** \$14.21

1 Multiply the price by the number: 2.03×7 .

2 Split it up: $2 \times 7 = 14$ and $0.03 \times 7 = 0.21$.

3 Add: $14 + 0.21 = 14.21$, so the cost is \$14.21.

TIP Estimate: \$2.03 is about \$2, and $2 \times 7 = 14$, so the answer is near \$14.

WHY NOT (A) adds instead of multiplying. (B) multiplies only the dollars and then tacks on the cents. (D) misplaces the decimal point.

8. **C** \$15.85

1 Total: $2.16 + 1.99 = 4.15$.

2 Change: $20.00 - 4.15 = 15.85$, so \$15.85.

TIP Count up from \$4.15: to the next dollar, then to \$20.

WHY NOT (A) is the total cost, not the change. (D) subtracts only the first item.

9. **A** 9.5

1 Make the divisor a whole number: move the decimal point 2 places to the right in *both* numbers.

2 $0.855 \div 0.09$ becomes $85.5 \div 9$.

3 Divide: $85.5 \div 9 = 9.5$. Check: $0.09 \times 9.5 = 0.855$.

WHY NOT (D) moves the decimal point one place too many.

10. **C** $\frac{71}{100}$

1 The last digit is in the hundredth place, so $0.71 = \frac{71}{100}$.

2 $\frac{71}{100}$ is already in lowest terms.

WHY NOT (A) reads the last digit as thousandths instead of hundredths. (D) reads the last digit as tenths instead of hundredths.

11. **B** 0.0102

- 1 Give every number the same number of decimal places by adding zeros: 0.2100, 0.0102, 0.0120, 0.2010.
- 2 Now compare them like whole numbers. In order from least to greatest: $0.0102 < 0.012 < 0.201 < 0.21$.
- 3 The least is 0.0102.

WHY NOT (A) is greater than 0.0102: in the tenths place it has 2, while 0.0102 has 0. (C) is greater than 0.0102: in the tenths place it has 2, while 0.0102 has 0. (D) is greater than 0.0102: in the thousandths place it has 2, while 0.0102 has 0.

12. **D** 0.78

- 1 A fraction bar means divide: $39 \div 50$.
- 2 Or scale to hundredths: $\frac{39}{50} = \frac{39 \times 2}{50 \times 2} = \frac{78}{100} = 0.78$.
- 3 So $\frac{39}{50} = 0.78$.

WHY NOT (A) puts an extra zero after the decimal point. (B) just writes the numerator and denominator after the decimal point.

13. **D** 0.035

- 1 Give every number the same number of decimal places by adding zeros: 0.305, 0.500, 0.350, 0.035.
- 2 Now compare them like whole numbers. In order from least to greatest: $0.035 < 0.305 < 0.35 < 0.5$.
- 3 The least is 0.035.

WHY NOT (A) is greater than 0.035: in the tenths place it has 5, while 0.035 has 0. (B) is greater than 0.035: in the tenths place it has 3, while 0.035 has 0. (C) is greater than 0.035: in the tenths place it has 3, while 0.035 has 0.

14. **A** 43

- 1 Make the divisor a whole number: move the decimal point 1 place to the right in *both* numbers.
- 2 $21.5 \div 0.5$ becomes $215 \div 5$.
- 3 Divide: $215 \div 5 = 43$. Check: $0.5 \times 43 = 21.5$.

WHY NOT (D) moves the decimal point one place too many.

15. **A** 1.39

- 1 Line up the decimal points and fill in zeros so both numbers have 2 decimal places: $50.00 - 48.61$.
- 2 Subtract column by column, keeping the point in place: $50.00 - 48.61 = 1.39$.

WHY NOT (B) borrows but forgets to reduce the ones digit it borrowed from. (C) subtracts the smaller digit from the larger in each column instead of borrowing. (D) adds instead of subtracting.

16. **B** 4.9

- 1 Ignore the decimal points and multiply: $14 \times 35 = 490$.
- 2 Count the decimal places: 1 in 1.4 and 1 in 3.5, 2 in all.
- 3 Put 2 decimal places in the product: 4.90, which is 4.9.

WHY NOT (A) puts too many decimal places in the answer. (C) puts too few decimal places in the answer. (D) counts the decimal places in only one of the numbers.

17. **C** \$4.45

- 1 First item: $3 \times 0.75 = 2.25$.
- 2 Second item: $2 \times 1.10 = 2.20$.
- 3 Total: $2.25 + 2.20 = 4.45$, so \$4.45.

WHY NOT (A) adds one of each price and ignores the quantities. (B) forgets to multiply the second price by 2. (D) charges every item at the higher price.

18. **B** \$0.91

- 1 Price per battery = total price $\div$ number of batteries.
- 2 $14.56 \div 16 = 0.91$. Check: $16 \times 0.91 = 14.56$.

WHY NOT (D) puts the decimal point one place too far right.

19. **A** \$0.02

- 1 Store A: $1.86 \div 2 = 0.93$ each.
- 2 Store B: $4.55 \div 5 = 0.91$ each.
- 3 Store B is cheaper by $0.93 - 0.91 = 0.02$, or \$0.02 per item.

WHY NOT (B) is the price per item at the cheaper store, not the difference. (C) is the price per item at the more expensive store. (D) compares the pack prices instead of the prices per item.

20. **D** \$534.00

- 1 Regular pay: $40 \times 12.00 = 480.00$.
- 2 Overtime rate: $1.5 \times 12.00 = 18.00$ per hour.
- 3 Overtime pay: $3 \times 18.00 = 54.00$.
- 4 Total: $480.00 + 54.00 = 534.00$, so \$534.

WHY NOT (A) leaves out the overtime pay. (B) pays the overtime hours at the regular rate. (C) adds 1.5 dollars to the rate instead of multiplying by 1.5.

21. **B** \$0.10

- 1 Store A: $3.72 \div 6 = 0.62$ each.
- 2 Store B: $5.76 \div 8 = 0.72$ each.
- 3 Store A is cheaper by $0.72 - 0.62 = 0.10$, or \$0.10 per item.

WHY NOT (C) is the price per item at the cheaper store, not the difference. (D) compares the pack prices instead of the prices per item.

22. **D** \$522.50

- 1 Regular pay: $40 \times 11.00 = 440.00$.
- 2 Overtime rate: $1.5 \times 11.00 = 16.50$ per hour.
- 3 Overtime pay: $5 \times 16.50 = 82.50$.
- 4 Total: $440.00 + 82.50 = 522.50$, so \$522.50.

WHY NOT (A) leaves out the overtime pay. (B) pays the overtime hours at the regular rate. (C) adds 1.5 dollars to the rate instead of multiplying by 1.5.

23. **D** 41 miles per gallon

- 1 Miles per gallon = miles $\div$ gallons: $131.2 \div 3.2$.
- 2 Estimate: 3.2 gallons is about 3, and $3 \times 41 = 123$, which is close to 131.2. So try 41.
- 3 Check: $3.2 \times 41 = 3 \times 41 + 0.2 \times 41 = 123 + 8.2 = 131.2$. It works exactly, so the answer is 41 miles per gallon.

WHY NOT (A) moves the decimal point in the gallons but not in the miles.

24. **A** \$29.35

- 1 First item: $3.5 \times 3.50 = 12.25$.
- 2 Second item: $3 \times 5.70 = 17.10$.
- 3 Total: $12.25 + 17.10 = 29.35$, so \$29.35.

TIP For 3.5×3.50 , think 3×3.50 plus half of \$3.50.

WHY NOT (C) misplaces the decimal point in 3.5×3.50 .

25. **A** \$11.66

- 1 Cost of each kind: $2 \times 2.70 = 5.40$ and $2 \times 1.47 = 2.94$.
- 2 Total: $5.40 + 2.94 = 8.34$.
- 3 Change: $20.00 - 8.34 = 11.66$, so \$11.66.

TIP Count up from \$8.34: to the next dollar, then to \$20.

WHY NOT (B) subtracts only the first item. (C) subtracts one of each price and ignores the quantities. (D) subtracts the smaller digit from the larger in each column instead of borrowing.

Percents

Tested on: Mathematics Knowledge • Arithmetic Reasoning

Key Concepts

Percent means “per hundred.” Percent questions appear on both math subtests, and most of them reduce to one sentence: **part = percent × whole**.

Converting between forms

- Percent → decimal: move the decimal point two places *left* ($35\% = 0.35$, $4\% = 0.04$, $150\% = 1.5$).
- Decimal → percent: move it two places *right* ($0.6 = 60\%$, $0.075 = 7.5\%$).
- Fraction → percent: divide, then convert ($\frac{3}{8} = 0.375 = 37.5\%$).

The three percent questions

Every basic percent problem gives you two of the three quantities in

$$\text{part} = \text{percent} \times \text{whole}$$

and asks for the third.

- **Find the part:** 25% of 64 is $0.25 \times 64 = 16$.
- **Find the percent:** 18 is what percent of 72? $\frac{18}{72} = 0.25 = 25\%$.
- **Find the whole:** 24 is 30% of what number? $24 \div 0.30 = 80$.

Percent change

$$\text{percent change} = \frac{\text{new} - \text{original}}{\text{original}} \times 100\%$$

Always divide by the *original* (starting) value.

WORKED EXAMPLE

A jacket costs \$60. Its price goes up 15%. What is the new price?

Solution. The increase is $0.15 \times 60 = 9$ dollars, so the new price is $60 + 9 = \$69$. *Shortcut:* a 15% increase means you pay 115%, and $1.15 \times 60 = 69$.

No-Calculator Tip Benchmark percents make mental math fast: 10% = move the decimal one place left, 5% = half of 10%, 1% = move it two places, 25% = $\frac{1}{4}$, 50% = $\frac{1}{2}$, 75% = $\frac{3}{4}$, 20% = $\frac{1}{5}$. So 35% of 80 is $30\% + 5\% = 24 + 4 = 28$.

Common Traps

- Answering the wrong question: if 35% *passed*, the number who *did not* pass uses 65%.
- Dividing by the new value in a percent-change problem.
- Adding successive percents: up 25% then down 20% is *not* a 5% increase.

Practice Set 6

Score: ____ / 25

Target time: 27 minutes

Problems 1–8 are warm-ups, 9–18 are test level, and 19–25 are challenge problems.

- What is 25% of 396?
A 99 B 297 C 495 D 990
- Which of the following is equal to 0.13?
A 1.3% B 13% C 17% D 130%
- What is 80% of 60?
A 12 B 48 C 108 D 480
- Which of the following is equal to 0.02?
A 0.02% B 0.2% C 2% D 20%
- A motor pool has 248 vehicles, and 75% of them are scheduled for maintenance this week. How many vehicles are *not* scheduled for maintenance?
A 62 B 72 C 173 D 186
- What is $\frac{9}{20}$ written as a percent?
A 4.5% B 9% C 29% D 45%
- A motor pool has 120 vehicles, and 10% of them are scheduled for maintenance this week. How many vehicles are *not* scheduled for maintenance?
A 108 B 110 C 118 D 138
- What is 75% of 280?
A 21 B 70 C 170 D 210
- 57 is what percent of 380?
A 0.15% B 1.5% C 7.5% D 15%
- If 25% of a number is 100, what is the number?
A 25 B 40 C 400 D 2,500
- What percent of 260 is 104?
A 0.4% B 4% C 40% D 250%
- 60 is 50% of what number?
A 30 B 120 C 160 D 3,000
- Last year the price of a tool kit was \$292. This year it is \$365. What is the percent increase?
A 10% B 20% C 25% D 73%
- If 40% of a number is 186, what is the number?
A 226 B 315 C 415 D 465
- Last year the price of a gallon of paint was \$80. This year it is \$96. What is the percent increase?
A 12% B 16% C 20% D 120%
- 53 is what percent of 212?
A 0.25% B 2.5% C 12.5% D 25%
- Which of the following is equal to 0.45?
A 45% B 55% C 90% D 450%
- What is 10% of 860?
A 66 B 86 C 774 D 860
- A store sold a power drill for \$80. It increased the price by 50%, then put the power drill on sale at 50% off the new price. What is the sale price?
A \$60 B \$70 C \$80 D \$120
- What is 75% of 60% of 600?
A 270 B 360 C 450 D 810

21. A camping tent was priced at \$340. The store raised the price by 20%, and a month later it marked the new price down by 25%. What is the final price?
A \$306 B \$323 C \$408 D \$612
22. A training battalion has 400 recruits. 80% of them are assigned to the morning shift, and $\frac{1}{5}$ of the morning-shift recruits are on kitchen duty. How many recruits are on kitchen duty?
A 64 B 80 C 256 D 320
23. A store sold a camping tent for \$140. It increased the price by 20%, then put the camping tent on sale at 50% off the new price. What is the sale price?
A \$70 B \$84 C \$98 D \$168
24. A training battalion has 250 recruits. 80% of them are assigned to the morning shift, and $\frac{2}{5}$ of the morning-shift recruits are on kitchen duty. How many recruits are on kitchen duty?
A 40 B 50 C 80 D 200
25. What is 80% of 25% of 120?
A 16 B 24 C 30 D 126

Answers & Explanations 6

1 A	6 D	11 C	16 D	21 A
2 B	7 A	12 B	17 A	22 A
3 B	8 D	13 C	18 B	23 B
4 C	9 D	14 D	19 A	24 C
5 A	10 C	15 C	20 A	25 B

1. **A** 99
 - 1 Change the percent to a decimal: $25\% = 0.25$.
 - 2 Multiply by the number: $0.25 \times 396 = 99$.

WHY NOT (B) is what is left after removing 25%, not 25% of the number. (C) adds 25% to the number. (D) moves the decimal point only one place.
2. **B** 13%
 - 1 To change a decimal to a percent, multiply by 100 (move the decimal point two places to the right).
 - 2 $0.13 \times 100 = 13$, so $0.13 = 13\%$.

WHY NOT (A) moves the decimal point only one place. (D) moves the decimal point three places.
3. **B** 48
 - 1 Change the percent to a decimal: $80\% = 0.8$.
 - 2 Multiply by the number: $0.8 \times 60 = 48$.

WHY NOT (A) is what is left after removing 80%, not 80% of the number. (C) adds 80% to the number. (D) moves the decimal point only one place.
4. **C** 2%
 - 1 To change a decimal to a percent, multiply by 100 (move the decimal point two places to the right).
 - 2 $0.02 \times 100 = 2$, so $0.02 = 2\%$.

WHY NOT (A) forgets to move the decimal point at all. (B) moves the decimal point only one place. (D) moves the decimal point three places.
5. **A** 62
 - 1 The question asks about the *other* group: $100\% - 75\% = 25\%$.
 - 2 Find that part: $0.25 \times 248 = 62$.

TIP Or find 75% first (186) and subtract: $248 - 186 = 62$.

WHY NOT (C) subtracts the percent as if it were a count. (D) is the number in the group described, not the number asked for.
6. **D** 45%
 - 1 Divide the numerator by the denominator: $9 \div 20 = 0.45$.
 - 2 Multiply by 100 to get a percent: $0.45 = 45\%$.

TIP Or scale the fraction to hundredths: $\frac{9}{20} = \frac{45}{100}$.

WHY NOT (A) moves the decimal point only one place. (B) just writes the numerator as if it were a percent.
7. **A** 108
 - 1 The question asks about the *other* group: $100\% - 10\% = 90\%$.
 - 2 Find that part: $0.9 \times 120 = 108$.

TIP Or find 10% first (12) and subtract: $120 - 12 = 108$.

WHY NOT (B) subtracts the percent as if it were a count.
8. **D** 210
 - 1 Change the percent to a decimal: $75\% = 0.75$.
 - 2 Multiply by the number: $0.75 \times 280 = 210$.

WHY NOT (A) moves the decimal point three places. (B) is what is left after removing 75%, not 75% of the number.
9. **D** 15%
 - 1 Write the part over the whole: $\frac{57}{380}$.
 - 2 Simplify and convert: $\frac{57}{380} = 0.15 = 15\%$.

WHY NOT (A) forgets to multiply by 100. (B) moves the decimal point only one place.
10. **C** 400
 - 1 Use part = percent $\times$ whole: $100 = 0.25 \times w$.
 - 2 Divide both sides by 0.25: $w = 100 \div 0.25 = 400$.
 - 3 Check: 25% of 400 is 100. ✓

WHY NOT (A) finds 25% of 100 instead. (B) slips the decimal point. (D) multiplies by the percent without converting it.
11. **C** 40%
 - 1 Write the part over the whole: $\frac{104}{260}$.
 - 2 Simplify and convert: $\frac{104}{260} = 0.4 = 40\%$.

WHY NOT (A) forgets to multiply by 100. (B) moves the decimal point only one place. (D) divides the whole by the part instead of the part by the whole.

12. **B** 120

- 1 Use part = percent $\times$ whole: $60 = 0.5 \times w$.
- 2 Divide both sides by 0.5: $w = 60 \div 0.5 = 120$.
- 3 Check: 50% of 120 is 60. $\checkmark$

WHY NOT (A) finds 50% of 60 instead. (D) multiplies by the percent without converting it.

13. **C** 25%

- 1 Find the amount of increase: $365 - 292 = 73$ dollars.
- 2 Divide by the *original* amount: $\frac{73}{292} = 0.25$.
- 3 Convert to a percent: $0.25 = 25\%$ increase.

TIP Hard division? Test the choices instead: 25% of \$292 is \$73, exactly the change.

WHY NOT (B) divides by the new amount instead of the original amount. (D) gives the dollar change instead of the percent change.

14. **D** 465

- 1 Use part = percent $\times$ whole: $186 = 0.4 \times w$.
- 2 Divide both sides by 0.4: $w = 186 \div 0.4 = 465$.
- 3 Check: 40% of 465 is 186. $\checkmark$

WHY NOT (A) adds the percent to the part.

15. **C** 20%

- 1 Find the amount of increase: $96 - 80 = 16$ dollars.
- 2 Divide by the *original* amount: $\frac{16}{80} = 0.2$.
- 3 Convert to a percent: $0.2 = 20\%$ increase.

TIP Hard division? Test the choices instead: 20% of \$80 is \$16, exactly the change.

WHY NOT (B) gives the dollar change instead of the percent change. (D) gives the new amount as a percent of the old amount.

16. **D** 25%

- 1 Write the part over the whole: $\frac{53}{212}$.
- 2 Simplify and convert: $\frac{53}{212} = 0.25 = 25\%$.

WHY NOT (A) forgets to multiply by 100. (B) moves the decimal point only one place.

17. **A** 45%

- 1 To change a decimal to a percent, multiply by 100 (move the decimal point two places to the right).
- 2 $0.45 \times 100 = 45$, so $0.45 = 45\%$.

WHY NOT (D) moves the decimal point three places.

18. **B** 86

- 1 Change the percent to a decimal: $10\% = 0.1$.
- 2 Multiply by the number: $0.1 \times 860 = 86$.

WHY NOT (C) is what is left after removing 10%, not 10% of the number. (D) moves the decimal point only one place.

19. **A** \$60

- 1 Raise the price by 50%: $80 \times 1.5 = 120$.
- 2 Take 50% off the *new* price: $120 \times 0.5 = 60$.
- 3 The final price is \$60.

WHY NOT (C) assumes the two changes cancel out. (D) stops after the increase.

20. **A** 270

- 1 Work from the inside out. First, 60% of 600: $0.6 \times 600 = 360$.

- 2 Then 75% of that result: $0.75 \times 360 = 270$.

TIP Percents of percents multiply: $0.75 \times 0.6 = 0.45$, and $0.45 \times 600 = 270$.

WHY NOT (B) stops after taking 60%. (C) takes only 75% of 600. (D) adds the two percents.

21. **A** \$306

- 1 Raise the price by 20%: $340 \times 1.2 = 408$.
- 2 Take 25% off the *new* price: $408 \times 0.75 = 306$.
- 3 The final price is \$306.

WHY NOT (B) combines the changes into a single 5% decrease. (C) stops after the increase.

22. **A** 64

- 1 Find the first group: $0.8 \times 400 = 320$.
- 2 Take $\frac{1}{5}$ of that group: $\frac{1}{5} \times 320 = 64$.

WHY NOT (B) takes $\frac{1}{5}$ of the whole total. (C) is the part of the group that is *not* included. (D) is the size of the first group only.

23. **B** \$84

- 1 Raise the price by 20%: $140 \times 1.2 = 168$.
- 2 Take 50% off the *new* price: $168 \times 0.5 = 84$.
- 3 The final price is \$84.

WHY NOT (A) applies only the decrease. (C) combines the changes into a single 30% decrease. (D) stops after the increase.

24. **C** 80

- 1 Find the first group: $0.8 \times 250 = 200$.
- 2 Take $\frac{2}{5}$ of that group: $\frac{2}{5} \times 200 = 80$.

WHY NOT (D) is the size of the first group only.

25. **B** 24

- 1 Work from the inside out. First, 25% of 120: $0.25 \times 120 = 30$.

- 2 Then 80% of that result: $0.8 \times 30 = 24$.

TIP Percents of percents multiply: $0.8 \times 0.25 = 0.2$, and $0.2 \times 120 = 24$.

WHY NOT (C) stops after taking 25%. (D) adds the two percents.

Ratios & Proportions

Tested on: Mathematics Knowledge • Arithmetic Reasoning

Key Concepts

A *ratio* compares two quantities; a *proportion* says that two ratios are equal. Together they handle recipes, maps, mixtures, unit rates, and every “how many of each” question on the test.

Writing and simplifying ratios

The ratio of a to b can be written $a : b$, “ a to b ,” or $\frac{a}{b}$. **Order matters:** the first quantity named goes first.

- Simplify a ratio like a fraction: divide both terms by their greatest common factor. $18 : 24 = 3 : 4$ (divide both by 6).
- Put both amounts in the *same unit* first: 2 feet to 8 inches is 24 inches to 8 inches, so the ratio is $24 : 8 = 3 : 1$.

Ratios as parts of a whole

If the ratio of boys to girls is $3 : 5$, think of the group as $3 + 5 = 8$ equal *parts*: boys are $\frac{3}{8}$ of the group and girls are $\frac{5}{8}$. With 40 people, one part is $40 \div 8 = 5$, so there are $3 \times 5 = 15$ boys and $5 \times 5 = 25$ girls.

Proportions, unit rates, and direct variation

- In a proportion $\frac{a}{b} = \frac{c}{d}$ the *cross products* are equal: $a \times d = b \times c$. To solve $\frac{x}{12} = \frac{15}{36}$, write $36x = 12 \times 15 = 180$, so $x = 180 \div 36 = 5$.
- A *unit rate* compares to one unit: 300 miles on 12 gallons is $300 \div 12 = 25$ miles per gallon.
- “ y varies directly with x ” means $y = kx$: the ratio $\frac{y}{x}$ is always the same number k .

WORKED EXAMPLE

A cookie recipe uses 3 cups of flour to make 24 cookies. How many cups of flour are needed to make 40 cookies?

Solution. Keep the same order in both ratios (cups over cookies): $\frac{3}{24} = \frac{x}{40}$. Cross-multiply: $24x = 3 \times 40 = 120$. Divide: $x = 120 \div 24 = 5$ cups.

No-Calculator Tip Simplify before you multiply. In the example, $\frac{3}{24} = \frac{1}{8}$: one cup makes 8 cookies, so 40 cookies need $40 \div 8 = 5$ cups. Smaller numbers mean fewer mistakes.

Common Traps

- Reversing the order: “girls to boys” is $5 : 3$, not $3 : 5$.

- Treating the ratio 3 : 5 as the fraction $\frac{3}{5}$ of the *total*; the total has $3 + 5 = 8$ parts.
- Adding instead of scaling: going from 24 to 40 cookies does *not* mean adding 16 cups of flour.
- Comparing amounts in different units (feet with inches, hours with minutes) without converting first.

Practice Set 7

Score: ____ / 25

Target time: 28 minutes

Problems 1–8 are warm-ups, 9–18 are test level, and 19–25 are challenge problems.

- A garden has 12 tomato plants and 14 pepper plants. What is the ratio of tomato plants to pepper plants, in simplest form?
A 6 : 7 B 7 : 6 C 9 : 7 D 12 : 7
- A team's ratio of wins to losses this season is 5 : 1, and it has had no ties. What fraction of its games this season were wins?
A $\frac{1}{6}$ B $\frac{1}{5}$ C $\frac{1}{3}$ D $\frac{5}{6}$
- If $\frac{20}{8} = \frac{x}{16}$, what is the value of x ?
A 28 B 35 C 40 D 320
- Leila typed a 201-word report in 3 minutes. How many words per minute did she type?
A 6.7 B 65 C 66 D 67
- Solve for x : $\frac{x}{18} = \frac{15}{27}$
A 6 B 7 C 10 D 270
- Emily typed a 590-word report in 10 minutes. How many words per minute did she type?
A 59 B 60 C 61 D 590
- The ratio of boys to girls in a class is 3 : 5. What fraction of the students in the class are girls?
A $\frac{1}{8}$ B $\frac{3}{8}$ C $\frac{5}{8}$ D $\frac{5}{3}$
- A used-car lot has 4 sedans and 24 pickup trucks for sale. What is the ratio of sedans to pickup trucks, in simplest form?
A 1 : 6 B 1 : 3 C 6 : 7 D 6 : 1
- Written in simplest form, what is the ratio of 40 seconds to 4 minutes?
A 1 : 10 B 1 : 6 C 6 : 1 D 10 : 1
- Solve for x : $\frac{6}{x} = \frac{24}{64}$
A 16 B 18 C 46 D 384
- What is the ratio of 3 feet to 14 inches, in simplest form?
A 3 : 14 B 7 : 18 C 18 : 7 D 20 : 7
- A recruiting station signed up 192 recruits last month, all for either the Army or the Navy. The ratio of Army recruits to Navy recruits was 7 : 9. How many Army recruits were there?
A 12 B 64 C 74 D 84
- A fruit stand sold 255 pieces of fruit on Saturday, all apples or oranges. The ratio of apples to oranges sold was 9 : 8. How many oranges were sold?
A 15 B 120 C 135 D 160
- A concrete mix uses 2 bags of cement for every 4 bags of sand. How many bags of cement are needed for 30 bags of sand?
A 9 B 13 C 15 D 28
- A mess hall recipe uses 9 pounds of rice to feed 40 soldiers. How many pounds of rice are needed to feed 60 soldiers?
A 13.5 B 14.5 C 22.5 D 540
- On a road map, 1 inch represents 50 miles. Two towns are $5\frac{1}{2}$ inches apart on the map. What is the actual distance between the towns?
A 275 miles B 300 miles
C 550 miles D 2,750 miles

17. A blueprint uses the scale $\frac{1}{4}$ inch = 1 foot. A garage is $3\frac{1}{2}$ inches long on the blueprint. How long is the actual garage, in feet?
A 14 B 18 C 22 D 140
18. If y is directly proportional to x , and $y = 9$ when $x = 3$, what is the value of y when $x = 20$?
A 1.35 B 3 C 26 D 60
19. A supply sergeant divides 264 cases of bottled water among Alpha, Bravo, and Charlie platoons in the ratio 3 : 2 : 7. How many cases does Alpha platoon receive?
A 44 B 66 C 88 D 154
20. Three employees share a \$10,800 bonus in the ratio 1 : 4 : 3, based on hours worked. How much larger is the largest share than the smallest share?
A \$1,350 B \$4,050 C \$5,400 D \$8,100
21. A theater sold adult and child tickets in the ratio 1 : 7. It sold 36 more child tickets than adult tickets. How many tickets did it sell in all?
A 48 B 63 C 96 D 288
22. A motor pool has only Humvees and trucks, in the ratio 2 : 11 (Humvees to trucks). There are 72 more trucks than Humvees. How many trucks are there?
A 16 B 44 C 83 D 88
23. On a floor plan, 1 inch represents 4 feet. A garage measures 4 inches by 3 inches on the plan. What is the actual area of the room, in square feet?
A 48 B 192 C 208 D 768
24. If y varies directly with x , and $y = 10$ when $x = 4$, which equation relates x and y ?
A $y = \frac{40}{x}$ B $y = \frac{5}{2}x$
C $y = x + 6$ D $y = \frac{2}{5}x$
25. A field kitchen's oatmeal recipe uses 2 cups of oats for every 8 soldiers. How many cups of oats are needed for 30 soldiers?
A $\frac{8}{15}$ B $5\frac{1}{2}$ C $7\frac{1}{2}$ D 60

Answers & Explanations 7

1 A	6 A	11 C	16 A	21 A
2 D	7 C	12 D	17 A	22 D
3 C	8 A	13 B	18 D	23 B
4 D	9 B	14 C	19 B	24 B
5 C	10 A	15 A	20 B	25 C

1. **A** 6 : 7

1 Write the amounts in the order asked, tomato plants first: 12 : 14.

2 Divide both terms by their greatest common factor, 2: $12 \div 2 = 6$ and $14 \div 2 = 7$. The ratio is 6 : 7.

WHY NOT (B) reverses the order; the question asks for tomato plants to pepper plants.

2. **D** $\frac{5}{6}$

1 Think in parts: the whole group is $5 + 1 = 6$ equal parts.

2 The wins are 5 of those 6 parts, so the fraction is $\frac{5}{6}$.

WHY NOT (A) is the fraction for the other group. (B) flips the ratio.

3. **C** 40

1 Cross-multiply: $8 \times x = 16 \times 20$, so $8x = 320$.

2 Divide both sides by 8: $x = 320 \div 8 = 40$.

TIP Simplify $\frac{20}{8}$ to $\frac{5}{2}$ first to work with smaller numbers: $2x = 16 \times 5 = 80$.

WHY NOT (A) adds the difference of the denominators ($16 - 8$) instead of scaling. (D) multiplies 16 by 20 but forgets to divide by 8.

4. **D** 67

1 *Per* means “for each one,” so divide the total by the number of units: $201 \div 3 = 67$.

2 The unit rate is 67 words per minute.

TIP Check by multiplying back: $67 \times 3 = 201$.

WHY NOT (A) misplaces the decimal point when dividing.

5. **C** 10

1 Cross-multiply: $27 \times x = 18 \times 15$, so $27x = 270$.

2 Divide both sides by 27: $x = 270 \div 27 = 10$.

TIP Simplify $\frac{15}{27}$ to $\frac{5}{9}$ first to work with smaller numbers: $9x = 18 \times 5 = 90$.

WHY NOT (A) adds the difference of the denominators ($18 - 27$) instead of scaling. (D) multiplies 18 by 15 but forgets to divide by 27.

6. **A** 59

1 *Per* means “for each one,” so divide the total by the number of units: $590 \div 10 = 59$.

2 The unit rate is 59 words per minute.

TIP Check by multiplying back: $59 \times 10 = 590$.

WHY NOT (D) misplaces the decimal point when dividing.

7. **C** $\frac{5}{8}$

1 Think in parts: the whole group is $3 + 5 = 8$ equal parts.

2 The girls are 5 of those 8 parts, so the fraction is $\frac{5}{8}$.

WHY NOT (A) is the size of just one part, not the whole group’s share. (B) is the fraction for the other group. (D) uses the ratio 5 : 3 as if it were the fraction of the total.

8. **A** 1 : 6

1 Write the amounts in the order asked, sedans first: 4 : 24.

2 Divide both terms by their greatest common factor, 4: $4 \div 4 = 1$ and $24 \div 4 = 6$. The ratio is 1 : 6.

WHY NOT (C) compares the pickup trucks to the total. (D) reverses the order; the question asks for sedans to pickup trucks.

9. **B** 1 : 6

1 Both amounts must be in the same unit. Change minutes to seconds: $4 \times 60 = 240$ seconds.

2 Now compare seconds to seconds, in the order asked: 40 : 240.

3 Divide both terms by their greatest common factor, 40: $40 \div 40 = 1$ and $240 \div 40 = 6$. The ratio is 1 : 6.

WHY NOT (A) uses 100 seconds in a minute instead of 60. (C) reverses the order of the ratio. (D) forgets to change minutes to seconds before comparing.

10. **A** 16

1 Cross-multiply: $24 \times x = 6 \times 64$, so $24x = 384$.

2 Divide both sides by 24: $x = 384 \div 24 = 16$.

3 Check: $\frac{6}{16}$ and $\frac{24}{64}$ both simplify to $\frac{3}{8}$. ✓

WHY NOT (C) adds the difference of the numerators instead of scaling. (D) multiplies 6 by 64 but forgets to divide by 24.

11. **C** 18 : 7
- Both amounts must be in the same unit. Change feet to inches: $3 \times 12 = 36$ inches.
 - Now compare inches to inches, in the order asked: $36 : 14$.
 - Divide both terms by their greatest common factor, 2: $36 \div 2 = 18$ and $14 \div 2 = 7$. The ratio is 18 : 7.
- WHY NOT** (A) forgets to change feet to inches before comparing. (B) reverses the order of the ratio.
12. **D** 84
- Add the terms of the ratio to get the number of equal parts: $7 + 9 = 16$.
 - Find the size of one part: $192 \div 16 = 12$.
 - The Army recruits make up 7 parts: $7 \times 12 = 84$.
- TIP** Check: $84 + 108 = 192$, and $84 : 108$ simplifies to 7 : 9.
- WHY NOT** (A) stops after finding the size of one part.
13. **B** 120
- Add the terms of the ratio to get the number of equal parts: $9 + 8 = 17$.
 - Find the size of one part: $255 \div 17 = 15$.
 - The oranges make up 8 parts: $8 \times 15 = 120$.
- TIP** Check: $120 + 135 = 255$, and $135 : 120$ simplifies to 9 : 8.
- WHY NOT** (A) stops after finding the size of one part. (C) is the number of apples.
14. **C** 15
- Set up a proportion with the same order on both sides (bags of cement over bags of sand): $\frac{2}{4} = \frac{x}{30}$.
 - Cross-multiply: $4x = 2 \times 30 = 60$.
 - Divide by 4: $x = 60 \div 4 = 15$ bags of cement.
- TIP** Unit rate: 1 bag is enough for 2 bags of sand, so $30 \div 2 = 15$ bags.
- WHY NOT** (B) finds only the extra amount needed and forgets the original 2 bags. (D) adds the 26 extra bags of sand instead of scaling.
15. **A** 13.5
- Set up a proportion with the same order on both sides (pounds over soldiers): $\frac{9}{40} = \frac{x}{60}$.
 - Cross-multiply: $40x = 9 \times 60 = 540$.
 - Divide by 40: $x = 540 \div 40 = 13.5$ pounds.
- WHY NOT** (D) multiplies by 60 but forgets to divide by 40.
16. **A** 275 miles
- Each inch on the map stands for 50 miles, so multiply the measurement by 50.
 - Split it up: $5 \times 50 = 250$ and $\frac{1}{2} \times 50 = 25$, so the total is $250 + 25 = 275$ miles.
- WHY NOT** (D) misplaces the decimal point in the product.
17. **A** 14
- Each $\frac{1}{4}$ inch stands for 1 foot, so each whole inch stands for 4 feet.
 - Multiply: $3\frac{1}{2} \times 4 = 14$ feet ($3 \times 4 = 12$ and $\frac{1}{2} \times 4 = 2$).
- WHY NOT** (D) misplaces the decimal point in the product.
18. **D** 60
- Direct variation means $y = kx$, where k is a constant.
 - Find k by dividing: $k = \frac{y}{x} = \frac{9}{3} = 3$.
 - Use it: $y = 3 \times 20 = 60$.
- WHY NOT** (A) uses inverse variation (constant product) instead of direct variation. (B) stops after finding the constant k . (C) adds the change in x instead of multiplying by the constant.
19. **B** 66
- Add the terms of the ratio: $3 + 2 + 7 = 12$ equal parts.
 - One part is $264 \div 12 = 22$ cases.
 - Alpha platoon's share is 3 parts: $3 \times 22 = 66$ cases.
- TIP** Check: $66 + 44 + 154 = 264$.
- WHY NOT** (A) is Bravo platoon's share. (C) splits the total evenly three ways. (D) is Charlie platoon's share.
20. **B** \$4,050
- Add the terms of the ratio: $1 + 4 + 3 = 8$ equal parts.
 - One part is $10,800 \div 8 = 1,350$ dollars.
 - The largest share is 4 parts and the smallest is 1 part, a difference of $4 - 1 = 3$ parts: $3 \times 1,350 = 4,050$ dollars.
- WHY NOT** (A) is the smallest share, not the difference. (C) is the largest share, not the difference.
21. **A** 48
- In parts, the child tickets are 7 parts and the adult tickets are 1 part, so the difference is $7 - 1 = 6$ parts.
 - Those 6 parts equal 36, so one part is $36 \div 6 = 6$.
 - The total is $1 + 7 = 8$ parts: $8 \times 6 = 48$.
- TIP** Check: $42 - 6 = 36$, and $6 : 42$ simplifies to 1 : 7.
- WHY NOT** (D) treats the difference of 36 as one part, but it is 6 parts.

22. **D** 88

- 1 In parts, the trucks are 11 parts and the Humvees are 2 parts, so the difference is $11 - 2 = 9$ parts.
- 2 Those 9 parts equal 72, so one part is $72 \div 9 = 8$.
- 3 The trucks are 11 parts: $11 \times 8 = 88$.

TIP Check: $88 - 16 = 72$, and $16 : 88$ simplifies to $2 : 11$.

WHY NOT (A) is the number of Humvees.

23. **B** 192

- 1 Convert each side to feet: $4 \times 4 = 16$ feet and $3 \times 4 = 12$ feet.
- 2 Area = length $\times$ width: $16 \times 12 = 192$ square feet.

TIP Areas scale by the *square* of the scale: the plan's 12 square inches times $4^2 = 16$ gives 192.

WHY NOT (A) multiplies the plan's area by 4 instead of converting each side. (D) converts the sides and then multiplies the area by 4 again.

24. **B** $y = \frac{5}{2}x$

- 1 Direct variation has the form $y = kx$.
- 2 Find k by dividing: $k = \frac{y}{x} = \frac{10}{4} = \frac{5}{2}$.
- 3 So the equation is $y = \frac{5}{2}x$. Check: $\frac{5}{2} \times 4 = 10$. ✓

WHY NOT (A) describes inverse variation. (C) describes a constant difference, not a constant ratio. (D) divides x by y instead of y by x .

25. **C** $7\frac{1}{2}$

- 1 Find the scale factor: 30 soldiers is $\frac{30}{8} = \frac{15}{4}$ times as many as 8 soldiers.

- 2 Multiply the amount by that factor: $2 \times \frac{15}{4} = \frac{30}{4} = 7\frac{1}{2}$ cups.

TIP Unit rate: each soldier needs $2 \div 8 = \frac{1}{4}$ cup, and $30 \times \frac{1}{4} = 7\frac{1}{2}$ cups.

WHY NOT (A) multiplies by $\frac{8}{30}$ instead of $\frac{30}{8}$. (B) finds only the change in the amount. (D) multiplies by 30 but forgets to divide by 8.

Exponents & Scientific Notation

Tested on: Mathematics Knowledge

Key Concepts

An *exponent* is shorthand for repeated multiplication: $2^5 = 2 \times 2 \times 2 \times 2 \times 2 = 32$. *Scientific notation* uses powers of 10 to write very large and very small numbers compactly. Both show up on the test as quick one-step questions.

Powers, signs, and special exponents

- b^n means n factors of b : $3^4 = 3 \times 3 \times 3 \times 3 = 81$.
- The exponent applies only to what it touches: $(-3)^2 = 9$, but $-3^2 = -(3^2) = -9$. An even power of a negative number is positive; an odd power is negative.
- Zero exponent: $b^0 = 1$ (for any $b \neq 0$). Negative exponent = reciprocal: $2^{-3} = \frac{1}{2^3} = \frac{1}{8}$, and $\left(\frac{2}{3}\right)^{-2} = \left(\frac{3}{2}\right)^2 = \frac{9}{4}$.

Rules for the same base

Multiply: add exponents	$a^m \cdot a^n = a^{m+n}$	$2^3 \cdot 2^4 = 2^7$
Divide: subtract exponents	$\frac{a^m}{a^n} = a^{m-n}$	$\frac{5^9}{5^3} = 5^6$
Power of a power: multiply	$(a^m)^n = a^{mn}$	$(3^2)^4 = 3^8$
Power of a fraction	$\left(\frac{p}{q}\right)^n = \frac{p^n}{q^n}$	$\left(\frac{2}{3}\right)^3 = \frac{8}{27}$

Scientific notation

A number is in scientific notation when it is written as $m \times 10^k$ with $1 \leq m < 10$. A positive k makes a big number (move the decimal point k places right); a negative k makes a small number (move it left): $4.5 \times 10^4 = 45,000$ and $3.6 \times 10^{-3} = 0.0036$.

WORKED EXAMPLE

Write $(3 \times 10^4)(4 \times 10^5)$ in scientific notation.

Solution. Multiply the first factors and add the exponents: $3 \times 4 = 12$ and $10^4 \times 10^5 = 10^9$, giving 12×10^9 . But 12 is not between 1 and 10: $12 = 1.2 \times 10^1$, so the answer is 1.2×10^{10} .

No-Calculator Tip Count the jumps. To write 0.0052 in scientific notation, the decimal point jumps 3 places right to make 5.2, so the exponent is -3 : 5.2×10^{-3} . For 52,000 it jumps 4 places left to make 5.2, so the exponent is $+4$. Small numbers get negative exponents; big numbers get positive ones.

Common Traps

- Multiplying the base by the exponent: 3^4 is 81, not 12.
- Thinking a negative exponent makes a negative number: $2^{-3} = \frac{1}{8}$, not -8 .
- Multiplying exponents when you should add them: $2^3 \cdot 2^4 = 2^7$, not 2^{12} .
- Leaving 12×10^9 unadjusted: the first factor must be less than 10.

Practice Set 8

Score: ____ / 25

Target time: 25 minutes

Problems 1–8 are warm-ups, 9–18 are test level, and 19–25 are challenge problems.

- Evaluate 10^4 .
A 40 B 1,000 C 5,000 D 10,000
- What is the value of 5^{-2} ?
A -25 B $\frac{1}{25}$ C $\frac{1}{10}$ D 25
- Which of the following is equal to $2^7 \times 2^4$?
A 2^{28} B 4^{28} C 2^{11} D 4^{11}
- Evaluate 4^{-2} .
A -16 B $\frac{1}{16}$ C $\frac{1}{8}$ D 16
- Simplify $8^5 \times 8^8$.
A 64^{40} B 8^{40} C 8^{13} D 64^{13}
- Evaluate $3^3 + 4^3$.
A 21 B 39 C 81 D 91
- What is 9.5×10^3 written in standard form?
A 0.0095 B 28.5 C 950 D 9,500
- Which of the following shows 47,000,000 in scientific notation?
A 4.7×10^7 B 47×10^6
C 4.7×10^8 D 4.7×10^6
- $(-5)^2$ is equal to which of the following?
A -125 B -25 C -10 D 25
- Evaluate $(1.3)^2$.
A 1.69 B 2.197 C 2.6 D 16.9
- What is 0.0933 written in scientific notation?
A 9.33×10^{-3} B 9.33×10^{-1}
C 9.33×10^2 D 9.33×10^{-2}
- Multiply: $(8 \times 10^4)(2.5 \times 10^8)$. Give the answer in scientific notation.
A 2×10^{12} B 1×10^{13} C 2×10^{13} D 2×10^{33}
- What is 43,400,000 written in scientific notation?
A 4.34×10^8 B 43.4×10^6
C 4.34×10^6 D 4.34×10^7
- What is $(1.5 \times 10^2)(2 \times 10^6)$ written in scientific notation?
A 3×10^8 B 3.5×10^8
C 3×10^9 D 3×10^{12}
- Evaluate $\left(\frac{2}{5}\right)^2$.
A $\frac{2}{25}$ B $\frac{4}{25}$ C $\frac{1}{5}$ D $\frac{4}{5}$
- What is the value of $(-3)^5$?
A -318 B -268 C -243 D 243
- Write 3.48×10^8 as an ordinary number.
A 27.84 B 34,800,000
C 348,000,000 D 3,480,000,000
- Simplify $(8^4)^4 \times 8^2$.
A 8^{18} B 8^{32} C 64^{18} D 8^{10}

19. Simplify $(4.5 \times 10^{10}) \div (2 \times 10^6)$. Give the answer in scientific notation.
A 2.25×10^4 B 2.5×10^4
C 4.25×10^4 D 2.25×10^{16}
20. The number 8×10^8 is how many times as large as 2×10^4 ?
A 2.5×10^{-5} B 4×10^3
C 4×10^4 D 4×10^{12}
21. If $(3^2)^n = 3^{12}$, what is the value of n ?
A 5 B 6 C 10 D 24
22. Which number below is less than 6.1×10^5 ?
A 3.9×10^7 B 5.6×10^5
C 3.9×10^6 D 9.1×10^5
23. If $(2^2)^n = 2^6$, what is the value of n ?
A 2 B 3 C 4 D 12
24. Simplify $(6 \times 10^{-2})(5 \times 10^6)$. Give the answer in scientific notation.
A 3×10^3 B 3×10^4 C 2×10^5 D 3×10^5
25. Evaluate $6^0 + 3^{-1}$.
A $1\frac{1}{3}$ B 2 C 4 D $6\frac{1}{3}$

Answers & Explanations 8

1 D	6 D	11 D	16 C	21 B
2 B	7 D	12 C	17 C	22 B
3 C	8 A	13 D	18 A	23 B
4 B	9 D	14 A	19 A	24 D
5 C	10 A	15 B	20 C	25 A

1. **D** 10,000

1 A power of 10 is 1 followed by as many zeros as the exponent: 10^4 has 4 zeros.

2 $10^4 = 10,000$.

WHY NOT (A) multiplies the base by the exponent (10×4). (B) uses only 3 factors of 10.

2. **B** $\frac{1}{25}$

1 A negative exponent means “take the reciprocal”: $5^{-2} = \frac{1}{5^2}$.

2 $5^2 = 25$, so the answer is $\frac{1}{25}$.

WHY NOT (A) makes the answer negative; a negative exponent means a reciprocal, not a negative number. (C) multiplies the base by the exponent before taking the reciprocal. (D) forgets to take the reciprocal.

3. **C** 2^{11}

1 When you multiply powers with the same base, keep the base and *add* the exponents.

2 $2^7 \times 2^4 = 2^{7+4} = 2^{11}$.

WHY NOT (A) multiplies the exponents instead of adding them. (B) multiplies both the bases and the exponents. (D) multiplies the bases as well as adding the exponents.

4. **B** $\frac{1}{16}$

1 A negative exponent means “take the reciprocal”: $4^{-2} = \frac{1}{4^2}$.

2 $4^2 = 16$, so the answer is $\frac{1}{16}$.

WHY NOT (A) makes the answer negative; a negative exponent means a reciprocal, not a negative number. (C) multiplies the base by the exponent before taking the reciprocal. (D) forgets to take the reciprocal.

5. **C** 8^{13}

1 When you multiply powers with the same base, keep the base and *add* the exponents.

2 $8^5 \times 8^8 = 8^{5+8} = 8^{13}$.

WHY NOT (A) multiplies both the bases and the exponents. (B) multiplies the exponents instead of adding them. (D) multiplies the bases as well as adding the exponents.

6. **D** 91

1 Exponents come before multiplying, adding, or subtracting, so find each power first.

2 $3^3 = 3 \times 3 \times 3 = 27$ and $4^3 = 4 \times 4 \times 4 = 64$.

3 Then $27 + 64 = 91$.

WHY NOT (A) multiplies each base by its exponent. (B) evaluates 3^3 correctly but multiplies 4 by 3.

7. **D** 9,500

1 Multiplying by 10^3 moves the decimal point 3 places to the right.

2 9.5 becomes 9,500 (fill the empty places with zeros).

WHY NOT (A) moves the decimal point the wrong way. (B) multiplies by the exponent instead of by a power of 10. (C) moves the decimal point one place too few.

8. **A** 4.7×10^7

1 Move the decimal point to get a number that is at least 1 but less than 10: 4.7.

2 The decimal point moved 7 places to the left, which made the number smaller, so multiply by 10^7 to keep the same value.

3 So $47,000,000 = 4.7 \times 10^7$.

WHY NOT (B) is equal to 47,000,000 but is not scientific notation: the first factor must be less than 10. (C) counts one place too many. (D) counts one place too few.

9. **D** 25

1 The parentheses make -5 the base: $(-5)^2 = (-5) \times (-5)$.

2 Count the negative signs: there are 2, and an even number of negative factors gives a positive product.

3 Since $5^2 = 25$, the answer is 25.

WHY NOT (A) uses 3 factors instead of 2. (B) gets the sign wrong: an even number of negative factors gives a positive product. (C) multiplies the base by the exponent.

10. **A** 1.69

- 1 Multiply the digits as whole numbers first: $13^2 = 169$.
- 2 Count decimal places: 1.3 has 1 decimal place, and it is used as a factor 2 times, so the answer has $1 \times 2 = 2$ decimal places: 1.69.

TIP Estimate: 1.3 is more than 1, so its square must be *larger* than 1.3.

WHY NOT (B) uses 3 factors instead of 2. (C) multiplies by the exponent instead of raising to the power. (D) puts too few decimal places in the answer (it needs 2).

11. **D** 9.33×10^{-2}

- 1 Move the decimal point to get a number that is at least 1 but less than 10: 9.33.
- 2 The decimal point moved 2 places to the right, which made the number larger, so multiply by 10^{-2} to keep the same value. (Numbers less than 1 always get a negative exponent.)
- 3 So $0.0933 = 9.33 \times 10^{-2}$.

WHY NOT (A) counts one place too many. (B) counts one place too few. (C) uses a positive exponent, but numbers less than 1 have negative exponents.

12. **C** 2×10^{13}

- 1 Multiply the first factors: $8 \times 2.5 = 20$.
- 2 Multiply the powers of 10 by adding exponents: $10^4 \times 10^8 = 10^{12}$.
- 3 20 is not less than 10, so write $20 = 2 \times 10^1$: the answer is 2×10^{13} .

WHY NOT (A) rewrites 20 as 2 but forgets to add 1 to the exponent. (D) multiplies the exponents instead of adding them.

13. **D** 4.34×10^7

- 1 Move the decimal point to get a number that is at least 1 but less than 10: 4.34.
- 2 The decimal point moved 7 places to the left, which made the number smaller, so multiply by 10^7 to keep the same value.
- 3 So $43,400,000 = 4.34 \times 10^7$.

WHY NOT (A) counts one place too many. (B) is equal to 43,400,000 but is not scientific notation: the first factor must be less than 10. (C) counts one place too few.

14. **A** 3×10^8

- 1 Multiply the first factors: $1.5 \times 2 = 3$.
- 2 Multiply the powers of 10 by adding exponents: $10^2 \times 10^6 = 10^8$.
- 3 3 is already between 1 and 10, so the answer is 3×10^8 .

WHY NOT (B) adds the first factors instead of multiplying them. (C) adds 1 to the exponent when no adjustment is needed. (D) multiplies the exponents instead of adding them.

15. **B** $\frac{4}{25}$

- 1 Raise the numerator and the denominator to the power 2: $\left(\frac{2}{5}\right)^2 = \frac{2^2}{5^2}$.
- 2 $2^2 = 4$ and $5^2 = 25$, so the answer is $\frac{4}{25}$.

WHY NOT (A) raises only the denominator to the power. (D) raises only the numerator to the power.

16. **C** -243

- 1 The parentheses make -3 the base: $(-3)^5 = (-3) \times (-3) \times (-3) \times (-3) \times (-3)$.
- 2 Count the negative signs: there are 5, and an odd number of negative factors gives a negative product.
- 3 Since $3^5 = 243$, the answer is -243.

WHY NOT (D) gets the sign wrong: an odd number of negative factors gives a negative product.

17. **C** 348,000,000

- 1 Multiplying by 10^8 moves the decimal point 8 places to the right.
- 2 3.48 becomes 348,000,000 (fill the empty places with zeros).

WHY NOT (A) multiplies by the exponent instead of by a power of 10. (B) moves the decimal point one place too few. (D) moves the decimal point one place too many.

18. **A** 8^{18}

- 1 Power of a power: multiply the exponents. $(8^4)^4 = 8^{16}$.
- 2 Product of powers: add the exponents. $8^{16} \times 8^2 = 8^{18}$.

WHY NOT (B) multiplies all the exponents. (C) multiplies the bases as well. (D) adds all the exponents; the power of a power should be multiplied.

19. **A** 2.25×10^4

- 1 Divide the first factors: $4.5 \div 2 = 2.25$.
- 2 Divide the powers of 10 by subtracting exponents: $10^{10} \div 10^6 = 10^4$.
- 3 2.25 is between 1 and 10, so the answer is 2.25×10^4 .

WHY NOT (B) subtracts the first factors instead of dividing them. (D) adds the exponents instead of subtracting them.

20. **C** 4×10^4

- 1 "How many times as large" means divide: $(8 \times 10^8) \div (2 \times 10^4)$.
- 2 Divide the first factors: $8 \div 2 = 4$. Subtract the exponents: $8 - 4 = 4$.
- 3 The answer is 4×10^4 , which is 40,000.

WHY NOT (A) divides the smaller number by the larger one. (D) adds the exponents instead of subtracting them.

21. **B** 6

- 1 A power of a power multiplies the exponents: $2 \cdot n = 12$.
- 2 Divide by 2: $n = 12 \div 2 = 6$.

WHY NOT (C) subtracts the exponents; a power of a power multiplies them. (D) multiplies instead of dividing.

22. **B** 5.6×10^5

- 1 Compare the powers of 10 first. Because every first factor is between 1 and 10, the number with the larger power of 10 is always the larger number.
- 2 5.6×10^5 has the same power of 10 as 6.1×10^5 , so compare the first factors: 5.6 is smaller than 6.1.
- 3 Each of the other choices is greater than 6.1×10^5 .

WHY NOT (A) has a smaller first factor, but its power of 10 is larger. (C) has a smaller first factor, but its power of 10 is larger, so it is greater than 6.1×10^5 . (D) has the same power of 10 but a larger first factor.

23. **B** 3

- 1 A power of a power multiplies the exponents: $2 \cdot n = 6$.
- 2 Divide by 2: $n = 6 \div 2 = 3$.

WHY NOT (C) subtracts the exponents; a power of a power multiplies them. (D) multiplies instead of dividing.

24. **D** 3×10^5

- 1 Multiply the first factors: $6 \times 5 = 30$.
- 2 Add the exponents: $-2 + 6 = 4$, giving 30×10^4 .
- 3 30 is not less than 10, so write it as 3×10^1 and add 1 to the exponent: the answer is 3×10^5 .

WHY NOT (B) rewrites 30 as 3 but forgets to add 1 to the exponent.

25. **A** $1\frac{1}{3}$

- 1 Any nonzero number to the zero power is 1: $6^0 = 1$.
- 2 A negative exponent means a reciprocal: $3^{-1} = \frac{1}{3^1} = \frac{1}{3}$.
- 3 Add: $1 + \frac{1}{3} = 1\frac{1}{3}$.

WHY NOT (C) forgets to take the reciprocal of 3^1 . (D) treats 6^0 as 6.

Square Roots & Radicals

Tested on: Mathematics Knowledge

Key Concepts

The *square root* of a number is the value that, multiplied by itself, gives the number: $\sqrt{49} = 7$ because $7 \times 7 = 49$. The symbol $\sqrt{}$ always means the positive root. Knowing the perfect squares by heart makes every problem in this chapter faster.

Perfect squares and cubes to memorize

n	1	2	3	4	5	6	7	8
n^2	1	4	9	16	25	36	49	64
n^3	1	8	27	64	125	216	343	512
n	9	10	11	12	13	14	15	20
n^2	81	100	121	144	169	196	225	400

The *cube root* undoes a cube: $\sqrt[3]{125} = 5$ because $5^3 = 125$.

Working with radicals

- Simplify: pull out the *largest* perfect-square factor. $\sqrt{72} = \sqrt{36 \cdot 2} = 6\sqrt{2}$.
- Add or subtract only *like* radicals (same number under the root): $3\sqrt{5} + 4\sqrt{5} = 7\sqrt{5}$.
- Multiply the numbers under the roots: $\sqrt{6} \cdot \sqrt{15} = \sqrt{90} = 3\sqrt{10}$.
- Rationalize a denominator: multiply top and bottom by the root. $\frac{6}{\sqrt{3}} = \frac{6\sqrt{3}}{3} = 2\sqrt{3}$.

Fractions, decimals, and equations

$\sqrt{\frac{49}{64}} = \frac{7}{8}$ (root of top over root of bottom), and $\sqrt{0.09} = \sqrt{\frac{9}{100}} = \frac{3}{10} = 0.3$. The equation $x^2 = 81$ has *two* solutions, $x = 9$ and $x = -9$; read the question to see whether it wants both or only the positive one.

WORKED EXAMPLE

Simplify $\sqrt{12} + \sqrt{27}$.

Solution. The radicals are not alike yet, so simplify each: $\sqrt{12} = \sqrt{4 \cdot 3} = 2\sqrt{3}$ and $\sqrt{27} = \sqrt{9 \cdot 3} = 3\sqrt{3}$. Now they are like radicals: $2\sqrt{3} + 3\sqrt{3} = 5\sqrt{3}$.

No-Calculator Tip Estimate a root by trapping it between perfect squares: $49 < 53 < 64$, so $\sqrt{53}$ is between 7 and 8, a little above 7.

Common Traps

- Splitting a root over addition: $\sqrt{9 + 16} = \sqrt{25} = 5$, not $3 + 4 = 7$.
- Adding unlike radicals: $\sqrt{2} + \sqrt{3}$ is *not* $\sqrt{5}$.
- Decimal places: $\sqrt{0.09} = 0.3$, not 0.03 (check: $0.3^2 = 0.09$).
- Forgetting the negative solution of $x^2 = 81$ when the question asks for all solutions.

Practice Set 9

Score: ____ / 25

Target time: 25 minutes

Problems 1–8 are warm-ups, 9–18 are test level, and 19–25 are challenge problems.

- What is the value of $\sqrt{100} - \sqrt{64}$?
A 1 B 2 C 18 D 36
- Between which two consecutive whole numbers does $\sqrt{195}$ lie?
A 13 and 14 B 14 and 15
C 15 and 16 D 16 and 17
- Which of the following is equal to $\sqrt{81} + \sqrt{121}$?
A 20 B 21 C 101 D 202
- Evaluate $\sqrt{\frac{144}{49}}$.
A $\frac{12}{7}$ B 2 C $\frac{15}{7}$ D $\frac{144}{7}$
- Evaluate $\sqrt{100} \times \sqrt{4}$.
A 20 B 21 C 200 D 400
- Evaluate $\sqrt{\frac{9}{16}}$.
A $\frac{3}{16}$ B $\frac{9}{32}$ C $\frac{3}{4}$ D $\frac{9}{4}$
- A square rug has an area of 45 square feet. Its side length, in feet, is between which two consecutive whole numbers?
A 6 and 7 B 7 and 8
C 8 and 9 D 9 and 10
- What is the positive solution of $y^2 - 25 = 0$?
A -5 B 5 C 6 D 25
- Simplify $\sqrt{384}$.
A 8 B $12\sqrt{2}$ C $8\sqrt{6}$ D $64\sqrt{6}$
- What is $5\sqrt{6} + 2\sqrt{6}$?
A 7 B $5\sqrt{6}$ C $6\sqrt{6}$ D $7\sqrt{6}$
- What is $\sqrt{45}$ in simplest radical form?
A $\sqrt{5}$ B 3 C $2\sqrt{5}$ D $3\sqrt{5}$
- What is $\sqrt{10} \cdot \sqrt{2}$ in simplest form?
A $2\sqrt{5}$ B $3\sqrt{5}$ C $4\sqrt{5}$ D 20
- Which of the following is equal to $\sqrt{375}$?
A 5 B $5\sqrt{15}$ C $15\sqrt{5}$ D $25\sqrt{15}$
- Simplify $\sqrt{18} \cdot \sqrt{5}$.
A $\sqrt{10}$ B $\sqrt{23}$ C $2\sqrt{10}$ D $3\sqrt{10}$
- Which of the following is equal to $6\sqrt{6} - 4\sqrt{6}$?
A 2 B $\sqrt{6}$ C $2\sqrt{6}$ D $24\sqrt{6}$
- Which of the following is equal to $\sqrt{1\frac{9}{16}}$?
A $1\frac{1}{4}$ B $1\frac{3}{4}$ C 2 D $6\frac{1}{4}$
- A square helipad has an area of 148 square feet. To the nearest foot, how long is each side?
A 11 B 12 C 13 D 74
- If $5x^2 = 500$ and $x > 0$, what is the value of x ?
A -10 B 5 C 10 D 100
- Simplify $\sqrt{150} - \sqrt{96}$.
A $\sqrt{6}$ B $2\sqrt{6}$ C $9\sqrt{6}$ D $20\sqrt{6}$
- Simplify $\frac{24}{\sqrt{2}}$ so that there is no radical in the denominator.
A 12 B $12\sqrt{2}$ C 24 D $24\sqrt{2}$

21. What is $\sqrt{63} + \sqrt{175}$ in simplest form?
A $\sqrt{238}$ B $8\sqrt{7}$ C $8\sqrt{14}$ D $15\sqrt{7}$
22. Which of the following is equal to $\frac{16}{\sqrt{6}}$?
A $\frac{8}{3}$ B $\frac{8\sqrt{6}}{3}$ C 16 D $16\sqrt{6}$
23. What is $(2\sqrt{6})(3\sqrt{15})$ in simplest form?
A $6\sqrt{21}$ B $9\sqrt{10}$ C $15\sqrt{10}$ D $18\sqrt{10}$
24. What are all the solutions of $5x^2 = 405$?
A 9 only B 81 and -81
C 9 and -9 D -9 only
25. What is the value of $\sqrt{400 + 225}$?
A 10 B 24 C 25 D 35

Answers & Explanations 9

1 B	6 C	11 D	16 A	21 B
2 A	7 A	12 A	17 B	22 B
3 A	8 B	13 B	18 C	23 D
4 A	9 C	14 D	19 A	24 C
5 A	10 D	15 C	20 B	25 C

1. **B** 2

- Find each square root: $\sqrt{100} = 10$ and $\sqrt{64} = 8$.
- Then $10 - 8 = 2$.

WHY NOT (C) divides by 2 instead of taking the square roots. (D) forgets to take the square roots.

2. **A** 13 and 14

- Find the perfect squares on either side of 195: $13^2 = 169$ and $14^2 = 196$.
- Since $169 < 195 < 196$, the square root is between 13 and 14.

3. **A** 20

- Find each square root: $\sqrt{81} = 9$ and $\sqrt{121} = 11$.
- Then $9 + 11 = 20$.

WHY NOT (C) divides by 2 instead of taking the square roots. (D) forgets to take the square roots.

4. **A** $\frac{12}{7}$

- Take the square root of the top and of the bottom: $\sqrt{\frac{144}{49}} = \frac{\sqrt{144}}{\sqrt{49}}$.
- $\sqrt{144} = 12$ and $\sqrt{49} = 7$, so the answer is $\frac{12}{7}$.

WHY NOT (D) takes the square root of the denominator only.

5. **A** 20

- Find each square root: $\sqrt{100} = 10$ and $\sqrt{4} = 2$.
- Then $10 \times 2 = 20$.

WHY NOT (C) divides by 2 instead of taking the square roots. (D) forgets to take the square roots.

6. **C** $\frac{3}{4}$

- Take the square root of the top and of the bottom: $\sqrt{\frac{9}{16}} = \frac{\sqrt{9}}{\sqrt{16}}$.
- $\sqrt{9} = 3$ and $\sqrt{16} = 4$, so the answer is $\frac{3}{4}$.

WHY NOT (A) takes the square root of the numerator only. (B) divides by 2 instead of taking the square root. (D) takes the square root of the denominator only.

7. **A** 6 and 7

- Find the perfect squares on either side of 45: $6^2 = 36$ and $7^2 = 49$.
- Since $36 < 45 < 49$, the square root is between 6 and 7.

8. **B** 5

- Add 25 to both sides: $y^2 = 25$.
- Undo the square by taking the square root: $y = \sqrt{25}$ or $y = -\sqrt{25}$.
- $\sqrt{25} = 5$, and the positive solution is $y = 5$.

WHY NOT (A) is the negative solution, but the question asks for the positive one. (D) forgets to take the square root.

9. **C** $8\sqrt{6}$

- Find the largest perfect square that divides 384: it is 64, because $384 = 64 \times 6$.
 - Split the root: $\sqrt{384} = \sqrt{64} \times \sqrt{6} = 8\sqrt{6}$.
- TIP** Check: $(8\sqrt{6})^2 = 64 \times 6 = 384$. ✓
- WHY NOT** (A) drops the leftover $\sqrt{6}$. (B) swaps the number outside the radical with the number inside. (D) pulls out 64 instead of its square root, 8.

10. **D** $7\sqrt{6}$

- Both terms have $\sqrt{6}$, so they are like radicals: combine the numbers in front, just like $5x + 2x$.
- $5 + 2 = 7$, so the answer is $7\sqrt{6}$. The $\sqrt{6}$ stays the same.

WHY NOT (A) drops the $\sqrt{6}$ after subtracting.

11. **D** $3\sqrt{5}$

- Find the largest perfect square that divides 45: it is 9, because $45 = 9 \times 5$.
 - Split the root: $\sqrt{45} = \sqrt{9} \times \sqrt{5} = 3\sqrt{5}$.
- TIP** Check: $(3\sqrt{5})^2 = 9 \times 5 = 45$. ✓

WHY NOT (B) drops the leftover $\sqrt{5}$.

12. **A** $2\sqrt{5}$

- Multiply the numbers under the radicals: $\sqrt{10} \cdot \sqrt{2} = \sqrt{10 \times 2} = \sqrt{20}$.
- Simplify: $\sqrt{20} = \sqrt{4 \cdot 5} = 2\sqrt{5}$.

WHY NOT (C) pulls out 4 instead of its square root, 2. (D) drops the radical sign.

13. **B** $5\sqrt{15}$
- Find the largest perfect square that divides 375: it is 25, because $375 = 25 \times 15$.
 - Split the root: $\sqrt{375} = \sqrt{25 \times 15} = 5\sqrt{15}$.
- TIP** Check: $(5\sqrt{15})^2 = 25 \times 15 = 375$. ✓
- WHY NOT** (A) drops the leftover $\sqrt{15}$. (C) swaps the number outside the radical with the number inside. (D) pulls out 25 instead of its square root, 5.
14. **D** $3\sqrt{10}$
- Multiply the numbers under the radicals: $\sqrt{18} \cdot \sqrt{5} = \sqrt{18 \times 5} = \sqrt{90}$.
 - Simplify: $\sqrt{90} = \sqrt{9 \cdot 10} = 3\sqrt{10}$.
- WHY NOT** (B) adds the numbers under the radicals instead of multiplying.
15. **C** $2\sqrt{6}$
- Both terms have $\sqrt{6}$, so they are like radicals: combine the numbers in front, just like $6x - 4x$.
 - $6 - 4 = 2$, so the answer is $2\sqrt{6}$. The $\sqrt{6}$ stays the same.
- WHY NOT** (A) drops the $\sqrt{6}$ after subtracting. (D) multiplies the numbers in front instead of combining them.
16. **A** $1\frac{1}{4}$
- Change the mixed number to an improper fraction: $1\frac{9}{16} = \frac{25}{16}$.
 - Take the root of the top and the bottom: $\sqrt{\frac{25}{16}} = \frac{5}{4}$.
 - As a mixed number, $\frac{5}{4} = 1\frac{1}{4}$. Check: $\frac{5}{4} \times \frac{5}{4} = \frac{25}{16}$. ✓
- WHY NOT** (B) takes the square root of the fraction part only. (D) changes to an improper fraction, then takes the root of the denominator only.
17. **B** 12
- Trap the root between perfect squares: $12^2 = 144$ and $13^2 = 169$, so $\sqrt{148}$ is between 12 and 13.
 - Compare distances: $148 - 144 = 4$ and $169 - 148 = 21$. 148 is closer to 144.
 - So $\sqrt{148}$ is closest to 12.
- WHY NOT** (C) is the other neighbor, but 148 is closer to 144. (D) divides 148 by 2 instead of taking the square root.
18. **C** 10
- Divide both sides by 5: $x^2 = 100$.
 - Take the square root: $x = 10$ or $x = -10$. Since $x > 0$, $x = 10$.
- WHY NOT** (A) is the negative solution, but the question says $x > 0$. (D) forgets to take the square root.
19. **A** $\sqrt{6}$
- The numbers under the radicals are different, so simplify each radical first.
 - $\sqrt{150} = \sqrt{25 \cdot 6} = 5\sqrt{6}$ and $\sqrt{96} = \sqrt{16 \cdot 6} = 4\sqrt{6}$.
 - Now they are like radicals: $5\sqrt{6} - 4\sqrt{6} = \sqrt{6}$.
- WHY NOT** (C) adds instead of subtracting. (D) multiplies the numbers in front instead of subtracting.
20. **B** $12\sqrt{2}$
- Multiply the top and bottom by $\sqrt{2}$: $\frac{24}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}} = \frac{24\sqrt{2}}{2}$, because $\sqrt{2} \cdot \sqrt{2} = 2$.
 - Simplify $\frac{24}{2}$: divide top and bottom by 2, giving $12\sqrt{2}$.
- WHY NOT** (A) multiplies only the bottom by $\sqrt{2}$. (C) cancels the 2 on the bottom against the $\sqrt{2}$ on top. (D) multiplies the top by $\sqrt{2}$ but forgets that the bottom becomes 2.
21. **B** $8\sqrt{7}$
- The numbers under the radicals are different, so simplify each radical first.
 - $\sqrt{63} = \sqrt{9 \cdot 7} = 3\sqrt{7}$ and $\sqrt{175} = \sqrt{25 \cdot 7} = 5\sqrt{7}$.
 - Now they are like radicals: $3\sqrt{7} + 5\sqrt{7} = 8\sqrt{7}$.
- WHY NOT** (A) adds the numbers under the radicals; roots of different numbers cannot be combined that way. (C) adds the numbers under the radicals after simplifying. (D) multiplies the numbers in front instead of adding.
22. **B** $\frac{8\sqrt{6}}{3}$
- Multiply the top and bottom by $\sqrt{6}$: $\frac{16}{\sqrt{6}} \times \frac{\sqrt{6}}{\sqrt{6}} = \frac{16\sqrt{6}}{6}$, because $\sqrt{6} \cdot \sqrt{6} = 6$.
 - Simplify $\frac{16}{6}$: divide top and bottom by 2, giving $\frac{8\sqrt{6}}{3}$.
- WHY NOT** (A) multiplies only the bottom by $\sqrt{6}$. (C) cancels the 6 on the bottom against the $\sqrt{6}$ on top. (D) multiplies the top by $\sqrt{6}$ but forgets that the bottom becomes 6.
23. **D** $18\sqrt{10}$
- Multiply the numbers in front and the numbers under the radicals separately: $2 \times 3 = 6$ and $\sqrt{6} \cdot \sqrt{15} = \sqrt{90}$.
 - Simplify the radical: $\sqrt{90} = \sqrt{9 \cdot 10} = 3\sqrt{10}$.
 - Multiply: $6 \times 3\sqrt{10} = 18\sqrt{10}$.
- WHY NOT** (A) adds the numbers under the radicals. (B) multiplies by only one of the numbers in front. (C) adds the numbers in front instead of multiplying them.

24. **C** 9 and -9

- 1 Divide both sides by 5: $x^2 = 81$.
- 2 Take the square root of both sides. A positive number has two square roots: $x = 9$ or $x = -9$.
- 3 Check: $5(9)^2 = 405$, and $(-9)^2$ is also 81. ✓

WHY NOT (A) forgets the negative solution. (B) forgets to take the square root. (D) keeps only the negative solution.

25. **C** 25

- 1 Work inside the root first: $400 + 225 = 625$.
- 2 Then $\sqrt{625} = 25$, because $25 \times 25 = 625$.

WHY NOT (D) takes the square root of each number and adds $(20 + 15)$; a root does not split over addition.

PART II

Algebra

Expressions, equations, inequalities, systems, polynomials, and quadratics: the algebra on the Mathematics Knowledge subtest.

- 10 Algebraic Expressions
- 11 Linear Equations
- 12 Inequalities
- 13 Systems of Equations
- 14 Polynomials & Exponent Rules
- 15 Factoring & Quadratic Equations

Algebraic Expressions

Tested on: Mathematics Knowledge • Arithmetic Reasoning

Key Concepts

An *algebraic expression* combines numbers and letters (variables) with operations, such as $3x + 5$ or $2a^2 - 3b$. This chapter covers the four skills the test checks most: plugging in values, combining like terms, distributing, and turning words into expressions.

Vocabulary

In $4x^2 - 7x + 9$ there are three *terms*. The number in front of a variable is its *coefficient* (4 and -7); 9 is the *constant*. *Like terms* have exactly the same variable part: $5x$ and $-2x$ are like terms, but $5x$ and $5x^2$ are not, and neither are $3x$ and $3y$.

Evaluating: substitute, then follow the order of operations

Replace each variable with its value *in parentheses*, then do powers, then multiplication, then addition and subtraction. For $a = -2$ and $b = 4$:

$$2a^2 - 3b = 2(-2)^2 - 3(4) = 2(4) - 12 = 8 - 12 = -4.$$

Note that $(-2)^2 = 4$, but $-x^2$ with $x = 2$ is $-(2^2) = -4$.

Simplifying

- Combine like terms by adding their coefficients: $5x + 3y - 2x + 7y = 3x + 10y$.
- Distribute: multiply *every* term inside the parentheses, signs included. $-2(x + 5) = -2x - 10$ and $-(x - 4) = -x + 4$.

Words into symbols

sum, more than, increased by	+
difference, less than, decreased by	-
product, times, twice	×
quotient, divided by, half of	÷

“5 less than twice a number” is $2n - 5$: *less than* reverses the order, so it is *not* $5 - 2n$.

WORKED EXAMPLE

Simplify $3(2x - 4) - 2(x + 5)$.

Solution. Distribute each number: $3(2x - 4) = 6x - 12$ and $-2(x + 5) = -2x - 10$. Combine like terms: $6x - 2x = 4x$ and $-12 - 10 = -22$. The result is $4x - 22$.

No-Calculator Tip Check a simplification by plugging in an easy number. With $x = 1$: $3(2 - 4) - 2(1 + 5) = -6 - 12 = -18$, and $4(1) - 22 = -18$ too.

Common Traps

- Squaring a negative without parentheses: $(-3)^2 = 9$, not -9 .
- Distributing a minus sign to only the first term: $-2(x + 5)$ is $-2x - 10$, not $-2x + 10$.
- Combining unlike terms: $3x + 4y$ cannot become $7xy$.

Practice Set 10

Score: ____ / 25

Target time: 26 minutes

Problems 1–8 are warm-ups, 9–18 are test level, and 19–25 are challenge problems.

- What is the value of $x^2 + 1$ when $x = 4$?
A 17 B 23 C 25 D 34
- Combine like terms: $4a + 8b + 5a - 2b$
A $9a + 10b$ B $9a + 6b$ C $10a + 6b$ D $15ab$
- What is the value of $8x - 15$ when $x = 9$?
A 2 B 47 C 57 D 74
- Simplify: $5a + 8b + a - 4b$
A $5a + 4b$ B $6a + 12b$ C $6a + 4b$ D $10ab$
- If n stands for a number, which expression represents “3 less than a number”?
A $3 - n$ B $3n$ C $n + 3$ D $n - 3$
- Which expression means “the quotient of a number and 12”?
A $n - 12$ B $12n$ C $\frac{12}{n}$ D $\frac{n}{12}$
- The perimeter of a rectangle is given by $P = 2l + 2w$. What is P when $l = 13$ and $w = 4$?
A 17 B 34 C 52 D 104
- A plumber charges $C = 75 + 50h$ dollars for a job that takes h hours. What is C when $h = 5$?
A 250 B 325 C 425 D 625
- What is the value of $4x^2 + 6x$ when $x = -2$?
A 4 B 6 C 28 D 52
- Which expression is equivalent to $4(2x - 9) - 2(2x + 7)$?
A $4x - 22$ B $12x - 22$
C $4x - 16$ D $4x - 50$
- If $x = -3$, what is the value of $5x^2 + 4x$?
A 33 B 43 C 57 D 213
- Simplify: $6(3x + 4) - 4(x - 6)$
A $14x + 48$ B $22x$
C $14x + 10$ D $14x$
- Combine like terms: $6x^2 + 8x - 3x^2 - 5x + 5$
A $9x^2 + 3x + 5$ B $6x^2 + 5$
C $3x^2 + 13x + 5$ D $3x^2 + 3x + 5$
- Which expression means “8 times the difference of a number and 14”?
A $8n - 14$ B $8(n - 14)$
C $8 - n - 14$ D $8(14 - n)$
- Which expression means “the square of a number, increased by 5”?
A $2n + 5$ B $5n^2$ C $n^2 + 5$ D $(n + 5)^2$
- Shirts cost \$11 each and hats cost \$8 each. Which expression gives the total cost, in dollars, of x shirts and y hats?
A $11x + 8y$ B $8x + 11y$
C $19xy$ D $19(x + y)$

17. A warehouse worker stacks p pallets with 48 boxes on each pallet and then adds 13 more boxes. Which expression gives the total number of boxes?
- A $48(p + 13)$ B $13p + 48$
C $61p$ D $48p + 13$
18. The formula $C = \frac{5}{9}(F - 32)$ converts a Fahrenheit temperature F to Celsius. What is 23°F in degrees Celsius?
- A -9 B -5 C -4 D -3
19. Which expression is equivalent to $x(x - 5) - 5(x + 7)$?
- A $x^2 - 10x - 35$ B $x^2 - 10x + 35$
C $x^2 - 35$ D $-8x - 35$
20. If $x = -3$, what is the value of $-x^2 - 3x + 2$?
- A -16 B -2 C 2 D 20
21. Which of the following is equal to $4(2x - 2) - (3x - 3)$?
- A $5x + 1$ B $11x - 11$
C $5x - 5$ D $5x - 11$
22. If $x = -1$ and $y = 3$, what is the value of $2x^2 - 6xy$?
- A -16 B 16 C 20 D 22
23. The area of a triangle is $A = \frac{1}{2}bh$. If $A = 80$ and $b = 16$, what is h ?
- A 2.5 B 5 C 6 D 10
24. A rectangle has a length of $2x + 6$ and a width of $3x + 2$. Which expression represents its perimeter?
- A $6x^2 + 22x + 12$ B $10x + 16$
C $10x + 8$ D $5x + 8$
25. Ryan has \$90. He buys x items that cost \$15 each. Which expression gives the amount of money, in dollars, he has left?
- A $90 - 15x$ B $15x - 90$
C $75x$ D $90 + 15x$

Answers & Explanations 10

1 A	6 D	11 A	16 A	21 C
2 B	7 B	12 A	17 D	22 C
3 C	8 B	13 D	18 B	23 D
4 C	9 A	14 B	19 A	24 B
5 D	10 D	15 C	20 C	25 A

- A** 17
 - Substitute $x = 4$: $(4)^2 + 1$.
 - Find the power first: $16 + 1$.
 - Then add or subtract: $16 + 1 = 17$.

WHY NOT (C) adds before squaring.
- B** $9a + 6b$
 - Group the like terms: a -terms $4a + 5a$ and b -terms $8b - 2b$.
 - Add the coefficients: $4 + 5 = 9$ and $8 - 2 = 6$.
 - The simplified expression is $9a + 6b$.

WHY NOT (A) ignores the minus sign in front of $2b$. (D) combines unlike terms; a -terms and b -terms cannot be added together.
- C** 57
 - Substitute $x = 9$: $8(9) - 15$.
 - Multiply first: $72 - 15$.
 - Then add or subtract: $72 - 15 = 57$.

WHY NOT (A) adds 8 and 9 instead of multiplying. (D) reads $8x$ as the two-digit number 89.
- C** $6a + 4b$
 - Group the like terms: a -terms $5a + a$ and b -terms $8b - 4b$.
 - Add the coefficients: $5 + 1 = 6$ and $8 - 4 = 4$.
 - The simplified expression is $6a + 4b$.

WHY NOT (A) treats a as 0; a variable by itself has coefficient 1. (B) ignores the minus sign in front of $4b$. (D) combines unlike terms; a -terms and b -terms cannot be added together.
- D** $n - 3$
 - "3 less than a number" means start with the number and take 3 away. The order is the reverse of the words.
 - With n for the number, the expression is $n - 3$.

WHY NOT (A) reverses the order; *less than* means subtract from the number. (C) adds instead of subtracting.
- D** $\frac{n}{12}$
 - "Quotient" means divide, in the order written: the number divided by 12.
 - With n for the number, the expression is $\frac{n}{12}$.

WHY NOT (B) multiplies instead of dividing. (C) reverses the order of the division.
- B** 34
 - Substitute: $P = 2(13) + 2(4)$.
 - Multiply, then add: $26 + 8 = 34$.

WHY NOT (A) adds one length and one width only. (C) finds the area instead of the perimeter.
- B** 325
 - Substitute: $C = 75 + 50(5)$.
 - Multiply first: $50 \times 5 = 250$. Then add: $75 + 250 = 325$.

WHY NOT (A) forgets to add the 75. (C) multiplies 75 by h instead of 50. (D) adds before multiplying.
- A** 4
 - Substitute, putting each value in parentheses: $4(-2)^2 + 6(-2)$.
 - Powers first: $(-2)^2 = 4$.
 - Multiply each term (watch the signs): $16 - 12$.
 - Add and subtract from left to right: $16 - 12 = 4$.

WHY NOT (D) squares $4x$ instead of only x .
- D** $4x - 50$
 - Distribute 4: $4(2x - 9) = 8x - 36$.
 - Distribute -2 (the minus sign comes with it): $-2(2x + 7) = -4x - 14$.
 - Combine like terms: $8x - 4x = 4x$ and $-36 - 14 = -50$. The result is $4x - 50$.

TIP Check with $x = 1$: the original and $4x - 50$ both equal -46 .

WHY NOT (A) multiplies 7 by 2 instead of -2 ; the minus sign goes to every term. (B) adds the second product instead of subtracting it. (C) multiplies only the first term in each set of parentheses.

11. **A** 33

- 1 Substitute, putting each value in parentheses: $5(-3)^2 + 4(-3)$.
- 2 Powers first: $(-3)^2 = 9$.
- 3 Multiply each term (watch the signs): $45 - 12$.
- 4 Add and subtract from left to right: $45 - 12 = 33$.

WHY NOT (D) squares $5x$ instead of only x .

 12. **A** $14x + 48$

- 1 Distribute 6: $6(3x + 4) = 18x + 24$.
- 2 Distribute -4 (the minus sign comes with it): $-4(x - 6) = -4x + 24$.
- 3 Combine like terms: $18x - 4x = 14x$ and $24 + 24 = 48$. The result is $14x + 48$.

TIP Check with $x = 1$: the original and $14x + 48$ both equal 62.

WHY NOT (B) adds the second product instead of subtracting it. (C) multiplies only the first term in each set of parentheses. (D) multiplies -6 by 4 instead of -4 ; the minus sign goes to every term.

 13. **D** $3x^2 + 3x + 5$

- 1 Like terms have the same variable part. The x^2 -terms are $6x^2 - 3x^2$; the x -terms are $8x - 5x$; the constant is 5.
- 2 Add the coefficients: $6 - 3 = 3$ and $8 - 5 = 3$. The exponent stays the same.
- 3 The simplified expression is $3x^2 + 3x + 5$.

WHY NOT (A) ignores the minus sign in front of $3x^2$. (B) treats x^2 and x as like terms. (C) ignores the minus sign in front of $5x$.

 14. **B** $8(n - 14)$

- 1 "The difference of a number and 14" is $n - 14$. It is one quantity, so keep it in parentheses and multiply it by 8.
- 2 With n for the number, the expression is $8(n - 14)$.

WHY NOT (A) multiplies only the number, not the whole difference. (D) reverses the order of the difference.

 15. **C** $n^2 + 5$

- 1 "The square of a number" is n^2 ; "increased by 5" adds 5 to that.
- 2 With n for the number, the expression is $n^2 + 5$.

WHY NOT (A) doubles the number instead of squaring it. (D) adds before squaring.

 16. **A** $11x + 8y$

- 1 Each shirt costs \$11, so x of them cost $11x$ dollars.
- 2 Each hat costs \$8, so y of them cost $8y$ dollars.
- 3 Add the two costs: $11x + 8y$.

WHY NOT (B) matches each price with the wrong item. (C) combines unlike terms. (D) charges both prices for every item.

 17. **D** $48p + 13$

- 1 p groups of 48 make $48p$.
- 2 Add the 13 extra: $48p + 13$.

WHY NOT (A) multiplies the 13 extra by 48 too. (B) swaps the two numbers. (C) multiplies the 13 extra by p too.

 18. **B** -5

- 1 Parentheses first: $23 - 32 = -9$.
- 2 Multiply by $\frac{5}{9}$: $-9 \div 9 = -1$, and $-1 \times 5 = -5$.
- 3 The temperature is -5°C .

WHY NOT (A) forgets to multiply by $\frac{5}{9}$.

 19. **A** $x^2 - 10x - 35$

- 1 Distribute x : $x(x - 5) = x^2 - 5x$.
- 2 Distribute -5 : $-5(x + 7) = -5x - 35$.
- 3 Combine the x -terms: $-5x - 5x = -10x$. The result is $x^2 - 10x - 35$.

TIP Check with $x = 1$: the original and $x^2 - 10x - 35$ both equal -44 .

WHY NOT (B) multiplies 7 by 5 instead of -5 . (C) adds $5x$ instead of subtracting it. (D) writes $x \cdot x$ as $2x$ instead of x^2 .

 20. **C** 2

- 1 Substitute, putting each value in parentheses: $-(-3)^2 - 3(-3) + 2$.
- 2 Powers first: $(-3)^2 = 9$.
- 3 Multiply each term (watch the signs): $-9 + 9 + 2$.
- 4 Add and subtract from left to right: $-9 + 9 + 2 = 2$.

WHY NOT (A) forgets that a negative times a negative is positive. (D) treats $(-3)^2$ as -9 .

 21. **C** $5x - 5$

- 1 Distribute 4: $4(2x - 2) = 8x - 8$.
- 2 A minus sign in front of parentheses means multiply by -1 : $-(3x - 3) = -3x + 3$. Both signs change.
- 3 Combine like terms: $5x - 5$.

TIP Check with $x = 1$: the original and $5x - 5$ both equal 0.

WHY NOT (A) multiplies only $2x$ by 4. (B) drops the parentheses without changing any signs. (D) changes the sign of only the first term inside the second parentheses.

22. **C** 20

- 1 Substitute, putting each value in parentheses: $2(-1)^2 - 6(-1)(3)$.
- 2 Powers first: $(-1)^2 = 1$.
- 3 Multiply each term (watch the signs): $2 + 18$.
- 4 Add and subtract from left to right: $2 + 18 = 20$.

WHY NOT (A) forgets that a negative times a negative is positive. (B) treats $(-1)^2$ as -1 . (D) squares $2x$ instead of only x .

23. **D** 10

- 1 Substitute: $80 = \frac{1}{2}(16)h$, so $80 = 8h$.
- 2 Divide by 8: $h = 10$.
- 3 Check: $\frac{1}{2} \times 16 \times 10 = 80$. ✓

WHY NOT (A) divides by 2 instead of multiplying by 2. (B) divides by 16 but forgets the $\frac{1}{2}$.

24. **B** $10x + 16$

- 1 Perimeter = $2(\text{length}) + 2(\text{width}) = 2(2x + 6) + 2(3x + 2)$.
- 2 Distribute: $4x + 12 + 6x + 4$.
- 3 Combine like terms: $10x + 16$.

WHY NOT (A) multiplies the length by the width, which gives the area. (C) doubles the x -terms but not the constants. (D) adds one length and one width only.

25. **A** $90 - 15x$

- 1 The items cost $15x$ dollars in all.
- 2 Subtract that from the starting amount: $90 - 15x$.

WHY NOT (B) subtracts in the wrong order. (C) subtracts the price from the starting amount first. (D) adds the cost instead of subtracting it.

Linear Equations

Tested on: Mathematics Knowledge • Arithmetic Reasoning

Key Concepts

An *equation* says that two expressions are equal. To *solve* it, find the value of the variable that makes both sides equal. The golden rule: **whatever you do to one side, do to the other side**, so the two sides stay balanced.

Undo operations in reverse order

Get the variable alone by undoing what was done to it, last operation first: addition $\leftrightarrow$ subtraction, multiplication $\leftrightarrow$ division.

$$3x - 7 = 11 \Rightarrow 3x = 18 \Rightarrow x = 6$$

(add 7 to both sides, then divide both sides by 3). **Check** by substituting: $3(6) - 7 = 18 - 7 = 11$. ✓

Clean up first

- **Parentheses:** distribute to *every* term inside: $4(x - 2) = 4x - 8$ and $-2(x + 5) = -2x - 10$.
- **Variables on both sides:** add or subtract to move all the x -terms to one side and all the plain numbers to the other.
- **Fractions:** multiply *every* term by the least common denominator (LCD). **Decimals:** multiply every term by 10 or 100.
- **Formulas:** to solve $d = rt$ for t , treat the other letters like numbers and divide: $t = \frac{d}{r}$.

Translating words into equations

is, equals, gives, the result is	=
more than, increased by, sum, plus	+
less than, decreased by, difference	-
times, twice, tripled, product	×
consecutive integers	$n, n + 1, n + 2, \dots$
consecutive even (or odd) integers	$n, n + 2, n + 4, \dots$

Careful: "5 less than x " is $x - 5$, *not* $5 - x$.

WORKED EXAMPLE

Solve $2(x + 3) = 5x - 9$.

Solution. Distribute: $2x + 6 = 5x - 9$. Subtract $2x$ from both sides: $6 = 3x - 9$. Add 9 to both sides: $15 = 3x$. Divide by 3: $x = 5$. Check: $2(5 + 3) = 16$ and $5(5) - 9 = 16$. ✓

No-Calculator Tip Multiple choice lets you *backsolve*: plug each choice into the equation; the one that makes both sides equal is the answer. When x -terms are on both sides, move them toward the side with the larger x -coefficient so that coefficient stays positive.

Common Traps

- Distributing to only the first term: $3(x + 4)$ is $3x + 12$, not $3x + 4$.
- Moving a term across the equal sign without changing its sign.
- Answering x when the question asks for something else, such as $2x + 1$.
- Reversing “less than”: “7 less than $3n$ ” is $3n - 7$.

Practice Set 11

Score: ____ / 25

Target time: 26 minutes

Problems 1–8 are warm-ups, 9–18 are test level, and 19–25 are challenge problems.

- If $\frac{x}{4} = 13$, what is the value of x ?
A $\frac{13}{4}$ B 52 C 53 D 54
- “7 more than six times a number is 91.” If n stands for the number, which equation represents this sentence?
A $6n + 7 = 91$ B $6n - 7 = 91$
C $7n + 6 = 91$ D $6(n + 7) = 91$
- Solve for x : $\frac{x}{8} = 9$.
A 1 B $\frac{9}{8}$ C 17 D 72
- When 2 is added to six times a number, the result is 80. What is the number?
A 11 B $\frac{34}{3}$ C 12 D 13
- What value of x makes $x - 21 = -12$ true?
A -33 B 8 C 9 D 33
- If 5 is subtracted from three times a number, the result is 37. What is the number?
A 14 B 15 C $\frac{52}{3}$ D 42
- “7 more than twice a number is 37.” If n stands for the number, which equation represents this sentence?
A $2n + 7 = 37$ B $2(n + 7) = 37$
C $7n + 2 = 37$ D $2n - 7 = 37$
- If $\frac{x}{2} + 8 = 16$, what is the value of x ?
A 4 B 16 C 24 D 48
- What value of x makes $9x + 6 = 3x - 30$ true?
A -36 B -6 C -3 D 6
- Specialist Murphy reserved a training range for a \$300 flat fee plus \$5 per soldier for ammunition. The total bill was \$450. How many soldiers trained?
A 30 soldiers B 32 soldiers
C 90 soldiers D 150 soldiers
- What value of x makes $7x + 2 = 3x + 10$ true?
A -2 B $\frac{4}{5}$ C 2 D 3
- A parking garage charges \$4 for the first hour and \$3 for each additional hour. Rosa paid \$16 to park. For how many hours did she park?
A 4 hours B 5 hours C 6 hours D 7 hours
- If $30 - 6x = 72$, what is the value of x ?
A -17 B -9 C -8 D -7
- If $7(x - 3) + 13 = 48$, what is the value of x ?
A 2 B $\frac{38}{7}$ C 7 D 8
- If $2.5x + 3.6 = 11.1$, what is the value of x ?
A 0.3 B 1 C 2 D 3

16. If $5x - y = 8$, which of the following expresses y in terms of x ?
- A $y = 5x + 8$ B $y = -5x + 8$
C $y = 5x - 8$ D $y = -5x - 8$
17. If $4x + 9 = 17$, then $2x - 8$ equals which of the following?
- A -4 B 2 C 4 D 5
18. The sum of three consecutive integers is 123. What is the middle of these integers?
- A 37 B 40 C 41 D 42
19. If $6(x - 3) - 4(x + 8) = -26$, what is the value of x ?
- A -20 B $\frac{12}{5}$ C 12 D 13
20. If $\frac{x}{4} - \frac{x}{10} = 12$, what is the value of x ?
- A 72 B 80 C 81 D 480
21. Solve $-9x + 5y = -15$ for y .
- A $y = -\frac{9}{5}x - 3$ B $y = \frac{9}{5}x - 3$
C $y = 9x - 15$ D $y = 9x - 3$
22. If $2x - y = -9$, what is the value of $8x - 4y$?
- A -40 B -36 C -9 D -5
23. If $9 - 5x = 2x + 16$, what is the value of x ?
- A -7 B $-\frac{25}{7}$ C $-\frac{7}{3}$ D -1
24. The sum of four consecutive even integers is 212. What is the largest of these integers?
- A 50 B 52 C 56 D 62
25. "A number divided by 9, plus 9, equals 20." If n stands for the number, which equation represents this sentence?
- A $\frac{n}{9} + 9 = 20$ B $\frac{9}{n} + 9 = 20$
C $9n + 9 = 20$ D $\frac{n+9}{9} = 20$

Answers & Explanations 11

1 B	6 A	11 C	16 C	21 B
2 A	7 A	12 B	17 A	22 B
3 D	8 B	13 D	18 C	23 D
4 D	9 B	14 D	19 C	24 C
5 C	10 A	15 D	20 B	25 A

1. **B** 52

1 $\frac{x}{4}$ means x divided by 4. Undo the division by multiplying both sides by 4.

2 $x = 13 \times 4 = 52$.

WHY NOT (A) divides by 4 instead of multiplying.

2. **A** $6n + 7 = 91$

1 Translate piece by piece. Let n be the number; “is,” “gives,” and “equals” all become “=.”

2 “Six times a number” is $6n$. “7 more than” that means add 7: $6n + 7$.

3 The equation is $6n + 7 = 91$.

WHY NOT (B) subtracts instead of adding. (C) mixes up which number multiplies n . (D) adds 7 before multiplying by 6.

3. **D** 72

1 $\frac{x}{8}$ means x divided by 8. Undo the division by multiplying both sides by 8.

2 $x = 9 \times 8 = 72$.

WHY NOT (A) subtracts 8 instead of multiplying by it. (B) divides by 8 instead of multiplying. (C) adds 8 instead of multiplying by it.

4. **D** 13

1 Let n be the number. Write the words as an equation: $6n + 2 = 80$.

2 Undo the addition: subtract 2 from both sides. $6n = 80 - 2 = 78$.

3 Divide both sides by 6: $n = \frac{78}{6} = 13$.

WHY NOT (B) divides 80 by 6 but forgets to divide 2.

5. **C** 9

1 The equation subtracts 21 from x . Undo that by adding 21 to both sides: $x = -12 + 21 = 9$.

2 Check: $9 - 21 = -12$. ✓

WHY NOT (A) subtracts 21 from both sides instead of adding it.

6. **A** 14

1 Let n be the number. Write the words as an equation: $3n - 5 = 37$.

2 Undo the subtraction: add 5 to both sides. $3n = 37 + 5 = 42$.

3 Divide both sides by 3: $n = \frac{42}{3} = 14$.

WHY NOT (C) divides 37 by 3 but forgets to divide 5. (D) forgets to divide by 3.

7. **A** $2n + 7 = 37$

1 Translate piece by piece. Let n be the number; “is,” “gives,” and “equals” all become “=.”

2 “Twice a number” is $2n$. “7 more than” that means add 7: $2n + 7$.

3 The equation is $2n + 7 = 37$.

WHY NOT (B) adds 7 before multiplying by 2. (C) mixes up which number multiplies n . (D) subtracts instead of adding.

8. **B** 16

1 Undo the addition: subtract 8 from both sides. $\frac{x}{2} = 16 - 8 = 8$.

2 Undo the division: multiply both sides by 2. $x = 8 \times 2 = 16$.

3 Check: $\frac{16}{2} + 8 = 8 + 8 = 16$. ✓

WHY NOT (A) divides by 2 instead of multiplying. (C) multiplies 16 by 2 but forgets to multiply 8. (D) moves 8 across without changing its sign.

9. **B** -6

1 Get the x -terms on the left: subtract $3x$ from both sides. $6x + 6 = -30$.

2 Undo the addition: subtract 6 from both sides. $6x = -30 - 6 = -36$.

3 Undo the multiplication: divide both sides by 6. $x = \frac{-36}{6} = -6$.

4 Check with $x = -6$: the left side is $9(-6) + 6 = -48$ and the right side is $3(-6) - 30 = -48$. ✓

WHY NOT (A) forgets to divide by 6. (C) adds the x -terms instead of subtracting one from the other. (D) makes a sign error when moving the terms.

10. **A** 30 soldiers

- Let s be the number of soldiers. The total is the fixed fee plus \$5 for each soldier: $300 + 5s = 450$.
- Subtract the fee from both sides: $5s = 450 - 300 = 150$.
- Divide by the rate: $s = 150 \div 5 = 30$ soldiers.

WHY NOT (C) ignores the fixed fee. (D) adds the fee instead of subtracting it.

 11. **C** 2

- Get the x -terms on the left: subtract $3x$ from both sides. $4x + 2 = 10$.
- Undo the addition: subtract 2 from both sides. $4x = 10 - 2 = 8$.
- Undo the multiplication: divide both sides by 4. $x = \frac{8}{4} = 2$.
- Check with $x = 2$: the left side is $7(2) + 2 = 16$ and the right side is $3(2) + 10 = 16$. ✓

WHY NOT (A) makes a sign error when moving the terms. (B) adds the x -terms instead of subtracting one from the other. (D) moves 2 across without changing its sign.

 12. **B** 5 hours

- Let h be the total number of hours. The first hour costs \$4; the other $h - 1$ hours cost \$3 each: $4 + 3(h - 1) = 16$.
- Subtract the first-hour charge: $3(h - 1) = 16 - 4 = 12$.
- Divide by 3: $h - 1 = 4$, so $h = 5$ hours.

WHY NOT (A) finds only the additional hours and forgets to count the first hour.

 13. **D** -7

- Move the 30 away from the x -term: subtract 30 from both sides. $-6x = 72 - 30 = 42$.
- The x -term is $-6x$, so divide both sides by -6 : $x = \frac{42}{-6} = -7$.
- Check: $30 - 6(-7) = 30 + 42 = 72$. ✓

TIP Dividing by a negative number changes the sign of the result.

WHY NOT (A) adds 30 to both sides instead of subtracting it.

 14. **D** 8

- Distribute 7 to *both* terms in the parentheses: $7 \cdot x = 7x$ and $7 \cdot (-3) = -21$. The equation becomes $7x - 21 + 13 = 48$.
- Combine the plain numbers: $-21 + 13 = -8$, so $7x - 8 = 48$.
- Undo the subtraction: add 8 to both sides. $7x = 48 + 8 = 56$.
- Undo the multiplication: divide both sides by 7. $x = \frac{56}{7} = 8$.

WHY NOT (A) writes 21 instead of -21 when distributing. (B) multiplies only the x by 7, not the -3 .

 15. **D** 3

- Clear the decimals: multiply *every* term by 10. $25x + 36 = 111$.
- Undo the addition: subtract 36 from both sides. $25x = 111 - 36 = 75$.
- Divide both sides by 25: $x = \frac{75}{25} = 3$.

TIP You can also work with the decimals directly: $2.5x = 7.5$, and $7.5 \div 2.5 = 3$.

WHY NOT (A) slips the decimal point.

 16. **C** $y = 5x - 8$

- Move the x -term to the right side: subtract $5x$ from both sides. $-y = -5x + 8$.
- Divide both sides by -1 : $y = \frac{-5x+8}{-1}$.
- Divide *each* term on top by -1 : $y = 5x - 8$.

WHY NOT (A) makes a sign error with the constant. (B) divides by 1 instead of -1 . (D) does not change the sign of $5x$ when moving it to the other side.

 17. **A** -4

- First solve for x . Undo the addition: subtract 9 from both sides. $4x = 17 - 9 = 8$.
- Undo the multiplication: divide both sides by 4. $x = \frac{8}{4} = 2$.
- The question asks for $2x - 8$, not x . Substitute: $2(2) - 8 = 4 - 8 = -4$.

TIP Always reread the question before choosing: finding x is only part of the job.

WHY NOT (B) gives the value of x , not of $2x - 8$. (C) forgets the -8 in $2x - 8$. (D) moves 9 across without changing its sign when solving.

 18. **C** 41

- Let n be the smallest. Consecutive integers go up by 1, so the numbers are $n, n + 1, n + 2$.
- Add them: $3n + 3 = 123$.
- Subtract 3: $3n = 120$. Divide by 3: $n = 40$.
- The numbers are 40, 41, 42. The middle is 41.

WHY NOT (B) is the smallest number, not the middle. (D) is the largest number, not the middle.

19. **C** 12

- 1 Distribute each number to both terms in its parentheses. The -4 multiplies *both* x and 8 : $-4(x + 8) = -4x - 32$.
- 2 So $6x - 18 - 4x - 32 = -26$.
- 3 Combine like terms: $2x - 50 = -26$.
- 4 Undo the subtraction: add 50 to both sides. $2x = -26 + 50 = 24$.
- 5 Undo the multiplication: divide both sides by 2. $x = \frac{24}{2} = 12$.

WHY NOT (A) writes $+32$ instead of -32 (does not distribute the minus sign). (B) adds $6x$ and $4x$ instead of subtracting.

20. **B** 80

- 1 Clear the fractions: multiply *every* term by the LCD of 4 and 10, which is 20.
- 2 $20 \cdot \frac{x}{4} = 5x$, $20 \cdot \frac{x}{10} = 2x$, and $20 \cdot 12 = 240$. So $5x - 2x = 240$.
- 3 Combine like terms: $3x = 240$.
- 4 Divide by 3: $x = \frac{240}{3} = 80$.
- 5 Check: $\frac{80}{4} - \frac{80}{10} = 20 - 8 = 12$. ✓

WHY NOT (A) subtracts the denominators, as if the result were $\frac{x}{6}$.

21. **B** $y = \frac{9}{5}x - 3$

- 1 Move the x -term to the right side: add $9x$ to both sides. $5y = 9x - 15$.
- 2 Divide both sides by 5: $y = \frac{9x-15}{5}$.
- 3 Divide *each* term on top by 5: $y = \frac{9}{5}x - 3$.

WHY NOT (A) does not change the sign of $-9x$ when moving it to the other side. (C) forgets to divide by 5. (D) divides only -15 by 5, not the x -term.

22. **B** -36

- 1 You cannot find x and y separately, and you don't need to. Compare the expressions: every coefficient in $8x - 4y$ is 4 times the one in $2x - y$.
- 2 So $8x - 4y = 4(2x - y)$.
- 3 Therefore $8x - 4y = 4 \times (-9) = -36$.

WHY NOT (C) gives the value of $2x - y$, not of $8x - 4y$. (D) adds 4 instead of multiplying by 4.

23. **D** -1

- 1 Get the x -terms on the right, where the coefficient is larger: add $5x$ to both sides. $9 = 7x + 16$.
- 2 Subtract 16 from both sides: $9 - 16 = 7x$, so $-7 = 7x$.
- 3 Undo the multiplication: divide both sides by 7. $x = \frac{-7}{7} = -1$.
- 4 Check with $x = -1$: the left side is $9 - 5(-1) = 14$ and the right side is $2(-1) + 16 = 14$. ✓

WHY NOT (A) forgets to divide by 7. (B) moves 9 across without changing its sign. (C) adds the x -terms instead of subtracting one from the other.

24. **C** 56

- 1 Let n be the smallest. Consecutive even integers go up by 2, so the numbers are $n, n + 2, n + 4, n + 6$.
- 2 Add them: $4n + 12 = 212$.
- 3 Subtract 12: $4n = 200$. Divide by 4: $n = 50$.
- 4 The numbers are 50, 52, 54, 56. The largest is 56.

WHY NOT (A) is the smallest number, not the largest. (B) is the second number, not the largest.

25. **A** $\frac{n}{9} + 9 = 20$

- 1 Translate piece by piece. Let n be the number; "is," "gives," and "equals" all become " $=$."
- 2 "A number divided by 9" is $\frac{n}{9}$. Then add 9: $\frac{n}{9} + 9$.
- 3 The equation is $\frac{n}{9} + 9 = 20$.

WHY NOT (B) divides in the wrong order. (C) multiplies by 9 instead of dividing. (D) adds 9 before dividing.

Inequalities

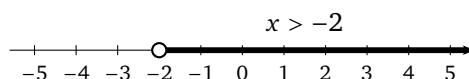
Tested on: Mathematics Knowledge • Arithmetic Reasoning

Key Concepts

An *inequality* compares two quantities that need not be equal. Its solution is usually a whole range of numbers, not a single value. You solve it almost exactly like an equation, with **one** extra rule.

Symbols and the number line

$x < 3$	x is less than 3	open circle at 3, shade left
$x > 3$	x is greater than 3	open circle at 3, shade right
$x \leq 3$	x is at most 3 (no more than 3)	closed dot at 3, shade left
$x \geq 3$	x is at least 3 (no less than 3)	closed dot at 3, shade right



Solving: one extra rule

Add or subtract any number on both sides, and multiply or divide both sides by a *positive* number, exactly as with equations. But when you **multiply or divide both sides by a negative number, flip the inequality sign**:

$$-2x < 6 \Rightarrow x > -3$$

Compound inequalities have three parts; do the same thing to all three: $-3 < 2x + 1 \leq 9 \Rightarrow -4 < 2x \leq 8 \Rightarrow -2 < x \leq 4$.

Absolute value

$|x - 3|$ is the distance between x and 3 on the number line.

- $|x - 3| < 5$: within 5 of 3, so $-2 < x < 8$ (one “between” piece).
- $|x - 3| > 5$: more than 5 away, so $x < -2$ or $x > 8$ (two pieces).

WORKED EXAMPLE

Solve $5 - 3x \geq 17$.

Solution. Subtract 5 from both sides: $-3x \geq 12$. Divide both sides by -3 and *flip* the sign: $x \leq -4$. Check with $x = -5$: $5 - 3(-5) = 20$, and $20 \geq 17$. ✓

No-Calculator Tip Test a point. Pick an easy number in your answer (such as 0, if it is included) and substitute it into the *original* inequality. If it makes a true statement, the direction of your sign is right.

Common Traps

- Forgetting to flip the sign after dividing by a negative number.
- Flipping when you only *subtracted* a number: $x - 5 > 2$ is simply $x > 7$.
- Mixing up $<$ and $\leq$: “at least 10” includes 10.
- Rounding the wrong way in word problems: to earn *at least* a goal, round the number of hours *up*.

Practice Set 12

Score: ____/25

Target time: 26 minutes

Problems 1–8 are warm-ups, 9–18 are test level, and 19–25 are challenge problems.

- Solve the inequality $9x + 7 < 97$.
A $x > 10$ B $x < 90$ C $x < 10$ D $x < \frac{104}{9}$
- Which inequality says: “Eight times a number n is a maximum of 57.”
A $8n < 57$ B $8n \geq 57$ C $8n \leq 57$ D $8n > 57$
- Solve the inequality $x + 1 \leq 8$.
A $x \leq 9$ B $x \leq 7$ C $x \geq 7$ D $x < 7$
- Which of the following is a solution of $2x + 9 > 7$?
A -3 B -2 C -1 D 2
- Which inequality describes all values of x that make $x - 6 \geq 2$ true?
A $x \geq 8$ B $x \geq -4$ C $x \leq 8$ D $x > 8$
- What is the largest integer x that satisfies $2x - 2 \leq 16$?
A 8 B 9 C 10 D 11
- “A number n decreased by 10 is a minimum of 27.” Which inequality represents this statement?
A $n - 10 \geq 27$ B $n - 10 \leq 27$
C $n - 10 < 27$ D $n - 10 > 27$
- How many integers x satisfy $-5 < x < 0$?
A 3 B 4 C 5 D 6
- Which of the following is the solution of $-3x - 4 \geq -34$?
A $x \leq \frac{38}{3}$ B $x \leq 10$ C $x \leq -10$ D $x \geq 10$
- Solve the inequality $11x - 14 \geq 6x + 16$.
A $x \geq \frac{2}{5}$ B $x \geq 6$ C $x \leq 6$ D $x \geq \frac{30}{17}$
- Which of the following is the solution of $-8 < 2x - 2 \leq 6$?
A $-5 < x \leq 2$ B $-3 \leq x < 4$
C $-3 < x \leq 4$ D $-6 < x \leq 4$
- Malik has \$120 saved and earns \$11 per hour at a part-time job. What is the least number of whole hours he must work to have at least \$1,100?
A 90 hours B 91 hours
C 100 hours D 111 hours
- Which inequality describes all values of x that make $11 \leq 4x + 3 \leq 27$ true?
A $8 \leq x \leq 6$ B $\frac{7}{2} \leq x \leq \frac{15}{2}$
C $2 \leq x \leq 6$ D $\frac{11}{4} \leq x \leq \frac{27}{4}$
- A trailer can carry at most 1,500 pounds. It already holds 220 pounds of tools. What is the greatest number of 70-pound ammunition crates that can be added?
A 18 crates B 19 crates
C 21 crates D 24 crates
- Which inequality describes all values of x that make $10x - 9 \geq 6x + 19$ true?
A $x \geq \frac{5}{2}$ B $x \geq \frac{7}{4}$ C $x \geq 7$ D $x \leq 7$
- Which of the following is the solution of $-3x - 1 < -25$?
A $x > \frac{26}{3}$ B $x < 8$ C $x > 8$ D $x > -8$

17. Which of the following is a solution of $-6 \leq 4x - 6 < 10$?
A 1 B 4 C 5 D 6
18. A trailer can carry at most 3,000 pounds. It is already loaded with 525 pounds of gear. Which inequality shows the additional weight w , in pounds, that can be loaded?
A $525 + w > 3,000$ B $525 + w \leq 3,000$
C $525 + w < 3,000$ D $525w \leq 3,000$
19. Solve $|x - 7| \leq 8$.
A $-15 \leq x \leq 1$ B $x \leq 15$
C $x < -1$ or $x > 15$ D $-1 \leq x \leq 15$
20. Car rental A costs \$35 per day plus \$0.25 per mile. Car rental B costs \$45 per day with unlimited miles. For a one-day rental, what is the greatest whole number of miles for which rental A costs less than rental B?
A 39 miles B 40 miles
C 180 miles D 320 miles
21. Which of the following is the solution of $|x + 6| \geq 7$?
A $x \geq 1$ B $x \leq -13$ or $x \geq 1$
C $x \leq -1$ or $x \geq 13$ D $-13 < x < 1$
22. A gym offers two options: pay \$7 per visit plus a \$20 monthly fee, or pay \$50 per month for unlimited visits. What is the greatest number of visits in a month for which paying per visit costs *less* than the unlimited plan?
A 4 visits B 5 visits C 7 visits D 10 visits
23. Which inequality describes all values of x that make $-15 \leq -7 - 2x < -3$ true?
A $-2 \leq x < 4$ B $-4 \leq x < 2$
C $5 < x \leq 11$ D $-2 < x \leq 4$
24. Solve the inequality $2x - 2 \leq 4x + 2$.
A $x \geq 2$ B $x \leq -2$ C $x \geq \frac{2}{3}$ D $x \geq -2$
25. How many integers x satisfy $-21 < 3x - 6 \leq -6$?
A 4 B 5 C 6 D 15

Answers & Explanations 12

1 C	6 B	11 C	16 C	21 B
2 C	7 A	12 A	17 A	22 A
3 B	8 B	13 C	18 B	23 D
4 D	9 B	14 A	19 D	24 D
5 A	10 B	15 C	20 A	25 B

1. **C** $x < 10$

- 1 Subtract 7 from both sides (adding or subtracting never changes the sign): $9x < 97 - 7$, so $9x < 90$.
- 2 Divide both sides by 9. Dividing by a positive number keeps the sign the same: $x < \frac{90}{9}$, so $x < 10$.
- 3 Check with a number in the answer, $x = 9$: $9(9) + 7 = 88$, and $88 < 97$ is true. ✓

WHY NOT (A) flips the sign, but you flip only when dividing by a negative number. (B) forgets to divide by 9. (D) moves 7 across without changing its sign.

2. **C** $8n \leq 57$

- 1 “A maximum of 57” means the quantity is less than or equal to 57: it includes 57 itself, so the symbol is $\leq$.
- 2 The quantity is $8n$, so the inequality is $8n \leq 57$.

WHY NOT (A) leaves out the boundary number, which is allowed. (B) points the sign the wrong way. (D) points the sign the wrong way and gets the boundary wrong.

3. **B** $x \leq 7$

- 1 Subtract 1 from both sides (adding or subtracting never changes the sign): $x \leq 8 - 1$, so $x \leq 7$.
- 2 Check with a number in the answer, $x = 6$: $6 + 1 = 7$, and $7 \leq 8$ is true. ✓

WHY NOT (A) adds 1 instead of subtracting it. (C) flips the sign, but you flip only when multiplying or dividing by a negative number. (D) uses $<$ instead of $\leq$.

4. **D** 2

- 1 Solve first. Subtract 9 from both sides (adding or subtracting never changes the sign): $2x > 7 - 9$, so $2x > -2$.
- 2 Divide both sides by 2 (positive, so the sign stays): $x > -1$.
- 3 Only 2 is greater than -1 . Check: $2(2) + 9 = 13$, which is greater than 7. ✓

WHY NOT (A) gives $2(-3) + 9 = 3$, which is not greater than 7. (B) gives $2(-2) + 9 = 5$, which is not greater than 7. (C) makes the left side equal to 7, but the inequality is strict.

5. **A** $x \geq 8$

- 1 Add 6 to both sides (adding or subtracting never changes the sign): $x \geq 2 + 6$, so $x \geq 8$.

2 Check with a number in the answer, $x = 9$: $9 - 6 = 3$, and $3 \geq 2$ is true. ✓

WHY NOT (B) subtracts 6 instead of adding it. (C) flips the sign, but you flip only when multiplying or dividing by a negative number. (D) uses $>$ instead of $\geq$.

6. **B** 9

- 1 Solve first. Add 2 to both sides (adding or subtracting never changes the sign): $2x \leq 16 + 2$, so $2x \leq 18$.

2 Divide both sides by 2 (positive, so the sign stays): $x \leq 9$.

3 9 itself is allowed ($\leq$), so the largest integer is 9.

WHY NOT (A) is a solution, but not the largest one. (C) gives $2(10) - 2 = 18$, which is not less than or equal to 16. (D) gives $2(11) - 2 = 20$, which is not less than or equal to 16.

7. **A** $n - 10 \geq 27$

- 1 “A minimum of 27” means the quantity is greater than or equal to 27: it includes 27 itself, so the symbol is $\geq$.

2 The quantity is $n - 10$, so the inequality is $n - 10 \geq 27$.

WHY NOT (B) points the sign the wrong way. (C) points the sign the wrong way and gets the boundary wrong. (D) leaves out the boundary number, which is allowed.

8. **B** 4

- 1 -5 is not included ($<$), and 0 is not included ($<$). So the integers run from -4 to -1 .

2 List them: $-4, -3, -2, -1$. That is 4 integers.

TIP Count from -4 to -1 with last $-$ first $+1$: $-1 - (-4) + 1 = 4$.

WHY NOT (C) just subtracts the endpoints, $0 - (-5) = 5$, without checking which ends are included. (D) counts both endpoints, but one or both are not included.

9. **B** $x \leq 10$

- 1 Add 4 to both sides (adding or subtracting never changes the sign): $-3x \geq -34 + 4$, so $-3x \geq -30$.
- 2 Divide both sides by -3 . Dividing by a *negative* number flips the sign: $\geq$ becomes $\leq$: $x \leq \frac{-30}{-3}$, so $x \leq 10$.
- 3 Check with a number in the answer, $x = 9$: $-3(9) - 4 = -31$, and $-31 \geq -34$ is true. ✓

WHY NOT (A) moves -4 across without changing its sign. (C) makes a sign error when dividing by -3 . (D) forgets to flip the sign when dividing by -3 .

 10. **B** $x \geq 6$

- 1 Collect the x -terms on the left: subtract $6x$ from both sides. $5x - 14 \geq 16$.
- 2 Add 14 to both sides (adding or subtracting never changes the sign): $5x \geq 16 + 14$, so $5x \geq 30$.
- 3 Divide both sides by 5. Dividing by a positive number keeps the sign the same: $x \geq \frac{30}{5}$, so $x \geq 6$.

WHY NOT (A) moves -14 across without changing its sign. (C) flips the sign, but no division by a negative number happened. (D) adds the x -terms instead of subtracting.

 11. **C** $-3 < x \leq 4$

- 1 Do the same thing to all three parts. Add 2 to each part: $-6 < 2x \leq 8$.
- 2 Divide each part by 2 (positive, so the signs stay the same): $-3 < x \leq 4$.
- 3 Check with $x = -2$: the middle is $2(-2) - 2 = -6$, and $-8 < -6 \leq 6$ is true. ✓

WHY NOT (A) subtracts 2 instead of adding it. (B) swaps which endpoint is included. (D) forgets to divide the left part by 2.

 12. **A** 90 hours

- 1 Let h be the number of hours. The total must be at least the goal: $120 + 11h \geq 1100$.
- 2 Subtract 120: $11h \geq 980$. Divide by 11: $h \geq \frac{980}{11} = 89\frac{1}{11}$.
- 3 h must be a whole number at least that big, so round *up*: 90. Check: $120 + 11(89) = 1099$ is too little, but $120 + 11(90) = 1110$ is enough.

WHY NOT (C) ignores the \$120 already saved. (D) adds the \$120 instead of subtracting it.

 13. **C** $2 \leq x \leq 6$

- 1 Do the same thing to all three parts. Subtract 3 from each part: $8 \leq 4x \leq 24$.
- 2 Divide each part by 4 (positive, so the signs stay the same): $2 \leq x \leq 6$.
- 3 Check with $x = 3$: the middle is $4(3) + 3 = 15$, and $11 \leq 15 \leq 27$ is true. ✓

WHY NOT (A) forgets to divide the left part by 4. (B) adds 3 instead of subtracting it. (D) divides by 4 without first removing the 3.

 14. **A** 18 crates

- 1 Let c be the number of crates. The total must stay at or under the limit: $220 + 70c \leq 1,500$.
- 2 Subtract 220: $70c \leq 1,280$. Divide by 70: $c \leq \frac{1,280}{70} = 18\frac{2}{7}$.
- 3 c must be a whole number no bigger than that, so round *down*: 18. Check: 19 would make 1,550, over the limit.

WHY NOT (B) rounds up, which goes over the limit. (C) forgets the 220 pounds already counted. (D) adds the starting amount instead of subtracting it.

 15. **C** $x \geq 7$

- 1 Collect the x -terms on the left: subtract $6x$ from both sides. $4x - 9 \geq 19$.
- 2 Add 9 to both sides (adding or subtracting never changes the sign): $4x \geq 19 + 9$, so $4x \geq 28$.
- 3 Divide both sides by 4. Dividing by a positive number keeps the sign the same: $x \geq \frac{28}{4}$, so $x \geq 7$.

WHY NOT (A) moves -9 across without changing its sign. (B) adds the x -terms instead of subtracting. (D) flips the sign, but no division by a negative number happened.

 16. **C** $x > 8$

- 1 Add 1 to both sides (adding or subtracting never changes the sign): $-3x < -25 + 1$, so $-3x < -24$.
- 2 Divide both sides by -3 . Dividing by a *negative* number flips the sign: $<$ becomes $>$: $x > \frac{-24}{-3}$, so $x > 8$.
- 3 Check with a number in the answer, $x = 9$: $-3(9) - 1 = -28$, and $-28 < -25$ is true. ✓

WHY NOT (A) moves -1 across without changing its sign. (B) forgets to flip the sign when dividing by -3 . (D) makes a sign error when dividing by -3 .

17. **A** 1

1 Solve first. Add 6 to all three parts: $0 \leq 4x < 16$, then divide all three by 4: $0 \leq x < 4$.

2 Only 1 is in this range. Check: $4(1) - 6 = -2$, and $-6 \leq -2 < 10$. ✓

WHY NOT (B) makes the middle equal to 10, but that end is not included. (C) gives $4(5) - 6 = 14$, which is outside the range. (D) gives $4(6) - 6 = 18$, which is outside the range.

18. **B** $525 + w \leq 3,000$

1 "At most 3,000" means the quantity is less than or equal to 3,000: it includes 3,000 itself, so the symbol is $\leq$.

2 The quantity is $525 + w$, so the inequality is $525 + w \leq 3,000$.

WHY NOT (A) points the sign the wrong way and gets the boundary wrong. (C) leaves out the boundary number, which is allowed. (D) multiplies instead of adding the weights.

19. **D** $-1 \leq x \leq 15$

1 $|x - 7| \leq 8$ means $x - 7$ is within 8 of zero, so it is *between* -8 and 8 : $-8 \leq x - 7 \leq 8$.

2 Add 7 to all three parts: $-1 \leq x \leq 15$.

3 Check with $x = 7$: $|7 - 7| = 0$, and $0 \leq 8$ is true. ✓

WHY NOT (A) uses the wrong sign: $x - 7$ is zero when $x = 7$, not $x = -7$. (B) keeps only the positive case; the inside must also be greater than -8 . (C) gives the "outside" set, which is for $>$.

20. **A** 39 miles

1 Let m be the number of miles. Rental A costs less when $35 + 0.25m < 45$.

2 Subtract 35: $0.25m < 10$.

3 Divide by 0.25: $m < 10 \div 0.25 = 40$ (dividing by 0.25 is the same as multiplying by 4).

4 At exactly 40 miles the two plans cost the same, so the greatest whole number that is *less* is 39.

WHY NOT (B) is where the plans cost the same or more, not less. (C) ignores the \$35 fee. (D) adds the fee instead of subtracting it.

21. **B** $x \leq -13$ or $x \geq 1$

1 $|x + 6| \geq 7$ means $x + 6$ is *more than* 7 away from zero (or exactly that far), so there are two pieces: $x + 6 \leq -7$ or $x + 6 \geq 7$.

2 Solve each piece: $x \leq -13$ or $x \geq 1$.

3 Check with $x = 2$: $|2 + 6| = 8$, and $8 \geq 7$ is true. ✓

WHY NOT (A) keeps only the positive case; the inside can also be very negative. (C) uses the wrong sign: $x + 6$ is zero when $x = -6$, not $x = 6$. (D) gives the "between" set, which is for $<$.

22. **A** 4 visits

1 Let v be the number of visits. Paying per visit costs less when $20 + 7v < 50$.

2 Subtract 20: $7v < 30$.

3 Divide by 7: $v < 30 \div 7 = 4\frac{2}{7}$.

4 The greatest whole number below that is 4.

WHY NOT (B) rounds up, which makes paying per visit cost more. (C) ignores the \$20 fee. (D) adds the fee instead of subtracting it.

23. **D** $-2 < x \leq 4$

1 Do the same thing to all three parts. Add 7 to each part: $-8 \leq -2x < 4$.

2 Divide each part by -2 . Dividing by a negative number flips *both* signs: $4 \geq x > -2$.

3 Rewrite it with the smaller number on the left: $-2 < x \leq 4$.

4 Check with $x = -1$: the middle is $-7 - 2(-1) = -5$, and $-15 \leq -5 < -3$ is true. ✓

WHY NOT (A) moves the numbers but leaves each symbol in its old place. (B) forgets that dividing by -2 changes the signs of the numbers. (C) subtracts 7 instead of adding it.

24. **D** $x \geq -2$

1 Collect the x -terms on the left: subtract $4x$ from both sides. $-2x - 2 \leq 2$.

2 Add 2 to both sides (adding or subtracting never changes the sign): $-2x \leq 2 + 2$, so $-2x \leq 4$.

3 Divide both sides by -2 . Dividing by a *negative* number flips the sign: $\leq$ becomes $\geq$: $x \geq \frac{4}{-2}$, so $x \geq -2$.

TIP To avoid dividing by a negative, collect the x -terms on the right instead: $-4 \leq 2x$, so $-2 \leq x$, which means the same as $x \geq -2$.

WHY NOT (A) makes a sign error when dividing by -2 . (B) forgets to flip the sign when dividing by -2 . (C) adds the x -terms instead of subtracting.

25. **B** 5

1 Solve first. Add 6 to all three parts: $-15 < 3x \leq 0$. Divide all three parts by 3: $-5 < x \leq 0$.

2 -5 is not included ($<$), and 0 is included ($\leq$). So the integers run from -4 to 0 .

3 List them: $-4, -3, -2, -1, 0$. That is 5 integers.

TIP Count from -4 to 0 with last $-$ first $+1$: $0 - (-4) + 1 = 5$.

WHY NOT (A) leaves out both endpoints. (C) counts both endpoints, but one or both are not included. (D) counts the integers before dividing by the coefficient of x .

Systems of Equations

Tested on: Mathematics Knowledge • Arithmetic Reasoning

Key Concepts

A *system of equations* is two equations that share the same variables. Its solution is the pair (x, y) that makes *both* equations true at the same time. On the ASVAB you solve systems by elimination or by substitution, and you set them up yourself in word problems with two unknowns.

Elimination: add or subtract the equations

Line the equations up and add (or subtract) them so one variable cancels.

$$\begin{array}{rcl} x + y & = & 10 \\ x - y & = & 4 \\ \hline 2x & = & 14 \end{array} \quad \Rightarrow \quad x = 7, \quad y = 10 - 7 = 3$$

If nothing cancels right away, first multiply one equation (*every* term, including the number on the right) so that a pair of coefficients match or are opposites.

Substitution

When one equation is already solved for a variable, such as $y = 2x - 1$, replace that variable in the other equation, *in parentheses*: $3x + 2y = 19$ becomes $3x + 2(2x - 1) = 19$, so $7x - 2 = 19$ and $x = 3$. Then $y = 2(3) - 1 = 5$.

Setting up word problems

Two unknowns need two equations: usually one counts the *items* and one adds up their *value*. For 200 tickets, adults \$12 and children \$5, total \$1,560:

$$a + c = 200 \quad 12a + 5c = 1,560$$

WORKED EXAMPLE

Solve the ticket system above for the number of adult tickets.

Solution. From the first equation, $c = 200 - a$. Substitute: $12a + 5(200 - a) = 1,560$, so $12a + 1,000 - 5a = 1,560$, which gives $7a = 560$ and $a = 80$. (Then $c = 120$. Check: $960 + 600 = 1,560$.)

No-Calculator Tip Read what is asked. If the question wants $x + y$, try adding or subtracting the equations first: you may get $x + y$ directly. You can also backsolve: a choice is correct only if it works in *both* equations.

Common Traps

- Answering with the wrong variable (y when the question asks for x).
- Subtracting equations but changing the sign of only the first term.
- Multiplying one side of an equation but not the other side.
- Dropping the parentheses in substitution: $3x - (2x - 1)$ is $3x - 2x + 1$.

Practice Set 13

Score: ____/25

Target time: 27 minutes

Problems 1–8 are warm-ups, 9–18 are test level, and 19–25 are challenge problems.

- What is the value of y in the solution of this system?

$$\begin{aligned} 2x + y &= 18 \\ x - y &= 3 \end{aligned}$$

 A 4 B 5 C 7 D 11
- Which ordered pair (x, y) satisfies both $x + 3y = 22$ and $2x - y = 2$?
 A (6, 4) B (7, 5) C (5, 8) D (4, 6)
- What is the value of x in the solution of this system?

$$\begin{aligned} 3x + y &= 37 \\ x - y &= 3 \end{aligned}$$

 A 3 B 7 C 10 D 40
- Which ordered pair (x, y) is the solution of this system?

$$\begin{aligned} 3x + 2y &= 13 \\ x - y &= 6 \end{aligned}$$

 A (5, -1) B (7, -4) C (-1, 5) D (6, 0)
- The sum of two numbers is 85, and their difference is 19. What is the larger number?
 A 33 B 52 C 66 D 104
- Which ordered pair (x, y) is the solution of this system?

$$\begin{aligned} 3x + y &= 2 \\ x - y &= -6 \end{aligned}$$

 A (0, 2) B (-1, 5) C (0, 6) D (5, -1)
- The sum of two numbers is 50, and their difference is 14. What is the larger number?
 A 18 B 32 C 36 D 64
- If $2x + y = 15$ and $x - y = -3$, what is the value of x ?
 A -4 B -3 C 4 D 7
- If $3x + y = 43$ and $5x + y = 65$, what is the value of $x + y$?
 A 10 B 11 C 21 D 22
- If $y = 2x - 1$ and $2x + y = -13$, what is the value of y ?
 A -10 B -8 C -7 D -3
- What is the value of y in the solution of this system?

$$\begin{aligned} 3x + 2y &= 6 \\ 5x + 2y &= -2 \end{aligned}$$

 A -13 B -4 C 5 D 9
- For Family Day on base, 120 barbecue tickets were sold. Adult tickets cost \$15 and child tickets cost \$6. Ticket sales totaled \$1,485. How many adult tickets were sold?
 A 35 B 51 C 60 D 85
- On a two-day ruck march, a platoon covered 30 miles. It marched 6 more miles on the first day than on the second. How many miles did it march on the second day?
 A 12 B 15 C 18 D 24

14. For Family Day on base, 100 barbecue tickets were sold. Adult tickets cost \$10 and child tickets cost \$5. Ticket sales totaled \$875. How many child tickets were sold?

A 15 B 25 C 50 D 75

15. Omar is 4 years older than Hannah. The sum of their ages is 28. How old is Omar?

A 12 B 13 C 14 D 16

16. If $3x + y = 13$ and $5x + y = 23$, what is the value of $x + y$?

A -3 B -2 C 2 D 3

17. Solve the system for x :

$$\begin{aligned} y &= 4x - 7 \\ 3x + 3y &= 99 \end{aligned}$$

A $\frac{26}{5}$ B $\frac{106}{15}$ C 8 D 25

18. A jar holds 48 coins, all dimes and nickels. The coins are worth \$4.15 in all. How many dimes are in the jar?

A 13 B 24 C 31 D 35

19. Solve the system below. What is the value of x ?

$$\begin{aligned} 3x + 2y &= 47 \\ 5x - y &= 35 \end{aligned}$$

A $\frac{82}{13}$ B 9 C 10 D 117

20. If $2x + y = -10$ and $x + 2y = -8$, what is the value of $x + y$?

A -6 B -5 C -4 D -2

21. A coffee shop mixes house coffee that costs \$6 per pound with premium coffee that costs \$14 per pound to make 12 pounds of a coffee blend that sells for \$12 per pound. How many pounds of house coffee are in the mix?

A 1 pound B 2 pounds
C 3 pounds D 9 pounds

22. If $5x + 2y = 44$ and $2x + 5y = 68$, what is the value of $x + y$?

A 4 B 16 C 17 D 112

23. A grocery store mixes peanuts that cost \$2 per pound with cashews that cost \$7 per pound to make 30 pounds of trail mix that sells for \$4 per pound. How many pounds of peanuts are in the mix?

A 12 pounds B 18 pounds
C 21 pounds D 45 pounds

24. What is the value of $x + y$ in the solution of this system?

$$\begin{aligned} 4x + 6y &= 26 \\ 3x + 2y &= 17 \end{aligned}$$

A 1 B 4 C 5 D 6

25. For a company party, the dining facility ordered 46 pizzas. Large pizzas cost \$12 each and small pizzas cost \$5 each. The bill was \$307. How many more small pizzas than large pizzas were ordered?

A 24 B 26 C 28 D 35

Answers & Explanations 13

1 A	6 B	11 D	16 D	21 C
2 D	7 B	12 D	17 C	22 B
3 C	8 C	13 A	18 D	23 B
4 A	9 C	14 B	19 B	24 D
5 B	10 C	15 D	20 A	25 A

1. **A** 4

- Add the equations. The y -terms cancel: $3x = 21$.
- Divide both sides by 3: $x = \frac{21}{3} = 7$.
- Substitute $x = 7$ into $2x + y = 18$: $2(7) + y = 18$, so $y = 4$.

WHY NOT (C) gives the value of x , not y . (D) gives the value of $x + y$, not y .

2. **D** (4, 6)

- A solution must make *both* equations true. Test (4, 6), that is, $x = 4$ and $y = 6$.
- First equation: $4 + 3(6) = 22$. ✓
- Second equation: $2(4) - 6 = 2$. ✓

WHY NOT (A) swaps the values of x and y . (B) works in the first equation but not the second. (C) works in the second equation but not the first.

3. **C** 10

- Add the equations. The y -terms cancel: $4x = 40$.
- Divide both sides by 4: $x = \frac{40}{4} = 10$.
- Substitute $x = 10$ into $3x + y = 37$: $3(10) + y = 37$, so $y = 7$.

WHY NOT (A) gives the value of $x - y$, not x . (B) gives the value of y , not x . (D) forgets to divide by 4.

4. **A** (5, -1)

- A solution must make *both* equations true. Test (5, -1), that is, $x = 5$ and $y = -1$.
- First equation: $3(5) + 2(-1) = 13$. ✓
- Second equation: $5 - (-1) = 6$. ✓

WHY NOT (B) works in the first equation but not the second. (C) swaps the values of x and y . (D) works in the second equation but not the first.

5. **B** 52

- Let L be the larger number and s be the smaller number. Then $L + s = 85$ and $L - s = 19$.
- Add the two equations; s cancels: $2L = 104$, so $L = 52$.
- Check: the other is $85 - 52 = 33$, and $52 - 33 = 19$. ✓

WHY NOT (A) gives the smaller number instead. (D) forgets to divide by 2.

6. **B** (-1, 5)

- A solution must make *both* equations true. Test (-1, 5), that is, $x = -1$ and $y = 5$.
- First equation: $3(-1) + 5 = 2$. ✓
- Second equation: $-1 - 5 = -6$. ✓

TIP You can also solve it directly: adding the equations cancels y .

WHY NOT (A) works in the first equation but not the second. (C) works in the second equation but not the first. (D) swaps the values of x and y .

7. **B** 32

- Let L be the larger number and s be the smaller number. Then $L + s = 50$ and $L - s = 14$.
- Add the two equations; s cancels: $2L = 64$, so $L = 32$.
- Check: the other is $50 - 32 = 18$, and $32 - 18 = 14$. ✓

WHY NOT (A) gives the smaller number instead. (D) forgets to divide by 2.

8. **C** 4

- Add the equations. The y -terms cancel: $3x = 12$.
- Divide both sides by 3: $x = \frac{12}{3} = 4$.
- Substitute $x = 4$ into $2x + y = 15$: $2(4) + y = 15$, so $y = 7$.

WHY NOT (A) makes a sign error. (B) gives the value of $x - y$, not x . (D) gives the value of y , not x .

9. **C** 21

- Subtract the first equation from the second (subtract *every* term). The y -terms cancel: $2x = 22$.
- Divide both sides by 2: $x = \frac{22}{2} = 11$.
- Substitute $x = 11$ into $3x + y = 43$: $3(11) + y = 43$, so $y = 10$.
- The question asks for $x + y$: $11 + 10 = 21$.

WHY NOT (A) gives the value of y , not $x + y$. (B) gives the value of x , not $x + y$.

10. **C** -7

- 1 The first equation tells you what y equals. Replace y in the second equation with $(2x-1)$: $2x + (2x-1) = -13$.
- 2 Distribute: $2x + 2x - 1 = -13$, so $4x - 1 = -13$.
- 3 Solve: $4x = -12$, so $x = -3$.
- 4 Now use the first equation: $y = 2(-3) - 1 = -7$.

WHY NOT (A) gives $x + y$. (B) makes a sign error with the -1 . (D) gives the value of x , not y .

 11. **D** 9

- 1 Subtract the first equation from the second (subtract *every* term). The y -terms cancel: $2x = -8$.
- 2 Divide both sides by 2: $x = \frac{-8}{2} = -4$.
- 3 Substitute $x = -4$ into $3x + 2y = 6$: $3(-4) + 2y = 6$, so $2y = 18$ and $y = 9$.

WHY NOT (A) gives the value of $x - y$, not y . (B) gives the value of x , not y . (C) gives the value of $x + y$, not y .

 12. **D** 85

- 1 Let a be the number of adult tickets. The rest are child tickets: $120 - a$.
 - 2 Add up the money: $15a + 6(120 - a) = 1,485$.
 - 3 Distribute and combine: $15a + 720 - 6a = 1,485$, so $9a + 720 = 1,485$.
 - 4 Subtract 720: $9a = 765$. Divide by 9: $a = 85$.
 - 5 Check: $15(85) + 6(35) = 1,275 + 210 = 1,485$. ✓
- WHY NOT** (A) gives the number of child tickets. (B) divides by \$15 instead of by the price difference \$9. (C) assumes the two kinds were sold in equal numbers.

 13. **A** 12

- 1 Let L be the first day's miles and s be the second day's miles. Then $L + s = 30$ and $L - s = 6$.
- 2 Add the two equations; s cancels: $2L = 36$, so $L = 18$.
- 3 Then $s = 30 - 18 = 12$.

WHY NOT (B) splits the total evenly and ignores the difference. (C) gives the first day's miles instead. (D) forgets to divide by 2.

 14. **B** 25

- 1 Let a be the number of adult tickets. The rest are child tickets: $100 - a$.
- 2 Add up the money: $10a + 5(100 - a) = 875$.
- 3 Distribute and combine: $10a + 500 - 5a = 875$, so $5a + 500 = 875$.
- 4 Subtract 500: $5a = 375$. Divide by 5: $a = 75$.
- 5 The question asks for child tickets: $100 - 75 = 25$.
- 6 Check: $10(75) + 5(25) = 750 + 125 = 875$. ✓

WHY NOT (C) assumes the two kinds were sold in equal numbers. (D) gives the number of adult tickets.

 15. **D** 16

- 1 Let L be Omar's age and s be Hannah's age. Then $L + s = 28$ and $L - s = 4$.
- 2 Add the two equations; s cancels: $2L = 32$, so $L = 16$.
- 3 Check: the other is $28 - 16 = 12$, and $16 - 12 = 4$. ✓

WHY NOT (A) gives Hannah's age instead. (C) splits the total evenly and ignores the difference.

 16. **D** 3

- 1 Subtract the first equation from the second (subtract *every* term). The y -terms cancel: $2x = 10$.
- 2 Divide both sides by 2: $x = \frac{10}{2} = 5$.
- 3 Substitute $x = 5$ into $3x + y = 13$: $3(5) + y = 13$, so $y = -2$.
- 4 The question asks for $x + y$: $5 + (-2) = 3$.

WHY NOT (A) makes a sign error. (B) gives the value of y , not $x + y$.

 17. **C** 8

- 1 The first equation tells you what y equals. Replace y in the second equation with $(4x-7)$: $3x + 3(4x-7) = 99$.
- 2 Distribute: $3x + 12x - 21 = 99$, so $15x - 21 = 99$.
- 3 Solve: $15x = 120$, so $x = 8$.

WHY NOT (A) makes a sign error with the -21 . (B) multiplies only $4x$ by 3, not -7 . (D) gives the value of y , not x .

 18. **D** 35

- 1 Work in cents so there are no decimals: a dime is 10 cents, a nickel is 5 cents, and \$4.15 is 415 cents.
- 2 Let b be the number of dimes; then there are $48 - b$ nickels. Total value: $10b + 5(48 - b) = 415$.
- 3 Distribute: $10b + 240 - 5b = 415$, so $5b = 415 - 240 = 175$.
- 4 Divide by 5: $b = 35$.

WHY NOT (A) gives the number of nickels. (B) assumes there are equally many of each.

 19. **B** 9

- 1 Multiply *every* term of the second equation by 2 so the y -terms will cancel: $10x - 2y = 70$.
- 2 Add the equations. The y -terms cancel: $13x = 117$.
- 3 Divide both sides by 13: $x = \frac{117}{13} = 9$.
- 4 Substitute $x = 9$ into $5x - y = 35$: $5(9) - y = 35$, so $-y = -10$ and $y = 10$.

WHY NOT (A) multiplies the left side of the second equation by 2 but not the right side. (C) gives the value of y , not x . (D) forgets to divide by 13.

20. **A** -6

1 Look at the coefficients: they are swapped. Add the two equations: $3x + 3y = -18$.

2 Every term has a factor of 3, so divide by 3: $x + y = \frac{-18}{3} = -6$.

TIP No need to find x and y separately (they are -4 and -2).

WHY NOT (C) gives the value of x , not $x + y$. (D) gives the value of y , not $x + y$.

21. **C** 3 pounds

1 Let h be the pounds of house coffee; then $12 - h$ pounds are premium coffee.

2 The value of the mix is $12 \times 12 = 144$ dollars, so $6h + 14(12 - h) = 144$.

3 Distribute and combine: $6h + 168 - 14h = 144$, so $-8h + 168 = 144$.

4 Subtract 168: $-8h = -24$. Divide by -8: $h = 3$ pounds of house coffee.

TIP Sense check: the mix price \$12 is closer to the price of premium coffee, so the mix must contain more premium coffee.

WHY NOT (D) gives the pounds of premium coffee.

22. **B** 16

1 Look at the coefficients: they are swapped. Add the two equations: $7x + 7y = 112$.

2 Every term has a factor of 7, so divide by 7: $x + y = \frac{112}{7} = 16$.

TIP No need to find x and y separately (they are 4 and 12).

WHY NOT (A) gives the value of x , not $x + y$. (D) forgets to divide by 7.

23. **B** 18 pounds

1 Let p be the pounds of peanuts; then $30 - p$ pounds are cashews.

2 The value of the mix is $30 \times 4 = 120$ dollars, so $2p + 7(30 - p) = 120$.

3 Distribute and combine: $2p + 210 - 7p = 120$, so $-5p + 210 = 120$.

4 Subtract 210: $-5p = -90$. Divide by -5: $p = 18$ pounds of peanuts.

TIP Sense check: the mix price \$4 is closer to the price of peanuts, so the mix must contain more peanuts.

WHY NOT (A) gives the pounds of cashews. (D) divides by \$2 instead of by the price difference \$5.

24. **D** 6

1 Multiply *every* term of the second equation by 3 so the y -terms will cancel: $9x + 6y = 51$.

2 Subtract the first equation from the new second (subtract *every* term). The y -terms cancel: $5x = 25$.

3 Divide both sides by 5: $x = \frac{25}{5} = 5$.

4 Substitute $x = 5$ into $3x + 2y = 17$: $3(5) + 2y = 17$, so $2y = 2$ and $y = 1$.

5 The question asks for $x + y$: $5 + 1 = 6$.

WHY NOT (A) gives the value of y , not $x + y$. (B) gives the value of $x - y$, not $x + y$. (C) gives the value of x , not $x + y$.

25. **A** 24

1 Let l be the number of large pizzas. The rest are small pizzas: $46 - l$.

2 Add up the money: $12l + 5(46 - l) = 307$.

3 Distribute and combine: $12l + 230 - 5l = 307$, so $7l + 230 = 307$.

4 Subtract 230: $7l = 77$. Divide by 7: $l = 11$.

5 So there were $46 - 11 = 35$ small pizzas. The question asks how many more: $35 - 11 = 24$.

6 Check: $12(11) + 5(35) = 132 + 175 = 307$. ✓

WHY NOT (D) gives the number of small pizzas, not the difference.

Polynomials & Exponent Rules

Tested on: Mathematics Knowledge

Key Concepts

A *polynomial* is a sum of terms such as $3x^2$, $-5x$, and 7 . Each term has a *coefficient* (the number in front) and a variable part. *Like terms* have exactly the same variable part: x^2 and $4x^2$ are like terms; x^2 and x^3 are not.

Exponent rules

$$x^a \cdot x^b = x^{a+b} \quad \text{add exponents} \quad x^0 = 1 \quad (x \neq 0)$$

$$\frac{x^a}{x^b} = x^{a-b} \quad \text{subtract exponents} \quad x^{-a} = \frac{1}{x^a} \quad \text{negative exponent}$$

$$(x^a)^b = x^{ab} \quad \text{multiply exponents} \quad (xy)^a = x^a y^a \quad \text{every factor}$$

A variable with no exponent has exponent 1: $x = x^1$.

Adding, subtracting, and vocabulary

Add or subtract polynomials by combining like terms: add the *coefficients* and keep the exponent ($2x^2 + 5x^2 = 7x^2$). To subtract, distribute the minus sign to *every* term: $-(x^2 - 4x + 1) = -x^2 + 4x - 1$. The *degree* is the highest exponent; the *leading coefficient* is the coefficient of that term. In $5x^3 - 2x^7 + x$ the degree is 7 and the leading coefficient is -2 .

Multiplying

Multiply every term by every term. For two binomials use FOIL (First, Outer, Inner, Last):

$$(x + 3)(x - 5) = x^2 - 5x + 3x - 15 = x^2 - 2x - 15$$

Special products worth memorizing:

$$(a + b)^2 = a^2 + 2ab + b^2 \quad (a - b)^2 = a^2 - 2ab + b^2 \quad (a + b)(a - b) = a^2 - b^2$$

WORKED EXAMPLE

Simplify $(2x^3y)^2 \cdot 3xy^4$.

Solution. Square every factor: $(2x^3y)^2 = 2^2x^{3 \cdot 2}y^2 = 4x^6y^2$. Then multiply coefficients and add exponents: $4x^6y^2 \cdot 3xy^4 = 12x^{6+1}y^{2+4} = 12x^7y^6$.

No-Calculator Tip Check with a number. Put $x = 2$ (and $y = 1$) into the original expression and into your answer: the two values must match. It takes seconds and catches most sign and exponent slips.

Common Traps

- $(x + 4)^2$ is *not* $x^2 + 16$: the middle term $8x$ is missing.
- $(2x^3)^2$ is $4x^6$, not $2x^6$: the coefficient is squared too.
- Adding exponents when you add like terms: $x^2 + x^2 = 2x^2$, not x^4 .
- Forgetting to change *every* sign when subtracting a polynomial.

Practice Set 14

Score: ____ / 25

Target time: 25 minutes

Problems 1–8 are warm-ups, 9–18 are test level, and 19–25 are challenge problems.

- What is the sum of $x^2 + 8x + 12$ and $7x^2 - 4x + 4$?
 A $8x^4 + 4x^2 + 16$ B $8x^2 + 12x + 16$
 C $12x^2 + 16$ D $8x^2 + 4x + 16$
- What is the product of $7x^3$ and $3x^4$?
 A $10x^7$ B $21x^7$ C $21x^{12}$ D $10x^{12}$
- Simplify: $\frac{8x^7}{2x^4}$
 A $4x^{11}$ B $4x^4$ C $6x^3$ D $4x^3$
- What is the leading coefficient of $8x^2 - 7x^4 + 5x^6$?
 A -5 B 5 C 6 D 8
- Simplify: $\frac{15x^3}{5x^2}$
 A $3x$ B $10x$ C $3x^2$ D $3x^5$
- What is the degree of the polynomial $8x^6 - 5x + 5x^2 - 5x^8$?
 A 4 B 6 C 8 D 17
- What is the product of $6x^6$ and $5x$?
 A $11x^6$ B $30x^6$ C $30x^7$ D $11x^7$
- Simplify: $(x^2 + 8x - 10) + (4x^2 + 8x + 8)$
 A $5x^2 - 2$ B $21x^2 - 2$
 C $5x^2 + 16x - 2$ D $5x^4 + 16x^2 - 2$
- What is $(5x^2 + x + 1) - (4x^2 + 6x - 10)$?
 A $x^2 + 7x - 9$ B $9x^2 + 7x - 9$
 C $x^2 - 5x + 11$ D $x^2 - 5x - 9$
- What is $(6x^3y)^2$?
 A $6x^6y^2$ B $12x^6y^2$ C $36x^5y^3$ D $36x^6y^2$
- Multiply: $(x + 2)(x - 8)$
 A $x^2 - 6x - 16$ B $x^2 + 6x - 16$
 C $x^2 - 16$ D $x^2 - 6x + 16$
- Multiply: $(3x - 7)^2$
 A $9x^2 + 49$ B $9x^2 - 42x + 49$
 C $9x^2 - 49$ D $9x^2 - 42x - 49$
- Multiply: $(x - 8)(x + 7)$
 A $x^2 - x + 56$ B $x^2 - x - 56$
 C $x^2 + x - 56$ D $x^2 - 56$
- Which expression is equal to $(2x - 9)(2x + 9)$?
 A $4x^2 - 36x + 81$ B $4x^2 - 81$
 C $4x^2 - 36x - 81$ D $4x^2 + 81$
- What is $(3x^5)^3$?
 A $3x^{15}$ B $27x^8$ C $9x^{15}$ D $27x^{15}$
- What is $(5x^2 + x + 3) - (7x^2 + 6x - 8)$?
 A $-2x^2 - 5x + 11$ B $-2x^2 + 7x - 5$
 C $12x^2 + 7x - 5$ D $-2x^2 - 5x - 5$
- Divide $16x^3 - 8x^2 - 8x$ by $4x$. (Assume $x \neq 0$.)
 A $4x^3 - 2x^2 - 2x$ B $4x^2 - 2x$
 C $4x^2 - 4x$ D $4x^2 - 2x - 2$
- What is the product of $-4x^4y^3$ and $7x^6y$?
 A $3x^{10}y^4$ B $-28x^{10}y^3$
 C $-28x^{10}y^4$ D $-28x^{24}y^3$
- When $(x - 6)(x^2 - 5x + 8)$ is multiplied out and simplified, what is the coefficient of x ?
 A -48 B 8 C 30 D 38

20. Rewrite $9x^{-2}$ using only positive exponents.

- A $\frac{9}{x^2}$ B $-9x^2$ C $\frac{1}{9x^2}$ D $9x^2$

21. Multiply: $(x - 5)(x^2 + 4x + 5)$

- A $x^3 - x^2 - 15x - 25$ B $x^3 - x^2 + 25x - 25$
C $x^3 + 4x^2 + 5x - 25$ D $x^3 - x^2 - 25$

22. Which expression is equal to $(4x^{-3})(3x^1)$?

- A $\frac{7}{x^2}$ B $\frac{12}{x^2}$ C $\frac{12}{x^3}$ D $12x^4$

23. What is $(-3xy)^3$?

- A $-3x^3y^3$ B $-27x^4y^4$
C $-27x^3y^3$ D $27x^3y^3$

24. Which expression is equal to $(4x + 2)(2x + 4)$?

- A $8x^2 + 20x + 8$ B $8x^2 + 20x - 8$
C $8x^2 - 20x + 8$ D $8x^2 + 8$

25. Which expression is equal to $(4x + 9)^2$?

- A $16x^2 + 72x + 81$ B $16x^2 + 36x + 81$
C $16x^2 + 81$ D $4x^2 + 72x + 81$

Answers & Explanations 14

1 D	6 C	11 A	16 A	21 A
2 B	7 C	12 B	17 D	22 B
3 D	8 C	13 B	18 C	23 C
4 B	9 C	14 B	19 D	24 A
5 A	10 D	15 D	20 A	25 A

1. **D** $8x^2 + 4x + 16$

1 Group like terms: $(1+7)x^2 + (8+(-4))x + (12+4)$.
 2 Add the coefficients and keep each exponent: $8x^2 + 4x + 16$.

WHY NOT (A) adds the exponents as well as the coefficients. (B) subtracts the x -terms instead of adding them. (C) combines unlike terms (x^2 -terms with x -terms).

2. **B** $21x^7$

1 Multiply the coefficients: $7 \cdot 3 = 21$.
 2 Multiply like bases by *adding* exponents: $x^3 \cdot x^4 = x^{3+4} = x^7$.
 3 So the product is $21x^7$.

WHY NOT (A) adds the coefficients instead of multiplying them. (C) multiplies the exponents instead of adding them. (D) adds the coefficients and multiplies the exponents.

3. **D** $4x^3$

1 Divide the coefficients: $8 \div 2 = 4$.
 2 Divide like bases by *subtracting* exponents: $\frac{x^7}{x^4} = x^{7-4} = x^3$.
 3 So the result is $4x^3$.

WHY NOT (A) adds the exponents instead of subtracting. (C) subtracts the coefficients instead of dividing.

4. **B** 5

1 Rewrite in standard form (highest exponent first): $5x^6 - 7x^4 + 8x^2$.

2 The leading coefficient is the number in front of the highest-power term, $5x^6$: it is 5, sign included.

WHY NOT (A) has the wrong sign; the sign is part of the coefficient. (C) gives the degree instead of the coefficient. (D) takes the coefficient of the first term written.

5. **A** $3x$

1 Divide the coefficients: $15 \div 5 = 3$.
 2 Divide like bases by *subtracting* exponents: $\frac{x^3}{x^2} = x^{3-2} = x$.
 3 So the result is $3x$.

WHY NOT (B) subtracts the coefficients instead of dividing. (D) adds the exponents instead of subtracting.

6. **C** 8

1 The degree is the *highest* exponent, no matter where the term is written.
 2 The exponents are 6, 1, 2, 8. The highest is 8.

WHY NOT (A) counts the terms. (B) gives the exponent of the first term written, not the highest exponent. (D) adds all the exponents.

7. **C** $30x^7$

1 Multiply the coefficients: $6 \cdot 5 = 30$.
 2 Multiply like bases by *adding* exponents: $x^6 \cdot x^1 = x^{6+1} = x^7$ (a letter with no exponent has exponent 1).
 3 So the product is $30x^7$.

WHY NOT (A) adds the coefficients and multiplies the exponents. (B) multiplies the exponents instead of adding them. (D) adds the coefficients instead of multiplying them.

8. **C** $5x^2 + 16x - 2$

1 Group like terms: $(1+4)x^2 + (8+8)x + (-10+8)$.
 2 Add the coefficients and keep each exponent: $5x^2 + 16x - 2$.

WHY NOT (A) subtracts the x -terms instead of adding them. (B) combines unlike terms (x^2 -terms with x -terms). (D) adds the exponents as well as the coefficients.

9. **C** $x^2 - 5x + 11$

- 1 Distribute the minus sign to *every* term of the second polynomial: $-(4x^2 + 6x - 10) = -4x^2 - 6x + 10$.
- 2 Combine like terms: $(5 - 4)x^2 + (1 - 6)x + (1 - (-10)) = x^2 - 5x + 11$.

WHY NOT (A) changes the sign of only the first term of the second polynomial. (B) adds the polynomials instead of subtracting. (D) forgets to change the sign of the last term.

 10. **D** $36x^6y^2$

- 1 Raise *every* factor inside the parentheses to the power 2.
- 2 Coefficient: $6^2 = 36$. Variables: multiply exponents, $(x^3)^2 = x^6$ and $(y)^2 = y^2$.
- 3 So the result is $36x^6y^2$.

WHY NOT (A) forgets to raise the coefficient 6 to the power. (B) multiplies the coefficient by 2 instead of raising it to the power 2. (C) adds the exponents instead of multiplying.

 11. **A** $x^2 - 6x - 16$

- 1 Use FOIL. First: $x \cdot x = x^2$. Outer: $x \cdot (-8) = -8x$. Inner: $2 \cdot x = 2x$. Last: $2 \cdot (-8) = -16$.
- 2 Add them and combine the two middle terms: $x^2 - 8x + 2x - 16 = x^2 - 6x - 16$.

WHY NOT (B) gets the sign of the middle term wrong. (C) multiplies only the First and Last terms and leaves out the middle. (D) gets the sign of the last term wrong.

 12. **B** $9x^2 - 42x + 49$

- 1 Use the pattern $(A - B)^2 = A^2 - 2AB + B^2$ with $A = 3x$ and $B = 7$.
- 2 $A^2 = (3x)^2 = 9x^2$, $-2AB = -2(3x)(7) = -42x$, and $B^2 = 7^2 = 49$.
- 3 So $(3x - 7)^2 = 9x^2 - 42x + 49$.

TIP Check with $x = 1$: the original and your answer must give the same number.

WHY NOT (A) squares each term but leaves out the middle term $-42x$. (C) squares each term, leaves out the middle term, and makes the last term negative. (D) makes the last term negative, but $(-7)^2 = 49$.

 13. **B** $x^2 - x - 56$

- 1 Use FOIL. First: $x \cdot x = x^2$. Outer: $x \cdot 7 = 7x$. Inner: $(-8) \cdot x = -8x$. Last: $(-8) \cdot 7 = -56$.
- 2 Add them and combine the two middle terms: $x^2 + 7x - 8x - 56 = x^2 - x - 56$.

WHY NOT (A) gets the sign of the last term wrong. (C) gets the sign of the middle term wrong. (D) multiplies only the First and Last terms and leaves out the middle.

 14. **B** $4x^2 - 81$

- 1 This is a sum times a difference of the same two terms, so use $(A + B)(A - B) = A^2 - B^2$ with $A = 2x$ and $B = 9$.
- 2 The middle terms cancel ($+18x$ and $-18x$), leaving $(2x)^2 - 9^2 = 4x^2 - 81$.

TIP Check with $x = 1$: the original and your answer must give the same number.

WHY NOT (A) treats it like a perfect square. (C) adds middle terms that actually cancel. (D) makes the last term positive, but a positive times a negative is negative.

 15. **D** $27x^{15}$

- 1 Raise *every* factor inside the parentheses to the power 3.
- 2 Coefficient: $3^3 = 27$. Variables: multiply exponents, $(x^5)^3 = x^{15}$.
- 3 So the result is $27x^{15}$.

WHY NOT (A) forgets to raise the coefficient 3 to the power. (B) adds the exponents instead of multiplying. (C) multiplies the coefficient by 3 instead of raising it to the power 3.

 16. **A** $-2x^2 - 5x + 11$

- 1 Distribute the minus sign to *every* term of the second polynomial: $-(7x^2 + 6x - 8) = -7x^2 - 6x + 8$.
- 2 Combine like terms: $(5 - 7)x^2 + (1 - 6)x + (3 - (-8)) = -2x^2 - 5x + 11$.

WHY NOT (B) changes the sign of only the first term of the second polynomial. (C) adds the polynomials instead of subtracting. (D) forgets to change the sign of the last term.

 17. **D** $4x^2 - 2x - 2$

- 1 Divide *each* term of the top by $4x$.
- 2 $\frac{16x^3}{4x} = 4x^2$, $\frac{-8x^2}{4x} = -2x$, and $\frac{-8x}{4x} = -2$.
- 3 So the result is $4x^2 - 2x - 2$.

WHY NOT (A) divides the coefficients but forgets to subtract the exponents. (B) drops the last term, but $\frac{-8x}{4x}$ is -2 , not 0. (C) leaves an x on the last term.

 18. **C** $-28x^{10}y^4$

- 1 Multiply the coefficients: $(-4) \cdot 7 = -28$.
- 2 Multiply like bases by *adding* exponents: $x^4 \cdot x^6 = x^{4+6} = x^{10}$ and $y^3 \cdot y^1 = y^{3+1} = y^4$ (a letter with no exponent has exponent 1).
- 3 So the product is $-28x^{10}y^4$.

WHY NOT (A) adds the coefficients instead of multiplying them. (B) treats a y with no exponent as y^0 instead of y^1 . (D) multiplies the exponents instead of adding them.

19. **D** 38

 1 Multiply each term of $(x - 6)$ by every term of $(x^2 - 5x + 8)$.

2 $x(x^2 - 5x + 8) = x^3 - 5x^2 + 8x$.

3 $(-6)(x^2 - 5x + 8) = -6x^2 + 30x - 48$.

4 Add and combine like terms: $x^3 - 5x^2 + 8x - 6x^2 + 30x - 48 = x^3 - 11x^2 + 38x - 48$.

WHY NOT (A) gives the constant term instead. (B) uses only the x -term from $x(x^2 - 5x + 8)$. (C) uses only the x -term from $(-6)(x^2 - 5x + 8)$.

 20. **A** $\frac{9}{x^2}$

 1 A negative exponent means “one over”: $x^{-2} = \frac{1}{x^2}$.

 2 The exponent applies only to x , not to the 9: $9x^{-2} = 9 \cdot \frac{1}{x^2} = \frac{9}{x^2}$.

WHY NOT (B) treats the negative exponent as a negative number. (C) moves the coefficient 9 to the bottom too, but the exponent belongs only to x . (D) just drops the negative sign.

 21. **A** $x^3 - x^2 - 15x - 25$

 1 Multiply each term of $(x - 5)$ by every term of $(x^2 + 4x + 5)$.

2 $x(x^2 + 4x + 5) = x^3 + 4x^2 + 5x$.

3 $(-5)(x^2 + 4x + 5) = -5x^2 - 20x - 25$.

4 Add and combine like terms: $x^3 + 4x^2 + 5x - 5x^2 - 20x - 25 = x^3 - x^2 - 15x - 25$.

WHY NOT (B) makes a sign error combining the x -terms. (C) multiplies -5 by only 5, the last term of the trinomial. (D) drops the x -terms.

 22. **B** $\frac{12}{x^2}$

 1 Multiply the coefficients: $4 \cdot 3 = 12$.

 2 Add the exponents: $x^{-3} \cdot x^1 = x^{-3+1} = x^{-2}$, and $x^{-2} = \frac{1}{x^2}$.

 3 So the product is $\frac{12}{x^2}$.

WHY NOT (A) adds the coefficients instead of multiplying them. (C) multiplies the exponents instead of adding them. (D) ignores the negative sign on the exponent.

 23. **C** $-27x^3y^3$

1 Raise every factor inside the parentheses to the power 3.

 2 Coefficient: $(-3)^3 = -27$. Variables: multiply exponents, $(x)^3 = x^3$ and $(y)^3 = y^3$.

 3 So the result is $-27x^3y^3$.

WHY NOT (A) forgets to raise the coefficient -3 to the power. (B) adds the exponents instead of multiplying. (D) loses the negative sign: a negative number to an odd power stays negative.

 24. **A** $8x^2 + 20x + 8$

 1 Use FOIL. First: $4x \cdot 2x = 8x^2$. Outer: $4x \cdot 4 = 16x$. Inner: $2 \cdot 2x = 4x$. Last: $2 \cdot 4 = 8$.

 2 Add them and combine the two middle terms: $8x^2 + 16x + 4x + 8 = 8x^2 + 20x + 8$.

WHY NOT (B) gets the sign of the last term wrong. (C) gets the sign of the middle term wrong. (D) multiplies only the First and Last terms and leaves out the middle.

 25. **A** $16x^2 + 72x + 81$

 1 Use the pattern $(A + B)^2 = A^2 + 2AB + B^2$ with $A = 4x$ and $B = 9$.

 2 $A^2 = (4x)^2 = 16x^2$, $2AB = 2(4x)(9) = 72x$, and $B^2 = 9^2 = 81$.

 3 So $(4x + 9)^2 = 16x^2 + 72x + 81$.

TIP Check with $x = 1$: the original and your answer must give the same number.

WHY NOT (B) forgets to double the middle term. (C) squares each term but leaves out the middle term $72x$. (D) does not square the coefficient 4.

Factoring & Quadratic Equations

Tested on: Mathematics Knowledge

Key Concepts

Factoring is multiplication in reverse: it rewrites a sum as a product. It is also the fastest way to solve a *quadratic equation* (an equation with an x^2 term) without a calculator.

Factoring patterns

- **Greatest common factor (GCF) first:** $6x^2 + 9x = 3x(2x + 3)$.
- **Trinomials** $x^2 + bx + c$: find two numbers that *multiply* to c and *add* to b . For $x^2 + 7x + 12$: $3 \cdot 4 = 12$ and $3 + 4 = 7$, so $(x + 3)(x + 4)$.
- **Difference of squares:** $a^2 - b^2 = (a + b)(a - b)$, so $x^2 - 49 = (x + 7)(x - 7)$.
- **Perfect squares:** $x^2 + 10x + 25 = (x + 5)^2$.
- **Leading coefficient not 1:** try factor pairs and check the middle term with FOIL: $2x^2 + 7x + 3 = (2x + 1)(x + 3)$.

Solving quadratic equations

1. Move everything to one side so the other side is 0.
2. Factor, then set *each* factor equal to 0 (a product is 0 only when one of its factors is 0).

If there is no x -term, take square roots and keep both signs: $x^2 = 49$ gives $x = 7$ or $x = -7$. When factoring is hard, use the quadratic formula for $ax^2 + bx + c = 0$:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Function notation

$f(x) = x^2 - 3x + 2$ is a rule; $f(-2)$ means “replace every x with (-2) ”: $f(-2) = (-2)^2 - 3(-2) + 2 = 4 + 6 + 2 = 12$.

WORKED EXAMPLE

Solve $x^2 - 5x = 14$.

Solution. Subtract 14: $x^2 - 5x - 14 = 0$. Two numbers that multiply to -14 and add to -5 are -7 and 2 , so $(x - 7)(x + 2) = 0$. Then $x - 7 = 0$ or $x + 2 = 0$: $x = 7$ or $x = -2$. Check: $49 - 35 = 14$ and $4 + 10 = 14$.
✓

No-Calculator Tip To find the pair for $x^2 + bx + c$, list the factor pairs of c . If c is positive, both numbers have the sign of b ; if c is negative, the signs differ and the larger number takes the sign of b .

Common Traps

- $(x + 3)(x - 5) = 0$ gives $x = -3$ or $x = 5$: use the *opposites* of the numbers in the factors.
- Forgetting the negative root of $x^2 = 49$.
- $x(x - 3) = 10$ does *not* mean $x = 10$ or $x - 3 = 10$; you need 0 on one side first.
- $(-3)^2 = 9$, but $-3^2 = -9$: keep the parentheses when you substitute.

Practice Set 15

Score: ____/25

Target time: 25 minutes

Problems 1–8 are warm-ups, 9–18 are test level, and 19–25 are challenge problems.

- What is the greatest common factor of the terms of $14x^2 + 12x$?
A $2x$ B $2x^2$ C $12x$ D $84x^2$
- Factor: $x^2 - 8x + 15$
A $(x - 9)(x + 1)$ B $(x + 3)(x + 5)$
C $(x - 15)(x - 1)$ D $(x - 5)(x - 3)$
- Solve $x(x + 2) = 0$.
A $x = 0$ or $x = 2$ B $x = -2$
C $x = -2$ or $x = 1$ D $x = -2$ or $x = 0$
- Factor: $x^2 - 17x + 72$
A $(x + 3)(x + 24)$ B $(x - 9)(x - 8)$
C $(x + 2)(x + 36)$ D $(x + 8)(x + 9)$
- What are the solutions of $(x - 8)(x - 7) = 0$?
A $x = 8$ B $x = -8$ or $x = 7$
C $x = 7$ or $x = 8$ D $x = -8$ or $x = -7$
- What is the greatest common factor of the terms of $16x^3 - 36x^2$?
A $144x^3$ B $4x^3$ C $4x^2$ D $16x^2$
- What are the solutions of $n^2 - 49 = 0$?
A $n = -7$ or $n = 7$ B $n = -49$ or $n = 49$
C $n = 7$ D $n = -8$ or $n = 8$
- Given $f(x) = x^2 + x - 3$, find $f(-5)$.
A -33 B 17 C 27 D 34
- Which is the factored form of $x^2 - 4x - 77$?
A $(x - 77)(x + 1)$ B $(x - 7)(x + 11)$
C $(x - 5)(x + 1)$ D $(x - 11)(x + 7)$
- Factor completely: $t^2 + 16t + 64$
A $(t - 8)(t + 8)$ B $(t + 1)(t + 64)$
C $(t + 8)^2$ D $(t - 8)^2$
- Which values of x satisfy $x^2 - 2x - 35 = 0$?
A $x = -5$ B $x = -7$ or $x = 5$
C $x = -5$ or $x = 7$ D $x = -1$ or $x = 35$
- Which of the following is a solution of $x^2 - 6x - 16 = 0$?
A -16 B -8 C 2 D 8
- Solve for x : $x^2 - 3x - 54 = 0$.
A $x = -6$ or $x = 9$ B $x = -9$ or $x = 6$
C $x = -1$ or $x = 54$ D $x = -2$ or $x = 27$
- Which value of x makes $x^2 + 15x + 54 = 0$ true?
A -6 B 6 C 9 D 54
- Which expression is equal to $m^2 - 6m + 9$?
A $(m - 9)(m - 1)$ B $(m + 3)^2$
C $(m - 3)(m + 3)$ D $(m - 3)^2$
- Factor: $x^2 + 4x - 45$
A $(x - 9)(x + 5)$ B $(x - 5)(x + 9)$
C $(x - 15)(x + 3)$ D $(x - 3)(x + 15)$
- Solve for x : $4x^2 = 484$.
A $x = -11$ or $x = 11$
B $x = -22$ or $x = 22$
C $x = -121$ or $x = 121$
D $x = 11$

18. Given $f(x) = -x^2 - 2x + 3$, find $f(-3)$.
 A -12 B -4 C -2 D 0
19. Which is the factored form of $5x^2 - 11x + 6$?
 A $(x - 6)(x - 1)$ B $(5x + 6)(x + 1)$
 C $(5x - 6)(x - 1)$ D $(5x - 1)(x - 6)$
20. What are the solutions of $6x^2 + x - 5 = 0$?
 A $x = -1$ or $x = \frac{5}{6}$ B $x = -\frac{5}{6}$ or $x = 1$
 C $x = -2$ or $x = \frac{5}{3}$ D $x = \frac{5}{6}$
21. Factor: $3x^2 + 11x - 42$
 A $(3x - 7)(x + 6)$ B $(3x + 7)(x - 6)$
 C $(3x + 6)(x - 7)$ D $(x - 7)(x + 6)$
22. Use the quadratic formula to solve $2x^2 + 13x + 15 = 0$.
 A $x = -10$ or $x = -3$ B $x = -5$ or $x = -\frac{3}{2}$
 C $x = \frac{3}{2}$ or $x = 5$ D $x = -\frac{15}{2}$ or $x = 1$
23. Which is the factored form of $25x^2 - 36$?
 A $(5x + 6)^2$ B $(5x - 6)(5x + 6)$
 C $(25x - 6)(25x + 6)$ D $(5x - 6)^2$
24. Which is the completely factored form of $5x^2 + 5x - 100$?
 A $5(x - 4)(x + 5)$ B $5(x - 5)(x + 4)$
 C $(x - 4)(x + 5)$ D $5(x - 20)(x + 1)$
25. Which values of x satisfy $x^2 = 10x - 21$?
 A $x = 7$ B $x = 3$ or $x = 7$
 C $x = -7$ or $x = -3$ D $x = 1$ or $x = 21$

Answers & Explanations 15

1 A	6 C	11 C	16 B	21 A
2 D	7 A	12 D	17 A	22 B
3 D	8 B	13 A	18 D	23 B
4 B	9 D	14 A	19 C	24 A
5 C	10 C	15 D	20 A	25 B

- A** $2x$
 - Coefficients: the largest number that divides both 14 and 12 is 2.
 - Variables: both terms contain x ; use the *lower* power, x .
 - So the GCF is $2x$. (Check: $14x^2 + 12x = 2x(7x + 6)$.)

WHY NOT (B) uses the higher power of x ; the GCF takes the lower power. (C) uses the smaller coefficient, which does not divide both terms. (D) gives the least common multiple, not the greatest common factor.
- D** $(x - 5)(x - 3)$
 - Two numbers that multiply to 15 and add to -8 are -3 and -5 : $-3 \cdot (-5) = 15$ and $-3 + (-5) = -8$.
 - So $x^2 - 8x + 15 = (x - 5)(x - 3)$.
 - Check with FOIL: $x^2 - 5x - 3x + 15 = x^2 - 8x + 15$. ✓

WHY NOT (A) uses 1 and -9 , which add to -8 but multiply to -9 , not 15. (B) has both signs backwards. (C) uses -15 and -1 , which multiply to 15 but add to -16 , not -8 .
- D** $x = -2$ or $x = 0$
 - A product is 0 only if one of its factors is 0, so set each factor equal to 0.
 - $x = 0$ or $x + 2 = 0$, which gives $x = -2$.
 - The solutions are $x = -2$ or $x = 0$.

WHY NOT (A) uses the number in the factor instead of its opposite. (B) forgets that the factor x gives the solution $x = 0$. (C) sets the factor x equal to 1 instead of 0.
- B** $(x - 9)(x - 8)$
 - Two numbers that multiply to 72 and add to -17 are -8 and -9 : $-8 \cdot (-9) = 72$ and $-8 + (-9) = -17$.
 - So $x^2 - 17x + 72 = (x - 9)(x - 8)$.
 - Check with FOIL: $x^2 - 9x - 8x + 72 = x^2 - 17x + 72$. ✓

WHY NOT (A) uses 3 and 24, which multiply to 72 but add to 27, not -17 . (C) uses 2 and 36, which multiply to 72 but add to 38, not -17 . (D) has both signs backwards.
- C** $x = 7$ or $x = 8$
 - A product is 0 only if one of its factors is 0, so set each factor equal to 0.
 - $x - 8 = 0$ gives $x = 8$; $x - 7 = 0$ gives $x = 7$.
 - The solutions are $x = 7$ or $x = 8$. (Notice they are the *opposites* of the numbers in the factors.)

WHY NOT (A) forgets the second factor. (B) gets the sign of one solution wrong. (D) uses the numbers in the factors instead of their opposites.
- C** $4x^2$
 - Coefficients: the largest number that divides both 16 and 36 is 4.
 - Variables: both terms contain x^2 ; use the *lower* power, x^2 .
 - So the GCF is $4x^2$. (Check: $16x^3 - 36x^2 = 4x^2(4x - 9)$.)

WHY NOT (A) gives the least common multiple, not the greatest common factor. (B) uses the higher power of x ; the GCF takes the lower power. (D) uses the smaller coefficient, which does not divide both terms.
- A** $n = -7$ or $n = 7$
 - Add 49 to both sides: $n^2 = 49$.
 - Take the square root of both sides and keep *both* signs: $n = 7$ or $n = -7$, because $7^2 = 49$ and $(-7)^2 = 49$.

WHY NOT (B) forgets to take the square root. (C) forgets the negative solution.
- B** 17
 - Replace every x with (-5) : $f(-5) = (-5)^2 + (-5) - 3$.
 - Square first (order of operations): $(-5)^2 = 25$. So $f(-5) = 25 - 5 - 3$.
 - Add: $f(-5) = 17$.

WHY NOT (A) treats $(-5)^2$ as -25 . (C) makes a sign error with x .

9. **D** $(x - 11)(x + 7)$

- Two numbers that multiply to -77 and add to -4 are 7 and -11 : $7 \cdot (-11) = -77$ and $7 + (-11) = -4$.
- So $x^2 - 4x - 77 = (x - 11)(x + 7)$.
- Check with FOIL: $x^2 - 11x + 7x - 77 = x^2 - 4x - 77$. ✓

WHY NOT (A) uses -77 and 1 , which multiply to -77 but add to -76 , not -4 . (B) has the signs switched, which makes the middle term $4x$. (C) uses 1 and -5 , which add to -4 but multiply to -5 , not -77 .

 10. **C** $(t + 8)^2$

- The first and last terms are perfect squares: $t^2 = (t)^2$ and $64 = 8^2$.
- The middle term is $2(t)(8) = 16t$, so it is a perfect square: $t^2 + 16t + 64 = (t + 8)^2$.

WHY NOT (A) is a difference of squares, which has no middle term. (B) uses 64 and 1 , which multiply correctly but give the wrong middle term. (D) gets the sign wrong; the middle term tells you the sign.

 11. **C** $x = -5$ or $x = 7$

- Factor: two numbers that multiply to -35 and add to -2 are 5 and -7 , so $(x + 5)(x - 7) = 0$.
- Set each factor equal to 0 : $x = -5$ or $x = 7$.

WHY NOT (A) finds only one of the two solutions. (B) uses the numbers in the factors instead of their opposites. (D) factors with -35 and 1 , which multiply to -35 but do not add to -2 .

 12. **D** 8

- Factor: two numbers that multiply to -16 and add to -6 are -8 and 2 , so $(x - 8)(x + 2) = 0$.
- The solutions are $x = 8$ and $x = -2$. Only 8 is among the choices.
- Check: $(8)^2 - 6(8) - 16 = 64 - 48 - 16 = 0$. ✓

WHY NOT (A) is the constant term, not a solution. (B) has the wrong sign: it is the number in a factor, not a solution. (C) is the opposite of the other solution.

 13. **A** $x = -6$ or $x = 9$

- Factor: two numbers that multiply to -54 and add to -3 are 6 and -9 , so $(x + 6)(x - 9) = 0$.
- Set each factor equal to 0 : $x = -6$ or $x = 9$.

WHY NOT (B) uses the numbers in the factors instead of their opposites. (C) factors with -54 and 1 , which multiply to -54 but do not add to -3 . (D) factors with -27 and 2 , which multiply to -54 but do not add to -3 .

 14. **A** -6

- Factor: two numbers that multiply to 54 and add to 15 are 6 and 9 , so $(x + 6)(x + 9) = 0$.
- The solutions are $x = -6$ and $x = -9$. Only -6 is among the choices.
- Check: $(-6)^2 + 15(-6) + 54 = 36 - 90 + 54 = 0$. ✓

WHY NOT (B) has the wrong sign: it is the number in a factor, not a solution. (C) is the opposite of the other solution. (D) is the constant term, not a solution.

 15. **D** $(m - 3)^2$

- The first and last terms are perfect squares: $m^2 = (m)^2$ and $9 = 3^2$.
- The middle term is $-2(m)(3) = -6m$, so it is a perfect square: $m^2 - 6m + 9 = (m - 3)^2$.

WHY NOT (A) uses 9 and 1 , which multiply correctly but give the wrong middle term. (B) gets the sign wrong; the middle term tells you the sign. (C) is a difference of squares, which has no middle term.

 16. **B** $(x - 5)(x + 9)$

- Two numbers that multiply to -45 and add to 4 are 9 and -5 : $9 \cdot (-5) = -45$ and $9 + (-5) = 4$.
- So $x^2 + 4x - 45 = (x - 5)(x + 9)$.
- Check with FOIL: $x^2 - 5x + 9x - 45 = x^2 + 4x - 45$. ✓

WHY NOT (A) has the signs switched, which makes the middle term $-4x$. (C) uses -15 and 3 , which multiply to -45 but add to -12 , not 4 . (D) uses -3 and 15 , which multiply to -45 but add to 12 , not 4 .

 17. **A** $x = -11$ or $x = 11$

- Divide both sides by 4 : $x^2 = 121$.
- Take the square root and keep both signs: $x = 11$ or $x = -11$.

WHY NOT (B) forgets to divide by 4 . (C) forgets to take the square root. (D) forgets the negative solution.

 18. **D** 0

- Replace every x with (-3) : $f(-3) = -(-3)^2 - 2(-3) + 3$.
- Square first (order of operations): $(-3)^2 = 9$. So $f(-3) = -9 + 6 + 3$.
- Add: $f(-3) = 0$.

WHY NOT (A) makes a sign error with $-2x$.

19. **C** $(5x - 6)(x - 1)$

- 1 The first terms must multiply to $5x^2$, so try $(5x + \underline{\quad})(x + \underline{\quad})$. The last numbers must multiply to 6.
- 2 Test pairs until the Outer + Inner products give $-11x$: with -6 and -1 , $5x \cdot (-1) + (-6) \cdot x = -5x - 6x = -11x$.
- 3 So $5x^2 - 11x + 6 = (5x - 6)(x - 1)$.

WHY NOT (A) ignores the leading coefficient 5. (B) has the signs backwards. (D) puts -6 and -1 in the wrong factors, so the middle term comes out $-31x$.

20. **A** $x = -1$ or $x = \frac{5}{6}$

- 1 Here $a = 6$, $b = 1$, $c = -5$.
- 2 Compute $b^2 - 4ac$: $1^2 - 4(6)(-5) = 1 + 120 = 121$, and $\sqrt{121} = 11$.
- 3 $x = \frac{-1 \pm 11}{12}$: either $x = \frac{-1 + 11}{12} = \frac{5}{6}$ or $x = \frac{-1 - 11}{12} = -1$.

TIP You can check by factoring: $6x^2 + x - 5 = (6x - 5)(x + 1)$.

WHY NOT (B) uses 1 instead of -1 in the formula. (C) divides by 6 instead of by $2a = 12$. (D) uses only the + sign of $\pm$.

21. **A** $(3x - 7)(x + 6)$

- 1 The first terms must multiply to $3x^2$, so try $(3x + \underline{\quad})(x + \underline{\quad})$. The last numbers must multiply to -42 .
- 2 Test pairs until the Outer + Inner products give $11x$: with -7 and 6 , $3x \cdot 6 + (-7) \cdot x = 18x - 7x = 11x$.
- 3 So $3x^2 + 11x - 42 = (3x - 7)(x + 6)$.

WHY NOT (B) has the signs backwards. (C) puts -7 and 6 in the wrong factors, so the middle term comes out $-15x$. (D) ignores the leading coefficient 3.

22. **B** $x = -5$ or $x = -\frac{3}{2}$

- 1 Here $a = 2$, $b = 13$, $c = 15$.
- 2 Compute $b^2 - 4ac$: $13^2 - 4(2)(15) = 169 - 120 = 49$, and $\sqrt{49} = 7$.
- 3 $x = \frac{-13 \pm 7}{4}$: either $x = \frac{-13 + 7}{4} = -\frac{3}{2}$ or $x = \frac{-13 - 7}{4} = -5$.

TIP You can check by factoring: $2x^2 + 13x + 15 = (2x + 3)(x + 5)$.

WHY NOT (A) divides by 2 instead of by $2a = 4$. (C) uses 13 instead of -13 in the formula. (D) adds $4ac$ instead of subtracting it.

23. **B** $(5x - 6)(5x + 6)$

- 1 $25x^2 - 36$ is a difference of squares: $(5x)^2 - 6^2$.
- 2 Use $A^2 - B^2 = (A + B)(A - B)$: $25x^2 - 36 = (5x - 6)(5x + 6)$.

WHY NOT (A) is a perfect square, not a difference of squares. (C) does not take the square root of 25. (D) is a perfect square, not a difference of squares.

24. **A** $5(x - 4)(x + 5)$

- 1 First take out the GCF, 5: $5x^2 + 5x - 100 = 5(x^2 + x - 20)$.
- 2 Now factor $x^2 + x - 20$: find two numbers that multiply to -20 and add to 1. They are 5 and -4 .
- 3 So $5x^2 + 5x - 100 = 5(x - 4)(x + 5)$.

WHY NOT (B) has the signs of both numbers backwards. (C) leaves out the common factor. (D) uses numbers that multiply to -20 but add to -19 , not 1.

25. **B** $x = 3$ or $x = 7$

- 1 Get 0 on one side: move every term to the left. $x^2 - 10x + 21 = 0$.
- 2 Factor: two numbers that multiply to 21 and add to -10 are -7 and -3 , so $(x - 7)(x - 3) = 0$.
- 3 Set each factor equal to 0: $x = 7$ or $x = 3$.

WHY NOT (A) finds only one of the two solutions. (C) uses the numbers in the factors instead of their opposites. (D) factors with -21 and -1 , which multiply to 21 but do not add to -10 .

PART III

Geometry

Angles, triangles, area, circles, volume, and the coordinate plane.

- 16 Lines & Angles
- 17 Triangles & the Pythagorean Theorem
- 18 Perimeter & Area
- 19 Circles
- 20 Volume & Surface Area
- 21 Coordinate Geometry

Lines & Angles

Tested on: Mathematics Knowledge

Key Concepts

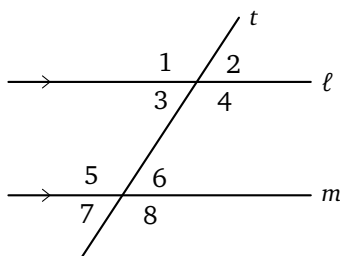
Angles are measured in degrees ($^{\circ}$). Almost every angle question on the ASVAB comes down to a few facts: a straight line is 180° , a full turn is 360° , and parallel lines create pairs of equal angles.

Angle vocabulary and angle pairs

acute less than 90° **straight** exactly 180°
right exactly 90° (marked with a small square) **reflex** between 180° and 360°
obtuse between 90° and 180°

- **Complementary** angles add up to 90° ; **supplementary** angles add up to 180° .
- Angles side by side on a straight line add up to 180° ; angles all the way around a point add up to 360° .
- When two lines cross, the angles directly across from each other are **vertical angles**, and they are *equal*.

Parallel lines cut by a transversal



When $\ell \parallel m$:

- **Corresponding** angles (same position at each crossing) are equal: $\angle 1 = \angle 5$, $\angle 4 = \angle 8$.
- **Alternate interior** angles are equal: $\angle 3 = \angle 6$, $\angle 4 = \angle 5$.
- **Alternate exterior** angles are equal: $\angle 1 = \angle 8$, $\angle 2 = \angle 7$.
- **Same-side interior** angles add up to 180° : $\angle 3 + \angle 5 = 180^{\circ}$.

Angles of a polygon (n = number of sides)

- Sum of the interior angles: $(n - 2) \times 180^{\circ}$ (triangle 180° , quadrilateral 360° , pentagon 540° , hexagon 720° , octagon $1,080^{\circ}$).
- In a *regular* polygon all angles are equal: each interior angle is $\frac{(n - 2) \times 180^{\circ}}{n}$.
- The exterior angles of any polygon add up to 360° , so each exterior angle of a regular polygon is $360^{\circ} \div n$.

WORKED EXAMPLE

Two angles form a straight line. One measures $(2x + 10)^{\circ}$ and the other $(3x - 5)^{\circ}$. What is x ?

Solution. Angles on a straight line add up to 180° : $(2x + 10) + (3x - 5) = 180$, so $5x + 5 = 180$, $5x = 175$, and $x = 35$. Check: the angles are 80° and 100° , and $80 + 100 = 180$.

No-Calculator Tip In a parallel-lines figure there are only *two* angle sizes: every small (acute) angle is equal to every other small angle, every large (obtuse) angle is equal to every other large angle, and small + large = 180° .

Common Traps

- Mixing up the totals: complementary comes before supplementary, just as 90 comes before 180.
- Treating same-side interior angles as equal. They add up to 180° .
- Finding x and stopping when the question asks for the *angle*: plug x back in.
- Using $n \times 180^\circ$ instead of $(n - 2) \times 180^\circ$ for a polygon.

Practice Set 16

Score: ____ / 25

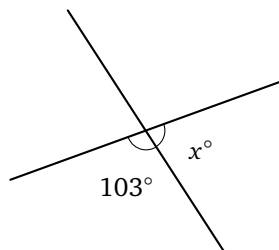
Target time: 25 minutes

Problems 1–8 are warm-ups, 9–18 are test level, and 19–25 are challenge problems.

1. Two angles are supplementary. One of them measures 46° . What is the measure of the other angle?

A 134° B 147° C 226° D 314°

2. The figure shows two intersecting straight lines and a 103° angle. What is the value of x ?

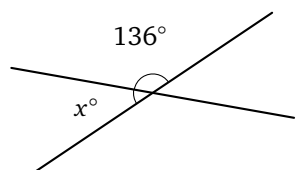


A 77 B 82 C 103 D 257

3. What is the measure of the complement of a 40° angle?

A 37° B 40° C 50° D 140°

4. The figure shows two intersecting straight lines and a 136° angle. What is the value of x ?

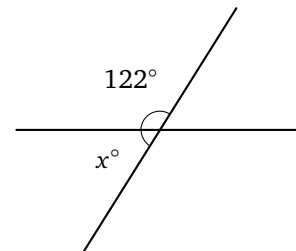


A 44 B 54 C 136 D 224

5. What type of angle has a measure of 180° ?

A acute B straight C reflex D obtuse

6. Two straight lines cross as shown. One angle measures 122° . What is the value of x ?



A 38 B 43 C 58 D 122

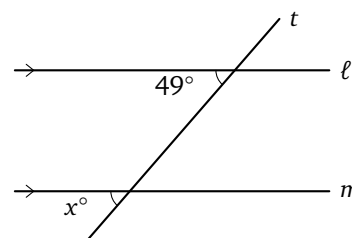
7. Which of the following is the measure of a reflex angle?

A 70° B 90° C 117° D 259°

8. What is the measure of the supplement of a 135° angle?

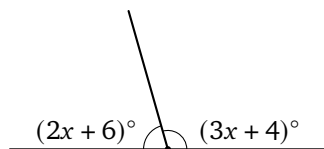
A 45° B 135° C 225° D 315°

9. Parallel lines ℓ and m are cut by transversal t , forming the 49° angle shown. What is the value of x ?



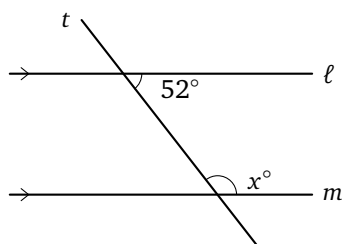
A 49 B 64 C 131 D 311

10. In the figure, a ray meets a straight line, forming angles of $(3x + 4)^\circ$ and $(2x + 6)^\circ$. What is the value of x ?



A 2 B 16 C 34 D 38

11. In the figure, lines ℓ and m are parallel, and one angle formed by transversal t measures 52° . What is the value of x ?

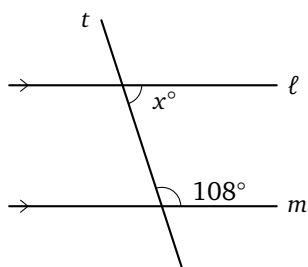


A 38 B 52 C 113 D 128

12. What is the sum of the measures of the interior angles of a decagon (10 sides)?

A 144° B 360° C $1,440^\circ$ D $1,800^\circ$

13. In the figure, lines ℓ and m are parallel, and one angle formed by transversal t measures 108° . What is the value of x ?



A 62 B 67 C 72 D 108

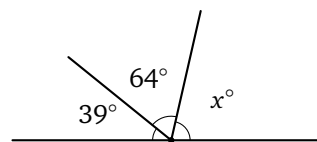
14. What is the sum of the measures of the interior angles of a hexagon?

A 720° B 900° C $1,080^\circ$ D $1,120^\circ$

15. $\angle P$ and $\angle Q$ are complementary. $\angle P$ measures $(x + 30)^\circ$, and $\angle Q$ measures $(2x + 24)^\circ$. What is the value of x ?

A 6 B 12 C 42 D 48

16. In the figure, three angles share a vertex on a straight line. Two of them measure 39° and 64° . What is the value of x ?

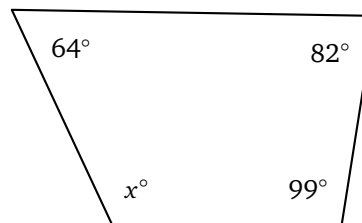


A 57 B 77 C 103 D 257

17. Leila is building a fountain base in the shape of a regular 12-sided polygon. What is the measure of each interior angle of the fountain base?

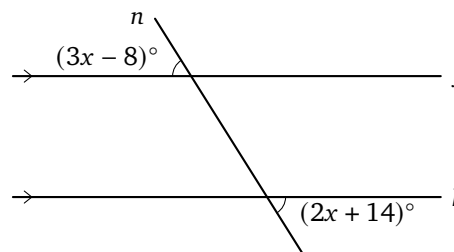
A 30° B 150° C 165° D $1,800^\circ$

18. In the quadrilateral shown, three of the interior angles measure 99° , 82° , and 64° . What is the value of x ?



A 90 B 115 C 125 D 295

19. In the figure, lines j and k are parallel. The marked angles measure $(2x + 14)^\circ$ and $(3x - 8)^\circ$. What is the measure of the $(2x + 14)^\circ$ angle?

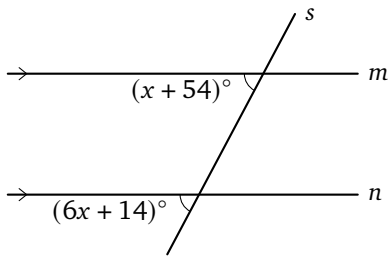


A 22° B 38° C 43° D 58°

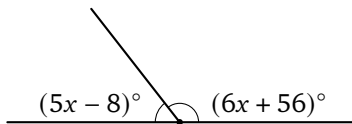
20. What is the measure of the smaller angle between the hour hand and the minute hand of a clock at 8:15?

A 142.5° B 150° C 152.5° D 157.5°

21. In the figure, lines m and n are parallel. The marked angles measure $(x + 54)^\circ$ and $(6x + 14)^\circ$. What is the measure of the $(6x + 14)^\circ$ angle?

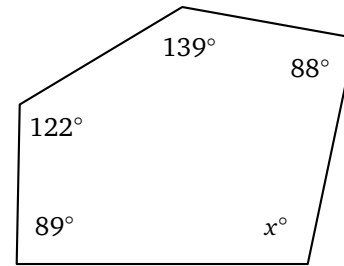


- A 8° B 52° C 57° D 62°
22. What is the measure of the smaller angle between the hour hand and the minute hand of a clock at 11:15?
- A 97.5° B 107.5° C 112.5° D 120°
23. In the figure, a ray meets a straight line, forming angles of $(6x + 56)^\circ$ and $(5x - 8)^\circ$. What is the measure of the $(6x + 56)^\circ$ angle?



- A 12° B 52° C 113° D 128°

24. In the pentagon shown, four of the interior angles measure 89° , 88° , 139° , and 122° . What is the value of x ?



- A 102 B 108 C 122 D 282
25. Sergeant Ruiz pilots a patrol boat around a course shaped like a regular polygon, turning 45° at each corner. How many sides does the course have?
- A 4 B 5 C 6 D 8

Answers & Explanations 16

1 A	6 C	11 D	16 B	21 D
2 A	7 D	12 C	17 B	22 C
3 C	8 A	13 C	18 B	23 D
4 A	9 A	14 A	19 D	24 A
5 B	10 C	15 B	20 D	25 D

1. **A** 134°

- Supplementary angles add up to 180° .
- Subtract the angle you know: $180^\circ - 46^\circ = 134^\circ$.
- Check: $46^\circ + 134^\circ = 180^\circ$. ✓

TIP Complementary comes before supplementary in the alphabet, just as 90 comes before 180.

WHY NOT (C) adds the angle to 180° instead of subtracting it. (D) subtracts from 360° , the total all the way around a point.

2. **A** 77

- The x° angle and the 103° angle sit side by side along one straight line, so together they make 180° .
- $x + 103 = 180$, so $x = 180 - 103 = 77$.

WHY NOT (C) treats the angles as vertical angles, but these two angles together form a straight line. (D) subtracts from 360° instead of 180° .

3. **C** 50°

- Complementary angles add up to 90° .
- Subtract the angle you know: $90^\circ - 40^\circ = 50^\circ$.
- Check: $40^\circ + 50^\circ = 90^\circ$. ✓

TIP Complementary comes before supplementary in the alphabet, just as 90 comes before 180.

WHY NOT (B) repeats the given angle, but complementary angles are not equal; they add up to 90° . (D) finds the supplement (total 180°) instead of the complement (total 90°).

4. **A** 44

- The x° angle and the 136° angle sit side by side along one straight line, so together they make 180° .
- $x + 136 = 180$, so $x = 180 - 136 = 44$.

WHY NOT (C) treats the angles as vertical angles, but these two angles together form a straight line. (D) subtracts from 360° instead of 180° .

5. **B** straight

- Compare with the benchmarks: acute is less than 90° , right is exactly 90° , obtuse is between 90° and 180° , straight is exactly 180° , and reflex is between 180° and 360° .

- 180° is exactly 180° , so the angle is *straight*.

WHY NOT (A) describes an angle smaller than 90° . (C) describes an angle between 180° and 360° . (D) describes an angle between 90° and 180° .

6. **C** 58

- The x° angle and the 122° angle sit side by side along one straight line, so together they make 180° .
- $x + 122 = 180$, so $x = 180 - 122 = 58$.

WHY NOT (D) treats the angles as vertical angles, but these two angles together form a straight line.

7. **D** 259°

- A reflex angle measures between 180° and 360° .
- Only 259° is between 180° and 360° .

WHY NOT (A) is an acute angle (less than 90°). (B) is a right angle (exactly 90°). (C) is an obtuse angle (between 90° and 180°).

8. **A** 45°

- Supplementary angles add up to 180° .
- Subtract the angle you know: $180^\circ - 135^\circ = 45^\circ$.
- Check: $135^\circ + 45^\circ = 180^\circ$. ✓

TIP Complementary comes before supplementary in the alphabet, just as 90 comes before 180.

WHY NOT (B) repeats the given angle, but supplementary angles are not equal; they add up to 180° . (C) subtracts from 360° , the total all the way around a point. (D) adds the angle to 180° instead of subtracting it.

9. **A** 49

1 The 49° angle and the x° angle are *corresponding* angles: they sit in the same position at the two crossings.

2 When the lines are parallel, corresponding angles are equal, so $x = 49$.

TIP Every acute angle in this figure is 49° and every obtuse angle is 131° . The x° angle is acute, so $x = 49$.

WHY NOT (C) adds the angles to 180° , but corresponding angles are equal. (D) subtracts from 360° .

10. **C** 34

1 Angles that form a straight line add up to 180° : $(3x + 4) + (2x + 6) = 180$.

2 Combine like terms: $5x + 10 = 180$.

3 Subtract 10 from both sides: $5x = 180 - 10 = 170$.

4 Divide both sides by 5: $x = 170 \div 5 = 34$.

TIP Check: with $x = 34$ the angles are 106° and 74° , and $106 + 74 = 180$.

WHY NOT (A) sets the two angles equal instead of adding them. (B) uses 90° instead of 180° . (D) makes a sign error when moving the constant term.

11. **D** 128

1 The 52° angle and the x° angle are *same-side interior* angles: they are between the parallel lines, on the same side of the transversal.

2 Same-side interior angles add up to 180° : $x = 180 - 52 = 128$.

TIP Every acute angle in this figure is 52° and every obtuse angle is 128° . The x° angle is obtuse, so $x = 128$.

WHY NOT (A) subtracts from 90° instead of 180° . (B) treats the angles as equal, but same-side interior angles add up to 180° .

12. **C** 1,440°

1 The polygon has $n = 10$ sides. Use: sum of interior angles $= (n - 2) \times 180^\circ$.

2 $(10 - 2) \times 180^\circ = 8 \times 180^\circ = 1,440^\circ$.

TIP Why it works: diagonals from one corner cut the polygon into $10 - 2 = 8$ triangles, and each triangle holds 180° .

WHY NOT (A) gives one angle of a regular polygon, not the sum of all the angles. (B) gives the sum of the exterior angles, which is always 360° . (D) multiplies by the number of sides instead of $n - 2$.

13. **C** 72

1 The 108° angle and the x° angle are *same-side interior* angles: they are between the parallel lines, on the same side of the transversal.

2 Same-side interior angles add up to 180° : $x = 180 - 108 = 72$.

TIP Every acute angle in this figure is 72° and every obtuse angle is 108° . The x° angle is acute, so $x = 72$.

WHY NOT (D) treats the angles as equal, but same-side interior angles add up to 180° .

14. **A** 720°

1 The polygon has $n = 6$ sides. Use: sum of interior angles $= (n - 2) \times 180^\circ$.

2 $(6 - 2) \times 180^\circ = 4 \times 180^\circ = 720^\circ$.

TIP Why it works: diagonals from one corner cut the polygon into $6 - 2 = 4$ triangles, and each triangle holds 180° .

WHY NOT (B) uses $n - 1$ instead of $n - 2$. (C) multiplies by the number of sides instead of $n - 2$.

15. **B** 12

1 Complementary angles add up to 90° : $(x + 30) + (2x + 24) = 90$.

2 Combine like terms: $3x + 54 = 90$.

3 Subtract 54 from both sides: $3x = 90 - 54 = 36$.

4 Divide both sides by 3: $x = 36 \div 3 = 12$.

TIP Check: with $x = 12$ the angles are 42° and 48° , and $42 + 48 = 90$.

WHY NOT (A) sets the two angles equal instead of adding them. (C) uses 180° instead of 90° . (D) makes a sign error when moving the constant term.

16. **B** 77

1 Angles that together form a straight line add up to 180° : $39 + 64 + x = 180$.

2 Add the known angles: $39 + 64 = 103$.

3 Subtract: $x = 180 - 103 = 77$.

WHY NOT (C) adds the two given angles and stops. (D) uses 360° instead of 180° .

17. **B** 150°

1 A 12-sided polygon has 12 sides. Sum of the interior angles: $(12 - 2) \times 180^\circ = 1,800^\circ$.

2 A regular polygon has 12 equal angles: $1,800^\circ \div 12 = 150^\circ$.

TIP Faster: each exterior angle is $360^\circ \div 12 = 30^\circ$, so each interior angle is $180^\circ - 30^\circ = 150^\circ$.

WHY NOT (A) is the exterior angle, not the interior angle. (C) uses $n - 1$ instead of $n - 2$. (D) is the sum of all the interior angles, not one angle.

18. **B** 115

- 1 A quadrilateral has 4 sides, so its interior angles add up to $(4 - 2) \times 180^\circ = 360^\circ$.
- 2 Add the known angles: $99 + 82 + 64 = 245$.
- 3 Subtract: $x = 360 - 245 = 115$.

WHY NOT (A) divides the angle sum by the number of angles, as if all the angles were equal. (D) uses $(n - 1) \times 180^\circ$ for the angle sum instead of $(n - 2) \times 180^\circ$.

19. **D** 58°

- 1 These are alternate exterior angles (they are outside the parallel lines, on opposite sides of the transversal). Alternate exterior angles are equal, so set the expressions equal: $3x - 8 = 2x + 14$.
- 2 Subtract $2x$ from both sides: $x - 8 = 14$.
- 3 Add 8 to both sides: $x = 14 + 8 = 22$.
- 4 Plug $x = 22$ into $2x + 14$: $2(22) + 14 = 58$, so the angle measures 58° .

WHY NOT (A) stops at the value of x instead of finding the angle.

20. **D** 157.5°

- 1 The minute hand moves $360^\circ \div 60 = 6^\circ$ per minute. At 15 minutes past the hour it is $15 \times 6^\circ = 90^\circ$ past the 12.
- 2 The hour hand moves 30° per hour plus 0.5° per minute. At 8:15 it is $8 \times 30^\circ + 15 \times 0.5^\circ = 240^\circ + 7.5^\circ = 247.5^\circ$ past the 12.
- 3 The hands are $247.5^\circ - 90^\circ = 157.5^\circ$ apart.

TIP Each hour mark on a clock face is $360^\circ \div 12 = 30^\circ$ from the next.

WHY NOT (B) forgets that the hour hand moves past the hour mark (0.5° per minute).

21. **D** 62°

- 1 These are corresponding angles (they sit in the same position at the two crossings). Corresponding angles are equal, so set the expressions equal: $6x + 14 = x + 54$.
- 2 Subtract x from both sides: $5x + 14 = 54$.
- 3 Subtract 14 from both sides: $5x = 54 - 14 = 40$.
- 4 Divide both sides by 5: $x = 40 \div 5 = 8$.
- 5 Plug $x = 8$ into $6x + 14$: $6(8) + 14 = 62$, so the angle measures 62° .

WHY NOT (A) stops at the value of x instead of finding the angle.

22. **C** 112.5°

- 1 The minute hand moves $360^\circ \div 60 = 6^\circ$ per minute. At 15 minutes past the hour it is $15 \times 6^\circ = 90^\circ$ past the 12.
- 2 The hour hand moves 30° per hour plus 0.5° per minute. At 11:15 it is $11 \times 30^\circ + 15 \times 0.5^\circ = 330^\circ + 7.5^\circ = 337.5^\circ$ past the 12.
- 3 The hands are $337.5^\circ - 90^\circ = 247.5^\circ$ apart.
- 4 That is more than 180° , so the smaller angle is $360^\circ - 247.5^\circ = 112.5^\circ$.

TIP Each hour mark on a clock face is $360^\circ \div 12 = 30^\circ$ from the next.

WHY NOT (D) forgets that the hour hand moves past the hour mark (0.5° per minute).

23. **D** 128°

- 1 Angles that form a straight line add up to 180° : $(6x + 56) + (5x - 8) = 180$.
- 2 Combine like terms: $11x + 48 = 180$.
- 3 Subtract 48 from both sides: $11x = 180 - 48 = 132$.
- 4 Divide both sides by 11: $x = 132 \div 11 = 12$.
- 5 Plug $x = 12$ into $6x + 56$: $6(12) + 56 = 128$, so the angle measures 128° .

WHY NOT (A) stops at the value of x instead of finding the angle. (B) is the measure of the other angle.

24. **A** 102

- 1 A pentagon has 5 sides, so its interior angles add up to $(5 - 2) \times 180^\circ = 540^\circ$.
- 2 Add the known angles: $89 + 88 + 139 + 122 = 438$.
- 3 Subtract: $x = 540 - 438 = 102$.

WHY NOT (B) divides the angle sum by the number of angles, as if all the angles were equal. (D) uses $(n - 1) \times 180^\circ$ for the angle sum instead of $(n - 2) \times 180^\circ$.

25. **D** 8

- 1 The exterior angles of any polygon add up to 360° ; the turn at each corner is an exterior angle.
 - 2 All n exterior angles are equal, so $n = 360 \div 45 = 8$.
- WHY NOT** (A) divides 180° instead of 360° .

Triangles & the Pythagorean Theorem

Tested on: Mathematics Knowledge • Arithmetic Reasoning

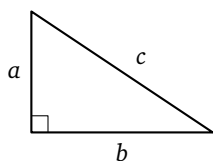
Key Concepts

A triangle has three sides and three angles. Two facts solve most triangle questions on the ASVAB: the angles always add up to 180° , and in a right triangle $a^2 + b^2 = c^2$.

Angles in a triangle

- The three angles add up to 180° . Missing angle = $180^\circ - (\text{the other two})$.
- **Isosceles** triangle: two equal sides, and the two angles opposite them (the *base angles*) are equal. **Equilateral**: three equal sides and three 60° angles. **Scalene**: no equal sides.
- By angles: **acute** (all angles less than 90°), **right** (one 90° angle), **obtuse** (one angle more than 90°).
- An **exterior angle** (made by extending one side) equals the sum of the two *remote* interior angles.
- **Triangle inequality**: any two sides must add up to *more* than the third side.

The Pythagorean theorem (right triangles only)



$$a^2 + b^2 = c^2$$

c is the **hypotenuse**: the longest side, across from the right angle. To find a leg, subtract: $a^2 = c^2 - b^2$. Memorize the common **Pythagorean triples** and their multiples: 3-4-5 (6-8-10, 9-12-15, ...), 5-12-13, 8-15-17, 7-24-25.

Special right triangles and similar triangles

- 45° - 45° - 90° : sides in the ratio $1 : 1 : \sqrt{2}$ (legs s , hypotenuse $s\sqrt{2}$).
- 30° - 60° - 90° : sides in the ratio $1 : \sqrt{3} : 2$ (short leg s , long leg $s\sqrt{3}$, hypotenuse $2s$).
- **Similar** triangles have the same shape: equal angles and proportional sides. Shadows cast at the same time of day make similar triangles.

WORKED EXAMPLE

A 13-foot ladder leans against a wall. Its foot is 5 feet from the wall. How high up the wall does the ladder reach?

Solution. The ladder is the hypotenuse: $5^2 + h^2 = 13^2$, so $h^2 = 169 - 25 = 144$ and $h = \sqrt{144} = 12$ feet.

No-Calculator Tip Spot the triple before you square anything. For legs 15 and 20, divide by 5 to get 3 and 4; the hypotenuse is $5 \times 5 = 25$.

Common Traps

- Forgetting the square root: $a^2 + b^2$ is c^2 , not c .
- Adding the squares when you need a leg; subtract from the hypotenuse instead.
- Mixing up $\sqrt{2}$ (45° - 45° - 90°) and $\sqrt{3}$ (30° - 60° - 90°).
- In similar triangles, *multiplying* by the scale factor, not *adding* the difference.

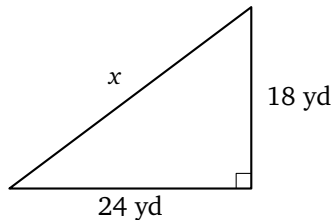
Practice Set 17

Score: ____ / 25

Target time: 26 minutes

Problems 1–8 are warm-ups, 9–18 are test level, and 19–25 are challenge problems.

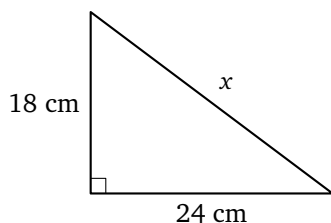
- Two angles of a triangle measure 36° and 60° . What is the measure of the third angle?
A 84° B 96° C 120° D 264°
- A right triangle has legs of 18 yards and 24 yards. What is the length of the hypotenuse?
- A triangle has angles of 73° and 75° . Which of the following describes the triangle?
A a right triangle
B an acute triangle
C an equilateral triangle
D an obtuse triangle



- A 30 yd B 31 yd C 42 yd D 900 yd

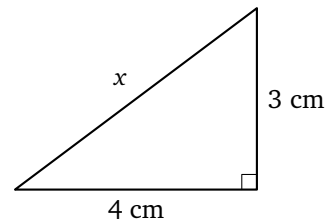
- Two angles of a triangle measure 72° and 41° . What is the measure of the third angle?
A 47° B 67° C 113° D 247°

- The legs of a right triangle measure 18 centimeters and 24 centimeters. How long is the hypotenuse?



- A 30 cm B 31 cm C 42 cm D 900 cm

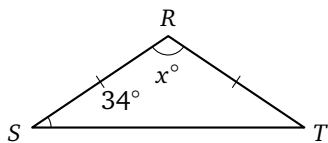
- The legs of a right triangle measure 3 centimeters and 4 centimeters. How long is the hypotenuse?



- A 2 cm B 3 cm C 5 cm D 7 cm

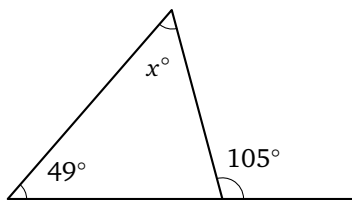
- Two angles of a triangle measure 17° and 45° . What kind of triangle is it?
A an equilateral triangle
B an obtuse triangle
C a right triangle
D an acute triangle
- Two angles of a triangle measure 44° and 61° . What is the measure of the third angle?
A 75° B 105° C 119° D 255°

9. In triangle RST , $RS = RT$ and $\angle S$ measures 34° . What is the value of x ?



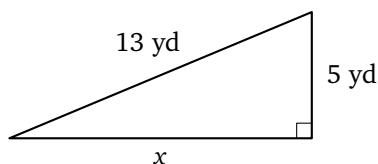
A 112 B 117 C 122 D 146

10. In the figure, one side of the triangle is extended to form a 105° exterior angle. One remote interior angle measures 49° . What is the value of x ?



A 36 B 41 C 56 D 154

11. A right triangle has a hypotenuse of 13 yards and a leg of 5 yards. How long is its other leg?



A 8 yd B 9 yd C 12 yd D 144 yd

12. Sergeant Kim's patrol moves 24 kilometers east, then 18 kilometers north. What is the straight-line distance from the patrol back to its starting point?

A 29 km B 30 km C 42 km D 900 km

13. Two sides of a triangle measure 5 and 7. Which of the following could be the length of the third side?

A 11 B 12 C 14 D 16

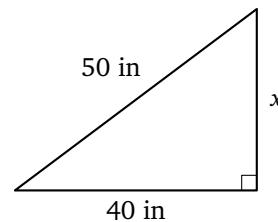
14. Seaman Carter's patrol moves 15 kilometers east, then 20 kilometers north. What is the straight-line distance from the patrol back to its starting point?

A 22 km B 24 km C 25 km D 35 km

15. Two sides of a triangle measure 6 and 19. Which describes all possible lengths x of the third side?

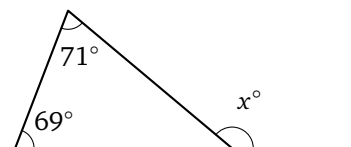
A $0 < x < 25$ B $13 < x < 25$
C $13 \leq x \leq 25$ D $6 < x < 19$

16. In the right triangle shown, the hypotenuse is 50 inches and one leg is 40 inches. What is the value of x ?



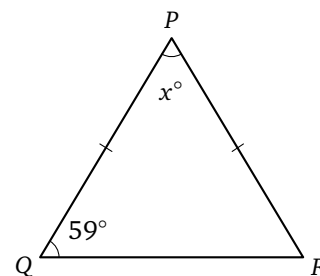
A 10 in B 27 in C 28 in D 30 in

17. A side of the triangle shown is extended to form an exterior angle of x° . The two remote interior angles measure 69° and 71° . What is x ?



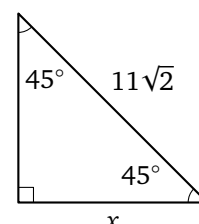
A 2 B 40 C 140 D 220

18. Each base angle of isosceles triangle PQR measures 59° . What is the value of x , the measure of the vertex angle?



A 42 B 52 C 59 D 62

19. In the 45° - 45° - 90° triangle shown, the hypotenuse measures $11\sqrt{2}$. What is the length x of each leg?

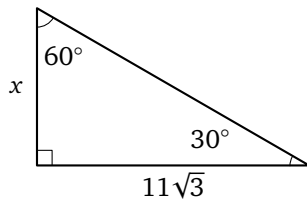


A $\frac{11\sqrt{2}}{2}$ B 11 C $11\sqrt{3}$ D 22

20. Petty Officer Lopez, who is 6 feet tall, casts a 12-foot shadow. At the same time, a radio tower on base casts a shadow 304 feet long. How tall is the radio tower?

A 142 ft B 147 ft C 152 ft D 298 ft

21. The figure shows a 30° - 60° - 90° triangle in which the longer leg is $11\sqrt{3}$. What is the value of x ?

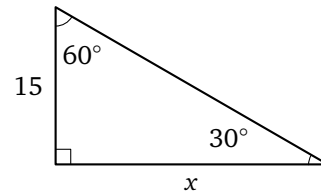


A 11 B $\frac{33}{2}$ C 22 D 33

22. A 3-foot fence post casts a shadow 4 feet long. At the same time, a radio tower casts a shadow 236 feet long. How tall is the radio tower?

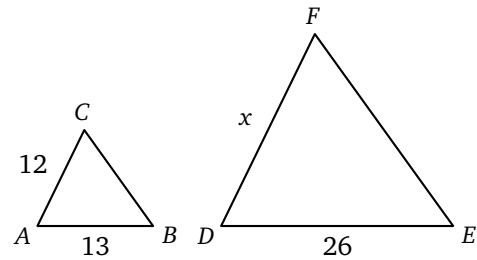
A 172 ft B 177 ft C 235 ft D 708 ft

23. In the 30° - 60° - 90° triangle shown, the shorter leg measures 15. What is the length x of the longer leg?



A $15\sqrt{2}$ B $15\sqrt{3}$ C 30 D $45\sqrt{3}$

24. $\triangle ABC \sim \triangle DEF$. If $AB = 13$, $AC = 12$, and $DE = 26$, what is the length x of DF ?



A 24 B 25 C 26 D 28

25. Corporal Okafor sets a 10-foot ladder against a barracks wall, 6 feet from the base. If the foot of the ladder is moved out to 8 feet from the wall, how many feet lower will the top of the ladder be?

A 2 ft B 4 ft C 6 ft D 8 ft

Answers & Explanations 17

1 A	6 C	11 C	16 D	21 A
2 A	7 B	12 B	17 C	22 B
3 B	8 A	13 A	18 D	23 B
4 A	9 A	14 C	19 B	24 A
5 B	10 C	15 B	20 C	25 A

1. **A** 84°

- The three angles of any triangle add up to 180° .
- Add the two known angles: $36 + 60 = 96$.
- Subtract from 180: $180 - 96 = 84$, so the third angle is 84° .

WHY NOT (B) adds the two given angles and stops. (C) subtracts only one of the given angles. (D) uses 360° instead of 180° for the angle sum.

2. **A** 30 yd

- Use the Pythagorean theorem: $a^2 + b^2 = c^2$, where c is the hypotenuse.
- Square the legs and add: $18^2 + 24^2 = 324 + 576 = 900$.
- Take the square root: $c = \sqrt{900} = 30$ yd.

TIP Spot the triple: 18-24-? is 3-4-5 times 6, so the hypotenuse is $5 \times 6 = 30$.

WHY NOT (C) adds the legs instead of using $a^2 + b^2 = c^2$. (D) forgets to take the square root.

3. **B** 67°

- The three angles of any triangle add up to 180° .
- Add the two known angles: $72 + 41 = 113$.
- Subtract from 180: $180 - 113 = 67$, so the third angle is 67° .

WHY NOT (C) adds the two given angles and stops. (D) uses 360° instead of 180° for the angle sum.

4. **A** 30 cm

- Use the Pythagorean theorem: $a^2 + b^2 = c^2$, where c is the hypotenuse.
- Square the legs and add: $18^2 + 24^2 = 324 + 576 = 900$.
- Take the square root: $c = \sqrt{900} = 30$ cm.

TIP Spot the triple: 18-24-? is 3-4-5 times 6, so the hypotenuse is $5 \times 6 = 30$.

WHY NOT (C) adds the legs instead of using $a^2 + b^2 = c^2$. (D) forgets to take the square root.

5. **B** an acute triangle

- Find the third angle: $180^\circ - 73^\circ - 75^\circ = 32^\circ$.
- The largest angle is 75° , which is less than 90° , so all three angles are acute. It is an acute triangle.

WHY NOT (A) would need a 90° angle, but the angles are 73° , 75° , and 32° . (C) would need all three angles to be 60° . (D) would need an angle greater than 90° , but the largest angle is 75° .

6. **C** 5 cm

- Use the Pythagorean theorem: $a^2 + b^2 = c^2$, where c is the hypotenuse.
- Square the legs and add: $3^2 + 4^2 = 9 + 16 = 25$.
- Take the square root: $c = \sqrt{25} = 5$ cm.

WHY NOT (D) adds the legs instead of using $a^2 + b^2 = c^2$.

7. **B** an obtuse triangle

- Find the third angle: $180^\circ - 17^\circ - 45^\circ = 118^\circ$.
- The largest angle is 118° , which is more than 90° . It is an obtuse triangle.

WHY NOT (A) would need all three angles to be 60° . (C) would need a 90° angle, but the angles are 17° , 45° , and 118° . (D) would need all three angles to be less than 90° , but one angle is 118° .

8. **A** 75°

- The three angles of any triangle add up to 180° .
- Add the two known angles: $44 + 61 = 105$.
- Subtract from 180: $180 - 105 = 75$, so the third angle is 75° .

WHY NOT (B) adds the two given angles and stops. (C) subtracts only one of the given angles. (D) uses 360° instead of 180° for the angle sum.

9. **A** 112

- The two base angles of an isosceles triangle are equal, so both measure 34° .
- The angles add up to 180: $x = 180 - 34 - 34 = 180 - 68 = 112$.

WHY NOT (D) subtracts only one base angle from 180° .

10. **C** 56

- 1 An exterior angle of a triangle equals the sum of the two remote interior angles.
- 2 So $49 + x = 105$, and $x = 105 - 49 = 56$.

WHY NOT (D) adds the angles instead of subtracting.

11. **C** 12 yd

- 1 The hypotenuse is the longest side, so use $a^2 + b^2 = c^2$ and solve for the missing leg: $\text{leg}^2 = c^2 - (\text{other leg})^2$.
- 2 $13^2 - 5^2 = 169 - 25 = 144$.
- 3 Take the square root: $\sqrt{144} = 12$ yd.

WHY NOT (A) subtracts the lengths instead of their squares. (D) forgets to take the square root.

12. **B** 30 km

- 1 Traveling east and then north makes a right angle, so the two legs of the trip and the straight-line distance form a right triangle.
- 2 $24^2 + 18^2 = 576 + 324 = 900$.
- 3 $\sqrt{900} = 30$, so the answer is 30 kilometers.

TIP Shortcut: divide by 6 and you get the 3-4-5 triple, so the hypotenuse is $5 \times 6 = 30$.

WHY NOT (C) adds the two distances instead of using the Pythagorean theorem. (D) forgets to take the square root.

13. **A** 11

- 1 Triangle inequality: any two sides must add up to *more* than the third side.
- 2 So the third side must be longer than $7 - 5 = 2$ and shorter than $5 + 7 = 12$: between 2 and 12, not equal to either.
- 3 Only 11 is in that range.

WHY NOT (B) equals $5 + 7$; the two given sides would lie flat along the third side. (C) is too long: $5 + 7 < 14$.

14. **C** 25 km

- 1 Traveling east and then north makes a right angle, so the two legs of the trip and the straight-line distance form a right triangle.
- 2 $15^2 + 20^2 = 225 + 400 = 625$.
- 3 $\sqrt{625} = 25$, so the answer is 25 kilometers.

TIP Shortcut: divide by 5 and you get the 3-4-5 triple, so the hypotenuse is $5 \times 5 = 25$.

WHY NOT (D) adds the two distances instead of using the Pythagorean theorem.

15. **B** $13 < x < 25$

- 1 Triangle inequality: any two sides must add up to *more* than the third side.
- 2 The third side must be shorter than the sum: $x < 6 + 19 = 25$.
- 3 It must be longer than the difference (otherwise $x + 6$ would not reach past 19): $x > 19 - 6 = 13$.
- 4 Together: $13 < x < 25$.

WHY NOT (A) forgets the lower limit (the difference of the sides). (C) includes 13 and 25, but then the sides would lie flat. (D) uses the two given sides as the limits.

16. **D** 30 in

- 1 The hypotenuse is the longest side, so use $a^2 + b^2 = c^2$ and solve for the missing leg: $\text{leg}^2 = c^2 - (\text{other leg})^2$.
- 2 $50^2 - 40^2 = 2,500 - 1,600 = 900$.
- 3 Take the square root: $\sqrt{900} = 30$ in.

TIP Shortcut: divide by 10 and you get the 3-4-5 triple, so the missing side is $3 \times 10 = 30$.

WHY NOT (A) subtracts the lengths instead of their squares.

17. **C** 140

- 1 An exterior angle of a triangle equals the sum of the two *remote* (non-adjacent) interior angles.
- 2 $x = 69 + 71 = 140$.
- 3 Check: the interior angle next to it is $180 - 140 = 40$, and $69 + 71 + 40 = 180$. ✓

WHY NOT (A) subtracts the two remote angles instead of adding them. (B) finds the interior angle next to the exterior angle, not the exterior angle. (D) subtracts from 360° .

18. **D** 62

- 1 The two base angles of an isosceles triangle are equal, so both measure 59° .
- 2 The angles add up to 180: $x = 180 - 59 - 59 = 180 - 118 = 62$.

WHY NOT (C) assumes the vertex angle equals the base angle.

19. **B** 11

- 1 In a 45° - 45° - 90° triangle the sides are in the ratio $1 : 1 : \sqrt{2}$ (leg : leg : hypotenuse).
- 2 Leg = hypotenuse $\div \sqrt{2} = 11\sqrt{2} \div \sqrt{2} = 11$.

WHY NOT (A) halves the hypotenuse, the 30° - 60° - 90° rule. (C) uses $\sqrt{3}$ instead of $\sqrt{2}$. (D) multiplies by $\sqrt{2}$ instead of dividing.

20. **C** 152 ft

1 Objects and their shadows at the same time of day form similar triangles, so $\frac{\text{height}}{\text{shadow}}$ is the same for both: $\frac{h}{304} = \frac{6}{12}$.

2 Cross-multiply: $12h = 6 \times 304 = 1,824$.

3 Divide by 12: $h = 1,824 \div 12 = 152$ feet.

TIP Shortcut: at this time of day every shadow is twice as long as the object is tall, so the radio tower is $304 \div 2 = 152$ feet tall.

WHY NOT (D) adds the difference between height and shadow instead of using a ratio.

21. **A** 11

1 In a 30° - 60° - 90° triangle the sides are in the ratio $1 : \sqrt{3} : 2$ (short leg : long leg : hypotenuse). The short leg is across from the 30° angle.

2 Short leg = long leg $\div \sqrt{3} = 11\sqrt{3} \div \sqrt{3} = 11$.

WHY NOT (C) gives the hypotenuse, not the short leg. (D) multiplies by $\sqrt{3}$ instead of dividing.

22. **B** 177 ft

1 Objects and their shadows at the same time of day form similar triangles, so $\frac{\text{height}}{\text{shadow}}$ is the same for both: $\frac{h}{236} = \frac{3}{4}$.

2 Cross-multiply: $4h = 3 \times 236 = 708$.

3 Divide by 4: $h = 708 \div 4 = 177$ feet.

WHY NOT (C) adds the difference between height and shadow instead of using a ratio. (D) multiplies by the height but forgets to divide by the shadow.

23. **B** $15\sqrt{3}$

1 In a 30° - 60° - 90° triangle the sides are in the ratio $1 : \sqrt{3} : 2$ (short leg : long leg : hypotenuse). The short leg is across from the 30° angle.

2 Long leg = short leg $\times \sqrt{3} = 15\sqrt{3}$.

WHY NOT (A) uses $\sqrt{2}$ instead of $\sqrt{3}$. (C) gives the hypotenuse, not the long leg.

24. **A** 24

1 Corresponding sides are proportional: $\frac{DF}{AC} = \frac{DE}{AB}$, so $\frac{x}{12} = \frac{26}{13}$.

2 The scale factor is $26 \div 13 = 2$.

3 Multiply: $x = 12 \times 2 = 24$.

WHY NOT (B) adds the difference between corresponding sides instead of multiplying by the scale factor.

25. **A** 2 ft

1 Before: height = $\sqrt{10^2 - 6^2} = \sqrt{64} = 8$ feet.

2 After: height = $\sqrt{10^2 - 8^2} = \sqrt{36} = 6$ feet.

3 The top slides down $8 - 6 = 2$ feet.

WHY NOT (C) gives the final height of the ladder. (D) gives the starting height of the ladder.

Perimeter & Area

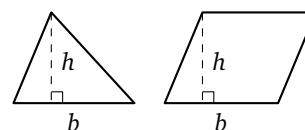
Tested on: Mathematics Knowledge • Arithmetic Reasoning

Key Concepts

Perimeter is the distance around a figure: add up the sides, and the answer is in plain units (feet, meters).
Area is the amount of surface inside: it is measured in *square* units (ft², or square feet).

Formulas

Shape	Perimeter	Area
Rectangle	$2l + 2w$	$l \times w$
Square	$4s$	s^2
Triangle	add the sides	$\frac{1}{2} \times b \times h$
Parallelogram	add the sides	$b \times h$
Trapezoid	add the sides	$\frac{1}{2}(b_1 + b_2) \times h$



The height h is always measured at a *right angle* to the base (the dashed lines). It is not a slanted side.

Composite shapes, units, and scaling

- Split an odd shape into rectangles and add the areas, or take a big rectangle and subtract the missing piece.
- 1 yard = 3 feet, so 1 square yard = $3 \times 3 = 9$ square feet. Likewise 1 square foot = $12 \times 12 = 144$ square inches.
- If every side is multiplied by k , the perimeter is multiplied by k and the area by k^2 : doubling the sides makes the area 4 times as large.

WORKED EXAMPLE

Carpet costs \$25 per square yard. How much does it cost to carpet a room that is 15 feet by 12 feet?

Solution. Area = $15 \times 12 = 180$ square feet. Change to square yards: $180 \div 9 = 20$ square yards. Cost = $20 \times \$25 = \500 .

No-Calculator Tip Convert lengths *before* you multiply when the numbers divide evenly: 15 ft $\times$ 12 ft is 5 yd $\times$ 4 yd = 20 square yards, with no large division at all.

Common Traps

- Mixing up perimeter (add the sides) and area (multiply).
- Forgetting the $\frac{1}{2}$ in the triangle and trapezoid formulas.
- Using a slanted side instead of the height.

- Dividing by 3 instead of 9 to change square feet to square yards.

Practice Set 18

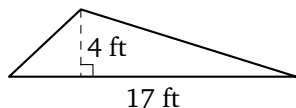
Score: ____ / 25

Target time: 26 minutes

Problems 1–8 are warm-ups, 9–18 are test level, and 19–25 are challenge problems.

- A square has sides of 10 centimeters. What is its area?
A 20 cm^2 B 40 cm^2 C 50 cm^2 D 100 cm^2
- A square has an area of 25 square inches. What is the length of one side?
A 1 in B 2 in C 5 in D 6.25 in

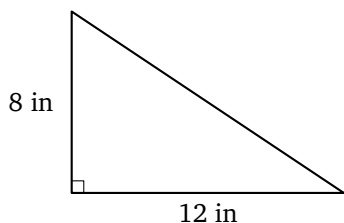
- What is the area of the triangle shown, with base 17 feet and height 4 feet?



- A 21 ft^2 B 34 ft^2 C 44 ft^2 D 68 ft^2

- A square has sides of 12 meters. What is its area?
A 24 m^2 B 48 m^2 C 144 m^2 D 148 m^2

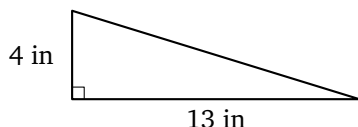
- A right triangle has legs of 12 inches and 8 inches. What is its area?



- A 20 in^2 B 24 in^2 C 48 in^2 D 96 in^2

- The area of a square is 16 square yards. How long is each side?
A 1 yd B 3 yd C 4 yd D 8 yd

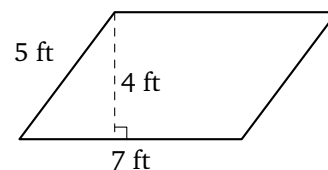
- What is the area of the right triangle shown, whose legs measure 13 inches and 4 inches?



- A 13 in^2 B 17 in^2 C 20 in^2 D 26 in^2

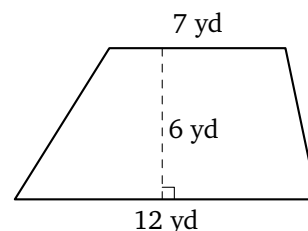
- A barracks room floor is a rectangle 18 feet long and 16 feet wide. What is its area?
A 34 ft^2 B 68 ft^2 C 144 ft^2 D 288 ft^2

- The parallelogram shown has a base of 7 feet, a height of 4 feet, and slanted sides of 5 feet. What is its area?



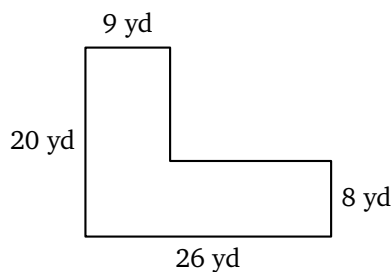
- A 14 ft^2 B 28 ft^2 C 34 ft^2 D 35 ft^2

- What is the area of a trapezoid with bases of 7 yards and 12 yards and a height of 6 yards?



- A 57 yd^2 B 72 yd^2 C 114 yd^2 D 252 yd^2

11. The figure shows a lawn. All corners are right angles. The sides shown measure 26 yards, 20 yards, 9 yards, and 8 yards. What is the area?

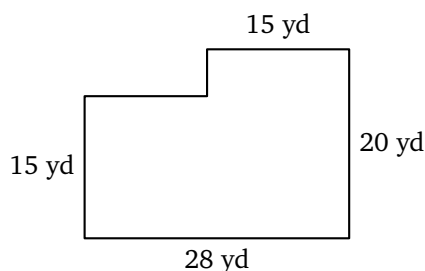


A 316 yd^2 B 320 yd^2 C 448 yd^2 D 520 yd^2

12. Jasmine is covering a rectangular lawn 55 feet by 40 feet with sod that costs \$0.50 per square foot. How much will the sod cost?

A \$47.50 B \$95.00
C \$1,075.00 D \$1,100.00

13. The figure shows a lawn. All corners are right angles. The sides shown measure 28 yards, 20 yards, 15 yards, and 15 yards. What is the area?

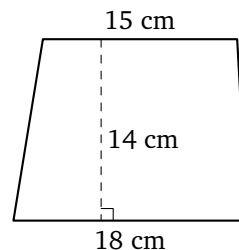


A 495 yd^2 B 499 yd^2 C 505 yd^2 D 560 yd^2

14. Priya is putting new baseboard trim along all four walls of a room that is 17 feet by 12 feet. Trim costs \$2.50 per foot. Ignoring the doorway, how much will the trim cost?

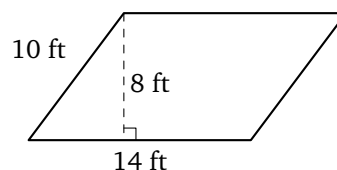
A \$58.00 B \$72.50 C \$145.00 D \$510.00

15. What is the area of a trapezoid with bases of 15 centimeters and 18 centimeters and a height of 14 centimeters?



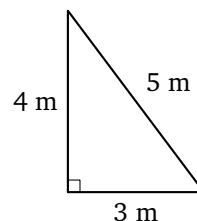
A 210 cm^2 B 231 cm^2 C 252 cm^2 D 462 cm^2

16. The parallelogram shown has a base of 14 feet, a height of 8 feet, and slanted sides of 10 feet. What is its area?



A 48 ft^2 B 56 ft^2 C 112 ft^2 D 140 ft^2

17. A right triangle has sides of 4 meters, 3 meters, and 5 meters. What is its area?



A 4 m^2 B 6 m^2 C 10 m^2 D 12 m^2

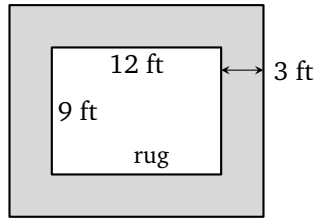
18. The area of a rectangle is 108 square feet, and the rectangle is 12 feet long. What is its perimeter?

A 9 ft B 21 ft C 38 ft D 42 ft

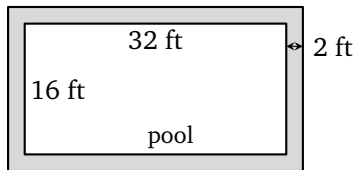
19. Grace is buying carpet for a recreation room that measures 12 feet by 18 feet. The carpet costs \$18 per square yard. What is the cost of the carpet?

A \$24 B \$432 C \$1,296 D \$3,888

20. A 12-foot by 9-foot rug is centered in a rectangular room, leaving a strip of bare floor 3 feet wide along every wall. What is the area of the bare floor alone?



- A 72 ft^2 B 162 ft^2 C 178 ft^2 D 270 ft^2
21. A square has an area of 81 square feet. If each side of the square is multiplied by 5, what is the area of the new square?
- A 180 ft^2 B 196 ft^2
C 405 ft^2 D $2,025 \text{ ft}^2$
22. A rectangular swimming pool is 32 feet long and 16 feet wide. A concrete walkway 2 feet wide surrounds the pool. What is the area of the walkway alone?



- A 100 ft^2 B 192 ft^2 C 200 ft^2 D 208 ft^2

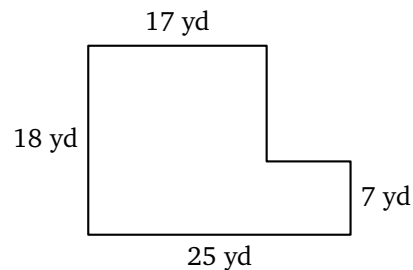
23. If the base and the height of a triangle are both multiplied by 10, the area of the triangle is multiplied by what number?

A 10 B 20 C 97 D 100

24. Brandon is buying carpet for a bedroom that measures 24 feet by 15 feet. The carpet costs \$32 per square yard. What is the cost of the carpet?

A \$1,280 B \$1,320 C \$3,840 D \$11,520

25. The figure shows a parking lot. All corners are right angles. The sides shown measure 25 yards, 18 yards, 17 yards, and 7 yards. What is the perimeter?



- A 86 yd B 94 yd C 124 yd D 362 yd

Answers & Explanations 18

1 D	6 D	11 A	16 C	21 D
2 B	7 C	12 D	17 B	22 D
3 C	8 D	13 A	18 D	23 D
4 C	9 B	14 C	19 B	24 A
5 C	10 A	15 B	20 B	25 A

1. **D** 100 cm²

1 Area of a square = side $\times$ side = s^2 .
 2 $10 \times 10 = 100$, so the area is 100 square centimeters.

WHY NOT (A) doubles the side instead of squaring it. (B) gives the perimeter, not the area.

2. **B** 34 ft²

1 The base and the height (the dashed line, at a right angle to the base) are given.
 2 Area of a triangle = $\frac{1}{2} \times$ base $\times$ height.
 3 $\frac{1}{2} \times 17 \times 4 = \frac{1}{2} \times 68 = 34$, so the area is 34 square feet.

WHY NOT (A) adds the base and height. (D) forgets to multiply by $\frac{1}{2}$.

3. **C** 144 m²

1 Area of a square = side $\times$ side = s^2 .
 2 $12 \times 12 = 144$, so the area is 144 square meters.

WHY NOT (A) doubles the side instead of squaring it. (B) gives the perimeter, not the area.

4. **C** 48 in²

1 In a right triangle the two legs are a base and a height.
 2 Area of a triangle = $\frac{1}{2} \times$ base $\times$ height.
 3 $\frac{1}{2} \times 12 \times 8 = \frac{1}{2} \times 96 = 48$, so the area is 48 square inches.

WHY NOT (A) adds the base and height. (B) takes half of the base and half of the height. (D) forgets to multiply by $\frac{1}{2}$.

5. **C** 4 yd

1 Area of a square = $s \times s$, so the side is the number that times itself gives the area.
 2 $4 \times 4 = 16$, so each side is 4 yards.

WHY NOT (D) divides the area by 2 instead of taking the square root.

6. **D** 26 in²

1 In a right triangle the two legs are a base and a height.

2 Area of a triangle = $\frac{1}{2} \times$ base $\times$ height.

3 $\frac{1}{2} \times 13 \times 4 = \frac{1}{2} \times 52 = 26$, so the area is 26 square inches.

WHY NOT (A) takes half of the base and half of the height. (B) adds the base and height.

7. **C** 5 in

1 Area of a square = $s \times s$, so the side is the number that times itself gives the area.

2 $5 \times 5 = 25$, so each side is 5 inches.

WHY NOT (D) divides the area by 4, as if it were the perimeter.

8. **D** 288 ft²

1 Area of a rectangle = length $\times$ width.

2 $18 \times 16 = 288$, so the area is 288 square feet.

WHY NOT (A) adds the length and width instead of multiplying. (B) gives the perimeter, not the area. (C) takes half, as if the shape were a triangle.

9. **B** 28 ft²

1 Area of a parallelogram = base $\times$ height, where the height is the dashed line at a right angle to the base (not the slanted side).

2 $7 \times 4 = 28$, so the area is 28 square feet.

TIP Why it works: cut the triangle off one end and slide it to the other end. You get a 7×4 rectangle.

WHY NOT (A) takes $\frac{1}{2}$, as for a triangle. (D) multiplies by the slanted side instead of the height.

10. **A** 57 yd²

1 Area of a trapezoid = $\frac{1}{2} \times (b_1 + b_2) \times h$: average the two parallel bases, then multiply by the height.

2 Add the bases: $12 + 7 = 19$.

3 $\frac{1}{2} \times 19 \times 6 = 19 \times 3 = 57$, so the area is 57 square yards.

WHY NOT (B) uses only the longer base. (C) forgets to multiply by $\frac{1}{2}$. (D) multiplies the two bases instead of adding them.

11. **A** 316 yd²

- Find the missing corner: it is $26 - 9 = 17$ yd wide and $20 - 8 = 12$ yd tall.
- Area of the whole 26×20 rectangle: $26 \times 20 = 520$.
- Subtract the missing corner: $520 - 17 \times 12 = 520 - 204 = 316$ square yards.

TIP Or split the shape into two rectangles: $9 \times 20 = 180$ and $17 \times 8 = 136$; $180 + 136 = 316$.

WHY NOT (C) subtracts the wrong rectangle. (D) forgets to subtract the missing corner.

12. **D** \$1,100.00

- Find the area: $55 \times 40 = 2,200$ square feet.
- Multiply by the price: $2,200 \times 0.5 = 1,100$, so the cost is \$1,100.

WHY NOT (A) adds the length and width instead of multiplying. (B) uses the perimeter instead of the area.

13. **A** 495 yd²

- Find the missing corner: it is $28 - 15 = 13$ yd wide and $20 - 15 = 5$ yd tall.
- Area of the whole 28×20 rectangle: $28 \times 20 = 560$.
- Subtract the missing corner: $560 - 13 \times 5 = 560 - 65 = 495$ square yards.

TIP Or split the shape into two rectangles: $15 \times 20 = 300$ and $13 \times 15 = 195$; $300 + 195 = 495$.

WHY NOT (D) forgets to subtract the missing corner.

14. **C** \$145.00

- Trim runs along the walls, so find the perimeter: $2 \times 17 + 2 \times 12 = 58$ feet.
- Multiply by the price: $58 \times 2.5 = 145$, so the trim costs \$145.

WHY NOT (A) finds the perimeter but forgets to multiply by the price. (B) covers only two of the four walls. (D) uses the area instead of the perimeter.

15. **B** 231 cm²

- Area of a trapezoid = $\frac{1}{2} \times (b_1 + b_2) \times h$: average the two parallel bases, then multiply by the height.
- Add the bases: $18 + 15 = 33$.
- $\frac{1}{2} \times 33 \times 14 = 33 \times 7 = 231$, so the area is 231 square centimeters.

WHY NOT (A) uses only the shorter base. (C) uses only the longer base. (D) forgets to multiply by $\frac{1}{2}$.

16. **C** 112 ft²

- Area of a parallelogram = base $\times$ height, where the height is the dashed line at a right angle to the base (not the slanted side).

- $14 \times 8 = 112$, so the area is 112 square feet.

TIP Why it works: cut the triangle off one end and slide it to the other end. You get a 14×8 rectangle.

WHY NOT (A) gives the perimeter, not the area. (B) takes $\frac{1}{2}$, as for a triangle. (D) multiplies by the slanted side instead of the height.

17. **B** 6 m²

- In a right triangle the two legs (the sides that form the right angle) are the base and the height. The longest side, 5 meters, is the hypotenuse and is not used.

- Area of a triangle = $\frac{1}{2} \times \text{base} \times \text{height}$.

- $\frac{1}{2} \times 3 \times 4 = \frac{1}{2} \times 12 = 6$, so the area is 6 square meters.

WHY NOT (C) uses the hypotenuse as the height. (D) forgets to multiply by $\frac{1}{2}$.

18. **D** 42 ft

- Find the width: $108 \div 12 = 9$ ft.

- Perimeter = $2 \times 12 + 2 \times 9 = 24 + 18 = 42$, so the perimeter is 42 feet.

WHY NOT (A) finds the width and stops. (B) adds only one length and one width.

19. **B** \$432

- Change each length to yards: $12 \div 3 = 4$ yd and $18 \div 3 = 6$ yd.

- Area in square yards: $4 \times 6 = 24$ square yards.

- Cost: $24 \times 18 = 432$, so the carpet costs \$432.

TIP Or find square feet first ($12 \times 18 = 216$) and divide by 9: $216 \div 9 = 24$ square yards.

WHY NOT (A) finds the number of square yards but forgets to multiply by the price. (C) divides by 3 instead of 9 (a square yard is 3 ft by 3 ft). (D) forgets to change square feet to square yards.

20. **B** 162 ft²

- The bare floor adds 3 feet on *both* ends of each side, so the outer rectangle is $12 + 2 \times 3 = 18$ by $9 + 2 \times 3 = 15$ feet.

- Outer area: $18 \times 15 = 270$ square feet.

- Rug area: $12 \times 9 = 108$ square feet.

- Bare floor alone: $270 - 108 = 162$ square feet.

WHY NOT (A) adds the bare floor width to only one end of each side. (D) gives the total area, including the rug.

21. **D** 2,025 ft²

- 1 The original side is 9 ft, because $9 \times 9 = 81$.
- 2 The new side is $9 \times 5 = 45$ ft.
- 3 New area: $45 \times 45 = 2,025$ square feet.

WHY NOT (A) gives the new perimeter instead of the new area. (B) adds 5 to the side instead of multiplying. (C) multiplies the area by 5 instead of 25.

22. **D** 208 ft²

- 1 The walkway adds 2 feet on *both* ends of each side, so the outer rectangle is $32 + 2 \times 2 = 36$ by $16 + 2 \times 2 = 20$ feet.
- 2 Outer area: $36 \times 20 = 720$ square feet.
- 3 Pool area: $32 \times 16 = 512$ square feet.
- 4 Walkway alone: $720 - 512 = 208$ square feet.

WHY NOT (A) adds the walkway width to only one end of each side. (B) multiplies the perimeter by the width and misses the four corner squares.

23. **D** 100

- 1 Area multiplies two lengths together, and each length is multiplied by 10.
- 2 So the area is multiplied by $10 \times 10 = 100$.

TIP Try numbers: a 1-by-1 square has area 1; a 10-by-10 square has area 100.

WHY NOT (A) assumes the area grows by the same factor as the sides. (B) doubles the scale factor instead of squaring it.

24. **A** \$1,280

- 1 Change each length to yards: $24 \div 3 = 8$ yd and $15 \div 3 = 5$ yd.
- 2 Area in square yards: $8 \times 5 = 40$ square yards.
- 3 Cost: $40 \times 32 = 1,280$, so the carpet costs \$1,280.

TIP Or find square feet first ($24 \times 15 = 360$) and divide by 9: $360 \div 9 = 40$ square yards.

WHY NOT (C) divides by 3 instead of 9 (a square yard is 3 ft by 3 ft). (D) forgets to change square feet to square yards.

25. **A** 86 yd

- 1 Find the missing corner: it is $25 - 17 = 8$ yd wide and $18 - 7 = 11$ yd tall.
- 2 Go around the shape and add all six sides: $25 + 7 + 8 + 11 + 17 + 18 = 86$ yards.

TIP Shortcut: the two inside edges together make up for the missing pieces of the top and side, so the perimeter equals that of a 25×18 rectangle: $2 \times (25 + 18) = 86$.

WHY NOT (C) counts the two inside edges twice. (D) gives the area, not the perimeter.

Circles

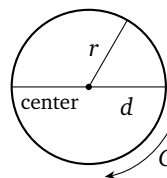
Tested on: Mathematics Knowledge • Arithmetic Reasoning

Key Concepts

Circle questions use just two formulas, and the ASVAB does *not* give them to you, so learn them by heart. Almost every mistake comes from mixing up the radius and the diameter, or the circumference and the area.

Parts of a circle

- The **radius** r runs from the center to the circle.
- The **diameter** d goes all the way across through the center:
 $d = 2r$ and $r = d \div 2$.
- The **circumference** C is the distance around the circle (its perimeter).
- The **area** A is the space inside, in square units.



The two formulas

$$C = 2\pi r = \pi d \qquad A = \pi r^2$$

$\pi \approx 3.14 \approx \frac{22}{7}$. Most answers are left “in terms of π ”: 36π simply means $36 \times \pi$. Use 3.14 or $\frac{22}{7}$ only when the question tells you to. In πr^2 only the radius is squared: if $r = 6$, then $A = \pi \times 36 = 36\pi$.

Parts of a circle's area and perimeter

- A **semicircle** is half a circle; a **quarter circle** is one fourth. Their perimeters also include the straight edges.
- A **sector** (a pie slice) with central angle a° is the fraction $\frac{a}{360}$ of the circle:

$$\text{arc length} = \frac{a}{360} \times 2\pi r \qquad \text{sector area} = \frac{a}{360} \times \pi r^2$$

- **Shaded regions:** area of the whole shape minus the area of the part that is cut out. A ring between two circles is $\pi R^2 - \pi r^2$.

WORKED EXAMPLE

A circular garden has a diameter of 10 feet. Using 3.14 for π , find its area and the length of fence needed to go around it.

Solution. The radius is $10 \div 2 = 5$ feet. Area: $A = \pi r^2 \approx 3.14 \times 5^2 = 3.14 \times 25 = 78.5$ square feet. Fence: $C = \pi d \approx 3.14 \times 10 = 31.4$ feet.

No-Calculator Tip To multiply by 3.14, multiply by 3 and by 0.14 and add: $3.14 \times 25 = 75 + 3.5 = 78.5$.
Scaling: if the radius is multiplied by k , the circumference is multiplied by k but the area by k^2 , so doubling the radius makes the area 4 times as large.

Common Traps

- Putting the diameter into πr^2 . Halve the diameter *first*.
- Mixing up the formulas: $2\pi r$ is the distance around, πr^2 is the space inside.
- Halving after squaring: with $d = 10$, the area is $\pi \times 5^2 = 25\pi$, not $\pi \times 100 \div 2 = 50\pi$.
- Leaving out the straight edges in the perimeter of a semicircle or quarter circle.

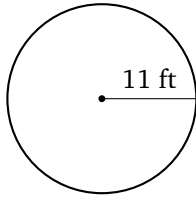
Practice Set 19

Score: ____ / 25

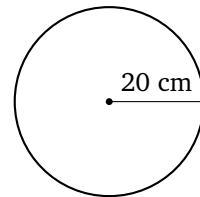
Target time: 26 minutes

Problems 1–8 are warm-ups, 9–18 are test level, and 19–25 are challenge problems.

1. In the figure, the radius of the circle is 11 feet. What is the area of the circle?

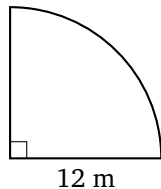


- A $11\pi \text{ ft}^2$ B $22\pi \text{ ft}^2$ C $121\pi \text{ ft}^2$ D $484\pi \text{ ft}^2$
2. What is the circumference of a circle with a radius of 9 centimeters?
 A $9\pi \text{ cm}$ B $14\pi \text{ cm}$ C $18\pi \text{ cm}$ D $81\pi \text{ cm}$
3. What is the area of a circle with a radius of 3 centimeters?
 A $6\pi \text{ cm}^2$ B $9\pi \text{ cm}^2$
 C $16\pi \text{ cm}^2$ D $36\pi \text{ cm}^2$
4. A round wall clock has a radius of 7 inches. What is the distance around the edge of the clock?
 A $14\pi \text{ in}$ B $16\pi \text{ in}$ C $28\pi \text{ in}$ D $49\pi \text{ in}$
5. A goat is tied to a stake in an open field with a rope 14 feet long. How many square feet of grass can the goat reach? Use $\frac{22}{7}$ for π .
 A 616 ft^2 B 638 ft^2
 C 660 ft^2 D $2,464 \text{ ft}^2$
6. What is the circumference of a circle with a diameter of 20 yards?
 A $10\pi \text{ yd}$ B $16\pi \text{ yd}$ C $18\pi \text{ yd}$ D $20\pi \text{ yd}$
7. Airman Patel marks the edge of a circular landing zone with a diameter of 70 yards using a rope. How many yards of rope are needed to go all the way around? Use $\frac{22}{7}$ for π .
 A 110 yd B 220 yd
 C 440 yd D 3,850 yd
8. In the figure, the radius of the circle is 20 centimeters. What is the area of the circle?



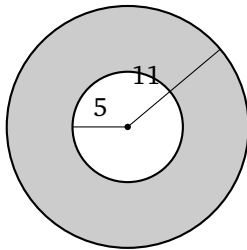
- A $40\pi \text{ cm}^2$ B $400\pi \text{ cm}^2$
 C $800\pi \text{ cm}^2$ D $1,600\pi \text{ cm}^2$
9. The circumference of a circle is 18π centimeters. What is the radius of the circle?
 A 4.5 cm B 7 cm C 9 cm D 18 cm

10. The figure shows one fourth of a circle. Each straight side is 12 meters long. What is the area of the region?



- A $6\pi \text{ m}^2$ B $36\pi \text{ m}^2$
C $72\pi \text{ m}^2$ D $144\pi \text{ m}^2$

11. The figure shows two circles with the same center. The large circle has a radius of 11 yards, and the small circle has a radius of 5 yards. What is the area of the shaded ring?

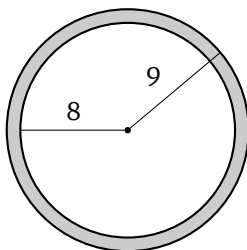


- A $96\pi \text{ yd}^2$ B $116\pi \text{ yd}^2$
C $121\pi \text{ yd}^2$ D $146\pi \text{ yd}^2$

12. Hannah is choosing between a pizza with a diameter of 6 inches and a pizza with a diameter of 18 inches. The area of the larger pizza is how many times the area of the smaller one?

- A 9 B 11 C 13 D 27

13. The figure shows two circles with the same center. The large circle has a radius of 9 feet, and the small circle has a radius of 8 feet. What is the area of the shaded ring?

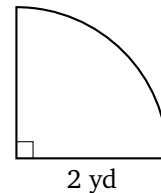


- A $\pi \text{ ft}^2$ B $2\pi \text{ ft}^2$ C $15\pi \text{ ft}^2$ D $17\pi \text{ ft}^2$

14. If the radius of a circle is multiplied by 4, the circumference of the circle is multiplied by what number?

- A 16 B 64 C 4 D 8

15. The figure shows one fourth of a circle. Each straight side is 2 yards long. What is the area of the region?

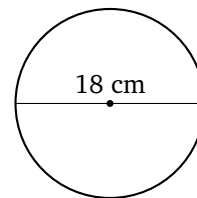


- A $\pi \text{ yd}^2$ B $2\pi \text{ yd}^2$ C $3\pi \text{ yd}^2$ D $4\pi \text{ yd}^2$

16. The circumference of a circle is 22π centimeters. What is the radius of the circle?

- A 5.5 cm B 10 cm C 11 cm D 22 cm

17. In the figure, the diameter of the circle is 18 centimeters. What is the area of the circle?

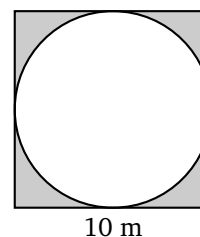


- A $81\pi \text{ cm}^2$ B $100\pi \text{ cm}^2$
C $162\pi \text{ cm}^2$ D $324\pi \text{ cm}^2$

18. A radar station covers a circular area with a diameter of 80 miles. How many square miles does it cover? Use 3.14 for π .

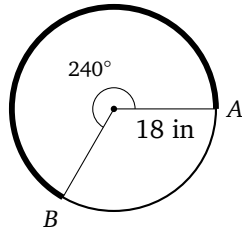
- A $5,024 \text{ mi}^2$ B $5,030.28 \text{ mi}^2$
C $10,048 \text{ mi}^2$ D $20,096 \text{ mi}^2$

19. In the figure, a circle fits exactly inside a square with sides of 10 meters, touching all four sides. What is the area of the shaded region?



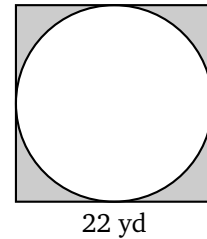
- A $(100 - 29\pi) \text{ m}^2$ B $(100 - 25\pi) \text{ m}^2$
C $(100 - 10\pi) \text{ m}^2$ D $25\pi \text{ m}^2$

20. In the circle shown, the radius is 18 inches and the central angle is 240° . What is the length of the arc from A to B shown in bold?

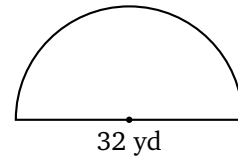


- A 12π in B 24π in C 36π in D 216π in
21. The diameter of a tire on Airman Jensen's supply trailer is 21 inches. About how many complete turns does the tire make in 3 miles? (1 mile = 5,280 feet; use $\frac{22}{7}$ for π .)
- A 240 B 1,440 C 2,160 D 2,880
22. A lawn sprinkler sprays water 30 feet and turns back and forth through an angle of 90° . What is the area of the lawn it waters?
- A 15π ft² B 225π ft² C 450π ft² D 900π ft²
23. The diameter of a tire on Corporal Okafor's supply trailer is 21 inches. About how many complete turns does the tire make in 3 miles? (1 mile = 5,280 feet; use $\frac{22}{7}$ for π .)
- A 2,880 B 3,600 C 4,320 D 5,760

24. In the figure, a circle fits exactly inside a square with sides of 22 yards, touching all four sides. What is the area of the shaded region?



- A $(484 - 121\pi)$ yd² B 121π yd²
 C $(484 - 22\pi)$ yd² D $(484 - 11\pi)$ yd²
25. In the figure, the straight side of the semicircle is 32 yards long. What is the perimeter of the region?



- A 16π yd B $(16\pi + 16)$ yd
 C $(16\pi + 32)$ yd D $(32\pi + 32)$ yd

Answers & Explanations 19

1 C	6 D	11 A	16 C	21 D
2 C	7 B	12 A	17 A	22 B
3 B	8 B	13 D	18 A	23 A
4 A	9 C	14 C	19 B	24 A
5 A	10 B	15 A	20 B	25 C

1. **C** $121\pi \text{ ft}^2$

1 The area of a circle is $A = \pi r^2$. Here the radius is 11 feet.

2 Square the radius: $11^2 = 121$.

3 So $A = \pi \times 121 = 121\pi$ square feet.

WHY NOT (A) forgets to square the radius. (B) uses the circumference formula $2\pi r$ instead of πr^2 . (D) uses the diameter in place of the radius.

2. **C** $18\pi \text{ cm}$

1 The circumference is the distance around: $C = 2\pi r$.

2 Substitute $r = 9$: $C = 2 \times \pi \times 9 = 18\pi$ centimeters.

WHY NOT (A) uses the radius in $C = \pi d$; the diameter is twice the radius. (D) finds the area, not the circumference.

3. **B** $9\pi \text{ cm}^2$

1 The area of a circle is $A = \pi r^2$. Here the radius is 3 centimeters.

2 Square the radius: $3^2 = 9$.

3 So $A = \pi \times 9 = 9\pi$ square centimeters.

WHY NOT (A) uses the circumference formula $2\pi r$ instead of πr^2 . (D) uses the diameter in place of the radius.

4. **A** $14\pi \text{ in}$

1 The circumference is the distance around: $C = 2\pi r$.

2 Substitute $r = 7$: $C = 2 \times \pi \times 7 = 14\pi$ inches.

WHY NOT (C) doubles the radius and then uses that diameter in $2\pi r$. (D) finds the area, not the circumference.

5. **A** 616 ft^2

1 The goat can reach every point within 14 feet of the stake, so the region is a circle with a radius of 14 feet.

2 The area of a circle is $A = \pi r^2$. Square the radius: $14^2 = 196$.

3 $A \approx \frac{22}{7} \times 196 = 22 \times 28 = 616$ square feet.

WHY NOT (D) uses the diameter in place of the radius.

6. **D** $20\pi \text{ yd}$

1 When you know the diameter, use $C = \pi d$.

2 Substitute $d = 20$: $C = \pi \times 20 = 20\pi$ yards.

WHY NOT (A) uses the radius in $C = \pi d$.

7. **B** 220 yd

1 The distance around a circle is its circumference: $C = \pi d$.

2 $C \approx \frac{22}{7} \times 70 = 22 \times 10 = 220$ yards.

WHY NOT (A) uses the radius instead of the diameter. (C) uses the diameter in $2\pi r$, which doubles the answer. (D) finds the area, not the distance around.

8. **B** $400\pi \text{ cm}^2$

1 The area of a circle is $A = \pi r^2$. Here the radius is 20 centimeters.

2 Square the radius: $20^2 = 400$.

3 So $A = \pi \times 400 = 400\pi$ square centimeters.

WHY NOT (A) uses the circumference formula $2\pi r$ instead of πr^2 . (D) uses the diameter in place of the radius.

9. **C** 9 cm

1 Use $C = 2\pi r$: $2\pi r = 18\pi$.

2 Divide both sides by π : $2r = 18$. (This is the diameter.)

3 Divide by 2: $r = 18 \div 2 = 9$ centimeters.

WHY NOT (A) divides by 4 instead of by 2. (D) gives the diameter; divide by 2 again to get the radius.

10. **B** $36\pi \text{ m}^2$

1 A whole circle with radius 12 has area $\pi \times 12^2 = 144\pi$.

2 A quarter circle is one fourth of it: $144\pi \div 4 = 36\pi$ square meters.

WHY NOT (A) finds the length of the curved edge, not the area. (C) takes half of the circle instead of one fourth. (D) finds the area of the whole circle.

11. **A** $96\pi \text{ yd}^2$

- 1 Shaded ring = large circle – small circle.
- 2 Large circle: $\pi \times 11^2 = 121\pi$. Small circle: $\pi \times 5^2 = 25\pi$.
- 3 Subtract: $121\pi - 25\pi = 96\pi$ square yards.

WHY NOT (C) finds the area of the large circle only. (D) adds the two areas instead of subtracting.

12. **A** 9

- 1 Radii: $6 \div 2 = 3$ and $18 \div 2 = 9$ inches.
- 2 Areas: $\pi \times 3^2 = 9\pi$ and $\pi \times 9^2 = 81\pi$ square inches.
- 3 Divide: $81\pi \div 9\pi = 9$.

TIP The diameter is 3 times as large, so the area is $3^2 = 9$ times as large.

WHY NOT (D) cubes the ratio.

13. **D** $17\pi \text{ ft}^2$

- 1 Shaded ring = large circle – small circle.
- 2 Large circle: $\pi \times 9^2 = 81\pi$. Small circle: $\pi \times 8^2 = 64\pi$.
- 3 Subtract: $81\pi - 64\pi = 17\pi$ square feet.

WHY NOT (A) subtracts the radii before squaring.

14. **C** 4

- 1 Circumference depends on the radius itself (not squared): $C = 2\pi r$.
- 2 Replace r with $4r$: $2\pi(4r) = 4 \times 2\pi r$.
- 3 So the circumference is multiplied by 4.

WHY NOT (A) squares the factor; only the area grows by the square. (B) cubes the factor. (D) doubles the factor because of the 2 in $2\pi r$.

15. **A** $\pi \text{ yd}^2$

- 1 A whole circle with radius 2 has area $\pi \times 2^2 = 4\pi$.
- 2 A quarter circle is one fourth of it: $4\pi \div 4 = \pi$ square yards.

WHY NOT (B) takes half of the circle instead of one fourth. (D) finds the area of the whole circle.

16. **C** 11 cm

- 1 Use $C = 2\pi r$: $2\pi r = 22\pi$.
- 2 Divide both sides by π : $2r = 22$. (This is the diameter.)
- 3 Divide by 2: $r = 22 \div 2 = 11$ centimeters.

WHY NOT (A) divides by 4 instead of by 2. (D) gives the diameter; divide by 2 again to get the radius.

17. **A** $81\pi \text{ cm}^2$

- 1 The formula $A = \pi r^2$ needs the radius, which is half the diameter: $r = 18 \div 2 = 9$ centimeters.
- 2 Square the radius: $9^2 = 81$.
- 3 So $A = \pi \times 81 = 81\pi$ square centimeters.

WHY NOT (C) squares the diameter and then halves it; halve first, then square. (D) uses the diameter in place of the radius.

18. **A** 5,024 mi^2

- 1 The area formula $A = \pi r^2$ needs the radius: $r = 80 \div 2 = 40$ miles.
- 2 Square the radius: $40^2 = 1,600$.
- 3 $A \approx 3.14 \times 1,600 = 5,024$ square miles.

TIP Multiply by 3 and by 0.14, then add: $3 \times 1,600 = 4,800$ and $0.14 \times 1,600 = 224$, so the total is 5,024.

WHY NOT (C) squares the diameter and then halves it; halve first, then square. (D) uses the diameter in place of the radius.

19. **B** $(100 - 25\pi) \text{ m}^2$

- 1 The circle touches all four sides, so its diameter equals the side of the square: $d = 10$, so $r = 5$.
- 2 Square: $10^2 = 100$. Circle: $\pi \times 5^2 = 25\pi$.
- 3 Shaded region = square – circle = $100 - 25\pi$ square meters.

TIP Sense check: 25π is a little more than 75, so the shaded corners are only a small part of the square's 100, just as the figure shows.

WHY NOT (C) subtracts the circumference instead of the area of the circle. (D) finds the area of the circle, not the shaded corners.

20. **B** $24\pi \text{ in}$

- 1 The arc is $\frac{240}{360} = \frac{2}{3}$ of the whole circle.
- 2 Circumference: $2\pi \times 18 = 36\pi$.
- 3 Arc length: $\frac{2}{3} \times 36\pi = 24\pi$ inches.

WHY NOT (A) uses πr instead of $2\pi r$ for the circumference. (C) finds the circumference of the whole circle. (D) finds the area of the sector, not the arc length.

21. **D** 2,880

- 1 One turn covers one circumference: $C = \pi d \approx \frac{22}{7} \times 21 = 22 \times 3 = 66$ inches.
- 2 Change to feet: $66 \div 12 = 5.5$ feet.
- 3 Change the distance to feet: $3 \times 5,280 = 15,840$ feet.
- 4 Turns = distance $\div$ circumference: $15,840 \div 5.5 = 2,880$.

TIP To divide by 5.5, double both numbers and divide by 11: $31,680 \div 11 = 2,880$.

WHY NOT (A) forgets to convert the circumference from inches to feet. (B) uses the diameter in $2\pi r$.

22. **B** $225\pi \text{ ft}^2$

- 1 The watered region is a sector of a circle with radius 30 feet and angle 90° .
- 2 That is $\frac{90}{360} = \frac{1}{4}$ of a whole circle. Whole circle: $\pi \times 30^2 = 900\pi$.
- 3 Sector: $\frac{1}{4} \times 900\pi = 225\pi$ square feet.

WHY NOT (A) finds the length of the arc, not the area. (C) divides the angle by 180° instead of 360° . (D) finds the area of the whole circle.

23. **A** 2,880

- 1 One turn covers one circumference: $C = \pi d \approx \frac{22}{7} \times 21 = 22 \times 3 = 66$ inches.
- 2 Change to feet: $66 \div 12 = 5.5$ feet.
- 3 Change the distance to feet: $3 \times 5,280 = 15,840$ feet.
- 4 Turns = distance $\div$ circumference: $15,840 \div 5.5 = 2,880$.

TIP To divide by 5.5, double both numbers and divide by 11: $31,680 \div 11 = 2,880$.

WHY NOT (D) uses the radius instead of the diameter.

24. **A** $(484 - 121\pi) \text{ yd}^2$

- 1 The circle touches all four sides, so its diameter equals the side of the square: $d = 22$, so $r = 11$.
- 2 Square: $22^2 = 484$. Circle: $\pi \times 11^2 = 121\pi$.
- 3 Shaded region = square – circle = $484 - 121\pi$ square yards.

TIP Sense check: 121π is a little more than 363, so the shaded corners are only a small part of the square's 484, just as the figure shows.

WHY NOT (B) finds the area of the circle, not the shaded corners. (C) subtracts the circumference instead of the area of the circle. (D) forgets to square the radius.

25. **C** $(16\pi + 32) \text{ yd}$

- 1 The distance around has two parts: the curved edge and the straight edge (the diameter, 32 yards).
- 2 The curved edge is half of the circumference: $\frac{1}{2} \times \pi \times 32 = 16\pi$ yards.
- 3 Add the straight edge: $16\pi + 32$ yards.

WHY NOT (A) forgets the straight edge (the diameter). (B) adds the radius instead of the diameter. (D) uses the whole circumference instead of half of it.

Volume & Surface Area

Tested on: Mathematics Knowledge • Arithmetic Reasoning

Key Concepts

Volume is the amount of space inside a solid, measured in cubic units (in^3 , ft^3). *Surface area* is the total area of all the faces you could paint or wrap, measured in square units (in^2 , ft^2).

Boxes and cubes

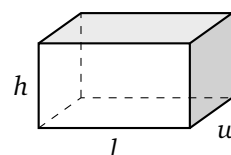
A rectangular box (prism) with length l , width w , and height h :

$$V = l \times w \times h \quad SA = 2lw + 2lh + 2wh$$

The surface area adds three *pairs* of matching faces: top and bottom, front and back, left and right.

A cube with edge s has six identical square faces:

$$V = s^3 \quad SA = 6s^2$$



Cylinders, cones, and spheres

Any solid with the same shape all the way up has $V = (\text{area of the base}) \times \text{height}$. For a cylinder the base is a circle, so

$$V = \pi r^2 h$$

A cone holds exactly $\frac{1}{3}$ as much as a cylinder with the same base and height: $V = \frac{1}{3}\pi r^2 h$. A sphere of radius r has $V = \frac{4}{3}\pi r^3$. (In this book, the cone and sphere formulas are given in the problem when you need them.)

Units

Use one unit for every length *before* you multiply: 6 inches = $\frac{1}{2}$ foot, 4 inches = $\frac{1}{3}$ foot.

- $1 \text{ ft}^3 = 12 \times 12 \times 12 = 1,728 \text{ in}^3$
- $1 \text{ yd}^3 = 3 \times 3 \times 3 = 27 \text{ ft}^3$ (concrete, gravel, and soil are sold by the cubic yard)
- 1 ft^3 of water is about 7.5 gallons

WORKED EXAMPLE

A fish tank is 3 feet long, 2 feet wide, and 2 feet tall. If 1 cubic foot holds 7.5 gallons, how many gallons does the tank hold?

Solution. $V = 3 \times 2 \times 2 = 12$ cubic feet, and $12 \times 7.5 = 90$ gallons.

No-Calculator Tip Multiply in the easiest order: $25 \times 7 \times 4 = (25 \times 4) \times 7 = 700$. Scaling: if every edge is doubled, the surface area is multiplied by $2^2 = 4$ and the volume by $2^3 = 8$.

Common Traps

- Adding the dimensions instead of multiplying them.
- Mixing inches and feet in one calculation.
- Counting only three faces for surface area (each face has a twin).
- Using the diameter instead of the radius in $\pi r^2 h$.
- Dividing cubic feet by 9 (or 3) instead of 27 to get cubic yards.

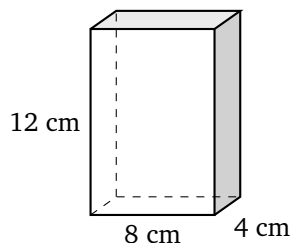
Practice Set 20

Score: ____ / 25

Target time: 26 minutes

Problems 1–8 are warm-ups, 9–18 are test level, and 19–25 are challenge problems.

1. The rectangular box shown is 8 centimeters long, 4 centimeters wide, and 12 centimeters high. What is its volume?



A 24 cm^3 B 32 cm^3 C 352 cm^3 D 384 cm^3

2. What is the surface area of a cube with an edge length of 7 feet?

A 42 ft^2 B 49 ft^2 C 196 ft^2 D 294 ft^2

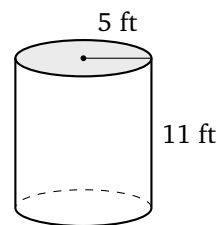
3. A rectangular box is 12 centimeters long, 10 centimeters wide, and 11 centimeters high. What is its volume?

A 33 cm^3 B $1,320 \text{ cm}^3$
C $1,430 \text{ cm}^3$ D $1,440 \text{ cm}^3$

4. What is the volume of a cube with an edge length of 9 feet?

A 27 ft^3 B 729 ft^3 C 810 ft^3 D 891 ft^3

5. What is the volume of the cylinder shown? The radius is 5 feet and the height is 11 feet.

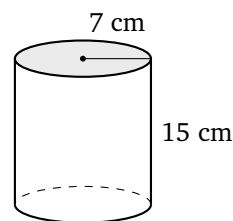


A $25\pi \text{ ft}^3$ B $55\pi \text{ ft}^3$ C $110\pi \text{ ft}^3$ D $275\pi \text{ ft}^3$

6. What is the volume of a cube with an edge length of 6 centimeters?

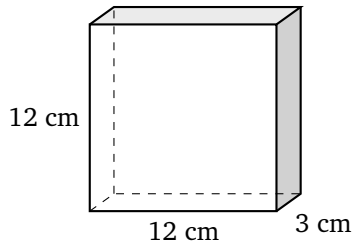
A 18 cm^3 B 36 cm^3 C 216 cm^3 D 288 cm^3

7. What is the volume of the cylinder shown? The radius is 7 centimeters and the height is 15 centimeters.

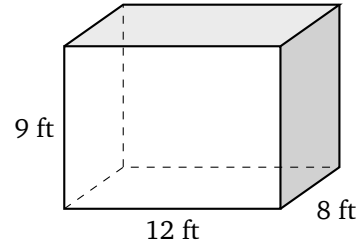


A $49\pi \text{ cm}^3$ B $105\pi \text{ cm}^3$
C $210\pi \text{ cm}^3$ D $735\pi \text{ cm}^3$

8. The rectangular box shown is 12 centimeters long, 3 centimeters wide, and 12 centimeters high. What is its volume? (Figure not drawn to scale.)



- A 27 cm³ B 36 cm³ C 216 cm³ D 432 cm³
9. Yuki is painting all six outside faces of a wooden crate that is 7 feet long, 2 feet wide, and 3 feet tall. How many square feet will be painted?
A 41 ft² B 42 ft² C 82 ft² D 84 ft²
10. An aquarium is 36 inches long and 10 inches wide. It holds 5,400 cubic inches of water. How deep is the water?
A 13 in B 15 in C 150 in D 360 in
11. The water tank on Sergeant Nguyen's field trailer is 6 feet long, 5 feet wide, and 2 feet deep. How many gallons of water does it hold when it is full? (1 cubic foot holds 7.5 gallons.)
A 8 gal B 60 gal C 450 gal D 780 gal
12. Airman Okafor is loading cube-shaped crates that measure 4 feet on each side into a cargo container that is 20 feet long, 8 feet wide, and 8 feet high. What is the greatest number of crates that will fit inside?
A 10 B 20 C 320 D 1,280
13. Grace's rectangular fish tank is 5 feet long, 2 feet wide, and 2 feet deep. How many gallons of water does it hold when it is full? (1 cubic foot holds 7.5 gallons.)
A 20 gal B 75 gal C 140 gal D 150 gal
14. Yuki is packing wooden blocks shaped like cubes, 3 inches on each side, into a toy chest that is 30 inches long, 15 inches wide, and 24 inches deep. How many blocks will fill the chest?
A 400 B 440 C 3,600 D 10,800
15. A cylinder has a radius of 3 inches and a volume of 45π cubic inches. What is its height?
A 3 in B 4 in C 5 in D 15 in
16. The closed rectangular box shown is 12 feet long, 8 feet wide, and 9 feet high. What is its total surface area?



- A 276 ft² B 552 ft² C 576 ft² D 864 ft²
17. A cube has a total surface area of 96 square inches. What is its volume?
A 64 in³ B 80 in³ C 96 in³ D 125 in³
18. A cylindrical water tank has a radius of 2 feet and a height of 5 feet. What is its volume? Use 3.14 for π .
A 62.8 ft³ B 63 ft³
C 63.4 ft³ D 251.2 ft³
19. Aisha is filling a sandbox that is 7 feet long and 7 feet wide with sand 6 inches deep. Sand is sold in bags that each hold half a cubic foot. How many bags does she need?
A 47 B 49 C 98 D 588
20. The height of a cylinder is doubled, and its radius stays the same. The volume of the new cylinder is how many times the volume of the original?
A 2 B 1 C 8 D 4
21. The volume of a sphere is $V = \frac{4}{3}\pi r^3$. A bowl is shaped like half of a sphere (a hemisphere) with radius 3 centimeters. How much does the bowl hold when full?
A 6π cm³ B 16π cm³
C 18π cm³ D 36π cm³
22. Omar has a box with a volume of 320 cubic centimeters. A smaller box has every dimension half as long as the first box. What is the volume of the smaller box?
A 20 cm³ B 35 cm³ C 40 cm³ D 160 cm³

23. The volume of a sphere is $V = \frac{4}{3}\pi r^3$. A bowl is shaped like half of a sphere (a hemisphere) with diameter 18 inches. How much does the bowl hold when full?
- A $486\pi \text{ in}^3$ B $972\pi \text{ in}^3$
C $1,458\pi \text{ in}^3$ D $3,888\pi \text{ in}^3$
24. Kevin is filling a sandbox that is 7 feet long and 7 feet wide with sand 6 inches deep. Sand is sold in bags that each hold half a cubic foot. How many bags does he need?
- A 49 B 53 C 98 D 588
25. Omar's backyard pond is 8 feet long, 6 feet wide, and 3 feet deep. It is being filled with a hose that delivers 40 gallons per minute. How many minutes will it take to fill it to a depth of 1 foot? (1 cubic foot holds 7.5 gallons.)
- A 1.2 min B 3 min C 4.5 min D 9 min

Answers & Explanations 20

1 D	6 C	11 C	16 B	21 C
2 D	7 D	12 B	17 A	22 C
3 B	8 D	13 D	18 A	23 A
4 B	9 C	14 A	19 B	24 A
5 D	10 B	15 C	20 A	25 D

1. **D** 384 cm^3

- The volume of a rectangular box is $V = l \times w \times h$.
- Multiply in a convenient order: $8 \times 4 = 32$, then $32 \times 12 = 384$.
- The volume is 384 cubic centimeters.

WHY NOT (A) adds the three dimensions instead of multiplying them. (B) finds the area of the base only and forgets the height. (C) finds the surface area, not the volume.

2. **D** 294 ft^2

- A cube has 6 identical square faces. One face has area $7^2 = 49$ square feet.
- All six faces: $6 \times 49 = 294$ square feet.

WHY NOT (A) forgets to square the edge. (B) finds the area of only one face. (C) counts only four of the six faces.

3. **B** $1,320 \text{ cm}^3$

- The volume of a rectangular box is $V = l \times w \times h$.
- Multiply in a convenient order: $10 \times 11 = 110$, then $110 \times 12 = 1,320$.
- The volume is 1,320 cubic centimeters.

WHY NOT (A) adds the three dimensions instead of multiplying them.

4. **B** 729 ft^3

- All three dimensions of a cube equal the edge: $V = s^3 = s \times s \times s$.
- $9 \times 9 = 81$, and $81 \times 9 = 729$.
- The volume is 729 cubic feet.

WHY NOT (A) multiplies the edge by 3 instead of cubing it.

5. **D** $275\pi \text{ ft}^3$

- The volume of a cylinder is (area of the base) $\times$ height: $V = \pi r^2 h$.
- Base: $\pi \times 5^2 = 25\pi$.
- Volume: $25\pi \times 11 = 275\pi$ cubic feet.

WHY NOT (A) finds the area of the base only. (B) forgets to square the radius. (C) uses $2\pi r$ instead of πr^2 for the base.

6. **C** 216 cm^3

- All three dimensions of a cube equal the edge: $V = s^3 = s \times s \times s$.
- $6 \times 6 = 36$, and $36 \times 6 = 216$.
- The volume is 216 cubic centimeters.

WHY NOT (A) multiplies the edge by 3 instead of cubing it. (B) squares the edge instead of cubing it.

7. **D** $735\pi \text{ cm}^3$

- The volume of a cylinder is (area of the base) $\times$ height: $V = \pi r^2 h$.
- Base: $\pi \times 7^2 = 49\pi$.
- Volume: $49\pi \times 15 = 735\pi$ cubic centimeters.

WHY NOT (A) finds the area of the base only. (B) forgets to square the radius. (C) uses $2\pi r$ instead of πr^2 for the base.

8. **D** 432 cm^3

- The volume of a rectangular box is $V = l \times w \times h$.
- Multiply in a convenient order: $12 \times 3 = 36$, then $36 \times 12 = 432$.
- The volume is 432 cubic centimeters.

WHY NOT (A) adds the three dimensions instead of multiplying them. (B) finds the area of the base only and forgets the height.

9. **C** 82 ft^2

- The box has three pairs of matching faces: top and bottom ($7 \times 2 = 14$), front and back ($7 \times 3 = 21$), and the two ends ($2 \times 3 = 6$).
- Add one face of each pair: $14 + 21 + 6 = 41$.
- Double it for the matching faces: $2 \times 41 = 82$ square feet.

WHY NOT (A) adds three faces but forgets that each face has a matching opposite face. (B) finds the volume, not the surface area. (D) assumes all six faces are the same size as the bottom.

10. **B** 15 in

- Use $V = l \times w \times h$: $5,400 = 36 \times 10 \times h$.
- Area of the base: $36 \times 10 = 360$.
- Divide: $h = 5,400 \div 360 = 15$ inches.

WHY NOT (C) divides by only one of the two given dimensions. (D) gives the area of the base.

11. **C** 450 gal

- 1 Volume: $6 \times 5 \times 2 = 60$ cubic feet.
- 2 Each cubic foot holds 7.5 gallons: $60 \times 7.5 = 450$ gallons.

TIP Multiplying by 7.5 is the same as multiplying by 15 and halving: $60 \times 15 = 900$, and $900 \div 2 = 450$.

WHY NOT (A) divides by 7.5 instead of multiplying. (B) finds the volume in cubic feet but forgets to change it to gallons. (D) uses the surface area instead of the volume.

 12. **B** 20

- 1 Count how many fit along each edge: $20 \div 4 = 5$, $8 \div 4 = 2$, and $8 \div 4 = 2$.
- 2 Multiply: $5 \times 2 \times 2 = 20$.

TIP Check with volumes: the container holds $20 \times 8 \times 8 = 1,280$ cubic feet, and one cube is $4^3 = 64$, so $1,280 \div 64 = 20$.

WHY NOT (A) counts only the bottom layer. (C) divides the volume by the edge length instead of by the volume of one cube. (D) finds the volume of the container but forgets to divide.

 13. **D** 150 gal

- 1 Volume: $5 \times 2 \times 2 = 20$ cubic feet.
- 2 Each cubic foot holds 7.5 gallons: $20 \times 7.5 = 150$ gallons.

TIP Multiplying by 7.5 is the same as multiplying by 15 and halving: $20 \times 15 = 300$, and $300 \div 2 = 150$.

WHY NOT (A) finds the volume in cubic feet but forgets to change it to gallons. (C) uses 7 gallons per cubic foot instead of 7.5.

 14. **A** 400

- 1 Count how many fit along each edge: $30 \div 3 = 10$, $15 \div 3 = 5$, and $24 \div 3 = 8$.
- 2 Multiply: $10 \times 5 \times 8 = 400$.

TIP Check with volumes: the container holds $30 \times 15 \times 24 = 10,800$ cubic inches, and one cube is $3^3 = 27$, so $10,800 \div 27 = 400$.

WHY NOT (C) divides the volume by the edge length instead of by the volume of one cube. (D) finds the volume of the container but forgets to divide.

 15. **C** 5 in

- 1 Use $V = \pi r^2 h$: $45\pi = \pi \times 3^2 \times h = 9\pi h$.
- 2 Divide both sides by 9π : $h = 45 \div 9 = 5$ inches.

WHY NOT (D) forgets to square the radius.

 16. **B** 552 ft²

- 1 The box has three pairs of matching faces: top and bottom ($12 \times 8 = 96$), front and back ($12 \times 9 = 108$), and the two ends ($8 \times 9 = 72$).
- 2 Add one face of each pair: $96 + 108 + 72 = 276$.
- 3 Double it for the matching faces: $2 \times 276 = 552$ square feet.

WHY NOT (A) adds three faces but forgets that each face has a matching opposite face. (C) assumes all six faces are the same size as the bottom. (D) finds the volume, not the surface area.

 17. **A** 64 in³

- 1 Each of the 6 faces has area $96 \div 6 = 16$ square inches.
- 2 A square with area 16 has side 4, so the edge is 4 inches.
- 3 Volume: $4^3 = 64$ cubic inches.

WHY NOT (C) gives the surface area, not the volume.

 18. **A** 62.8 ft³

- 1 Use $V = \pi r^2 h$. The base: $\pi r^2 \approx 3.14 \times 2^2 = 3.14 \times 4 = 12.56$.
- 2 Multiply by the height: $12.56 \times 5 = 62.8$ cubic feet.

WHY NOT (D) uses the diameter in place of the radius.

 19. **B** 49

- 1 Change the depth to feet: $6 \text{ in} = \frac{6}{12} \text{ ft} = \frac{1}{2} \text{ ft}$.
- 2 Volume: $7 \times 7 \times \frac{1}{2} = 24.5$ cubic feet.
- 3 Each bag holds $\frac{1}{2}$ cubic foot, so it takes 2 bags for every cubic foot: $24.5 \times 2 = 49$ bags.

WHY NOT (C) leaves out the depth. (D) uses the depth in inches as if it were feet.

 20. **A** 2

- 1 In $V = \pi r^2 h$, the height is not squared.
- 2 Replace h with $2h$: $\pi r^2 (2h) = 2\pi r^2 h$, so the volume is multiplied by 2.

WHY NOT (C) cubes the factor. (D) squares the factor; only the radius is squared in the formula.

 21. **C** $18\pi \text{ cm}^3$

- 1 Cube the radius: $3^3 = 3 \times 3 \times 3 = 27$.
- 2 Substitute: $V = \frac{4}{3} \times \pi \times 27 = 36\pi$. Half of the sphere: $36\pi \div 2 = 18\pi$ cubic centimeters.

TIP Divide before you multiply: $27 \div 3 = 9$, then $4 \times 9 = 36$.

WHY NOT (A) squares the radius instead of cubing it. (D) finds the volume of the whole sphere, not half.

22. **C** 40 cm^3

- 1 Each of the three dimensions is multiplied by $\frac{1}{2}$, so the volume is multiplied by $\frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} = \frac{1}{8}$.
- 2 New volume: $320 \div 8 = 40$ cubic centimeters.

WHY NOT (D) halves the volume instead of dividing by 8.

23. **A** $486\pi \text{ in}^3$

- 1 The radius is half the diameter: $18 \div 2 = 9$.
- 2 Cube the radius: $9^3 = 9 \times 9 \times 9 = 729$.
- 3 Substitute: $V = \frac{4}{3} \times \pi \times 729 = 972\pi$. Half of the sphere: $972\pi \div 2 = 486\pi$ cubic inches.

TIP Divide before you multiply: $729 \div 3 = 243$, then $4 \times 243 = 972$.

WHY NOT (B) finds the volume of the whole sphere, not half. (C) forgets the $\frac{1}{3}$. (D) uses the diameter in place of the radius.

24. **A** 49

- 1 Change the depth to feet: $6 \text{ in} = \frac{6}{12} \text{ ft} = \frac{1}{2} \text{ ft}$.
- 2 Volume: $7 \times 7 \times \frac{1}{2} = 24.5$ cubic feet.
- 3 Each bag holds $\frac{1}{2}$ cubic foot, so it takes 2 bags for every cubic foot: $24.5 \times 2 = 49$ bags.

WHY NOT (C) leaves out the depth. (D) uses the depth in inches as if it were feet.

25. **D** 9 min

- 1 Volume of water: $8 \times 6 \times 1 = 48$ cubic feet (use the water depth, not the tank's full depth).
- 2 Gallons: $48 \times 7.5 = 360$.
- 3 Time: $360 \div 40 = 9$ minutes.

WHY NOT (A) forgets to change cubic feet to gallons.

Coordinate Geometry

Tested on: Mathematics Knowledge

Key Concepts

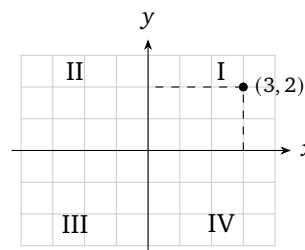
The coordinate plane is a map: every point has an address (x, y) . The x -coordinate tells you how far to go right (+) or left (-); the y -coordinate tells you how far to go up (+) or down (-). Always x first, then y .

Points and quadrants

The axes split the plane into four **quadrants**, numbered counterclockwise starting at the upper right:

- Quadrant I: $(+, +)$ Quadrant II: $(-, +)$
- Quadrant III: $(-, -)$ Quadrant IV: $(+, -)$

A point on an axis is in no quadrant. The point $(3, 2)$ is 3 units right and 2 units up from the origin $(0, 0)$.



Slope

Slope measures steepness: **rise over run**.

$$m = \frac{\text{rise}}{\text{run}} = \frac{y_2 - y_1}{x_2 - x_1}$$

A positive slope goes up to the right, a negative slope goes down to the right, a horizontal line has slope 0, and a vertical line has no slope (undefined). **Parallel** lines have equal slopes. **Perpendicular** lines have slopes that are negative reciprocals: flip the fraction *and* change the sign ($\frac{2}{3}$ and $-\frac{3}{2}$).

Equations of lines

In **slope-intercept form** $y = mx + b$, m is the slope and b is the y -intercept (where the line crosses the y -axis). To read the slope of an equation like $2x + 3y = 12$, solve for y first: $3y = -2x + 12$, so $y = -\frac{2}{3}x + 4$. To find the x -intercept, set $y = 0$; for the y -intercept, set $x = 0$. A point is on a line exactly when its coordinates make the equation true.

Distance and midpoint

For points (x_1, y_1) and (x_2, y_2) :

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \quad M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

The distance formula is the Pythagorean theorem: the horizontal and vertical changes are the legs of a right triangle.

WORKED EXAMPLE

Find the equation of the line through (1, 2) and (3, 8).

Solution. Slope: $m = \frac{8-2}{3-1} = \frac{6}{2} = 3$. Put (1, 2) into $y = 3x + b$: $2 = 3 + b$, so $b = -1$. The line is $y = 3x - 1$.
Check with (3, 8): $3 \times 3 - 1 = 8$. ✓

No-Calculator Tip Check a line by plugging in a point. For distances, look for Pythagorean triples: 3-4-5, 6-8-10, 5-12-13, 8-15-17. If the legs are 6 and 8, the distance is 10, with no square roots needed.

Common Traps

- Putting the run on top: slope is $\frac{\text{change in } y}{\text{change in } x}$.
- Subtracting in different orders on top and bottom (sign error).
- Perpendicular slope: changing the sign *or* flipping, instead of both.
- Reading $y = 4 - 2x$ as slope 4; the slope is the number with x , so it is -2 .

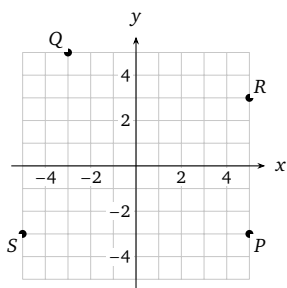
Practice Set 21

Score: ____ / 25

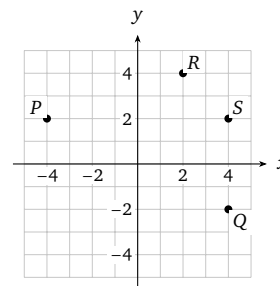
Target time: 25 minutes

Problems 1–8 are warm-ups, 9–18 are test level, and 19–25 are challenge problems.

- The point $(-2, 1)$ is plotted on a coordinate plane. In which quadrant is it?
A Quadrant I B Quadrant II
C Quadrant III D Quadrant IV
- Which point in the figure has coordinates $(5, -3)$? (Each grid square is 1 unit.)
- What is the slope of the line that passes through $(4, 3)$ and $(7, 7)$?
A $-\frac{4}{3}$ B $\frac{4}{3}$ C 4 D $\frac{3}{4}$
- What is the slope of the line $y = 12 - 3x$?
A 3 B -3 C $-\frac{1}{3}$ D 12
- Which point in the figure has coordinates $(4, 2)$? (Each grid square is 1 unit.)



A point S B point Q C point R D point P



A point Q B point R C point S D point P

- A line passes through the points $(5, 0)$ and $(8, 5)$. What is its slope?
A 5 B $\frac{3}{5}$ C $\frac{5}{3}$ D $-\frac{5}{3}$
- What is the slope of the line $y = -2x + 3$?
A $-\frac{1}{2}$ B 2 C 3 D -2
- Point P has a negative x -coordinate and a negative y -coordinate. In which quadrant is point P?
A Quadrant I B Quadrant II
C Quadrant III D Quadrant IV

9. What is the slope of the line that passes through (6, 7) and (-6, -1)?

A $\frac{3}{2}$ B $\frac{2}{3}$ C $-\frac{2}{3}$ D -8

10. Find the midpoint between the points (-3, -8) and (-5, 8).

A (-4, 0) B (-8, 0) C (-1, 8) D (0, -4)

11. What is the distance between the points (-2, -1) and (10, 8)?

A 15 B 16 C 21 D 225

12. What is the x -intercept of the line $y = 3x + 15$?

A -5 B 5 C $\frac{1}{5}$ D 15

13. Which of the following points lies on the line $y = -2x - 1$?

A (7, -4) B (-4, -7) C (-4, 9) D (-4, 7)

14. At what value of y does the line $2x + 3y = -18$ cross the y -axis?

A 6 B -9 C -6 D -18

15. Which ordered pair is a solution of the equation $y = 2x + 7$?

A (1, -3) B (-3, 1)
C (-3, -13) D (-3, 6)

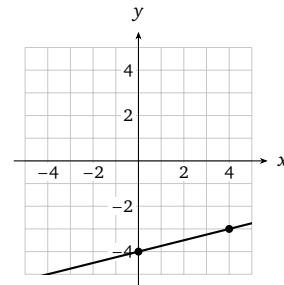
16. How far apart are the points (6, -5) and (-6, 11)?

A 4 B 14 C 19 D 20

17. The midpoint of segment AB is $M(2, 1)$. If point A is (-8, -3), what are the coordinates of point B ?

A (12, 5) B (-6, -2)
C (-18, -7) D (10, 4)

18. The line in the figure passes through the points (4, -3) and (0, -4). What is the slope of the line?



A -1 B $-\frac{1}{4}$ C $\frac{1}{4}$ D 4

19. A line passes through the points (3, 3) and (5, 7). What is its equation?

A $y = 2x - 3$ B $y = \frac{1}{2}x + \frac{3}{2}$
C $y = -2x + 9$ D $y = 2x + 3$

20. What is the slope of a line that is perpendicular to the line $x + 2y = 12$?

A $-\frac{1}{2}$ B $\frac{1}{2}$ C -2 D 2

21. Which equation describes the line that passes through (1, -4) and (3, -10)?

A $y = -3x - 4$ B $y = -3x - 1$
C $y = -3x + 1$ D $y = -\frac{1}{3}x - \frac{11}{3}$

22. What is the slope of the line $x + 2y = 22$?

A -2 B $-\frac{1}{2}$ C $\frac{1}{2}$ D 1

23. A line passes through the points (2, -3) and (4, -13). What is its equation?

A $y = -5x + 7$ B $y = -5x - 3$
C $y = -5x - 7$ D $y = -\frac{1}{5}x - \frac{13}{5}$

24. What is the slope of the line $4x + 3y = 6$?

A $\frac{4}{3}$ B 4 C $-\frac{3}{4}$ D $-\frac{4}{3}$

25. What is the slope of a line that is perpendicular to the line $y = \frac{2}{5}x - 7$?

A $-\frac{5}{2}$ B $\frac{2}{5}$ C $\frac{5}{2}$ D $-\frac{2}{5}$

Answers & Explanations 21

1 B	6 B	11 A	16 D	21 B
2 D	7 C	12 A	17 A	22 B
3 C	8 C	13 D	18 C	23 A
4 D	9 B	14 C	19 A	24 D
5 B	10 A	15 B	20 D	25 A

1. **B** Quadrant II

1 The x -coordinate -2 is negative, so the point is left of the y -axis. The y -coordinate 1 is positive, so it is above the x -axis.

2 That is the upper left region. The quadrants are numbered counterclockwise from the upper right (I, II, III, IV), so this is Quadrant II.

WHY NOT (A) reads the sign of the x -coordinate backwards. (C) reads the sign of the y -coordinate backwards. (D) swaps the x - and y -coordinates.

2. **D** point P

1 The x -coordinate comes first: go 5 units right from the origin.

2 Then go 3 units down. That is point P .

WHY NOT (A) reads the sign of the x -coordinate backwards. (B) swaps the x - and y -coordinates. (C) reads the sign of the y -coordinate backwards.

3. **C** $\frac{5}{3}$

1 Use $m = \frac{y_2 - y_1}{x_2 - x_1}$ with $(5, 0)$ and $(8, 5)$.

2 Rise: $5 - 0 = 5$. Run: $8 - 5 = 3$.

3 Slope: $\frac{5}{3}$.

TIP Quick check: as x increases, y increases, so the slope must be positive.

WHY NOT (A) finds the rise but forgets to divide by the run. (B) divides the change in x by the change in y (run over rise). (D) subtracts the coordinates in opposite orders on the top and bottom.

4. **D** -2

1 In $y = mx + b$, the slope is m , the number multiplied by x .

2 Here that number is -2 , so the slope is -2 .

WHY NOT (A) flips the slope upside down. (B) drops the sign of the slope. (C) gives the y -intercept (the number without x).

5. **B** $\frac{4}{3}$

1 Use $m = \frac{y_2 - y_1}{x_2 - x_1}$ with $(4, 3)$ and $(7, 7)$.

2 Rise: $7 - 3 = 4$. Run: $7 - 4 = 3$.

3 Slope: $\frac{4}{3}$.

TIP Quick check: as x increases, y increases, so the slope must be positive.

WHY NOT (A) subtracts the coordinates in opposite orders on the top and bottom. (C) finds the rise but forgets to divide by the run. (D) divides the change in x by the change in y (run over rise).

6. **B** -3

1 Rewrite the equation in the order $y = mx + b$: $y = -3x + 12$.

2 In $y = mx + b$, the slope is m , the number multiplied by x .

3 Here that number is -3 , so the slope is -3 .

WHY NOT (A) drops the sign of the slope. (C) flips the slope upside down. (D) takes the first number in the equation; the slope is the number multiplied by x .

7. **C** point S

1 The x -coordinate comes first: go 4 units right from the origin.

2 Then go 2 units up. That is point S .

WHY NOT (A) reads the sign of the y -coordinate backwards. (B) swaps the x - and y -coordinates. (D) reads the sign of the x -coordinate backwards.

8. **C** Quadrant III

1 A negative x -coordinate means left of the y -axis; a negative y -coordinate means below the x -axis.

2 That is the lower left region. The quadrants are numbered counterclockwise from the upper right (I, II, III, IV), so this is Quadrant III.

WHY NOT (A) reverses both signs. (B) reads the sign of the y -coordinate backwards. (D) reads the sign of the x -coordinate backwards.

9. **B** $\frac{2}{3}$

 1 Use $m = \frac{y_2 - y_1}{x_2 - x_1}$ with (6, 7) and (-6, -1).

 2 Rise: $-1 - 7 = -8$. Run: $-6 - 6 = -12$.

 3 Slope: $\frac{-8}{-12} = \frac{2}{3}$.

TIP Quick check: as x increases, y increases, so the slope must be positive.

WHY NOT (A) divides the change in x by the change in y (run over rise). (C) subtracts the coordinates in opposite orders on the top and bottom. (D) finds the rise but forgets to divide by the run.

 10. **A** (-4, 0)

 1 Average the x -coordinates: $\frac{-3 + (-5)}{2} = \frac{-8}{2} = -4$.

 2 Average the y -coordinates: $\frac{-8 + 8}{2} = \frac{0}{2} = 0$.

3 The midpoint is (-4, 0).

WHY NOT (B) adds the coordinates but forgets to divide by 2. (C) subtracts the coordinates instead of adding them. (D) swaps the x - and y -coordinates.

 11. **A** 15

 1 Horizontal change: $|10 - (-2)| = 12$. Vertical change: $|8 - (-1)| = 9$.

 2 These are the legs of a right triangle, and the distance is the hypotenuse: $d = \sqrt{12^2 + 9^2} = \sqrt{144 + 81} = \sqrt{225} = 15$.

TIP The legs 9 and 12 belong to the Pythagorean triple 9-12-15.

WHY NOT (C) adds the horizontal and vertical distances instead of using the Pythagorean theorem. (D) forgets to take the square root.

 12. **A** -5

 1 The x -intercept is where the line crosses the x -axis, so set $y = 0$: $0 = 3x + 15$.

 2 $3x = -15$, so $x = -15 \div 3 = -5$.

WHY NOT (B) makes a sign error when solving for x . (C) divides the slope by the y -intercept (the division is upside down). (D) gives the y -intercept instead of the x -intercept.

 13. **D** (-4, 7)

 1 A point is on the line if its coordinates make the equation true. Substitute each x and see whether you get the matching y .

 2 For (-4, 7): $y = -2(-4) - 1 = 8 - 1 = 7$. It matches, so (-4, 7) is on the line.

 3 Each of the other points gives a different y when you substitute its x .

WHY NOT (A) swaps the x - and y -coordinates of a point on the line. (B) adds the slope to x instead of multiplying. (C) uses the wrong sign for the y -intercept.

 14. **C** -6

 1 On the y -axis, $x = 0$. Substitute: $2(0) + 3y = -18$, so $3y = -18$.

 2 Divide by 3: $y = -18 \div 3 = -6$.

WHY NOT (A) makes a sign error. (B) finds the x -intercept instead (sets $y = 0$). (D) forgets to divide by the coefficient of y .

 15. **B** (-3, 1)

 1 A point is on the line if its coordinates make the equation true. Substitute each x and see whether you get the matching y .

 2 For (-3, 1): $y = 2(-3) + 7 = -6 + 7 = 1$. It matches, so (-3, 1) is on the line.

 3 Each of the other points gives a different y when you substitute its x .

WHY NOT (A) swaps the x - and y -coordinates of a point on the line. (C) uses the wrong sign for the y -intercept. (D) adds the slope to x instead of multiplying.

 16. **D** 20

 1 Horizontal change: $|-6 - 6| = 12$. Vertical change: $|11 - (-5)| = 16$.

 2 These are the legs of a right triangle, and the distance is the hypotenuse: $d = \sqrt{12^2 + 16^2} = \sqrt{144 + 256} = \sqrt{400} = 20$.

TIP The legs 12 and 16 belong to the Pythagorean triple 12-16-20.

WHY NOT (A) subtracts the two distances.

 17. **A** (12, 5)

 1 From A to M , x changes by $2 - (-8) = 10$ and y changes by $1 - (-3) = 4$.

 2 B is the same distance past M : $x = 2 + 10 = 12$ and $y = 1 + 4 = 5$.

 3 Check: the midpoint of (-8, -3) and (12, 5) is $(\frac{4}{2}, \frac{2}{2}) = (2, 1)$. ✓

WHY NOT (B) adds A and M without doubling M . (C) goes from M through A instead of from A through M . (D) finds only the change from A to M , not the endpoint.

 18. **C** $\frac{1}{4}$

 1 Use $m = \frac{y_2 - y_1}{x_2 - x_1}$ with (4, -3) and (0, -4).

 2 Rise: $-4 - (-3) = -1$. Run: $0 - 4 = -4$.

 3 Slope: $\frac{-1}{-4} = \frac{1}{4}$.

TIP Quick check: as x increases, y increases, so the slope must be positive.

WHY NOT (A) finds the rise but forgets to divide by the run. (B) subtracts the coordinates in opposite orders on the top and bottom. (D) divides the change in x by the change in y (run over rise).

19. **A** $y = 2x - 3$

- 1 Slope: $m = \frac{7-3}{5-3} = \frac{4}{2} = 2$.
- 2 Substitute the point (3, 3) to find b : $3 = 2(3) + b = 6 + b$, so $b = 3 - 6 = -3$.
- 3 The equation is $y = 2x - 3$.
- 4 Check with (5, 7): $2(5) - 3 = 7$. ✓

WHY NOT (B) uses run over rise for the slope. (C) makes a sign error in the slope. (D) gets the sign of the y -intercept wrong.

20. **D** 2

- 1 Solve for y to find the slope: $2y = -x + 12$, so $y = -\frac{1}{2}x + 6$ and the slope is $-\frac{1}{2}$.
- 2 A perpendicular slope is the negative reciprocal: flip $-\frac{1}{2}$ and change its sign to get 2.
- 3 Check: $(-\frac{1}{2}) \times 2 = -1$. ✓

WHY NOT (A) gives the slope of a parallel line. (B) changes the sign but forgets to flip the fraction. (C) flips the fraction but forgets to change the sign.

21. **B** $y = -3x - 1$

- 1 Slope: $m = \frac{-10-(-4)}{3-1} = \frac{-6}{2} = -3$.
- 2 Substitute the point (1, -4) to find b : $-4 = -3(1) + b = -3 + b$, so $b = -4 - (-3) = -1$.
- 3 The equation is $y = -3x - 1$.
- 4 Check with (3, -10): $-3(3) - 1 = -10$. ✓

WHY NOT (A) uses -4, the y -coordinate of a point, as the y -intercept. (C) gets the sign of the y -intercept wrong. (D) uses run over rise for the slope.

22. **B** $-\frac{1}{2}$

- 1 Solve for y . Move the x -term to the other side: $2y = -x + 22$.
- 2 Divide every term by 2: $y = -\frac{1}{2}x + 11$.
- 3 The slope is the coefficient of x : $-\frac{1}{2}$.

WHY NOT (A) flips the fraction (puts the x -coefficient on the bottom). (C) forgets to change the sign when moving the x -term. (D) uses the coefficient of x without solving for y .

23. **A** $y = -5x + 7$

- 1 Slope: $m = \frac{-13-(-3)}{4-2} = \frac{-10}{2} = -5$.
- 2 Substitute the point (2, -3) to find b : $-3 = -5(2) + b = -10 + b$, so $b = -3 - (-10) = 7$.
- 3 The equation is $y = -5x + 7$.
- 4 Check with (4, -13): $-5(4) + 7 = -13$. ✓

WHY NOT (B) uses -3, the y -coordinate of a point, as the y -intercept. (C) gets the sign of the y -intercept wrong. (D) uses run over rise for the slope.

24. **D** $-\frac{4}{3}$

- 1 Solve for y . Move the x -term to the other side: $3y = -4x + 6$.
- 2 Divide every term by 3: $y = -\frac{4}{3}x + 2$.
- 3 The slope is the coefficient of x : $-\frac{4}{3}$.

WHY NOT (A) forgets to change the sign when moving the x -term. (B) uses the coefficient of x without solving for y . (C) flips the fraction (puts the x -coefficient on the bottom).

25. **A** $-\frac{5}{2}$

- 1 The line $y = \frac{2}{5}x - 7$ has slope $\frac{2}{5}$ (the coefficient of x).
- 2 A perpendicular slope is the negative reciprocal: flip $\frac{2}{5}$ and change its sign to get $-\frac{5}{2}$.
- 3 Check: $\frac{2}{5} \times (-\frac{5}{2}) = -1$. ✓

WHY NOT (B) gives the slope of a parallel line. (C) flips the fraction but forgets to change the sign. (D) changes the sign but forgets to flip the fraction.

PART IV

Word Problems

The heart of the Arithmetic Reasoning subtest: money, percents, rates, work, mixtures, statistics, probability, and multi-step problems.

- 22 Money: Shopping, Discounts, Tax & Wages
- 23 Interest, Commission & Percent Change
- 24 Rate, Distance & Time
- 25 Work Rates & Mixtures
- 26 Averages, Statistics & Probability
- 27 Measurement, Units & Multi-Step Problems

Money: Shopping, Discounts, Tax & Wages

Tested on: Arithmetic Reasoning

Key Concepts

Money problems are the most common word problems on the Arithmetic Reasoning test. They use only a handful of ideas (add up a bill, take a percent off, add a percent on, figure out pay), but they chain two or three of them together. Label every number you write with what it means, and always re-read what the question actually asks for.

Shopping

- **Total cost** = quantity $\times$ price for each item, added up. **Change** = amount paid – total cost.
- **Sale price**: subtract the discount, or multiply by the percent you still pay. 25% off means you pay 75%: $0.75 \times \$80 = \60 .
- **Sales tax**: add the tax on. A 6% tax means you pay 106%: $1.06 \times \$50 = \53 .
- **Discount and tax together**: take the discount first, then figure the tax on the *sale* price.
- **Unit price** = price $\div$ number of units. The *better buy* is the one with the lower unit price.

Wages and paychecks

Hourly pay = hours $\times$ rate. **Overtime** at “time-and-a-half” pays 1.5 $\times$ the regular rate, but *only* for the hours over 40 in a week. A salary is split evenly over the paychecks in a year:

Paid ...	Paychecks per year
weekly	52
every two weeks (biweekly)	26
twice a month (semimonthly)	24
monthly	12

A full-time year is $40 \times 52 = 2,080$ hours.

Rounding to fit the situation

“How many can you buy?” rounds *down*: you cannot buy part of an item. “How many weeks until you can afford it?” rounds *up*: after a partial week you are still short.

WORKED EXAMPLE

Kevin earns \$18 per hour and gets time-and-a-half for hours over 40. Last week he worked 46 hours. How much did he earn?

Solution. Regular pay: $40 \times \$18 = \720 . Overtime rate: $1.5 \times \$18 = \27 per hour, and $46 - 40 = 6$ overtime hours earn $6 \times \$27 = \162 . Total: $\$720 + \$162 = \$882$.

No-Calculator Tip Build percents from 10% (move the decimal point one place left). For \$64: $10\% = \$6.40$, so $5\% = \$3.20$, $15\% = \$9.60$, and $20\% = \$12.80$. To multiply by 1.5, add half: $1.5 \times \$18 = \$18 + \$9 = \27 .

Common Traps

- Stopping a step early: giving the discount instead of the sale price, or the tax instead of the total.
- Paying time-and-a-half for *all* the hours instead of only the hours over 40.
- Reading “every two weeks” as twice a month: that is 26 paychecks a year, not 24.
- Figuring the tax on the original price when the item is on sale.

Practice Set 22

Score: ____ / 40

Target time: 48 minutes

Problems 1–12 are warm-ups, 13–28 are test level, and 29–40 are challenge problems.

- At the base exchange, Corporal Santos buys a pair of boot socks for \$11.25 and a tin of boot polish for \$6.50. She pays with a \$50 bill. How much change does she get back?
A \$17.75 B \$32.25 C \$33.25 D \$38.75
- Shopping for school supplies, Aisha buys a pack of pens for \$3.25 and a set of highlighters for \$7.25. She pays with a \$20 bill. How much change does she get back?
A \$9.50 B \$10.50 C \$12.50 D \$16.75
- At a sporting-goods store, Victor buys a can of tennis balls for \$4.75 and a water bottle for \$12.49. He pays with a \$50 bill. How much change should he receive?
A \$17.24 B \$27.76 C \$32.00 D \$32.76
- A store marks down a pair of hiking boots from its regular price of \$80 by 40%. What is the new price?
A \$32.00 B \$48.00 C \$76.80 D \$112.00
- A microwave oven regularly sells for \$65. This week it is on sale for 40% off. What is the sale price?
A \$25 B \$26 C \$34 D \$39
- The base exchange is having a 50%-off sale. Seaman Walker buys a hydration backpack that regularly costs \$55. How much does she pay? (There is no sales tax at the exchange.)
A \$5.00 B \$12.50 C \$22.50 D \$27.50
- Mei buys a lawn mower online for \$280. The sales tax rate is 5%. What is the total cost, including tax?
A \$14 B \$266 C \$285 D \$294
- Emily buys a laptop at a department store for \$670. The sales tax rate is 8%. What is the total cost, including tax?
A \$53.60 B \$723.60
C \$1,023.60 D \$1,206.00
- A coach has \$200 to spend on new soccer balls that cost \$18 each. How many soccer balls can the coach buy?
A 11 B 12 C 13 D 15
- A unit's morale fund has \$230 to spend on large pizzas for a platoon party. Each pizza costs \$12. What is the greatest number of pizzas the unit can buy?
A 17 B 18 C 19 D 20

11. Derek takes home \$2,760 a month. He pays $\frac{1}{8}$ of it toward a car loan. How much does he pay toward a car loan each month?
A \$245 B \$295 C \$345 D \$2,415
12. Kevin takes home \$3,000 a month. He spends $\frac{1}{4}$ of it on rent. How much of the month's pay is left?
A \$750 B \$2,250 C \$2,450 D \$2,850
13. Shopping for school supplies, Maria buys 3 sets of highlighters at \$6.49 each and 3 packs of index cards at \$2.50 each. She pays with a \$100 bill. How much change should she receive?
A \$26.97 B \$70.00 C \$73.03 D \$74.03
14. At a sporting-goods store, Jamal buys 4 water bottles at \$8.50 each and a yoga mat for \$27.75. He pays with a \$100 bill. How much change should he receive?
A \$23.25 B \$28.25 C \$36.00 D \$38.25
15. At the base exchange, a box of 15 protein bars costs \$20.10, and a box of 20 protein bars costs \$24.20. Seaman Reyes wants the better buy, so he compares the prices per bar. What is the lower price per bar?
A \$0.13 B \$1.08 C \$1.21 D \$1.34
16. Online, a 50-ounce bottle of laundry detergent costs \$7, and a 64-ounce bottle of laundry detergent costs \$9.60. What is the price per ounce of the better buy?
A \$0.14 B \$0.15 C \$0.24 D \$2.60
17. Corporal Okafor buys 2 cans of paint at \$34 each and a paint roller for \$12 in town on a weekend pass. The sales tax rate is 8%. What is the total cost, including tax?
A \$6.40 B \$73.60 C \$86.40 D \$88.00
18. Carlos works as a security guard and earns \$15 per hour, with time-and-a-half for every hour over 40 in a week. Last week he worked 48 hours. How much did he earn last week?
A \$180 B \$660 C \$720 D \$780
19. Malik works as a forklift operator at an Army supply depot and earns \$24 per hour, with time-and-a-half for every hour over 40 in a week. Last week he worked 47 hours. How much did he earn last week?
A \$1,128 B \$1,212 C \$1,296 D \$1,692
20. Leila works as a bank teller for a salary of \$39,000 a year and is paid biweekly (every two weeks). How much is each paycheck before taxes and other deductions?
A \$1,500 B \$1,625 C \$1,900 D \$3,250
21. Technical Sergeant Novak's yearly base pay is \$54,000. Service members are paid twice a month, on the 1st and the 15th. How much is each paycheck before taxes and other deductions?
A \$1,850 B \$2,250 C \$2,650 D \$4,500
22. After a long hike, 3 friends eat at a diner near the trailhead. The bill is \$126, and they leave a 15% tip. They split the total evenly. How much does each friend pay?
A \$42.00 B \$48.30 C \$60.90 D \$144.90
23. After a bowling league night, the 5 members of a team share a meal. The bill is \$80, and they leave a 20% tip. If they split the total evenly, how much does each team member pay?
A \$19.20 B \$21.20 C \$32.00 D \$96.00
24. Jasmine is saving for a gaming computer, which costs \$1,400. She has already saved \$490 and can save \$110 a week. How many weeks will it take her to have enough money?
A 5 weeks B 7 weeks C 8 weeks D 9 weeks
25. Petty Officer Novak is saving for a plane ticket home for holiday leave, which costs \$725. He has already saved \$290 and can save \$30 a week. How many weeks will it take him to have enough money?
A 14 weeks B 15 weeks
C 25 weeks D 34 weeks

26. Tyrone takes home \$2,760 a month. He puts $\frac{1}{10}$ of it into savings and spends $\frac{1}{5}$ of it on groceries. How much is left each month for everything else?
A \$1,932 B \$2,208 C \$2,392 D \$2,484
27. A squad has \$125 to spend before a field exercise. It buys a \$35 first-aid kit and spends the rest on boxes of hand warmers at \$8 each. How many boxes can it buy?
A 11 B 12 C 15 D 20
28. At a sporting-goods store, energy bars cost \$1.75 per bar, or \$20.88 for a box of 18. How much is saved by buying a box of 18 instead of 18 single bars?
A \$0.59 B \$1.16 C \$7.62 D \$10.62
29. Seaman Carter is furnishing an apartment off base. A television that regularly costs \$440 is on sale for 20% off. A sales tax of 8% is charged on the sale price. How much does she pay in all?
A \$352.00 B \$380.16 C \$387.20 D \$475.20
30. A mattress is priced at \$520. It is on sale for 20% off, and a 7% sales tax is charged on the sale price. What is the total cost?
A \$386.88 B \$416.00 C \$445.12 D \$452.40
31. Leila buys a recliner during a 15%-off sale. The regular price is \$580, and 8% sales tax is added to the sale price. How much does she pay, including tax?
A \$453.56 B \$482.44 C \$493.00 D \$532.44
32. Tyrone works as a security guard and earns \$15 per hour, with time-and-a-half for every hour over 40 in a week. Last week he worked 10 hours a day Monday through Thursday and 6 hours on Saturday. How much did he earn for the week?
A \$735 B \$780 C \$935 D \$1,035
33. Grace works as a forklift operator at an Army supply depot and earns \$20 per hour, with time-and-a-half for every hour over 40 in a week. Last week her pay was \$860. How many hours did she work last week?
A 2 hours B 40 hours
C 42 hours D 43 hours
34. At a warehouse store, bottled water comes in three sizes: a case of 12 bottles of water for \$3.24, a case of 24 bottles of water for \$7.44, and a case of 40 bottles of water for \$14. Which size is the best buy?
A All three cost the same per bottle
B the case of 12
C the case of 24
D the case of 40
35. At a grocery store, coffee comes in three sizes: a 12-ounce bag of coffee for \$3.96, a 24-ounce bag of coffee for \$8.40, and a 32-ounce bag of coffee for \$13.12. Which size is the best buy?
A the 32-ounce bag
B the 24-ounce bag
C the 12-ounce bag
D All three cost the same per ounce
36. Private Kim's take-home pay is \$2,640 a month. He pays $\frac{1}{8}$ of the pay toward a car loan, and then puts $\frac{1}{10}$ of what is left into savings. How much money does he put into savings?
A \$171 B \$191 C \$231 D \$264
37. Kevin works as an electrician's apprentice for a salary of \$46,800 a year and is paid biweekly (every two weeks). Of each paycheck, 25% is withheld for taxes. How much does he take home from each paycheck?
A \$1,350.00 B \$1,462.50
C \$1,800.00 D \$2,925.00
38. Ethan has two job offers. Job A pays \$16 per hour, and Job B pays a salary of \$37,440 a year. At either job he would work 40 hours a week, 52 weeks a year. How much more per year does the better-paying job pay?
A \$80 B \$4,160 C \$6,720 D \$33,280

39. Specialist Ruiz is leaving the Army after four years and has two civilian job offers. Job A pays \$19.50 per hour, and Job B pays a salary of \$37,440 a year. At either job she would work 40 hours a week, 52 weeks a year. How much more per year does the better-paying job pay?
A \$3,120 B \$3,620 C \$4,120 D \$40,560
40. At a grocery store, bottled water costs \$0.75 per bottle, or \$12.48 for a case of 24. A youth-league coach needs 64 bottles. The plan is to buy as many full cases as possible and the rest as single bottles. What is the total cost?
A \$24.96 B \$33.28 C \$36.96 D \$48.00

Answers & Explanations 22

1 B	9 A	17 C	25 B	33 C
2 A	10 C	18 D	26 A	34 B
3 D	11 C	19 B	27 A	35 C
4 B	12 B	20 A	28 D	36 C
5 D	13 C	21 B	29 B	37 A
6 D	14 D	22 B	30 C	38 B
7 D	15 C	23 A	31 D	39 A
8 B	16 A	24 D	32 A	40 C

1. **B** \$32.25

1 Add the prices to find the total cost: $\$11.25 + \$6.50 = \$17.75$.

2 Subtract the total from the amount paid: $\$50.00 - \$17.75 = \$32.25$.

TIP Check by counting up: $\$17.75 + \$32.25 = \$50.00$.

WHY NOT (A) is the total cost, not the change. (C) forgets to borrow a dollar when subtracting the cents. (D) leaves out the price of the tin of boot polish.

2. **A** \$9.50

1 Add the prices to find the total cost: $\$3.25 + \$7.25 = \$10.50$.

2 Subtract the total from the amount paid: $\$20.00 - \$10.50 = \$9.50$.

TIP Check by counting up: $\$10.50 + \$9.50 = \$20.00$.

WHY NOT (B) is the total cost, not the change. (D) leaves out the price of the set of highlighters.

3. **D** \$32.76

1 Add the prices to find the total cost: $\$4.75 + \$12.49 = \$17.24$.

2 Subtract the total from the amount paid: $\$50.00 - \$17.24 = \$32.76$.

TIP Check by counting up: $\$17.24 + \$32.76 = \$50.00$.

WHY NOT (A) is the total cost, not the change. (C) rounds each price up to a whole dollar instead of using the exact prices.

4. **B** \$48.00

1 Find the discount: 40% of $\$80 = 0.4 \times 80 = \32 .

2 Subtract it from the regular price: $\$80 - \$32 = \$48$.

TIP Shortcut: 40% off means you pay 100% – 40% = 60%, and $0.6 \times \$80 = \48 .

WHY NOT (A) is the amount of the discount, not the sale price. (C) moves the decimal point one place too far when finding the discount. (D) adds the discount instead of subtracting it.

5. **D** \$39

1 Find the discount: 40% of $\$65 = 0.4 \times 65 = \26 .

2 Subtract it from the regular price: $\$65 - \$26 = \$39$.

TIP Shortcut: 40% off means you pay 100% – 40% = 60%, and $0.6 \times \$65 = \39 .

WHY NOT (A) subtracts the percent as if it were dollars. (B) is the amount of the discount, not the sale price.

6. **D** \$27.50

1 Find the discount: 50% of $\$55 = 0.5 \times 55 = \27.50 .

2 Subtract it from the regular price: $\$55 - \$27.50 = \$27.50$.

TIP Shortcut: 50% off means you pay 100% – 50% = 50%, and $0.5 \times \$55 = \27.50 .

WHY NOT (A) subtracts the percent as if it were dollars.

7. **D** \$294

1 Find the tax: 5% of $\$280 = 0.05 \times 280 = \14 .

2 Add the tax to the price: $\$280 + \$14 = \$294$.

TIP In one step: with tax you pay 105%, so multiply by 1.05.

WHY NOT (A) is the tax alone, not the total cost. (B) subtracts the tax instead of adding it. (C) adds the tax rate as if it were dollars.

8. **B** \$723.60

1 Find the tax: 8% of $\$670 = 0.08 \times 670 = \53.60 .

2 Add the tax to the price: $\$670 + \$53.60 = \$723.60$.

TIP In one step: with tax you pay 108%, so multiply by 1.08.

WHY NOT (A) is the tax alone, not the total cost. (D) moves the decimal point only one place when finding the tax.

9. **A** 11

1 Divide by the price of one: $\$200 \div \18 is 11 with some money left over, because $11 \times \$18 = \198 .

2 One more would cost \$216, which is more than \$200. Round *down*: 11 soccer balls.

TIP Left over: $\$200 - \$198 = \$2$, not enough for one more.

WHY NOT (B) rounds up, but 12 soccer balls would cost \$216.

10. **C** 19

1 Divide by the price of one: $\$230 \div \12 is 19 with some money left over, because $19 \times \$12 = \228 .

2 One more would cost \$240, which is more than \$230. Round *down*: 19 pizzas.

TIP Left over: $\$230 - \$228 = \$2$, not enough for one more.

WHY NOT (D) rounds up, but 20 pizzas would cost \$240.

11. **C** \$345

1 *Of* means multiply: $\frac{1}{8} \times \$2,760$. To find $\frac{1}{8}$ of an amount, divide by 8.

2 $\$2,760 \div 8 = \345 .

TIP Check: $8 \times \$345 = \$2,760$.

WHY NOT (D) is the amount left over, not the amount paid on the car loan.

12. **B** \$2,250

1 Find $\frac{1}{4}$ of the pay: $\$3,000 \div 4 = \750 . That is the amount spent on rent.

2 Subtract it from the pay: $\$3,000 - \$750 = \$2,250$.

TIP Or: $1 - \frac{1}{4} = \frac{3}{4}$ of the pay is left, and $\frac{3}{4} \times \$3,000 = \$2,250$.

WHY NOT (A) is the amount spent on rent, not the amount left.

13. **C** \$73.03

1 Find the cost of each kind of item: $3 \times \$6.49 = \19.47 , and $3 \times \$2.50 = \7.50 .

2 Add to get the total cost: $\$19.47 + \$7.50 = \$26.97$.

3 Subtract the total from the amount paid: $\$100.00 - \$26.97 = \$73.03$.

WHY NOT (A) is the total cost, not the change. (B) rounds each price up to a whole dollar instead of using the exact prices. (D) forgets to borrow a dollar when subtracting the cents.

14. **D** \$38.25

1 Find the cost of each kind of item: $4 \times \$8.50 = \34 , and the yoga mat costs \$27.75.

2 Add to get the total cost: $\$34 + \$27.75 = \$61.75$.

3 Subtract the total from the amount paid: $\$100.00 - \$61.75 = \$38.25$.

WHY NOT (C) rounds each price up to a whole dollar instead of using the exact prices.

15. **C** \$1.21

1 Divide each price by its number of bars. Smaller size: $\$20.10 \div 15 = \1.34 per bar.

2 Larger size: $\$24.20 \div 20 = \1.21 per bar.

3 The better buy is the one with the lower price per bar: the box of 20, at \$1.21 per bar.

WHY NOT (A) is the difference between the two prices per bar. (D) is the price per bar of the worse buy.

16. **A** \$0.14

1 Divide each price by its number of ounces. Smaller size: $\$7 \div 50 = \0.14 per ounce.

2 Larger size: $\$9.60 \div 64 = \0.15 per ounce.

3 The better buy is the one with the lower price per ounce: the 50-ounce bottle, at \$0.14 per ounce.

WHY NOT (B) is the price per ounce of the worse buy. (D) compares the total prices instead of the prices per ounce.

17. **C** \$86.40

1 Find the cost before tax: $2 \times \$34 + \$12 = \$68 + \$12 = \$80$.

2 Find the tax: $8\% \text{ of } \$80 = 0.08 \times 80 = \6.40 .

3 Add the tax to the price: $\$80 + \$6.40 = \$86.40$.

WHY NOT (A) is the tax alone, not the total cost. (B) subtracts the tax instead of adding it. (D) adds the tax rate as if it were dollars.

18. **D** \$780

1 Only hours over 40 are overtime: $48 - 40 = 8$ overtime hours.

2 Regular pay: $40 \times \$15 = \600 . Overtime rate: $1.5 \times \$15 = \22.50 per hour.

3 Overtime pay: $8 \times \$22.50 = \180 .

4 Total: $\$600 + \$180 = \$780$.

WHY NOT (A) is the overtime pay only. (B) pays the overtime hours only the extra half rate. (C) pays every hour at the regular rate.

19. **B** \$1,212

- 1 Only hours over 40 are overtime: $47 - 40 = 7$ overtime hours.
- 2 Regular pay: $40 \times \$24 = \960 . Overtime rate: $1.5 \times \$24 = \36 per hour.
- 3 Overtime pay: $7 \times \$36 = \252 .
- 4 Total: $\$960 + \$252 = \$1,212$.

WHY NOT (A) pays every hour at the regular rate. (C) pays double time instead of time-and-a-half. (D) pays time-and-a-half for all the hours, not just the hours over 40.

20. **A** \$1,500

- 1 Every two weeks means $52 \div 2 = 26$ paychecks a year (*not* 24).
- 2 Divide the yearly pay by the number of paychecks: $\$39,000 \div 26 = \$1,500$.

TIP Check: $26 \times \$1,500 = \$39,000$.

WHY NOT (B) uses 24 paychecks, but every two weeks gives 26 paychecks a year. (D) divides by 12, which gives one month's pay.

21. **B** \$2,250

- 1 Twice a month means $12 \times 2 = 24$ paychecks a year.
- 2 Divide the yearly pay by the number of paychecks: $\$54,000 \div 24 = \$2,250$.

TIP Check: $24 \times \$2,250 = \$54,000$.

WHY NOT (D) divides by 12, which gives one month's pay.

22. **B** \$48.30

- 1 Find the tip: $0.15 \times \$126 = \18.90 .
- 2 Total with tip: $\$126 + \$18.90 = \$144.90$.
- 3 Split it 3 ways: $\$144.90 \div 3 = \48.30 .

WHY NOT (A) forgets to add the tip. (C) adds the whole tip to one friend's share. (D) is the total for the whole group, not one friend's share.

23. **A** \$19.20

- 1 Find the tip: $0.2 \times \$80 = \16 .
- 2 Total with tip: $\$80 + \$16 = \$96$.
- 3 Split it 5 ways: $\$96 \div 5 = \19.20 .

WHY NOT (C) adds the whole tip to one team member's share. (D) is the total for the whole group, not one team member's share.

24. **D** 9 weeks

- 1 Money still needed: $\$1,400 - \$490 = \$910$.
- 2 Divide by the weekly savings: $910 \div 110$ is 8 with 30 left over. After 8 weeks she has saved $8 \times \$110 = \880 , still \$30 short.
- 3 So she needs one more week: $8 + 1 = 9$ weeks.

WHY NOT (C) rounds down, but after 8 weeks she is still \$30 short.

25. **B** 15 weeks

- 1 Money still needed: $\$725 - \$290 = \$435$.
- 2 Divide by the weekly savings: $435 \div 30$ is 14 with 15 left over. After 14 weeks he has saved $14 \times \$30 = \420 , still \$15 short.
- 3 So he needs one more week: $14 + 1 = 15$ weeks.

WHY NOT (A) rounds down, but after 14 weeks he is still \$15 short. (C) ignores the \$290 already saved. (D) adds the money already saved to the price instead of subtracting it.

26. **A** \$1,932

- 1 Amount saved: $\frac{1}{10} \times \$2,760 = \276 .
- 2 Amount spent on groceries: $\frac{1}{5} \times \$2,760 = \552 .
- 3 What is left: $\$2,760 - \$276 - \$552 = \$1,932$.

WHY NOT (B) subtracts only the amount spent on groceries. (C) adds the fractions by adding the tops and the bottoms. (D) subtracts only the amount saved.

27. **A** 11

- 1 First take out the fixed cost: $\$125 - \$35 = \$90$ is left to spend.
- 2 Divide by the price of one: $\$90 \div \8 is 11 with some money left over, because $11 \times \$8 = \88 .
- 3 One more would cost \$96, which is more than \$90. Round *down*: 11 boxes.

TIP Left over: $\$90 - \$88 = \$2$, not enough for one more.

WHY NOT (B) rounds up, but 12 boxes would cost \$96. (C) forgets to subtract the \$35 first. (D) adds the \$35 instead of subtracting it.

28. **D** \$10.62

- 1 Cost of 18 single bars: $18 \times \$1.75 = \31.50 .
- 2 Savings: $\$31.50 - \$20.88 = \$10.62$.

WHY NOT (A) is the savings on one bar, not on the whole box. (B) is the price per bar in the box.

29. **B** \$380.16

- 1 Find the discount: $0.2 \times \$440 = \88 .
- 2 Sale price: $\$440 - \$88 = \$352$.
- 3 Tax on the *sale* price: $0.08 \times \$352 = \28.16 .
- 4 Total: $\$352 + \$28.16 = \$380.16$.

TIP Or chain the multipliers: $\$440 \times 0.8 \times 1.08 = \380.16 .

WHY NOT (A) forgets to add the sales tax. (C) figures the tax on the regular price instead of the sale price. (D) adds the tax but forgets the discount.

30. **C** \$445.12

- 1 Find the discount: $0.2 \times \$520 = \104 .
- 2 Sale price: $\$520 - \$104 = \$416$.
- 3 Tax on the *sale* price: $0.07 \times \$416 = \29.12 .
- 4 Total: $\$416 + \$29.12 = \$445.12$.

TIP Or chain the multipliers: $\$520 \times 0.8 \times 1.07 = \445.12 .

WHY NOT (A) subtracts the tax instead of adding it. (B) forgets to add the sales tax. (D) figures the tax on the regular price instead of the sale price.

31. **D** \$532.44

- 1 Find the discount: $0.15 \times \$580 = \87 .
- 2 Sale price: $\$580 - \$87 = \$493$.
- 3 Tax on the *sale* price: $0.08 \times \$493 = \39.44 .
- 4 Total: $\$493 + \$39.44 = \$532.44$.

TIP Or chain the multipliers: $\$580 \times 0.85 \times 1.08 = \532.44 .

WHY NOT (A) subtracts the tax instead of adding it. (C) forgets to add the sales tax.

32. **A** \$735

- 1 Total hours: $10 \times 4 + 6 = 46$ hours.
- 2 Only hours over 40 are overtime: $46 - 40 = 6$ overtime hours.
- 3 Regular pay: $40 \times \$15 = \600 . Overtime rate: $1.5 \times \$15 = \22.50 per hour.
- 4 Overtime pay: $6 \times \$22.50 = \135 .
- 5 Total: $\$600 + \$135 = \$735$.

WHY NOT (B) pays double time instead of time-and-a-half. (D) pays time-and-a-half for all the hours, not just the hours over 40.

33. **C** 42 hours

- 1 Pay for the first 40 hours: $40 \times \$20 = \800 .
- 2 The rest of the pay is overtime: $\$860 - \$800 = \$60$.
- 3 Overtime rate: $1.5 \times \$20 = \30 per hour. Overtime hours: $\$60 \div \$30 = 2$.
- 4 Total hours: $40 + 2 = 42$ hours.

WHY NOT (A) gives only the overtime hours, not the total. (D) divides the overtime pay by the regular rate instead of the overtime rate.

34. **B** the case of 12

- 1 Find each price per bottle (price $\div$ number of bottles): $\$3.24 \div 12 = \0.27 ; $\$7.44 \div 24 = \0.31 ; $\$14 \div 40 = \0.35 .

- 2 The lowest price per bottle is \$0.27, so the case of 12 is the best buy.

TIP The biggest package is not always the best deal, so always compare unit prices.

WHY NOT (A) is not true: the prices per bottle are \$0.27, \$0.31, and \$0.35. (C) costs \$0.31 per bottle, more than \$0.27. (D) is the largest size, but it costs \$0.35 per bottle, more than \$0.27.

35. **C** the 12-ounce bag

- 1 Find each price per ounce (price $\div$ number of ounces): $\$3.96 \div 12 = \0.33 ; $\$8.40 \div 24 = \0.35 ; $\$13.12 \div 32 = \0.41 .

- 2 The lowest price per ounce is \$0.33, so the 12-ounce bag is the best buy.

TIP The biggest package is not always the best deal, so always compare unit prices.

WHY NOT (A) is the largest size, but it costs \$0.41 per ounce, more than \$0.33. (B) costs \$0.35 per ounce, more than \$0.33. (D) is not true: the prices per ounce are \$0.33, \$0.35, and \$0.41.

36. **C** \$231

- 1 First amount (paid on the car loan): $\frac{1}{8} \times \$2,640 = \330 .
- 2 What is left: $\$2,640 - \$330 = \$2,310$.
- 3 Second amount is $\frac{1}{10}$ of *that*: $\frac{1}{10} \times \$2,310 = \231 .

WHY NOT (D) takes $\frac{1}{10}$ of the whole pay instead of $\frac{1}{10}$ of what is left.

37. **A** \$1,350.00

- 1 Every two weeks means $52 \div 2 = 26$ paychecks a year (*not* 24).
- 2 Divide the yearly pay by the number of paychecks: $\$46,800 \div 26 = \$1,800$.
- 3 Taxes withheld: $0.25 \times \$1,800 = \450 .
- 4 Take-home pay: $\$1,800 - \$450 = \$1,350$.

WHY NOT (B) uses the wrong number of paychecks per year. (C) forgets to subtract the taxes withheld. (D) divides by 12 months instead of the number of paychecks.

38. **B** \$4,160

- 1 Hours in a year: $40 \times 52 = 2,080$ hours.
- 2 Job A per year: $2,080 \times \$16 = \$33,280$.
- 3 Compare with Job B's \$37,440: $\$37,440 - \$33,280 = \$4,160$. Job B pays \$4,160 more.

WHY NOT (A) is the difference for one week, not for a year. (C) assumes 4 weeks in every month (48 weeks), but a year has 52 weeks. (D) is Job A's yearly pay, not the difference.

39. **A** \$3,120

- 1 Hours in a year: $40 \times 52 = 2,080$ hours.
- 2 Job A per year: $2,080 \times \$19.50 = \$40,560$. (Think $2,080 \times 19 = \$39,520$ plus $2,080 \times 0.50 = \$1,040$.)
- 3 Compare with Job B's \$37,440: $\$40,560 - \$37,440 = \$3,120$. Job A pays \$3,120 more.

WHY NOT (D) is Job A's yearly pay, not the difference.

40. **C** \$36.96

- 1 Full cases: $64 \div 24$ is 2 with 16 left over, so buy 2 cases and 16 single bottles.
- 2 Cases: $2 \times \$12.48 = \24.96 . Singles: $16 \times \$0.75 = \12 .
- 3 Total: $\$24.96 + \$12 = \$36.96$.

WHY NOT (A) forgets the 16 single bottles. (B) pays for part of a case at the case price, but stores sell whole cases. (D) buys every bottle singly.

Interest, Commission & Percent Change

Tested on: Arithmetic Reasoning

Key Concepts

Chapter 6 covered the basic percent questions. This chapter puts percents to work with money: interest on savings and loans, commission and markup, raises, and the two kinds of problems that trip up the most test takers, *chains of percents* and *working backward* from a percent.

Simple interest

$$I = P \times r \times t$$

P = principal (the amount saved or borrowed), r = yearly rate as a decimal, t = time *in years*. The balance (or the amount owed) is $P + I$. Change months to years by dividing by 12: 6 months = $\frac{6}{12} = \frac{1}{2}$ year, 8 months = $\frac{8}{12} = \frac{2}{3}$ year, 18 months = $1\frac{1}{2}$ years.

Commission, markup, and raises

- **Commission** = rate $\times$ sales. Total pay = base pay + commission.
- **Markup** = rate $\times$ cost; selling price = cost + markup; profit = selling price – cost.
- **Raise:** new pay = old pay + rate $\times$ old pay = $(1 + \text{rate}) \times \text{old pay}$.

Chains of percents and working backward

- Percent changes in a row **multiply**: 20% off and then 10% off leaves $0.80 \times 0.90 = 0.72$ of the price, a 28% discount, not 30%.
- **Compound interest** works the same way: each year's interest is added before the next year's interest is figured.
- **Working backward:** if the price *after* 20% off is \$64, then \$64 is 80% of the original. Divide: $64 \div 0.80 = \$80$.

WORKED EXAMPLE

After a 25% raise, a store manager earns \$20 per hour. What did she earn before the raise?

Solution. The new wage is 125% of the old wage: $1.25 \times \text{old} = 20$, so $\text{old} = 20 \div 1.25 = \16 . Check: 25% of \$16 is \$4, and $\$16 + \$4 = \$20$. (Taking 25% off \$20 gives \$15, which is wrong: the raise was 25% of the *old* wage.)

No-Calculator Tip Turn friendly decimals into fractions: dividing by 0.80 is multiplying by $\frac{5}{4}$, and dividing by 1.25 is multiplying by $\frac{4}{5}$. For interest, find one year's interest first, then multiply by the number of years.

Common Traps

- Using months as if they were years in $I = Prt$.
- Working backward by adding the percent to the *new* amount (\$64 after 20% off is not $64 + 12.80$).
- Adding two discounts in a row, or adding a raise and a cut that follow each other.
- Paying commission on *all* sales when it is earned only above a set amount.

Practice Set 23

Score: ____ / 35

Target time: 42 minutes

Problems 1–10 are warm-ups, 11–24 are test level, and 25–35 are challenge problems.

- Carlos borrows \$5,000 from a credit union to buy a used car. The loan charges 4% simple interest per year. How much interest will he pay in 4 years?
A \$200 B \$600 C \$800 D \$5,800
- Specialist Walker puts a \$5,000 enlistment bonus into a savings account at the base credit union. The account pays 2% simple interest per year. How much interest will he earn in 4 years?
A \$400 B \$4,000 C \$5,400 D \$40,000
- Mei buys a certificate of deposit (CD) for \$5,500. The CD pays 3% simple interest per year. How much interest will she earn in 2 years?
A \$165 B \$280 C \$330 D \$5,830
- Grace, an Army veteran, sells insurance and earns a 5% commission on the yearly premiums she sells. This month she sold \$29,000 in premiums. What is her commission?
A \$1,450 B \$14,500
C \$27,550 D \$145,000
- Emily sells cars and earns a 4% commission on the price of each car she sells. What is her commission on a car that sells for \$22,500?
A \$90 B \$800 C \$900 D \$9,000
- When Sergeant Carter is promoted, her monthly base pay goes up 6%. Before the promotion it was \$2,950. What is it after the promotion?
A \$177 B \$2,773 C \$2,956 D \$3,127
- Tyrone earns a salary of \$70,000 a year. He gets a 6% raise. What is his new yearly salary?
A \$4,200 B \$74,200
C \$104,200 D \$112,000
- Specialist Kim's monthly base pay is \$2,950. This year's military pay raise is 3%. What will his new monthly base pay be?
A \$88.50 B \$3,038.50
C \$3,835.00 D \$4,538.50
- A battalion's goal for its charity drive is \$2,500. So far it has raised \$1,000. What percent of the goal has it reached?
A 40% B 50% C 60% D 250%
- A recruiting station's goal is 40 enlistments this quarter. So far it has signed 22 recruits. What percent of its goal has it met?
A 0.55% B 40% C 45% D 55%
- Mei deposits \$3,100 in a savings account that pays 5% simple interest per year. How much interest will she earn in 18 months?
A \$232.50 B \$279.00
C \$2,790.00 D \$3,332.50
- Hannah borrows \$11,000 from a credit union to buy a used car. The loan charges 6% simple interest per year. How much interest will she pay in 10 months?
A \$55 B \$550 C \$660 D \$6,600

13. Jamal deposits \$500 in a savings account that pays 5% simple interest per year. How much money will be in the account after 6 months?
A \$512.50 B \$525.00 C \$625.00 D \$650.00
14. Seaman Novak borrows \$5,500 from the base credit union to furnish his first apartment off base. The loan charges 6% simple interest per year. What is the total amount he must pay back at the end of 3 years?
A \$6,490 B \$7,490 C \$8,490 D \$15,400
15. Brandon works at a home-security company and earns a base salary of \$500 per week plus a 5% commission on his sales. His sales this week total \$6,100. What is his total pay for the week?
A \$305 B \$805 C \$3,550 D \$6,600
16. Derek works at a furniture store and earns a base salary of \$1,500 per month plus a 2% commission on his sales. His sales this month total \$75,000. What is his total pay for the month?
A \$3,000 B \$4,000 C \$16,500 D \$76,500
17. After a promotion, Sergeant Patel's monthly base pay went up 12% to \$3,248. What was his monthly base pay before the promotion?
A \$389.76 B \$2,858.24
C \$2,900.00 D \$3,236.00
18. A unit's blood drive collected 96 pints this year, 50% more than last year. How many pints did it collect last year?
A 48 B 64 C 65 D 69
19. A food truck spends \$2 to make each burrito and marks up the cost by 150%. What price is put on each burrito?
A \$2.30 B \$3.00 C \$4.00 D \$5.00
20. A vendor at the base's Fourth of July fair buys T-shirts for \$8 each and marks up the cost by 50%. What price is put on each T-shirt?
A \$12 B \$13 C \$14 D \$58
21. A car dealer lowers the price of a \$9,500 used car by 40%. When the car does not sell, the dealer cuts the new price by another 10%. What is the final price of the used car?
A \$4,750 B \$5,130 C \$5,700 D \$8,550
22. A store takes 50% off a \$150 winter coat. Service members get an extra 10% military discount on the sale price. What is the price after both discounts?
A \$7.50 B \$57.50 C \$60.00 D \$67.50
23. On a weekend pass, a group of airmen order a meal that costs \$116. They pay a 5% sales tax and leave a 20% tip, both figured on the \$116. What is the total cost?
A \$121.80 B \$139.20 C \$145.00 D \$146.16
24. Private Reyes wants to run 100 miles this month to get ready for a PT test. She has completed 40% of that distance. How many more miles must she run?
A 40 miles B 60 miles C 62 miles D 64 miles
25. Private Santos deposited \$4,000 in an account that pays simple interest. After 2 years the account had earned \$360 in interest. What is the yearly interest rate?
A 0.045% B 2.25% C 4.5% D 9%
26. Tyrone buys a certificate of deposit (CD) for \$6,000. The CD pays 5% simple interest per year. How many years will it take for the account to earn \$1,200 in interest?
A 0.04 years B 2 years
C 3 years D 4 years
27. Ana works at a home-security company. She earns \$350 per week plus a 12% commission on all sales over \$3,000 for the week. Her sales this week total \$11,200. What is her total pay for the week?
A \$984 B \$1,334 C \$1,344 D \$1,694
28. Jasmine earns \$550 per week plus a 10% commission on her sales at a furniture store. How much must she sell in a week to earn a total of \$1,050?
A \$500 B \$5,000 C \$6,000 D \$10,500
29. Sofia bought a bicycle on sale for 20% off. With an 8% sales tax added to the sale price, she paid \$397.44. What was the original price of the bicycle?
A \$310 B \$368 C \$410 D \$460

30. Brandon bought a backpack on sale for 40% off. With an 8% sales tax added to the sale price, he paid \$71.28. What was the original price of the backpack?
A \$66.00 B \$100.00 C \$110.00 D \$118.80
31. A furniture store discounts a recliner by 40%. A coupon gives 10% off the discounted price. The two discounts together are the same as a single discount of what percent?
A 4% B 36% C 46% D 50%
32. A military surplus store buys used rucksacks for \$35 each and marks up the cost by 60%. Later, the rucksack goes on sale for 30% off the marked price. What is the sale price?
A \$39.20 B \$45.50 C \$49.20 D \$56.00
33. Victor deposits \$8,000 in an account that pays 6% interest per year, compounded once a year. How much will be in the account after 2 years?
A \$8,480.00 B \$8,960.00
C \$8,988.80 D \$10,988.80
34. Aisha deposits \$4,000 in an account that pays 4% yearly interest, compounded every six months (so 2% is added every six months). How much will be in the account after 1 year?
A \$161.60 B \$4,080.00
C \$4,160.00 D \$4,161.60
35. Petty Officer Murphy takes out a loan of \$4,800 to buy furniture for his apartment off base. The loan charges 5% simple interest per year, and he will repay the principal plus interest in equal monthly payments over 4 years. How much is each monthly payment?
A \$100 B \$105 C \$110 D \$120

Answers & Explanations 23

1 C	8 B	15 B	22 D	29 D
2 A	9 A	16 A	23 C	30 C
3 C	10 D	17 C	24 B	31 C
4 A	11 A	18 B	25 C	32 A
5 C	12 B	19 D	26 D	33 C
6 D	13 A	20 A	27 B	34 D
7 B	14 A	21 B	28 B	35 D

- C** \$800
 - Use $I = P \times r \times t$ with $r = 4\% = 0.04$ and $t = 4$ years.
 - One year's interest: $\$5,000 \times 0.04 = \200 .
 - For 4 years: $\$200 \times 4 = \800 .

WHY NOT (A) finds the interest for only one year. (D) is the total with interest, not the interest alone.
- A** \$400
 - Use $I = P \times r \times t$ with $r = 2\% = 0.02$ and $t = 4$ years.
 - One year's interest: $\$5,000 \times 0.02 = \100 .
 - For 4 years: $\$100 \times 4 = \400 .

WHY NOT (B) writes 2% as 0.2 instead of 0.02. (C) is the total with interest, not the interest alone. (D) forgets to change the percent to a decimal.
- C** \$330
 - Use $I = P \times r \times t$ with $r = 3\% = 0.03$ and $t = 2$ years.
 - One year's interest: $\$5,500 \times 0.03 = \165 .
 - For 2 years: $\$165 \times 2 = \330 .

WHY NOT (A) finds the interest for only one year. (D) is the total with interest, not the interest alone.
- A** \$1,450
 - Commission = rate $\times$ sales. Change the rate to a decimal: $5\% = 0.05$.
 - Multiply: $0.05 \times \$29,000 = \$1,450$.

WHY NOT (B) writes 5% as 0.5 instead of 0.05. (C) subtracts the commission from the sale price. (D) forgets to change the percent to a decimal.
- C** \$900
 - Commission = rate $\times$ sales. Change the rate to a decimal: $4\% = 0.04$.
 - Multiply: $0.04 \times \$22,500 = \900 .

WHY NOT (A) moves the decimal point three places instead of two. (D) writes 4% as 0.4 instead of 0.04.
- D** \$3,127
 - Find the raise: $0.06 \times \$2,950 = \177 .
 - Add it to the old pay: $\$2,950 + \$177 = \$3,127$.

TIP In one step: $1.06 \times \$2,950 = \$3,127$.

WHY NOT (A) is the amount of the raise, not the new monthly base pay. (B) subtracts the change instead of adding it. (C) adds the percent as if it were dollars.
- B** \$74,200
 - Find the raise: $0.06 \times \$70,000 = \$4,200$.
 - Add it to the old pay: $\$70,000 + \$4,200 = \$74,200$.

TIP In one step: $1.06 \times \$70,000 = \$74,200$.

WHY NOT (A) is the amount of the raise, not the new salary. (D) moves the decimal point only one place.
- B** \$3,038.50
 - Find the raise: $0.03 \times \$2,950 = \88.50 .
 - Add it to the old pay: $\$2,950 + \$88.50 = \$3,038.50$.

TIP In one step: $1.03 \times \$2,950 = \$3,038.50$.

WHY NOT (A) is the amount of the raise, not the new monthly base pay. (C) moves the decimal point only one place.
- A** 40%
 - Write the part done over the goal: $\frac{1,000}{2,500}$.
 - Divide and change to a percent: $1,000 \div 2,500 = 0.4 = 40\%$.

WHY NOT (C) is the percent still to go, not the percent done. (D) divides the goal by the amount done instead of the amount done by the goal.
- D** 55%
 - Write the part done over the goal: $\frac{22}{40}$.
 - Divide and change to a percent: $22 \div 40 = 0.55 = 55\%$.

WHY NOT (A) forgets to multiply by 100. (C) is the percent still to go, not the percent done.

11. **A** \$232.50

- 1 Time must be in years: $18 \text{ months} = \frac{18}{12} = 1\frac{1}{2} = 1.5 \text{ years}$.
- 2 One year's interest: $\$3,100 \times 0.05 = \155 .
- 3 For $1\frac{1}{2}$ years: $\$155 \times 1\frac{1}{2} = \232.50 .

WHY NOT (B) writes 18 months as 1.8 years. (C) uses 18 years instead of 18 months. (D) is the total with interest, not the interest alone.

12. **B** \$550

- 1 Time must be in years: $10 \text{ months} = \frac{10}{12} = \frac{5}{6} \text{ year}$.
- 2 One year's interest: $\$11,000 \times 0.06 = \660 .
- 3 For $\frac{5}{6}$ of a year: $\$660 \times \frac{5}{6} = \550 .

WHY NOT (A) finds the interest for just one month. (C) finds the interest for a full year. (D) uses 10 years instead of 10 months.

13. **A** \$512.50

- 1 Change the time to years: $6 \text{ months} = \frac{6}{12} = \frac{1}{2} = 0.5 \text{ year}$.
- 2 One year's interest: $\$500 \times 0.05 = \25 .
- 3 Interest for the whole time: $\$25 \times \frac{1}{2} = \12.50 .
- 4 Add it to the principal: $\$500 + \$12.50 = \$512.50$.

WHY NOT (B) adds only one year of interest. (C) writes 5% as 0.5 instead of 0.05. (D) uses 6 years instead of 6 months.

14. **A** \$6,490

- 1 One year's interest: $\$5,500 \times 0.06 = \330 .
- 2 Interest for the whole time: $\$330 \times 3 = \990 .
- 3 Add it to the principal: $\$5,500 + \$990 = \$6,490$.

WHY NOT (D) writes 6% as 0.6 instead of 0.06.

15. **B** \$805

- 1 Commission: $0.05 \times \$6,100 = \305 .
- 2 Add the base salary: $\$500 + \$305 = \$805$.

WHY NOT (A) is the commission only; it leaves out the base salary. (C) writes 5% as 0.5. (D) adds the sales instead of the commission.

16. **A** \$3,000

- 1 Commission: $0.02 \times \$75,000 = \$1,500$.
- 2 Add the base salary: $\$1,500 + \$1,500 = \$3,000$.

WHY NOT (C) writes 2% as 0.2. (D) adds the sales instead of the commission.

17. **C** \$2,900.00

- 1 The new pay is 112% of the old pay: $1.12 \times \text{old pay} = \$3,248$.
- 2 Divide: $\$3,248 \div 1.12 = \$2,900$.

TIP Check: $1.12 \times \$2,900 = \$3,248$.

WHY NOT (A) finds 12% of the new pay instead of removing the raise. (B) takes 12% off the new pay, but the raise was 12% of the old pay. (D) subtracts the percent as if it were dollars.

18. **B** 64

- 1 The new number is 150% of the old number: $1.5 \times \text{old} = 96$.
- 2 Divide: $96 \div 1.5 = 64$.

TIP Check: 50% of 64 is 32, and $64 + 32 = 96$.

WHY NOT (A) takes 50% off this year's number, but the increase was 50% of the earlier number.

19. **D** \$5.00

- 1 Markup: $1.5 \times \$2 = \3 .
- 2 Selling price: $\$2 + \$3 = \$5$.

TIP In one step: $2.5 \times \$2 = \5 .

WHY NOT (A) moves the decimal point one place too far when finding the markup. (B) is the markup, not the selling price.

20. **A** \$12

- 1 Markup: $0.5 \times \$8 = \4 .
- 2 Selling price: $\$8 + \$4 = \$12$.

TIP In one step: $1.5 \times \$8 = \12 .

WHY NOT (D) adds the percent as if it were dollars.

21. **B** \$5,130

- 1 First discount: $\$9,500 - 0.4 \times \$9,500 = \$9,500 - \$3,800 = \$5,700$.
- 2 The second discount is taken from the *new* price: $\$5,700 - 0.1 \times \$5,700 = \$5,700 - \$570 = \$5,130$.

TIP In one line: $\$9,500 \times 0.6 \times 0.9 = \$5,130$.

WHY NOT (A) adds the discounts to get 50% off. (C) stops after the first discount. (D) applies only the second discount.

22. **D** \$67.50

- 1 First discount: $\$150 - 0.5 \times \$150 = \$150 - \$75 = \$75$.
- 2 The second discount is taken from the *new* price: $\$75 - 0.1 \times \$75 = \$75 - \$7.50 = \$67.50$.

TIP In one line: $\$150 \times 0.5 \times 0.9 = \67.50 .

WHY NOT (A) multiplies the two discounts together. (C) adds the discounts to get 60% off.

23. **C** \$145.00

- 1 Tax: $0.05 \times \$116 = \5.80 .
- 2 Tip: $0.2 \times \$116 = \23.20 .
- 3 Total: $\$116 + \$5.80 + \$23.20 = \145 .

TIP Both percents are of the same amount, so add them first: $5\% + 20\% = 25\%$, and $1.25 \times \$116 = \145 .

WHY NOT (A) forgets the tip. (B) forgets the tax. (D) figures the tip on the total with tax, but the tip is on the food cost only.

24. **B** 60 miles

- 1 Amount done: $0.4 \times 100 = 40$.
- 2 Amount still needed: $100 - 40 = 60$.

TIP Or: $100\% - 40\% = 60\%$ is left, and $0.6 \times 100 = 60$.

WHY NOT (A) is the amount already done, not the amount still needed.

25. **C** 4.5%

- 1 Interest for one year: $\$360 \div 2 = \180 .
- 2 Divide by the principal: $\$180 \div \$4,000 = 0.045$.
- 3 Change to a percent: $0.045 = 4.5\%$.

TIP Check: $\$4,000 \times 0.045 \times 2 = \360 .

WHY NOT (A) forgets to change the decimal to a percent. (B) divides by 2 twice. (D) forgets to divide by the 2 years.

26. **D** 4 years

- 1 One year's interest: $\$6,000 \times 0.05 = \300 .
- 2 Number of years: $\$1,200 \div \$300 = 4$.

TIP Check: $\$300 \times 4 = \$1,200$.

WHY NOT (A) forgets to change the percent to a decimal.

27. **B** \$1,334

- 1 Sales that earn commission: $\$11,200 - \$3,000 = \$8,200$.
- 2 Commission: $0.12 \times \$8,200 = \984 .
- 3 Total pay: $\$350 + \$984 = \$1,334$.

WHY NOT (A) is the commission only; it leaves out the base pay. (C) pays commission on all the sales and leaves out the base pay. (D) pays commission on all the sales, not just the sales over \$3,000.

28. **B** \$5,000

- 1 Commission needed: $\$1,050 - \$550 = \$500$.
- 2 That commission is 10% of sales: $0.1 \times \text{sales} = \500 .
- 3 Divide: $\$500 \div 0.1 = \$5,000$.

TIP Check: $0.1 \times \$5,000 = \500 , and $\$550 + \$500 = \$1,050$.

WHY NOT (A) is the commission needed, not the sales needed. (D) forgets to subtract the base pay first.

29. **D** \$460

- 1 Remove the tax first. The amount paid is 108% of the sale price: $\$397.44 \div 1.08 = \368 .
- 2 The sale price is 80% of the original: $\$368 \div 0.8 = \460 .

TIP Check: $\$460 \times 0.8 = \368 , and $\$368 \times 1.08 = \397.44 .

WHY NOT (B) is the sale price, not the original price.

30. **C** \$110.00

- 1 Remove the tax first. The amount paid is 108% of the sale price: $\$71.28 \div 1.08 = \66 .
- 2 The sale price is 60% of the original: $\$66 \div 0.6 = \110 .

TIP Check: $\$110 \times 0.6 = \66 , and $\$66 \times 1.08 = \71.28 .

WHY NOT (A) is the sale price, not the original price. (D) forgets to remove the sales tax.

31. **C** 46%

- 1 After 40% off you pay 60% of the price, and after another 10% off you pay 90% of *that*.
- 2 Multiply: $0.6 \times 0.9 = 0.54$, so you pay 54% of the original price.
- 3 The single discount is $100\% - 54\% = 46\%$.

TIP Test it on \$100: 40% off gives \$60, and 10% off that gives \$54, which is \$46 off.

WHY NOT (A) multiplies the two percents. (D) adds the two discounts, but the second is taken from a smaller price.

32. **A** \$39.20

- 1 Marked price: $\$35 + 0.6 \times \$35 = \$35 + \$21 = \$56$.
- 2 Take 30% off the *marked* price: $\$56 - 0.3 \times \$56 = \$56 - \$16.80 = \$39.20$.

TIP In one line: $\$35 \times 1.6 \times 0.7 = \39.20 .

WHY NOT (B) combines 60% up and 30% off into one 30% markup. (D) is the marked price; it forgets the sale.

33. **C** \$8,988.80

- 1 Rate for each period: 6%.
- 2 After the first year: $\$8,000 + 0.06 \times \$8,000 = \$8,000 + \$480 = \$8,480$.
- 3 After the second: $\$8,480 + 0.06 \times \$8,480 = \$8,480 + \$508.80 = \$8,988.80$.

TIP The second year earns more (\$508.80 vs. \$480) because it earns interest on the interest.

WHY NOT (A) stops after the first period. (B) uses simple interest; it leaves out the interest earned on the first interest.

34. **D** \$4,161.60

- 1 Rate for each period: 2% (half of the yearly rate).
- 2 After the first six months: $\$4,000 + 0.02 \times \$4,000 = \$4,000 + \$80 = \$4,080$.
- 3 After the second: $\$4,080 + 0.02 \times \$4,080 = \$4,080 + \$81.60 = \$4,161.60$.

TIP The second period earns more (\$81.60 vs. \$80) because it earns interest on the interest.

WHY NOT (A) is the interest only, not the balance. (B) stops after the first period. (C) uses simple interest; it leaves out the interest earned on the first interest.

35. **D** \$120

- 1 Interest: $\$4,800 \times 0.05 \times 4 = \960 .
- 2 Total to repay: $\$4,800 + \$960 = \$5,760$.
- 3 Number of payments: $4 \times 12 = 48$ months.
- 4 Each payment: $\$5,760 \div 48 = \120 .

WHY NOT (A) forgets to add the interest. (B) adds only one year of interest.

Rate, Distance & Time

Tested on: Arithmetic Reasoning

Key Concepts

Anything that moves at a steady rate (a car, a convoy, a runner, a printer turning out pages) follows one formula. The skill is in setting it up with matching units and in knowing which rates to add, subtract, or leave alone.

The distance formula

$$d = r \times t \quad r = \frac{d}{t} \quad t = \frac{d}{r}$$

d = distance, r = rate (speed), t = time. The units must match: with a speed in miles per hour, the time must be in *hours*. The same formula works for any rate: amount = rate $\times$ time (words typed, pages printed, gallons used).

Minutes and hours

Divide minutes by 60 to get hours: 15 min = $\frac{1}{4}$ h = 0.25 h, 20 min = $\frac{1}{3}$ h, 30 min = 0.5 h, 45 min = 0.75 h. Going the other way, multiply by 60: 0.4 h = 24 min. Beware: 2.5 hours is 2 hours 30 minutes, not 2 hours 50 minutes.

Two movers

- Moving **toward** each other or in **opposite** directions: the gap changes at the *sum* of the speeds.
- Moving in the **same** direction (one catching up): the gap changes at the *difference* of the speeds.
- **Average speed** = $\frac{\text{total distance}}{\text{total time}}$. It is *not* the average of the speeds unless the times are equal.

WORKED EXAMPLE

A convoy drives 120 miles to a range at 40 mph and returns at 60 mph. What is its average speed for the round trip?

Solution. Out: $120 \div 40 = 3$ hours. Back: $120 \div 60 = 2$ hours. Total: 240 miles in 5 hours, so the average speed is $240 \div 5 = 48$ mph, not 50 mph: the convoy spends more time at the slower speed.

No-Calculator Tip For pace and speed, use 60: a pace of 8 minutes per mile is $60 \div 8 = 7.5$ mph, and 10 mph is $60 \div 10 = 6$ minutes per mile.

Common Traps

- Multiplying a speed in miles per hour by a time in *minutes*.
- Writing 45 minutes as 0.45 hour (it is 0.75 hour).

- Averaging two speeds instead of dividing total distance by total time.
- Adding speeds when one vehicle chases another (subtract them).

Practice Set 24

Score: ____ / 40

Target time: 48 minutes

Problems 1–12 are warm-ups, 13–28 are test level, and 29–40 are challenge problems.

1. A platoon marches at an average speed of 4 miles per hour for 3 hours. How far does the platoon march?
A 7 miles B 11 miles C 12 miles D 16 miles
2. A Black Hawk helicopter flies at an average speed of 150 miles per hour for 2 hours. How far does the helicopter fly?
A 75 miles B 152 miles
C 275 miles D 300 miles
3. A passenger jet flies at an average speed of 425 miles per hour for 4 hours. How far does the jet fly?
A 429 miles B 1,500 miles
C 1,700 miles D 2,125 miles
4. Jasmine rides a bicycle at an average speed of 17 miles per hour for 2 hours. How far does she ride?
A 19 miles B 24 miles C 29 miles D 34 miles
5. A helicopter flies at 120 miles per hour for 40 minutes. How many miles does the helicopter fly?
A 48 miles B 60 miles
C 80 miles D 4,800 miles
6. Maria jogs at 6 miles per hour for 50 minutes. How many miles does she jog?
A 5 miles B 6 miles
C 7 miles D 300 miles
7. Diego's car used 11 gallons of gas to travel 363 miles. How many miles per gallon did it get?
A 30 B 33 C 37 D 352
8. During a field exercise, a Humvee used 19 gallons of diesel to travel 171 miles. How many miles per gallon did it get?
A 6 B 7 C 9 D 152
9. Ana's motorcycle used 3 gallons of gas to travel 162 miles. How many miles per gallon did it get?
A 50 B 54 C 59 D 159
10. Tasha types 60 words per minute. How many minutes will it take her to type a 600-word report?
A 8 minutes B 10 minutes
C 15 minutes D 540 minutes
11. An office printer prints 18 pages per minute. How many minutes will it take to print 216 pages?
A 9 minutes B 12 minutes
C 13 minutes D 198 minutes
12. Yuki reads about 25 pages per hour. How many hours will it take her to read a 125-page book?
A 5 hours B 7 hours
C 10 hours D 100 hours
13. Private Brooks's supply convoy travels at 36 miles per hour for 2 hours 30 minutes. How many miles does the convoy travel?
A 82.8 miles B 90 miles
C 108 miles D 120 miles
14. A train covers 18 miles in 20 minutes. What is the train's average speed in miles per hour?
A 0.9 mph B 54 mph
C 90 mph D 360 mph

15. A shuttle bus travels 15 miles between two gates of a large base at 36 miles per hour. How many minutes does the trip take?
A 25 minutes B 35 minutes
C 41 minutes D 144 minutes
16. Two supply trucks start 400 miles apart and travel toward each other on the same road. One goes 40 miles per hour and the other goes 60 miles per hour. In how many hours will they meet?
A 1 hour B 2 hours
C 4 hours D 20 hours
17. Two trains leave a town at the same time and travel in the same direction on parallel tracks. One goes 55 miles per hour and the other goes 60 miles per hour. How far apart are they after 4 hours?
A 20 miles B 22 miles
C 240 miles D 460 miles
18. Two cars start 210 miles apart and travel toward each other on the same road. One goes 45 miles per hour and the other goes 60 miles per hour. In how many hours will they meet?
A 2 hours B 3.5 hours
C 4 hours D 14 hours
19. Malik's car gets 30 miles per gallon. Gas costs \$3.75 per gallon. How much will gas cost for a road trip of 120 miles?
A \$15.00 B \$17.00 C \$112.50 D \$450.00
20. Marcus's pickup truck gets 17 miles per gallon. Gas costs \$3.90 per gallon. How much will gas cost for a road trip of 153 miles?
A \$9.00 B \$35.10 C \$66.30 D \$596.70
21. James jogs 4 miles in 48 minutes. What is his average speed in miles per hour?
A 5 mph B 6 mph
C 12 mph D 192 mph
22. Omar runs at a steady 7.5 miles per hour. How many minutes does it take him to run 5 miles?
A 8 minutes B 25 minutes
C 30 minutes D 40 minutes
23. Ethan walks 3 miles in 45 minutes. What is his average speed in miles per hour?
A 1 mph B 3 mph C 4 mph D 15 mph
24. Diego drives 105 miles on the highway at 70 miles per hour, then 40 miles on country roads at 40 miles per hour. How many hours does the whole trip take?
A 1.5 hours B 2.5 hours
C 3.5 hours D 4 hours
25. A convoy travels 40 miles on paved road at 40 miles per hour, then 30 miles on a dirt road at 15 miles per hour. How many hours does the whole trip take?
A 1 hour B 1.75 hours
C 3 hours D 5 hours
26. Kevin leaves home at 10:55 a.m. and drives 84 miles at an average speed of 70 miles per hour. At what time does he arrive?
A 11:55 a.m. B 12:15 p.m.
C 12:07 p.m. D 1:07 p.m.
27. A train leaves the station at 5:15 a.m. and travels 336 miles at 80 miles per hour. At what time does it arrive?
A 9:15 a.m. B 9:27 a.m.
C 9:35 a.m. D 10:27 a.m.
28. A water purification unit at a field camp produces 1,200 gallons of drinking water per hour. How many gallons does it produce in 45 minutes?
A 540 B 700 C 800 D 900
29. A convoy drives 120 miles to a training range at 60 miles per hour and returns on the same road at 20 miles per hour. What is the average speed for the whole round trip?
A 15 mph B 20 mph C 25 mph D 30 mph
30. A boat travels 48 miles upstream at 8 miles per hour and returns downstream at 12 miles per hour. What is the average speed for the round trip?
A 4.8 mph B 6.6 mph C 8.6 mph D 9.6 mph

31. Leila drives 120 miles to a job site at 60 miles per hour in morning traffic and drives back at 40 miles per hour in the evening. What is the average speed for the round trip?
A 24 mph B 33 mph C 48 mph D 50 mph
32. Keisha leaves home on a bicycle at 8 miles per hour. Half an hour later, her sister follows on a scooter at 16 miles per hour. How many hours after the sister leaves will she catch up?
A 0.25 hours B 0.5 hours
C 1 hour D 1.5 hours
33. A freight train leaves a station at 49 miles per hour. Two hours later, Hannah boards a passenger train that leaves the same station on a parallel track at 77 miles per hour. How many hours after the passenger train leaves will it catch up?
A 2 hours B 3.5 hours
C 5 hours D 5.5 hours
34. A moving truck leaves Jamal's old house at 48 miles per hour. One hour later, Jamal follows the same route in his car at 72 miles per hour. How far from the start will the car catch up?
A 144 miles B 164 miles
C 204 miles D 216 miles
35. A pitcher throws a fastball at 90 miles per hour. What is this speed in feet per second? (1 mile = 5,280 feet and 1 hour = 3,600 seconds.)
A 112 B 132 C 7,920 D 475,200
36. A speedboat cruises at 60 miles per hour. How many feet per second is this? (1 mile = 5,280 feet and 1 hour = 3,600 seconds.)
A 82 B 84 C 88 D 5,280
37. Leila bikes 21 miles on a flat trail at 14 miles per hour, then 3 miles uphill at 6 miles per hour. What is the average speed for the whole trip?
A 12 mph B 16 mph C 20 mph D 48 mph
38. A convoy travels 150 miles on paved road at 50 miles per hour, then 30 miles on a dirt road at 15 miles per hour. What is the average speed for the whole trip?
A 31 mph B 32.5 mph
C 36 mph D 65 mph
39. Specialist Kim is driving 450 miles to a new duty station and can take either a car that gets 30 miles per gallon or a truck that gets 18 miles per gallon. Gas costs \$3.10 per gallon. How much will he save on gas by taking the car?
A \$31.00 B \$37.20 C \$46.50 D \$77.50
40. Yuki leaves home at 8:35 a.m. and drives 75 miles at an average speed of 50 miles per hour. After a 30-minute stop, she drives another 48 miles at 48 miles per hour. At what time does she arrive?
A 11:50 a.m. B 11:35 a.m.
C 11:05 a.m. D 12:35 p.m.

Answers & Explanations 24

1 C	9 B	17 A	25 C	33 B
2 D	10 B	18 A	26 C	34 A
3 C	11 B	19 A	27 B	35 B
4 D	12 A	20 B	28 D	36 C
5 C	13 B	21 A	29 D	37 A
6 A	14 B	22 D	30 D	38 C
7 B	15 A	23 C	31 C	39 A
8 C	16 C	24 B	32 B	40 B

1. **C** 12 miles

1 Use $d = r \times t$: distance = speed $\times$ time.
 2 $d = 4 \times 3 = 12$ miles.

WHY NOT (A) adds the speed and the time instead of multiplying.

2. **D** 300 miles

1 Use $d = r \times t$: distance = speed $\times$ time.
 2 $d = 150 \times 2 = 300$ miles.

WHY NOT (A) divides the speed by the time instead of multiplying. (B) adds the speed and the time instead of multiplying.

3. **C** 1,700 miles

1 Use $d = r \times t$: distance = speed $\times$ time.
 2 $d = 425 \times 4 = 1,700$ miles.

WHY NOT (A) adds the speed and the time instead of multiplying.

4. **D** 34 miles

1 Use $d = r \times t$: distance = speed $\times$ time.
 2 $d = 17 \times 2 = 34$ miles.

WHY NOT (A) adds the speed and the time instead of multiplying.

5. **C** 80 miles

1 Change the time to hours: $40 \text{ min} = \frac{40}{60} = \frac{2}{3}$ hour.
 2 Distance: $120 \times \frac{2}{3} = 80$ miles.

TIP Or think per minute: 120 miles per hour is $120 \div 60 = 2$ miles per minute, and $2 \times 40 = 80$.

WHY NOT (A) treats 40 minutes as 0.4 hour. (D) uses the minutes as if they were hours.

6. **A** 5 miles

1 Change the time to hours: $50 \text{ min} = \frac{50}{60} = \frac{5}{6}$ hour.
 2 Distance: $6 \times \frac{5}{6} = 5$ miles.

TIP Or think per minute: 6 miles per hour is $6 \div 60 = 0.1$ mile per minute, and $0.1 \times 50 = 5$.

WHY NOT (B) gives the distance for a full hour. (D) uses the minutes as if they were hours.

7. **B** 33

1 Miles per gallon = miles $\div$ gallons.
 2 $363 \div 11 = 33$ miles per gallon.

TIP Check: $33 \times 11 = 363$.

WHY NOT (D) subtracts the gallons from the miles.

8. **C** 9

1 Miles per gallon = miles $\div$ gallons.
 2 $171 \div 19 = 9$ miles per gallon.

TIP Check: $9 \times 19 = 171$.

WHY NOT (D) subtracts the gallons from the miles.

9. **B** 54

1 Miles per gallon = miles $\div$ gallons.
 2 $162 \div 3 = 54$ miles per gallon.

TIP Check: $54 \times 3 = 162$.

WHY NOT (D) subtracts the gallons from the miles.

10. **B** 10 minutes

1 Time = amount $\div$ rate.
 2 $600 \div 60 = 10$ minutes.

TIP Check: $60 \times 10 = 600$ words.

WHY NOT (D) subtracts the rate from the total.

11. **B** 12 minutes

1 Time = amount $\div$ rate.
 2 $216 \div 18 = 12$ minutes.

TIP Check: $18 \times 12 = 216$ pages.

WHY NOT (D) subtracts the rate from the total.

12. **A** 5 hours

1 Time = amount $\div$ rate.
 2 $125 \div 25 = 5$ hours.

TIP Check: $25 \times 5 = 125$ pages.

WHY NOT (D) subtracts the rate from the total.

13. **B** 90 miles

- 1 Change the minutes to hours: $30 \text{ min} = \frac{30}{60} = \frac{1}{2}$ hour, so the time is $2\frac{1}{2}$ hours.
- 2 Distance: $36 \times 2\frac{1}{2} = 36 \times 2 + 36 \times \frac{1}{2} = 72 + 18 = 90$ miles.

WHY NOT (A) treats 2 hours 30 minutes as 2.3 hours. (C) rounds the time up to a whole hour.

14. **B** 54 mph

- 1 Change the time to hours: $20 \text{ min} = \frac{20}{60} = \frac{1}{3}$ hour.
 - 2 Speed: $18 \div \frac{1}{3} = 18 \times 3 = 54$ miles per hour.
- TIP** Or scale up to an hour: 20 minutes goes into 60 minutes 3 times, and $18 \times 3 = 54$.

WHY NOT (A) gives miles per minute, not miles per hour. (C) treats 20 minutes as 0.2 hour. (D) multiplies the distance by the minutes.

15. **A** 25 minutes

- 1 Time in hours: $t = d \div r = 15 \div 36 = \frac{15}{36} = \frac{5}{12}$ hour.
- 2 Change to minutes: $\frac{5}{12} \times 60 = 25$ minutes.

WHY NOT (C) reads 0.41 hour as 41 minutes. (D) divides the speed by the distance before multiplying by 60.

16. **C** 4 hours

- 1 Moving toward each other, the gap closes at the sum of the speeds: $40 + 60 = 100$ miles per hour.
 - 2 Time to close 400 miles: $400 \div 100 = 4$ hours.
- TIP** Check: in 4 hours they cover $160 + 240 = 400$ miles.

WHY NOT (D) subtracts the speeds, but the two are closing the gap together.

17. **A** 20 miles

- 1 In the same direction, the faster one pulls ahead at the difference of the speeds: $60 - 55 = 5$ miles per hour.
- 2 After 4 hours: $5 \times 4 = 20$ miles apart.

WHY NOT (C) gives the faster one's distance, not the distance between them. (D) adds the speeds, but the two travel in the same direction.

18. **A** 2 hours

- 1 Moving toward each other, the gap closes at the sum of the speeds: $45 + 60 = 105$ miles per hour.
 - 2 Time to close 210 miles: $210 \div 105 = 2$ hours.
- TIP** Check: in 2 hours they cover $90 + 120 = 210$ miles.

WHY NOT (B) uses only one speed. (C) divides by the average of the two speeds instead of their sum. (D) subtracts the speeds, but the two are closing the gap together.

19. **A** \$15.00

- 1 Gallons needed: $120 \div 30 = 4$ gallons.
- 2 Cost: $4 \times \$3.75 = \15 .

WHY NOT (C) multiplies the miles per gallon by the price. (D) multiplies the miles by the price per gallon.

20. **B** \$35.10

- 1 Gallons needed: $153 \div 17 = 9$ gallons.
- 2 Cost: $9 \times \$3.90 = \35.10 .

WHY NOT (A) gives the number of gallons, not the cost. (C) multiplies the miles per gallon by the price. (D) multiplies the miles by the price per gallon.

21. **A** 5 mph

- 1 Change the time to hours: $48 \text{ min} = \frac{48}{60} = \frac{4}{5}$ hour.
- 2 Speed: $4 \div \frac{4}{5} = 4 \times \frac{5}{4} = 5$ miles per hour.

TIP Or: the pace is $48 \div 4 = 12$ minutes per mile, and $60 \div 12 = 5$ mph.

WHY NOT (C) gives the pace in minutes per mile, not the speed. (D) multiplies the distance by the time.

22. **D** 40 minutes

- 1 Time in hours: $5 \div 7.5 = \frac{2}{3}$ hour.
- 2 Change to minutes: $\frac{2}{3} \times 60 = 40$ minutes.

TIP Or: at 7.5 mph one mile takes $60 \div 7.5 = 8$ minutes, and $5 \times 8 = 40$.

WHY NOT (A) finds the minutes for one mile only.

23. **C** 4 mph

- 1 Change the time to hours: $45 \text{ min} = \frac{45}{60} = \frac{3}{4}$ hour.
- 2 Speed: $3 \div \frac{3}{4} = 3 \times \frac{4}{3} = 4$ miles per hour.

TIP Or: the pace is $45 \div 3 = 15$ minutes per mile, and $60 \div 15 = 4$ mph.

WHY NOT (D) gives the pace in minutes per mile, not the speed.

24. **B** 2.5 hours

- 1 First part: $105 \div 70 = 1.5$ hours.
- 2 Second part: $40 \div 40 = 1$ hour.
- 3 Total: $1.5 + 1 = 2.5$ hours.

WHY NOT (A) finds the time for the first part only.

25. **C** 3 hours

- 1 First part: $40 \div 40 = 1$ hour.
- 2 Second part: $30 \div 15 = 2$ hours.
- 3 Total: $1 + 2 = 3$ hours.

WHY NOT (A) finds the time for the first part only. (B) uses the first speed for the whole trip.

26. **C** 12:07 p.m.

- 1 Travel time: $84 \div 70 = 1\frac{1}{5}$ hours.
- 2 Change the fraction of an hour to minutes: $\frac{1}{5} \times 60 = 12$ minutes, so the trip takes 1 hour 12 minutes.
- 3 Add to the start time: 10:55 a.m. plus 1 hour 12 minutes is 12:07 p.m.

WHY NOT (A) leaves out the part of an hour. (B) reads 1.2 hours as 1 hour 20 minutes.

27. **B** 9:27 a.m.

- 1 Travel time: $336 \div 80 = 4\frac{1}{5}$ hours.
- 2 Change the fraction of an hour to minutes: $\frac{1}{5} \times 60 = 12$ minutes, so the trip takes 4 hours 12 minutes.
- 3 Add to the start time: 5:15 a.m. plus 4 hours 12 minutes is 9:27 a.m.

WHY NOT (A) leaves out the part of an hour. (C) reads 4.2 hours as 4 hours 20 minutes.

28. **D** 900

- 1 Change the time to hours: 45 min = $\frac{45}{60} = \frac{3}{4}$ hour.
- 2 Amount: $1,200 \times \frac{3}{4} = 900$ gallons.

TIP Or find the rate per minute: $1,200 \div 60 = 20$, and $20 \times 45 = 900$.

WHY NOT (A) treats 45 minutes as 0.45 hour.

29. **D** 30 mph

- 1 Time going: $120 \div 60 = 2$ hours. Time returning: $120 \div 20 = 6$ hours.
- 2 Total distance: $2 \times 120 = 240$ miles. Total time: $2 + 6 = 8$ hours.
- 3 Average speed: $240 \div 8 = 30$ miles per hour.

TIP Shortcut for equal distances: $2ab \div (a + b) = 2 \times 60 \times 20 \div 80 = 30$.

WHY NOT (A) uses the one-way distance instead of the round-trip distance.

30. **D** 9.6 mph

- 1 Time going: $48 \div 8 = 6$ hours. Time returning: $48 \div 12 = 4$ hours.
- 2 Total distance: $2 \times 48 = 96$ miles. Total time: $6 + 4 = 10$ hours.
- 3 Average speed: $96 \div 10 = 9.6$ miles per hour.

TIP Shortcut for equal distances: $2ab \div (a + b) = 2 \times 8 \times 12 \div 20 = 9.6$.

WHY NOT (A) uses the one-way distance instead of the round-trip distance.

31. **C** 48 mph

- 1 Time going: $120 \div 60 = 2$ hours. Time returning: $120 \div 40 = 3$ hours.
- 2 Total distance: $2 \times 120 = 240$ miles. Total time: $2 + 3 = 5$ hours.
- 3 Average speed: $240 \div 5 = 48$ miles per hour.

TIP Shortcut for equal distances: $2ab \div (a + b) = 2 \times 60 \times 40 \div 100 = 48$.

WHY NOT (A) uses the one-way distance instead of the round-trip distance. (D) averages the two speeds, but more time is spent at the slower speed.

32. **B** 0.5 hours

- 1 Head start: in 0.5 hours the first one goes $8 \times 0.5 = 4$ miles.
- 2 The gap closes at the difference of the speeds: $16 - 8 = 8$ miles per hour.
- 3 Time to close it: $4 \div 8 = 0.5$ hours.

TIP Check: in 0.5 hours the chaser goes $16 \times 0.5 = 8$ miles, and the first one has gone $8 \times 1 = 8$ miles.

WHY NOT (A) divides the head start by the faster speed instead of the difference. (C) counts the time from when the first one left.

33. **B** 3.5 hours

- 1 Head start: in 2 hours the first one goes $49 \times 2 = 98$ miles.
- 2 The gap closes at the difference of the speeds: $77 - 49 = 28$ miles per hour.
- 3 Time to close it: $98 \div 28 = 3.5$ hours.

TIP Check: in 3.5 hours the chaser goes $77 \times 3.5 = 269.5$ miles, and the first one has gone $49 \times 5.5 = 269.5$ miles.

WHY NOT (D) counts the time from when the first one left.

34. **A** 144 miles

- 1 Head start: $48 \times 1 = 48$ miles.
- 2 The gap closes at $72 - 48 = 24$ miles per hour, so it takes $48 \div 24 = 2$ hours.
- 3 Distance from the start: $72 \times 2 = 144$ miles.

TIP Check: the slower one has gone $48 \times 3 = 144$ miles too.

WHY NOT (D) counts the car's time from when the first one left.

35. **B** 132

1 Change miles to feet: $90 \times 5,280 = 475,200$ feet per hour.

2 Change hours to seconds: $475,200 \div 3,600 = 132$ feet per second.

TIP Simplify first: $5,280 \div 3,600 = \frac{22}{15}$, so $90 \times \frac{22}{15} = 132$.

WHY NOT (C) gives feet per minute, not feet per second. (D) gives feet per hour, not feet per second.

36. **C** 88

1 Change miles to feet: $60 \times 5,280 = 316,800$ feet per hour.

2 Change hours to seconds: $316,800 \div 3,600 = 88$ feet per second.

TIP Simplify first: $5,280 \div 3,600 = \frac{22}{15}$, so $60 \times \frac{22}{15} = 88$.

WHY NOT (D) gives feet per minute, not feet per second.

37. **A** 12 mph

1 First part: $21 \div 14 = 1.5$ hours.

2 Second part: $3 \div 6 = 0.5$ hours.

3 Total distance: $21 + 3 = 24$ miles. Total time: $1.5 + 0.5 = 2$ hours.

4 Average speed: $24 \div 2 = 12$ miles per hour.

WHY NOT (B) divides the total distance by the first part's time only. (C) adds the two speeds. (D) divides the total distance by the second part's time only.

38. **C** 36 mph

1 First part: $150 \div 50 = 3$ hours.

2 Second part: $30 \div 15 = 2$ hours.

3 Total distance: $150 + 30 = 180$ miles. Total time: $3 + 2 = 5$ hours.

4 Average speed: $180 \div 5 = 36$ miles per hour.

WHY NOT (B) averages the two speeds, but more time is spent at one of them. (D) adds the two speeds.

39. **A** \$31.00

1 Gas for the truck: $450 \div 18 = 25$ gallons. Gas for the car: $450 \div 30 = 15$ gallons.

2 Gallons saved: $25 - 15 = 10$.

3 Money saved: $10 \times \$3.10 = \31 .

WHY NOT (B) subtracts the miles per gallon and multiplies by the price. (C) is the cost for the car, not the savings. (D) is the cost for the truck, not the savings.

40. **B** 11:35 a.m.

1 First part: $75 \div 50 = 1.5$ hours (1 hour 30 minutes). Second part: $48 \div 48 = 1$ hour.

2 Total time: 1 hour 30 minutes + 1 hour + 30 minutes = 3 hours.

3 Add to the start time: 8:35 a.m. plus 3 hours is 11:35 a.m.

WHY NOT (C) forgets the 30-minute stop.

Work Rates & Mixtures

Tested on: Arithmetic Reasoning

Key Concepts

Work and mixture problems look different, but both are about *rates and totals*: how much of a job gets done each hour, or how much of an ingredient is in each gallon. Set up the totals carefully and the arithmetic stays short.

Work rates

If a job takes t hours, the worker does $\frac{1}{t}$ of the job per hour.

- Working together, **rates add**: $\frac{1}{a} + \frac{1}{b} = \frac{1}{T}$, where T is the time together. Then flip: time = $1 \div$ rate.
- Two-worker shortcut (product over sum): $T = \frac{a \times b}{a + b}$. For 3 h and 6 h: $\frac{18}{9} = 2$ h.
- A drain or a leak works against you: **subtract** its rate.
- To find one worker's time from the time together, subtract rates: $\frac{1}{b} = \frac{1}{T} - \frac{1}{a}$.

Equal workers: worker-hours

Total work = number of workers $\times$ time. If 6 workers need 8 days, the job is $6 \times 8 = 48$ worker-days, so 4 workers need $48 \div 4 = 12$ days. *More* workers means *less* time (an inverse proportion).

Mixtures

amount of pure ingredient = percent $\times$ total amount

- When two mixtures are combined, the pure amounts add and the totals add.
- Adding water raises the total but does *not* change the pure amount.
- Price mixtures work the same way: total cost $\div$ total pounds = price per pound.

WORKED EXAMPLE

How many liters of a 50% antifreeze solution must be added to 10 liters of a 20% solution to make a 30% solution?

Solution. Let x be the liters of 50% solution. The pure antifreeze going in equals the pure antifreeze in the mixture: $0.20(10) + 0.50x = 0.30(10 + x)$, so $2 + 0.5x = 3 + 0.3x$. Then $0.2x = 1$ and $x = 5$ liters. *Check:* $2 + 2.5 = 4.5$ liters of antifreeze in 15 liters, and $4.5 \div 15 = 0.30 = 30\%$.

No-Calculator Tip Sanity checks save points. Two workers together must be *faster* than the faster one alone (3 h and 6 h together: less than 3 h). A mixture's percent or price must land *between* the two ingredients, closer to the one you use more of.

Common Traps

- Averaging or adding the times: 3 h and 6 h together is 2 h, not 4.5 h or 9 h.
- Treating workers as a direct proportion: fewer workers need *more* time, not less.
- Taking the simple average of two percents or prices when the amounts are different.
- Answering with the final amount when the question asks how much water (or how much of one ingredient) to *add*.

Practice Set 25

Score: ____ / 35

Target time: 42 minutes

Problems 1–10 are warm-ups, 11–24 are test level, and 25–35 are challenge problems.

- An older printer can print the programs for a graduation ceremony in 24 minutes. A newer printer can print them in 12 minutes. If the two printers share the job, how long will it take?
A 6 minutes B 7 minutes
C 8 minutes D 18 minutes
- For a barracks cleanup, a soldier mixes 8 gallons of floor-cleaning solution that is 40% concentrate and the rest water. How many gallons of concentrate are in the solution?
A 3.2 gallons B 4.8 gallons
C 32 gallons D 40 gallons
- At a recruiting station, one clerk can enter a stack of enlistment forms into the computer in 6 hours. A second clerk can enter the same stack in 12 hours. How long will it take the two clerks working together?
A 3 hours B 4 hours
C 9 hours D 18 hours
- A medic's 600-milliliter cleaning solution contains 30 milliliters of bleach. What percent of the solution is bleach?
A 5% B 30% C 95% D 2,000%
- A crew of 6 landscapers can plant the trees for a new park in 9 days. At the same rate, how many days would it take 3 landscapers to do the job?
A 9 days B 12 days C 16 days D 18 days
- Combat engineers mix concrete for a bunker floor using cement, sand, and gravel in the ratio 1 : 3 : 4 by volume. How many cubic feet of sand are in 40 cubic feet of concrete mix?
A 5 cubic feet B 15 cubic feet
C 24 cubic feet D 25 cubic feet
- A team of 10 mechanics can service all of a battalion's trucks in 20 days. At the same rate, how many days would it take 5 mechanics to do the job?
A 10 days B 20 days C 25 days D 40 days
- To touch up vehicles, a motor pool mixes tan paint and green paint in the ratio 4 : 1. How many gallons of tan paint are in 35 gallons of mixed paint?
A 7 gallons B 28 gallons
C 32 gallons D 140 gallons
- A 128-ounce bottle of fruit punch contains 12.8 ounces of real fruit juice. What percent of the punch is real juice?
A 10% B 12.8% C 90% D 1,000%
- Luis can paint the outside of a house in 18 hours. Keisha can paint the same house in 9 hours. If they work together, how long will it take them to paint the house?
A 6 hours
B 6 hours 45 minutes
C 13 hours 30 minutes
D 27 hours

11. Aisha can mow a large lawn in 70 minutes, and Emily can mow it in 28 minutes. If they use two mowers and work together, how long will it take them to mow the lawn?
A 14 minutes B 16 minutes
C 18 minutes D 20 minutes
12. Two fuel pumps working together can fill the storage tanks at a forward operating base in 6 hours. One of the pumps alone would take 15 hours. How long would the other pump take alone?
A 10 hours B 12 hours
C 13 hours D 21 hours
13. A kitchen faucet fills a large sink in 6 minutes. The drain, if left open, empties a full sink in 7 minutes. If the drain is left open, how long will it take the faucet to fill the empty sink?
A 1 minute B 6 minutes
C 42 minutes D 57 minutes
14. A detail of 8 soldiers can fill all the sandbags for a checkpoint in 9 hours. How many soldiers working at the same rate are needed to finish the job in 6 hours?
A 9 B 10 C 11 D 12
15. A lab technician mixes 6 liters of a 25% acid solution with 9 liters of a 90% acid solution. What percent of the mixture is acid?
A 57.5% B 64% C 70% D 115%
16. A base dining facility mixes 36 pounds of 80%-lean ground beef costing \$4 per pound with 54 pounds of 93%-lean ground beef costing \$8 per pound. What is the cost per pound of the mixture?
A \$6.00 B \$6.40 C \$288.00 D \$576.00
17. A cafeteria worker mixes 4 gallons of a punch that is 10% fruit juice with 2 gallons of a punch that is 40% fruit juice. What percent of the mixture is fruit juice?
A 1.2% B 20% C 25% D 30%
18. A coffee shop mixes 4 pounds of Colombian coffee costing \$10 per pound with 8 pounds of Kona blend coffee costing \$19 per pound. What is the cost per pound of the mixture?
A \$16.00 B \$16.50 C \$96.00 D \$192.00
19. A crew of 2 painters can paint an apartment building in 25 days. How many painters working at the same rate are needed to finish the job in 10 days?
A 2 B 4 C 5 D 50
20. A pump can fill an empty water tank at a base camp in 3 hours. The tank has a leak that would empty a full tank in 4 hours. If the pump fills the empty tank while it leaks, how long will it take to fill?
A 1 hour B 3 hours
C 7 hours D 12 hours
21. Ana and Sofia are stuffing envelopes for a charity mailing. Together they can finish in 4 hours. Alone, Ana would need 6 hours. How many hours would Sofia need alone?
A 2 hours B 8 hours
C 10 hours D 12 hours
22. One pump can empty a flooded basement in 15 hours. A second pump can empty it in 3 hours. How long will it take to empty the basement if both pumps run at the same time?
A 24 minutes B 1 hour 30 minutes
C 2 hours 30 minutes D 9 hours
23. A chemistry teacher has 12 liters of a 20% acid solution. How many liters of water must be added to dilute it to a 10% solution?
A 6 liters B 12 liters C 15 liters D 24 liters
24. A foundry makes bronze from copper and tin in the ratio 4 : 1 by weight. How many pounds of bronze can be made if 9 pounds of tin are used?
A 13 pounds B 45 pounds
C 50 pounds D 60 pounds

25. A road crew of 28 workers can repave a road in 30 days. After 12 days, 14 more workers join the crew. How many days in all will the job take?
A 24 days B 30 days C 32 days D 39 days
26. A medic at an aid station has 400 milliliters of a 50% alcohol solution. How many milliliters of a 70% alcohol solution must be added to make a mixture that is 65% alcohol?
A 400 milliliters B 1,200 milliliters
C 1,500 milliliters D 1,600 milliliters
27. A lab has 16 liters of a 25% acid solution. How many liters of pure acid must be added to make a 40% acid solution?
A 1 liter B 2 liters C 4 liters D 6 liters
28. The base exchange has 34 pounds of raisins priced at \$4 per pound. How many pounds of almonds priced at \$9 per pound must be mixed in to make a blend worth \$7 per pound?
A 51 pounds B 56 pounds
C 85 pounds D 102 pounds
29. A cook has 150 gallons of sugar syrup that is 10% sugar. How many gallons of water must be boiled off to make the syrup 15% sugar?
A 35 gallons B 40 gallons
C 50 gallons D 100 gallons
30. A candy store has 20 pounds of gummy bears priced at \$4 per pound. How many pounds of chocolate drops priced at \$9 per pound must be mixed in to make a blend worth \$7 per pound?
A 20 pounds B 24 pounds
C 30 pounds D 50 pounds
31. A mechanic has 7 quarts of a 70% antifreeze solution. How many quarts of a 30% antifreeze solution must be added to make a mixture that is 65% antifreeze?
A 1 quart B 7 quarts
C 8 quarts D 49 quarts
32. A crew of 8 workers can frame a house in 24 days. After 12 days, 2 workers leave for another job. How many more days will it take to finish?
A 9 days B 12 days C 16 days D 32 days
33. Three pumps are set up to empty a flooded basement. Working alone, they would take 2 hours, 12 hours, and 4 hours. How long will it take to empty the basement with all three pumps running?
A 36 minutes B 40 minutes
C 50 minutes D 1 hour 12 minutes
34. A large landscaping crew works three times as fast as a small crew. Together the two crews can plant all the trees in a new park in 7 hours. How long would the large crew take working alone?
A 7 hours B 8 hours 20 minutes
C 8 hours 40 minutes D 9 hours 20 minutes
35. A ship's ballast tank can be filled by either of two pumps: one alone takes 3 hours and the other 4 hours. A stuck valve lets water out and would empty a full tank in 2 hours. With both pumps on, how long will it take to fill the empty tank?
A 5 minutes B 3 hours 30 minutes
C 9 hours D 12 hours

Answers & Explanations 25

1 C	8 B	15 B	22 C	29 C
2 A	9 A	16 B	23 B	30 C
3 B	10 A	17 B	24 B	31 A
4 A	11 D	18 A	25 A	32 C
5 D	12 A	19 C	26 B	33 D
6 B	13 C	20 D	27 C	34 D
7 D	14 D	21 D	28 A	35 D

1. **C** 8 minutes

1 Turn each time into a rate (the part of the job done in one minute): the newer printer does $\frac{1}{12}$ of the job per minute, and the older printer does $\frac{1}{24}$.

2 Add the rates (common denominator 24): $\frac{1}{12} + \frac{1}{24} = \frac{2}{24} + \frac{1}{24} = \frac{3}{24} = \frac{1}{8}$ of the job per minute.

3 Time = 1 job $\div$ rate: $1 \div \frac{1}{8} = 8$ minutes.

TIP Product over sum: $\frac{12 \times 24}{12 + 24} = \frac{288}{36} = 8$. The answer must be less than the faster time, 12 minutes.

WHY NOT (A) halves the faster time, as if both worked as fast as the faster one. (D) averages the two times.

2. **A** 3.2 gallons

1 Amount of concentrate = percent $\times$ total. Change the percent to a decimal: $40\% = 0.4$.

2 Multiply: $0.4 \times 8 = 3.2$ gallons.

WHY NOT (B) is the amount of water, not concentrate. (C) moves the decimal point only one place when changing the percent. (D) uses the percent as if it were the amount.

3. **B** 4 hours

1 Turn each time into a rate (the part of the job done in one hour): the first clerk does $\frac{1}{6}$ of the job per hour, and the second clerk does $\frac{1}{12}$.

2 Add the rates (common denominator 12): $\frac{1}{6} + \frac{1}{12} = \frac{2}{12} + \frac{1}{12} = \frac{3}{12} = \frac{1}{4}$ of the job per hour.

3 Time = 1 job $\div$ rate: $1 \div \frac{1}{4} = 4$ hours.

TIP Product over sum: $\frac{6 \times 12}{6 + 12} = \frac{72}{18} = 4$. The answer must be less than the faster time, 6 hours.

WHY NOT (A) halves the faster time, as if both worked as fast as the faster one. (C) averages the two times. (D) adds the two times instead of adding the two rates.

4. **A** 5%

1 Percent = part $\div$ whole: $\frac{30}{600}$.

2 Divide and change to a percent: $\frac{30}{600} = 0.05 = 5\%$.

WHY NOT (B) gives the amount, not the percent. (C) is the percent that is water. (D) divides the whole by the part.

5. **D** 18 days

1 Find the total amount of work in worker-days: $6 \times 9 = 54$ worker-days.

2 Share that work among 3 landscapers: $54 \div 3 = 18$ days.

TIP Fewer workers means more time, so the answer must be more than 9 days.

WHY NOT (A) assumes the time does not change. (B) changes the time by one day for each worker added or removed.

6. **B** 15 cubic feet

1 Add the parts: $1 + 3 + 4 = 8$ parts in all.

2 Find the size of one part: $40 \div 8 = 5$ cubic feet.

3 Sand is 3 parts: $3 \times 5 = 15$ cubic feet.

WHY NOT (A) is the size of just one part. (C) divides by the other parts instead of by the total parts. (D) is the amount of everything except the sand.

7. **D** 40 days

1 Find the total amount of work in worker-days: $10 \times 20 = 200$ worker-days.

2 Share that work among 5 mechanics: $200 \div 5 = 40$ days.

TIP Fewer workers means more time, so the answer must be more than 20 days.

WHY NOT (A) sets up a direct proportion; fewer workers should take more time, not less. (B) assumes the time does not change. (C) changes the time by one day for each worker added or removed.

8. **B** 28 gallons

1 Add the parts: $4 + 1 = 5$ parts in all.

2 Find the size of one part: $35 \div 5 = 7$ gallons.

3 Tan paint is 4 parts: $4 \times 7 = 28$ gallons.

WHY NOT (A) is the amount of everything except the tan paint. (D) divides by the other parts instead of by the total parts.

9. **A** 10%

- Percent = part $\div$ whole: $\frac{12.8}{128}$.
- Divide and change to a percent: $\frac{12.8}{128} = 0.1 = 10\%$.

WHY NOT (B) gives the amount, not the percent. (C) is the percent that is water and sweetener. (D) divides the whole by the part.

 10. **A** 6 hours

- Turn each time into a rate (the part of the job done in one hour): Luis does $\frac{1}{18}$ of the job per hour, and Keisha does $\frac{1}{9}$.
- Add the rates (common denominator 18): $\frac{1}{18} + \frac{1}{9} = \frac{1}{18} + \frac{2}{18} = \frac{3}{18} = \frac{1}{6}$ of the job per hour.
- Time = 1 job $\div$ rate: $1 \div \frac{1}{6} = 6$ hours.

TIP Product over sum: $\frac{9 \times 18}{9 + 18} = \frac{162}{27} = 6$. The answer must be less than the faster time, 9 hours.

WHY NOT (B) halves the average of the two times. (C) averages the two times. (D) adds the two times instead of adding the two rates.

 11. **D** 20 minutes

- Turn each time into a rate (the part of the job done in one minute): Aisha does $\frac{1}{70}$ of the job per minute, and Emily does $\frac{1}{28}$.
- Add the rates (common denominator 140): $\frac{1}{70} + \frac{1}{28} = \frac{2}{140} + \frac{5}{140} = \frac{7}{140} = \frac{1}{20}$ of the job per minute.
- Time = 1 job $\div$ rate: $1 \div \frac{1}{20} = 20$ minutes.

TIP The answer must be less than the faster time, 28 minutes.

WHY NOT (A) halves the faster time, as if both worked as fast as the faster one.

 12. **A** 10 hours

- Rates: together they do $\frac{1}{6}$ of the job per hour, and the first pump alone does $\frac{1}{15}$ of the job per hour.
- The other worker's rate is the difference: $\frac{1}{6} - \frac{1}{15} = \frac{5}{30} - \frac{2}{30} = \frac{3}{30} = \frac{1}{10}$ of the job per hour.
- Doing $\frac{1}{10}$ of the job each hour, the other pump needs 10 hours alone.

TIP Check: $\frac{1}{15} + \frac{1}{10} = \frac{1}{6}$, the rate together. $\checkmark$

WHY NOT (B) assumes the two work at the same speed. (D) adds the two times.

 13. **C** 42 minutes

- Rates: the faucet fills $\frac{1}{6}$ of the sink per minute, and the drain empties $\frac{1}{7}$ of it per minute.
- The drain works against the fill, so *subtract*: $\frac{1}{6} - \frac{1}{7} = \frac{7}{42} - \frac{6}{42} = \frac{1}{42}$ of the sink per minute.
- Time = 1 sink $\div$ rate = $1 \div \frac{1}{42} = 42$ minutes.

TIP Shortcut for fill and drain: $\frac{6 \times 7}{7 - 6} = \frac{42}{1} = 42$.

WHY NOT (A) subtracts the two times. (B) ignores the drain.

 14. **D** 12

- Total work: $8 \times 9 = 72$ worker-hours.
- To finish in 6 hours: $72 \div 6 = 12$ soldiers are needed.

WHY NOT (C) adds one worker for each hour saved.

 15. **B** 64%

- Pure acid in each: $0.25 \times 6 = 1.5$ and $0.9 \times 9 = 8.1$ liters.
- Totals: $1.5 + 8.1 = 9.6$ liters of acid in $6 + 9 = 15$ liters of mixture.
- Percent: $\frac{9.6}{15} = 0.64 = 64\%$.

TIP Check: 64% is between 25% and 90%, closer to the percent of the larger amount.

WHY NOT (A) averages the two percents without weighting by the amounts. (D) adds the two percents.

 16. **B** \$6.40

- Cost of each part: $36 \times 4 = 144$ and $54 \times 8 = 432$ dollars.
- Totals: \$576 for $36 + 54 = 90$ pounds.
- Price per pound: $576 \div 90 = 6.40$, or \$6.40 per pound.

TIP Check: \$6.40 is between \$4 and \$8, closer to the price of the item with more pounds.

WHY NOT (A) averages the two prices without weighting by the pounds. (C) divides the total cost by 2 instead of by the total pounds. (D) is the total cost of the mixture, not the price per pound.

 17. **B** 20%

- Pure fruit juice in each: $0.1 \times 4 = 0.4$ and $0.4 \times 2 = 0.8$ gallons.
- Totals: $0.4 + 0.8 = 1.2$ gallons of fruit juice in $4 + 2 = 6$ gallons of mixture.
- Percent: $\frac{1.2}{6} = 0.2 = 20\%$.

TIP Check: 20% is between 10% and 40%, closer to the percent of the larger amount.

WHY NOT (A) gives the amount of pure fruit juice, not the percent. (C) averages the two percents without weighting by the amounts. (D) weights each percent by the wrong amount.

18. **A** \$16.00

- Cost of each part: $4 \times 10 = 40$ and $8 \times 19 = 152$ dollars.
- Totals: \$192 for $4 + 8 = 12$ pounds.
- Price per pound: $192 \div 12 = 16.00$, or \$16 per pound.

TIP Check: \$16 is between \$10 and \$19, closer to the price of the item with more pounds.

WHY NOT (C) divides the total cost by 2 instead of by the total pounds. (D) is the total cost of the mixture, not the price per pound.

19. **C** 5

- Total work: $2 \times 25 = 50$ worker-days.
- To finish in 10 days: $50 \div 10 = 5$ painters are needed.

WHY NOT (D) gives the total number of worker-days, not the number of workers.

20. **D** 12 hours

- Rates: the pump fills $\frac{1}{3}$ of the tank per hour, and the leak empties $\frac{1}{4}$ of it per hour.
- The drain works against the fill, so *subtract*: $\frac{1}{3} - \frac{1}{4} = \frac{4}{12} - \frac{3}{12} = \frac{1}{12}$ of the tank per hour.
- Time = 1 tank $\div$ rate = $1 \div \frac{1}{12} = 12$ hours.

TIP Shortcut for fill and drain: $\frac{3 \times 4}{4 - 3} = \frac{12}{1} = 12$.

WHY NOT (A) subtracts the two times. (B) ignores the drain. (C) adds the two times.

21. **D** 12 hours

- Rates: together they do $\frac{1}{4}$ of the job per hour, and Ana alone does $\frac{1}{6}$ of the job per hour.
- The other worker's rate is the difference: $\frac{1}{4} - \frac{1}{6} = \frac{3}{12} - \frac{2}{12} = \frac{1}{12}$ of the job per hour.
- Doing $\frac{1}{12}$ of the job each hour, Sofia needs 12 hours alone.

TIP Check: $\frac{1}{6} + \frac{1}{12} = \frac{1}{4}$, the rate together. ✓

WHY NOT (A) subtracts the times instead of the rates. (B) assumes the two work at the same speed. (C) adds the two times.

22. **C** 2 hours 30 minutes

- Turn each time into a rate (the part of the job done in one hour): the first pump does $\frac{1}{15}$ of the job per hour, and the second pump does $\frac{1}{3}$.
- Add the rates (common denominator 15): $\frac{1}{15} + \frac{1}{3} = \frac{1}{15} + \frac{5}{15} = \frac{6}{15} = \frac{2}{5}$ of the job per hour.
- Time = 1 job $\div$ rate: $1 \div \frac{2}{5} = \frac{5}{2} = 2\frac{1}{2}$ hours.
- Change the fraction of an hour to minutes: $\frac{1}{2} \times 60 = 30$ minutes. So they need 2 hours 30 minutes.

TIP Product over sum: $\frac{3 \times 15}{3 + 15} = \frac{45}{18} = 2\frac{1}{2}$. The answer must be less than the faster time, 3 hours.

WHY NOT (A) stops at the combined rate and forgets to flip it into a time. (B) halves the faster time, as if both worked as fast as the faster one. (D) averages the two times.

23. **B** 12 liters

- Pure acid now: $0.2 \times 12 = 2.4$ liters. Adding water does not change this amount.
- In the new mixture these 2.4 liters must be 10% of the total: $2.4 \div 0.1 = 24$ liters in all.
- Water to add: $24 - 12 = 12$ liters.

WHY NOT (A) sets up the ratio upside down. (D) is the final amount of the mixture, not the water added.

24. **B** 45 pounds

- Tin is 1 part, so one part is 9 pounds.
- The whole batch is 5 parts: $5 \times 9 = 45$ pounds.

WHY NOT (A) adds the difference in parts instead of using a multiple.

25. **A** 24 days

- After 12 days, the original 28 workers still had 18 days' worth of work left: $28 \times 18 = 504$ worker-days.
- Now there are $28 + 14 = 42$ workers, so the rest takes $504 \div 42 = 12$ days.
- Add the days already passed: $12 + 12 = 24$ days in all.

WHY NOT (B) ignores the new arrivals. (C) forgets the 12 days already used. (D) sets up a direct proportion instead of an inverse one.

26. **B** 1,200 milliliters

1 Let y be the milliliters added. The pure alcohol going in must equal the pure alcohol in the mixture: $0.5(400) + 0.7y = 0.65(400 + y)$.

2 Multiply out: $200 + 0.7y = 260 + 0.65y$.

3 Collect the y terms: $0.05y = 60$, so $y = 1,200$ milliliters.

TIP Balance shortcut: a 50% alcohol solution is 15 points from 65% and a 70% alcohol solution is 5 points away, so $400 \times 15 = y \times 5$, which gives $y = 1,200$.

WHY NOT (A) assumes equal amounts of the two solutions. (D) gives the total amount of the mixture, not the amount added.

 27. **C** 4 liters

1 Let y be the liters of pure acid added (pure means 100%). Pure acid before and after: $0.25(16) + y = 0.4(16 + y)$.

2 Multiply out: $4 + y = 6.4 + 0.4y$.

3 Collect terms: $0.6y = 2.4$, so $y = 4$ liters.

WHY NOT (D) forgets that the added liquid also raises the total.

 28. **A** 51 pounds

1 Let y be the pounds of almonds. The money must balance: $4(34) + 9y = 7(34 + y)$.

2 Multiply out: $136 + 9y = 238 + 7y$.

3 Collect terms: $9y - 7y = 238 - 136$, so $2y = 102$ and $y = 51$ pounds.

TIP Balance shortcut: each pound of raisins is \$3 away from \$7, each pound of almonds is \$2 away; the totals must match: $34 \times 3 = 51 \times 2$.

WHY NOT (C) gives the total weight of the blend.

 29. **C** 50 gallons

1 Pure sugar: $0.1 \times 150 = 15$ gallons. Removing water does not change this amount.

2 After evaporating, 15 gallons must be 15% of the total: $15 \div 0.15 = 100$ gallons left.

3 Water removed: $150 - 100 = 50$ gallons.

WHY NOT (D) is the amount left after evaporating, not the amount removed.

 30. **C** 30 pounds

1 Let y be the pounds of chocolate drops. The money must balance: $4(20) + 9y = 7(20 + y)$.

2 Multiply out: $80 + 9y = 140 + 7y$.

3 Collect terms: $9y - 7y = 140 - 80$, so $2y = 60$ and $y = 30$ pounds.

TIP Balance shortcut: each pound of gummy bears is \$3 away from \$7, each pound of chocolate drops is \$2 away; the totals must match: $20 \times 3 = 30 \times 2$.

WHY NOT (A) assumes equal amounts, which only works if the target is halfway between the prices. (D) gives the total weight of the blend.

 31. **A** 1 quart

1 Let y be the quarts added. The pure antifreeze going in must equal the pure antifreeze in the mixture: $0.7(7) + 0.3y = 0.65(7 + y)$.

2 Multiply out: $4.9 + 0.3y = 4.55 + 0.65y$.

3 Collect the y terms: $0.35 = 0.35y$, so $y = 1$ quart.

TIP Balance shortcut: a 70% antifreeze solution is 5 points from 65% and a 30% antifreeze solution is 35 points away, so $7 \times 5 = y \times 35$, which gives $y = 1$.

WHY NOT (B) assumes equal amounts of the two solutions. (C) gives the total amount of the mixture, not the amount added. (D) swaps the two differences.

 32. **C** 16 days

1 After 12 days, the original 8 workers still had 12 days' worth of work left: $8 \times 12 = 96$ worker-days.

2 Now there are $8 - 2 = 6$ workers, so the rest takes $96 \div 6 = 16$ days.

WHY NOT (A) sets up a direct proportion instead of an inverse one. (B) ignores the smaller group. (D) forgets the 12 days already used.

 33. **D** 1 hour 12 minutes

1 Each rate is 1 job divided by the time alone: $\frac{1}{2}$, $\frac{1}{12}$, and $\frac{1}{4}$ of the job per hour.

2 Add the three rates (common denominator 12): $\frac{1}{2} + \frac{1}{12} + \frac{1}{4} = \frac{6}{12} + \frac{1}{12} + \frac{3}{12} = \frac{10}{12} = \frac{5}{6}$ of the job per hour.

3 Time = 1 job $\div$ rate: $1 \div \frac{5}{6} = \frac{6}{5} = 1\frac{1}{5}$ hours.

4 Change the fraction of an hour to minutes: $\frac{1}{5} \times 60 = 12$ minutes. So they need 1 hour 12 minutes.

TIP Check: the answer must be less than the fastest time alone, 2 hours.

WHY NOT (B) divides the fastest time by 3, as if all three were that fast. (C) stops at the combined rate and forgets to flip it into a time.

34. **D** 9 hours 20 minutes

1 Let the small crew do r of the job per hour. Then the large crew does $3r$, and together they do $r + 3r = 4r$ per hour.

2 Together they finish in 7 hours, so $4r = \frac{1}{7}$ and $r = \frac{1}{28}$. So the small crew alone needs 28 hours.

3 Working three times as fast, the large crew needs one third of that time: $28 \div 3 = 9\frac{1}{3}$ hours, which is 9 hours 20 minutes.

TIP Check: rates $\frac{3}{28} + \frac{1}{28} = \frac{1}{7}$ of the job per hour. ✓

WHY NOT (A) gives the time for both working together.

35. **D** 12 hours

1 Rates per hour: the pumps add $\frac{1}{3}$ and $\frac{1}{4}$ of the tank; the stuck valve takes away $\frac{1}{2}$.

2 Net rate (common denominator 12): $\frac{1}{3} + \frac{1}{4} - \frac{1}{2} = \frac{4}{12} + \frac{3}{12} - \frac{6}{12} = \frac{1}{12}$ of the tank per hour.

3 Time = 1 tank $\div$ rate = $1 \div \frac{1}{12} = 12$ hours.

WHY NOT (A) stops at the net rate and forgets to flip it into a time. (B) averages the two fill times and ignores the outflow. (C) adds the three times.

Averages, Statistics & Probability

Tested on: Arithmetic Reasoning

Key Concepts

Averages and probability questions are short, but they hide traps: the wrong count in a division, a list that was never put in order, or two chances that were added when they should have been multiplied. Work with *totals* and *counts*, and these problems become simple arithmetic.

Mean, median, mode, range

- **Mean** (average) = sum of the values $\div$ number of values.
- **Median** = the middle value *after* putting the list in order. With an even number of values, average the two middle ones.
- **Mode** = the value that appears most often. **Range** = largest value – smallest value.

Think in totals

$$\text{sum} = \text{mean} \times \text{number of values}$$

To find a missing score, a new average, or the average of two groups, turn every average into a total first, do the adding or subtracting with totals, then divide by the new count. Two groups of different sizes: add the two totals and divide by the total number of people.

Probability and counting

- $P(\text{event}) = \frac{\text{number of favorable outcomes}}{\text{total number of outcomes}}$, a number from 0 to 1. $P(\text{not } A) = 1 - P(A)$.
- Independent events (“and”): **multiply**. A coin and a die: $P(\text{heads and } 6) = \frac{1}{2} \times \frac{1}{6} = \frac{1}{12}$.
- Without replacement, the second draw has one fewer item in the bag.
- Counting principle: multiply the number of choices at each step. Ordering n things: $n \times (n-1) \times \cdots \times 1$. Choosing a group of 2 when order doesn’t matter: $\frac{n(n-1)}{2}$.
- Expected count = probability $\times$ number of tries.

WORKED EXAMPLE

Maria scored 82, 88, and 91 on three tests. What must she score on the fourth test to average 88 on all four?

Solution. For an average of 88 on 4 tests she needs a total of $4 \times 88 = 352$ points. She has $82 + 88 + 91 = 261$. She needs $352 - 261 = 91$ on the fourth test.

No-Calculator Tip Check probability answers for size: a probability can never be more than 1, and “not” events are usually the larger ones. For averages, the answer must lie between the smallest and largest values.

Common Traps

- Finding the median without first putting the numbers in order.
- Averaging two averages when the groups have different sizes.
- Answering with the target average when the question asks what score is *needed*.
- Adding probabilities for “and” events, or forgetting that a drawn item is not put back.

Practice Set 26

Score: ____/40

Target time: 48 minutes

Problems 1–12 are warm-ups, 13–28 are test level, and 29–40 are challenge problems.

- The daily high temperatures at a training base for six days were 67, 64, 78, 61, 87, and 75 degrees Fahrenheit. What is the mean (average) of these temperatures?
A 72 B 74 C 75 D 432
- The scores of six students on a quiz were 72, 87, 77, 81, 81, 69. What is the mode of the scores?
A 2 B 81 C 87 D 91
- A drawer holds 8 gray, 6 blue, and 10 white socks. If one sock is picked at random, what is the probability that it is gray?
A $\frac{1}{24}$ B $\frac{1}{6}$ C $\frac{1}{3}$ D $\frac{1}{2}$
- The ages of the seven recruits in a training group are 24, 23, 19, 18, 21, 25, 26. What is the median of their ages?
A 18 B 23 C 24 D 29
- For a raffle at a school fundraiser, 250 tickets are sold and one winning ticket is drawn. Luis bought 4 tickets. What is the probability that he wins?
A $\frac{1}{250}$ B $\frac{2}{125}$ C $\frac{2}{123}$ D $\frac{123}{125}$
- Mei spent \$119, \$142, \$99, and \$140 on groceries over four weeks. What is the mean (average) amount she spent per week?
A \$120.50 B \$125.00 C \$129.50 D \$500.00
- A pizza shop offers 3 sizes, 2 kinds of crust, and 5 toppings. How many different one-topping pizzas can be ordered?
A 6 B 10 C 30 D 125
- Leila scored 95, 85, 95, and 97 on four practice tests. What is the mean (average) of her scores?
A 93 B 95 C 124 D 372
- A new pickup truck comes in 5 colors, 4 cab styles, and 2 engine sizes. How many different versions of the truck are possible?
A 11 B 40 C 49 D 125
- Cards numbered 1 through 24 are placed in a box, and one card is drawn at random. What is the probability that the number on the card is a multiple of 5?
A $\frac{1}{6}$ B $\frac{1}{5}$ C $\frac{5}{24}$ D $\frac{5}{6}$
- The times, in minutes, of five soldiers on a two-mile run were 15, 20, 14, 16, 17. What is the median of the times?
A 6 B 10 C 14 D 16
- A survey found that 10% of high school seniors plan to join the military. In a group of 600 seniors, how many would you expect to plan to join?
A 10 B 60 C 540 D 600

13. While training for a marathon, Rosa ran 14, 14, and 12 miles in the first three weeks. How many miles must she run in the next week to average 17 miles per week over four weeks?
A 17 B 28 C 30 D 34
14. The average age of the 10 players on a softball team is 24 years. A new player who is 46 years old joins the team. What is the new average age of the players?
A 24 B 26 C 32 D 35
15. Brandon scored 82, 84, and 76 on three math tests. What score does he need on the next test to average 78 on all four tests?
A 64 B 70 C 73 D 78
16. The heights, in inches, of the six players on a basketball team are 67, 71, 70, 73, 77, 77. What is the median of their heights?
A 72 B 72.5 C 73 D 73.5
17. On a fitness test, the 39 soldiers of 1st Platoon averaged 73 points and the 26 soldiers of 2nd Platoon averaged 78 points. What is the average score of all the soldiers in the two platoons?
A 75 B 76 C 78 D 4,875
18. A motor pool has 30 vehicles, and 7 of them are down for maintenance. If one vehicle is picked at random for a convoy, what is the probability that it is *not* down for maintenance?
A $\frac{7}{30}$ B $\frac{7}{23}$ C $\frac{23}{30}$ D $\frac{23}{7}$
19. A patrol carries two radios. Each radio works independently with probability $\frac{5}{6}$. What is the probability that *both* radios work?
A $\frac{25}{36}$ B $\frac{7}{9}$ C $\frac{5}{6}$ D $\frac{35}{36}$
20. A fair six-sided die is rolled once. What is the probability that the number rolled is *not* less than 5?
A $\frac{1}{6}$ B $\frac{1}{3}$ C $\frac{1}{2}$ D $\frac{2}{3}$
21. A spinner has 5 equal sections, each a different color, one of them yellow. The spinner is spun once and a six-sided die is rolled once. What is the probability that the spinner lands on yellow and the die shows an even number?
A $\frac{1}{30}$ B $\frac{1}{20}$ C $\frac{1}{10}$ D $\frac{7}{10}$
22. On a two-mile run, the 15 airmen in A Flight averaged 16 minutes and the 30 airmen in B Flight averaged 19 minutes. What was the average time for all the airmen?
A 12 minutes B 14 minutes
C 17 minutes D 18 minutes
23. The daily high temperatures (in degrees Fahrenheit) for six days were 62, 67, 87, 87, 93, 68. What is the median of the temperatures?
A 68 B 77 C 77.5 D 87
24. On four practice fitness tests, Private Lopez did 66, 54, 36, and 52 push-ups. How many push-ups must Lopez do on the next test to average 56 over all five tests?
A 54 B 56 C 60 D 72
25. The average height of the 9 players on a basketball team is 72 inches. A player who is 82 inches tall joins the team. What is the new average height?
A 64 inches B 72 inches
C 73 inches D 77 inches
26. At a Navy boot camp division, about 10% of recruits do not complete training. If 1,300 recruits start, about how many would you expect to graduate?
A 0 B 130 C 1,170 D 1,290
27. Priya's hiking club has 9 members. One person will be chosen as president and a different person as vice president. In how many ways can the two positions be filled?
A 17 B 36 C 72 D 81
28. A platoon makes radio call signs from two letters followed by one digit. The letters must come from the first 3 letters of the alphabet (repeats allowed), and each digit can be 0 to 9. How many call signs are possible?
A 16 B 30 C 60 D 90
29. A box contains 7 glazed donuts and 6 chocolate donuts. Two donuts are taken at random, one after the other. What is the probability that the first donut is glazed and the second is chocolate?
A $\frac{7}{52}$ B $\frac{35}{156}$ C $\frac{42}{169}$ D $\frac{7}{26}$

30. The average high temperature for the first 6 days of a training exercise was 61 degrees. Including the next day, the average for all 7 days is 63 degrees. What was the high temperature on that day?
A 75 B 78 C 84 D 441
31. A squad has 6 privates and 9 specialists. The squad leader puts all of their names in a helmet. Two names are drawn at random for weekend guard duty. What is the probability that the first name is a private and the second is a specialist?
A $\frac{6}{25}$ B $\frac{9}{35}$ C $\frac{11}{35}$ D $\frac{18}{35}$
32. In an Army mechanic course, the unit tests count for 50% of the final grade and the final exam counts for 50%. Kevin has a test average of 89. What score does he need on the final exam to earn a final grade of 76?
A 54 B 57 C 63 D 76
33. Seven students, including Malik, will give speeches one at a time. Malik must speak last. In how many different orders can the speeches be given?
A 720 B 5,039 C 5,040 D 46,656
34. At a sandwich shop, Andre may choose any 3 different toppings from a list of 7. How many different groups of 3 toppings are possible?
A 21 B 35 C 105 D 210
35. A science fair has 6 projects, including Kevin's. In how many ways can the judges award first, second, and third place?
A 20 B 120 C 130 D 216
36. At a meeting, Jamal and the other 11 people each shake hands once with every other person. How many handshakes take place?
A 24 B 55 C 66 D 132
37. In a class of 27 students, the 18 boys averaged 86 on a test, and the whole class averaged 84. What was the average score of the girls?
A 80 B 82 C 84 D 720
38. The 12 finalists in a base raffle include 3 soldiers from Alpha Company; the rest are from other companies. Two different winners are drawn at random. What is the probability that both winners are from Alpha Company?
A $\frac{1}{24}$ B $\frac{1}{22}$ C $\frac{1}{16}$ D $\frac{3}{44}$
39. Sergeant Novak's average score on 10 rifle-qualification tables is 42 out of 50. If Novak's lowest score is thrown out, the average of the other 9 is 43. What was the lowest score?
A 33 B 39 C 41 D 42
40. On a rifle range, Sergeant Patel hit 28, 33, 37, and 30 targets out of 40 on four practice rounds. To average 30 hits over six rounds, how many hits does Patel need on each of the next two rounds, if both are the same?
A 20 B 22 C 24 D 26

Answers & Explanations 26

1 A	9 B	17 A	25 C	33 A
2 B	10 A	18 C	26 C	34 B
3 C	11 D	19 A	27 C	35 B
4 B	12 B	20 B	28 D	36 C
5 B	13 B	21 C	29 D	37 A
6 B	14 B	22 D	30 A	38 B
7 C	15 B	23 C	31 B	39 A
8 A	16 A	24 D	32 C	40 D

1. **A** 72

1 Add the 6 values: $67 + 64 + 78 + 61 + 87 + 75 = 432$.

2 Divide by the number of values, 6: $432 \div 6 = 72$.

TIP Check: the mean must be between the smallest value (61) and the largest (87).

WHY NOT (B) averages only the highest and lowest values. (D) is the total, not the average.

2. **B** 81

1 Put the values in order to spot repeats: 69, 72, 77, 81, 81, 87.

2 The value 81 appears twice, more than any other value, so the mode is 81.

WHY NOT (A) gives how many times the mode appears, not the mode itself. (C) gives the largest value, not the most frequent one.

3. **C** $\frac{1}{3}$

1 Count all the socks: $8 + 6 + 10 = 24$.

2 Favorable outcomes: 8. Probability = $\frac{8}{24} = \frac{1}{3}$.

WHY NOT (A) counts only one of the favorable items. (D) compares the favorable outcomes to the unfavorable ones instead of to the total.

4. **B** 23

1 Put the 7 values in order: 18, 19, 21, 23, 24, 25, 26.

2 The middle value is the 4th one (3 values on each side), so the median is 23.

WHY NOT (A) takes the middle number of the list without putting it in order first.

5. **B** $\frac{2}{125}$

1 Favorable outcomes: his 4 tickets. Total outcomes: 250 tickets.

2 Probability = $\frac{4}{250} = \frac{2}{125}$.

WHY NOT (A) uses only one ticket. (C) compares the tickets bought to the tickets not bought instead of to all tickets. (D) is the probability of *not* winning.

6. **B** \$125.00

1 Add the 4 values: $119 + 142 + 99 + 140 = 500$.

2 Divide by the number of values, 4: $500 \div 4 = 125$.

TIP Check: the mean must be between the smallest value (99) and the largest (142).

WHY NOT (A) averages only the highest and lowest values. (C) is the median (the middle value), not the mean. (D) is the total, not the average.

7. **C** 30

1 Each choice can be paired with every other choice, so multiply the number of options at each step (the counting principle).

2 $3 \times 2 \times 5 = 30$.

WHY NOT (A) leaves out one of the choices. (B) adds the choices instead of multiplying them.

8. **A** 93

1 Add the 4 values: $95 + 85 + 95 + 97 = 372$.

2 Divide by the number of values, 4: $372 \div 4 = 93$.

TIP Check: the mean must be between the smallest value (85) and the largest (97).

WHY NOT (B) is the median (the middle value), not the mean. (C) divides by one less than the number of values. (D) is the total, not the average.

9. **B** 40

1 Each choice can be paired with every other choice, so multiply the number of options at each step (the counting principle).

2 $5 \times 4 \times 2 = 40$.

WHY NOT (A) adds the choices instead of multiplying them.

10. **A** $\frac{1}{6}$

- 1 The multiples of 5 from 1 to 24 are 5, 10, 15, 20: 4 cards.
- 2 There are 24 cards in all, so the probability is $\frac{4}{24} = \frac{1}{6}$.

WHY NOT (B) compares the favorable cards to the unfavorable ones instead of to all the cards. (C) miscounts the favorable cards by one. (D) is the probability that the card is *not* a multiple of 5.

11. **D** 16

- 1 Put the 5 values in order: 14, 15, 16, 17, 20.
- 2 The middle value is the 3rd one (2 values on each side), so the median is 16.

WHY NOT (A) is the range, not the median. (C) takes the middle number of the list without putting it in order first.

12. **B** 60

- 1 Expected count = probability $\times$ number of tries: $0.1 \times 600 = 60$.

WHY NOT (A) uses the percent as the count. (C) is the expected number not planning to join. (D) moves the decimal point only one place.

13. **B** 28

- 1 To average 17 over 4 weeks, the total must be $17 \times 4 = 68$.
- 2 The total so far is $14 + 14 + 12 = 40$.
- 3 Subtract: $68 - 40 = 28$.

TIP Check: $14 + 14 + 12 + 28 = 68$, and $68 \div 4 = 17$. ✓

WHY NOT (A) is the target average, not the score needed.

14. **B** 26

- 1 Turn the average into a total: $10 \times 24 = 240$.
- 2 Add the value: $240 + 46 = 286$.
- 3 Divide by the new count, $10 + 1 = 11$: $286 \div 11 = 26$.

WHY NOT (A) assumes the average does not change. (D) averages the old average with the new value.

15. **B** 70

- 1 To average 78 over 4 tests, the total must be $78 \times 4 = 312$.
- 2 The total so far is $82 + 84 + 76 = 242$.
- 3 Subtract: $312 - 242 = 70$.

TIP Check: $82 + 84 + 76 + 70 = 312$, and $312 \div 4 = 78$. ✓

WHY NOT (D) is the target average, not the score needed.

16. **A** 72

- 1 Put the 6 values in order: 67, 70, 71, 73, 77, 77.
- 2 With an even number of values, the median is the average of the two middle ones (the 3rd and 4th): $(71 + 73) \div 2 = 144 \div 2 = 72$.

WHY NOT (B) is the mean, not the median. (C) uses only one of the two middle values.

17. **A** 75

- 1 Turn each average into a total: $39 \times 73 = 2,847$ and $26 \times 78 = 2,028$.
- 2 Add the totals and the counts: $2,847 + 2,028 = 4,875$ for $39 + 26 = 65$ soldiers.
- 3 Divide: $4,875 \div 65 = 75$.

TIP Check: the answer is closer to the average of the larger group (39 soldiers).

WHY NOT (B) weights each average by the other group's size. (D) is the combined total, not the average.

18. **C** $\frac{23}{30}$

- 1 Vehicles not in maintenance: $30 - 7 = 23$.
- 2 Probability = $\frac{23}{30}$.

WHY NOT (A) is the probability that the vehicle is down for maintenance. (D) compares good vehicles to broken ones instead of to the total.

19. **A** $\frac{25}{36}$

- 1 Each radio has probability $\frac{5}{6}$, and the events are independent, so multiply.
- 2 $\frac{5}{6} \times \frac{5}{6} = \frac{25}{36}$.

WHY NOT (C) gives the probability for only one radio. (D) finds the probability of at least one, not all.

20. **B** $\frac{1}{3}$

- 1 The outcomes that are less than 5: 1, 2, 3, 4 (4 of them).
- 2 The other 2 outcomes are favorable: $\frac{2}{6} = \frac{1}{3}$.

WHY NOT (A) counts only one outcome. (C) compares favorable to unfavorable outcomes. (D) is the probability that the number is less than 5.

21. **C** $\frac{1}{10}$

- 1 $P(\text{yellow on the spinner}) = \frac{1}{5}$. $P(\text{die}) = \frac{3}{6} = \frac{1}{2}$.
- 2 The two results do not affect each other, so multiply: $\frac{1}{5} \times \frac{1}{2} = \frac{1}{10}$.

WHY NOT (A) counts only one favorable combination. (D) adds the probabilities instead of multiplying them.

22. **D** 18 minutes

- 1 Turn each average into a total: $15 \times 16 = 240$ and $30 \times 19 = 570$.
- 2 Add the totals and the counts: $240 + 570 = 810$ for $15 + 30 = 45$ airmen.
- 3 Divide: $810 \div 45 = 18$.

TIP Check: the answer is closer to the average of the larger group (30 airmen).

WHY NOT (C) weights each average by the other group's size.

 23. **C** 77.5

- 1 Put the 6 values in order: 62, 67, 68, 87, 87, 93.
- 2 With an even number of values, the median is the average of the two middle ones (the 3rd and 4th): $(68 + 87) \div 2 = 155 \div 2 = 77.5$.

WHY NOT (A) uses only one of the two middle values. (D) uses only one of the two middle values.

 24. **D** 72

- 1 To average 56 over 5 tests, the total must be $56 \times 5 = 280$.
- 2 The total so far is $66 + 54 + 36 + 52 = 208$.
- 3 Subtract: $280 - 208 = 72$.

TIP Check: $66 + 54 + 36 + 52 + 72 = 280$, and $280 \div 5 = 56$. ✓

WHY NOT (A) averages the target with the current average. (B) is the target average, not the score needed. (C) makes up the shortfall for only one test, not for all of them.

 25. **C** 73 inches

- 1 Turn the average into a total: $9 \times 72 = 648$.
- 2 Add the value: $648 + 82 = 730$.
- 3 Divide by the new count, $9 + 1 = 10$: $730 \div 10 = 73$.

WHY NOT (B) assumes the average does not change. (D) averages the old average with the new value.

 26. **C** 1,170

- 1 If 10% do not finish, then $100\% - 10\% = 90\%$ graduate.
- 2 Expected graduates: $0.9 \times 1,300 = 1,170$.

WHY NOT (A) moves the decimal point only one place. (B) is the number expected *not* to graduate. (D) subtracts the percent as if it were a number of recruits.

 27. **C** 72

- 1 President: 9 choices. Vice president: 8 choices (anyone except the president).
- 2 Multiply: $9 \times 8 = 72$.

WHY NOT (A) adds the choices instead of multiplying them. (B) ignores the difference between the two positions. (D) lets the same person hold both positions.

 28. **D** 90

- 1 Count the choices for each position: 3, 3, 10.
- 2 Multiply (counting principle): $3 \times 3 \times 10 = 90$.

WHY NOT (A) adds the choices instead of multiplying them. (B) counts only one letter. (C) does not let the letters repeat.

 29. **D** $\frac{7}{26}$

- 1 First pick: $\frac{7}{13}$. After one glazed donut is taken, 12 donuts are left, and all 6 chocolate donuts are still there: $\frac{6}{12}$.
- 2 Multiply: $\frac{7}{13} \times \frac{6}{12} = \frac{42}{156} = \frac{7}{26}$.

WHY NOT (B) wrongly removes one of the chocolate donuts before the second pick. (C) puts the first one back (uses the same total for both picks).

 30. **A** 75

- 1 Total for the first 6: $6 \times 61 = 366$.
- 2 Total for all 7: $7 \times 63 = 441$.
- 3 The new value is the difference: $441 - 366 = 75$.

WHY NOT (D) is the total for all 7, not the last value.

 31. **B** $\frac{9}{35}$

- 1 First pick: $\frac{6}{15}$. After one private is taken, 14 names are left, and all 9 specialists are still there: $\frac{9}{14}$.
- 2 Multiply: $\frac{6}{15} \times \frac{9}{14} = \frac{54}{210} = \frac{9}{35}$.

WHY NOT (A) puts the first one back (uses the same total for both picks). (D) counts both orders, but the order is fixed here.

 32. **C** 63

- 1 The tests contribute $0.5 \times 89 = 44.5$ points to the grade.
- 2 The final must contribute $76 - 44.5 = 31.5$ points.
- 3 The final counts 50%, so the exam score is $31.5 \div 0.5 = 63$.

TIP Check: $0.5(89) + 0.5(63) = 44.5 + 31.5 = 76$. ✓

WHY NOT (D) is the final grade wanted, not the exam score needed.

33. **A** 720

- 1 Malik's place is fixed, so only the other 6 people need to be arranged.
- 2 $6 \times 5 \times 4 \times 3 \times 2 \times 1 = 720$.

WHY NOT (C) ignores the rule about Malik. (D) lets the same person fill more than one place.

34. **B** 35

- 1 If order mattered: $7 \times 6 \times 5 = 210$.
- 2 Each group of 3 can be ordered in $3 \times 2 \times 1 = 6$ ways, and all of these are the same group. Divide: $210 \div 6 = 35$.

WHY NOT (C) divides by 2 instead of by 6. (D) counts each group several times (order does not matter here).

35. **B** 120

- 1 First position: 6 choices; second: 5; third: 4 (no one can repeat).
- 2 Multiply: $6 \times 5 \times 4 = 120$.

WHY NOT (A) counts groups of three instead of ordered positions. (D) lets the same one fill more than one position.

36. **C** 66

- 1 Each of the 12 can be paired with 11 others: $12 \times 11 = 132$.
- 2 That counts every pair twice (A with B and B with A), so divide by 2: $132 \div 2 = 66$.

WHY NOT (B) leaves one person out. (D) counts each pair twice.

37. **A** 80

- 1 Total for everyone: $27 \times 84 = 2,268$.
- 2 Total for the first group: $18 \times 86 = 1,548$.
- 3 The girls have $2,268 - 1,548 = 720$ points in all, so their average is $720 \div 9 = 80$.

WHY NOT (B) assumes the two groups are the same size. (C) gives the average of the whole group. (D) is the total for the girls, not their average.

38. **B** $\frac{1}{22}$

- 1 First pick: $\frac{3}{12}$. After one Alpha Company soldier is taken, 2 of the 11 names left are Alpha Company soldiers: $\frac{2}{11}$.

- 2 Multiply: $\frac{3}{12} \times \frac{2}{11} = \frac{6}{132} = \frac{1}{22}$.

WHY NOT (A) lowers the count of the kind picked but not the total. (C) puts the first one back (uses the same counts for both picks). (D) forgets to remove the first one from the count of Alpha Company soldiers.

39. **A** 33

- 1 Total of all 10: $10 \times 42 = 420$.
- 2 Total of the other 9: $9 \times 43 = 387$.
- 3 The dropped score is the difference: $420 - 387 = 33$.

WHY NOT (C) assumes the dropped score was as far below the average as the average went up.

40. **D** 26

- 1 To average 30 over 6 rounds, the total must be $30 \times 6 = 180$.
- 2 The total so far is $28 + 33 + 37 + 30 = 128$.
- 3 The next two must add up to $180 - 128 = 52$, so each one is $52 \div 2 = 26$.

TIP Check: $28 + 33 + 37 + 30 + 26 + 26 = 180$, and $180 \div 6 = 30$. ✓

WHY NOT (B) plans for only one more round instead of two.

Measurement, Units & Multi-Step Problems

Tested on: Arithmetic Reasoning

Key Concepts

Many Arithmetic Reasoning questions are easy arithmetic wrapped in units: feet and inches, quarts and gallons, a 24-hour clock. The rule that saves you: **convert first, then calculate**, and write down each step.

US customary units (memorize these)

Length	Capacity	Weight and time
12 inches = 1 foot	8 fluid ounces = 1 cup	16 ounces = 1 pound
3 feet = 1 yard	2 cups = 1 pint	2,000 pounds = 1 ton
36 inches = 1 yard	2 pints = 1 quart	60 minutes = 1 hour
5,280 feet = 1 mile	4 quarts = 1 gallon	7 days = 1 week
Big unit → small unit: multiply . Small unit → big unit: divide . (There are <i>more</i> inches than feet, so the number gets bigger.)		

Metric units

kilo = 1,000 centi = $\frac{1}{100}$ milli = $\frac{1}{1,000}$. So 1 km = 1,000 m, 1 m = 100 cm = 1,000 mm, 1 kg = 1,000 g, 1 L = 1,000 mL. Converting is just moving the decimal point: 2.5 km = 2,500 m. US–metric factors (such as 1 mile $\approx$ 1.6 km) are given in the problem.

Military (24-hour) time and elapsed time

Hours run from 0000 (midnight) to 2359. For p.m. times add 12 to the hour: 3:45 p.m. = 1545. To find elapsed time, subtract hours and minutes separately, and when you borrow an hour, it is worth **60** minutes, not 100. Across midnight: time until 2400, plus time after 0000.

WORKED EXAMPLE

A convoy leaves at 0745 and arrives at 1420. How long is the trip?

Solution. 20 minutes minus 45 minutes won't go, so borrow 1 hour: 14:20 becomes 13 hours 80 minutes. Then $13 - 7 = 6$ hours and $80 - 45 = 35$ minutes. The trip takes 6 hours 35 minutes.

No-Calculator Tip Decide how to round from the question. “How many trips (boxes, tents) are *needed*?” rounds *up*: 75 soldiers in 6-person tents need 13 tents. “How many *full* pieces (or items you can afford)?” rounds *down*.

Common Traps

- Subtracting clock times like ordinary numbers: $1420 - 0745 = 675$ is not 6 hours 75 minutes.

- Fence posts: a 60-foot fence with posts every 6 feet needs $60 \div 6 + 1 = 11$ posts (one at each end). Cutting a board into 5 pieces takes only 4 cuts.
- “ $\frac{1}{3}$ of the *remainder*” is a fraction of what is left, not of the original amount.
- Using the wrong conversion factor (3 feet in a yard, but 12 inches in a foot).

Practice Set 27

Score: ____ / 35

Target time: 42 minutes

Problems 1–10 are warm-ups, 11–24 are test level, and 25–35 are challenge problems.

- A hallway in the office building where Mei works is 21 feet long. How many yards of carpet runner are needed to cover its length?
A 1.75 yards B 7 yards
C 18 yards D 63 yards
- Andre mails a package that weighs 152 ounces. How many pounds does it weigh?
A 8.5 pounds B 9.5 pounds
C 136 pounds D 2,432 pounds
- A hydration pack holds 3,750 milliliters of water. How many liters is that?
A 0.375 liters B 3.65 liters
C 3.75 liters D 37.5 liters
- Derek starts a hike at 11:35 a.m. and hikes for 2 hours 40 minutes. At what time does he finish?
A 1:15 p.m. B 2:15 p.m.
C 2:25 p.m. D 2:35 p.m.
- A bottle of water holds 0.25 liters. How many milliliters is that?
A 25 milliliters B 250 milliliters
C 300 milliliters D 2,500 milliliters
- A weekend first-aid class runs from 11:55 a.m. to 1:10 p.m. How long is the class?
A 1 hour 15 minutes
B 1 hour 55 minutes
C 2 hours
D 10 hours 45 minutes
- Leila buys a bag of flour that weighs 4 pounds. How many ounces does it weigh?
A 0.25 ounces B 20 ounces
C 48 ounces D 64 ounces
- Jasmine needs a piece of trim 21 feet long. How many inches long is the trim?
A 1.75 inches B 33 inches
C 63 inches D 252 inches
- A ruck march has two legs. The first takes 6 hr 52 min and the second takes 1 hr 31 min. What is the total time?
A 7 hr 23 min B 8 hr 17 min
C 8 hr 21 min D 8 hr 23 min
- A farm packs eggs in cartons of 12. How many cartons are needed to pack 833 eggs?
A 5 B 60 C 69 D 70
- Aisha's luggage weighs 66 pounds. About how many kilograms is that? (1 kilogram $\approx$ 2.2 pounds)
A 30 kilograms B 33 kilograms
C 63.8 kilograms D 145.2 kilograms
- A Navy cook works in the ship's galley from 1235 to 2205. How long is the cook's shift?
A 9 hours 10 minutes
B 9 hours 30 minutes
C 10 hours 10 minutes
D 10 hours 30 minutes

13. A roll of fabric holds 20 yd 1 ft. Ethan cuts off 1 yd 2 ft. How much fabric is left on the roll?
A 17 yd 2 ft B 18 yd 2 ft
C 19 yd 1 ft D 19 yd 2 ft
14. Sofia pours juice from a 2-liter bottle into 350-milliliter cups. How many cups can be filled completely?
A 3 B 4 C 5 D 6
15. Mei has a board that is 4 ft 3 in long and cuts off a piece 1 ft 9 in long. How long is the board that is left?
A 2 ft 6 in B 2 ft 8 in C 3 ft D 3 ft 6 in
16. In the field, each soldier needs 6 quarts of water per day. How many soldiers can a full 40-gallon water tank supply for one day?
A 4 B 13 C 24 D 26
17. Sergeant Ruiz begins guard duty at 3:45 p.m. The shift lasts 2 hours 25 minutes. At what time does the shift end, in military time?
A 0610 B 1710 C 1745 D 1810
18. A picture frame is 4 inches long. How many centimeters is that? (1 inch = 2.54 centimeters)
A 1.016 centimeters B 6.54 centimeters
C 10.16 centimeters D 101.6 centimeters
19. A running track on base is 220 yards around. How many laps must a soldier run to cover 1 mile? (1 mile = 1,760 yards)
A 8 B 10 C 24 D 1,760
20. Sergeant Murphy's squad has ten full 5-gallon water cans. Each canteen holds 1 quart. How many canteens can the squad fill?
A 50 B 100 C 200 D 400
21. A track is 500 meters around. How many laps does Tyrone need to run to complete 10 kilometers?
A 2 B 14 C 20 D 10,000
22. A water tank holds 420 gallons. Soldiers use $\frac{2}{3}$ of the water in the morning and $\frac{1}{4}$ of what is left in the afternoon. How many gallons are left?
A 35 B 105 C 140 D 280
23. In a stadium row, seats are numbered in order. A group bought seats 39 through 98. How many seats did they buy?
A 58 B 59 C 60 D 61
24. In 6 years, Brandon will be three times as old as his daughter will be then. The daughter is 7 now. How old is Brandon now?
A 21 B 23 C 27 D 33
25. It takes 6 minutes to make one cut through a log. How long will it take to cut a 35-foot log into 5-foot pieces?
A 6 minutes B 31 minutes
C 36 minutes D 42 minutes
26. A training course starts on a Saturday (day 1) and runs every day without a break. On what day of the week is day 129?
A Tuesday B Monday
C Thursday D Sunday
27. The sum of the ages of Maria and her daughter is 56. In 4 years, Maria will be three times as old as the daughter. How old is Maria now?
A 44 B 48 C 54 D 59
28. Of the recruits who started a training course, $\frac{2}{5}$ were sent to other units in the first week, and $\frac{1}{2}$ of the rest were sent away in the second week. 48 recruits remain. How many recruits started the course?
A 80 B 140 C 160 D 480
29. A flight carrying troops leaves at 2155 and the flight takes 5 hours 15 minutes. At what time (in military time, same time zone) does it land?
A 0140 B 0210 C 0255 D 0310
30. Grace won a cash prize. She spent $\frac{1}{6}$ of it on a car repair and saved $\frac{2}{5}$ of the rest. She has \$900 left to spend. How much was the prize?
A \$1,080 B \$1,470 C \$1,800 D \$1,900

31. A convoy leaves at 2025 and reaches its destination at 0745 the next morning. How long is the drive?
- A 7 hours 45 minutes
B 11 hours 20 minutes
C 12 hours
D 13 hours 20 minutes
32. Yuki is three times as old as her daughter. Eight years ago, Yuki was five times as old as the daughter. How old is Yuki now?
- A 16 B 48 C 56 D 80
33. Today is Sunday. Diego enlisted exactly 108 days ago. On what day of the week did he enlist?
- A Wednesday B Saturday
C Friday D Thursday
34. Trees are planted on *both* sides of a straight 240-foot road, one every 15 feet, with a tree at each end on each side. How many trees are planted?
- A 17 B 32 C 33 D 34
35. One gallon of paint covers 350 square feet. Paint is sold only in whole gallons at \$42 per gallon. How much will it cost to paint 1,000 square feet of walls?
- A \$84 B \$120 C \$126 D \$168

Answers & Explanations 27

1 B	8 D	15 A	22 B	29 D
2 B	9 D	16 D	23 C	30 C
3 C	10 D	17 D	24 D	31 B
4 B	11 A	18 C	25 C	32 B
5 B	12 B	19 A	26 B	33 D
6 A	13 B	20 C	27 A	34 D
7 D	14 C	21 C	28 C	35 C

1. **B** 7 yards
 - 1 1 yard = 3 feet. Going from a small unit to a big unit, divide: $21 \div 3 = 7$ yards.
 - WHY NOT** (A) uses 12 instead of 3 feet in a yard. (C) subtracts 3 instead of dividing by it. (D) multiplies instead of dividing.
2. **B** 9.5 pounds
 - 1 1 pound = 16 ounces. From a small unit to a big unit, divide: $152 \div 16 = 9.5$ pounds.
 - WHY NOT** (C) subtracts 16 instead of dividing by it. (D) multiplies instead of dividing.
3. **C** 3.75 liters
 - 1 1 liter = 1,000 milliliters. Divide by 1,000 (move the decimal point 3 places to the left): $3,750 \div 1,000 = 3.75$ liters.
 - WHY NOT** (A) moves the decimal point one place too far. (D) uses 100 instead of 1,000.
4. **B** 2:15 p.m.
 - 1 Add the minutes: $35 + 40 = 75$ minutes, which is 1 hour 15 minutes. Write down 15 minutes and carry 1 hour.
 - 2 Add the hours: $11 + 2 + 1 = 14$, and past 12 the clock starts over.
 - 3 The time is 2:15 p.m.
 - WHY NOT** (A) forgets to carry the extra hour from the minutes.
5. **B** 250 milliliters
 - 1 1 liter = 1,000 milliliters. Multiply by 1,000 (move the decimal point 3 places to the right): $0.25 \times 1,000 = 250$ milliliters.
 - WHY NOT** (A) uses 100 instead of 1,000. (D) moves the decimal point one place too far.
6. **A** 1 hour 15 minutes
 - 1 From 11:55 a.m. to 12 noon is 5 minutes.
 - 2 From 12 noon to 1:10 p.m. is 1 hour 10 minutes.
 - 3 Add: 5 minutes + 1 hour 10 minutes = 1 hour 15 minutes.
 - WHY NOT** (B) subtracts the clock times like ordinary numbers (borrows 100 minutes instead of 60). (C) counts only the change in the hour. (D) forgets that the clock starts over at 12 noon.
7. **D** 64 ounces
 - 1 1 pound = 16 ounces. From a big unit to a small unit, multiply: $4 \times 16 = 64$ ounces.
 - WHY NOT** (A) divides instead of multiplying. (B) adds 16 instead of multiplying by it. (C) uses 12 ounces in a pound instead of 16.
8. **D** 252 inches
 - 1 1 foot = 12 inches. Going from a big unit to a small unit, multiply: $21 \times 12 = 252$ inches.
 - TIP** Check the size: inches are smaller than feet, so there should be more of them.
 - WHY NOT** (A) divides instead of multiplying. (B) adds 12 instead of multiplying by it. (C) uses 3 instead of 12 inches in a foot.
9. **D** 8 hr 23 min
 - 1 Add the minutes: $52 + 31 = 83$ min. Since 60 min = 1 hr, 83 min = 1 hr 23 min.
 - 2 Add the hours: $6 + 1 + 1 = 8$ hr.
 - 3 Total: 8 hr 23 min.
 - WHY NOT** (A) changes 83 min to 23 min but forgets to add 1 to the hours.
10. **D** 70
 - 1 Divide: $833 \div 12$ is 69 with a remainder of 5.
 - 2 With only 69, 5 would be left over, so one more is needed: $69 + 1 = 70$.
 - WHY NOT** (A) gives the leftover amount, not the number asked for. (C) rounds down, which leaves some left over.

11. **A** 30 kilograms

1 A kilogram is heavier than a pound, so there are fewer kilograms. Divide by 2.2: $66 \div 2.2 = 660 \div 22 = 30$ kilograms.

WHY NOT (B) divides by 2 instead of 2.2. (C) subtracts 2.2 instead of dividing. (D) multiplies by 2.2 instead of dividing.

12. **B** 9 hours 30 minutes

1 5 minutes minus 35 minutes won't go, so borrow 1 hour: 2205 becomes 21 hours 65 minutes.

2 Subtract: $21 - 12 = 9$ hours and $65 - 35 = 30$ minutes.

3 The time is 9 hours 30 minutes.

WHY NOT (C) subtracts $2205 - 1235$ like ordinary numbers (borrows 100 minutes instead of 60). (D) subtracts the smaller number of minutes from the larger instead of borrowing.

13. **B** 18 yd 2 ft

1 You can't take 2 ft from 1 ft, so borrow 1 yd: 20 yd 1 ft becomes 19 yd 4 ft.

2 Subtract: $4 - 2 = 2$ ft and $19 - 1 = 18$ yd.

3 What is left: 18 yd 2 ft.

WHY NOT (C) subtracts the smaller number of feet from the larger instead of borrowing. (D) borrows 1 yd but forgets to take it away from the yards.

14. **C** 5

1 Change liters to milliliters: $2 \times 1,000 = 2,000$ milliliters.

2 Divide: $2,000 \div 350$ is 5 with a remainder of 250.

3 Only full ones count, so round down to 5.

WHY NOT (D) rounds up, but there is not enough for one more full piece.

15. **A** 2 ft 6 in

1 You can't take 9 in from 3 in, so borrow 1 ft: 4 ft 3 in becomes 3 ft 15 in.

2 Subtract: $15 - 9 = 6$ in and $3 - 1 = 2$ ft.

3 What is left: 2 ft 6 in.

WHY NOT (D) subtracts the smaller number of inches from the larger instead of borrowing.

16. **D** 26

1 Change gallons to quarts: $40 \times 4 = 160$ quarts.

2 Divide: $160 \div 6$ is 26 with a remainder of 4.

3 Only full ones count, so round down to 26.

WHY NOT (A) gives the leftover amount, not the number asked for. (B) uses 2 quarts in a gallon instead of 4.

17. **D** 1810

1 Change to military time: 3:45 p.m. is $3 + 12 = 15$ hours, so 1545.

2 Add the minutes: $45 + 25 = 70$ minutes = 1 hour 10 minutes.

3 Add the hours: $15 + 2 + 1 = 18$. The shift ends at 1810.

WHY NOT (A) forgets to change the p.m. time to 24-hour time. (B) forgets to carry the extra hour from the minutes. (C) adds only the hours.

18. **C** 10.16 centimeters

1 Each inch is 2.54 centimeters, so multiply: $4 \times 2.54 = 10.16$ centimeters.

WHY NOT (A) slips the decimal point. (B) adds 2.54 instead of multiplying. (D) slips the decimal point.

19. **A** 8

1 Distance in yards: $1 \times 1,760 = 1,760$ yards.

2 Divide by one lap: $1,760 \div 220 = 8$ laps.

WHY NOT (C) uses feet in a mile with a track measured in yards. (D) gives the distance in yards, not the number of laps.

20. **C** 200

1 Total water: $10 \times 5 = 50$ gallons.

2 1 gallon = 4 quarts: $50 \times 4 = 200$ quarts, so 200 canteens.

WHY NOT (A) forgets to change gallons to quarts. (B) uses 2 quarts in a gallon instead of 4. (D) changes gallons to pints instead of quarts.

21. **C** 20

1 Change kilometers to meters: $10 \times 1,000 = 10,000$ meters.

2 Divide by one lap: $10,000 \div 500 = 20$ laps.

WHY NOT (A) uses 100 meters in a kilometer instead of 1,000. (D) gives the distance in meters, not the number of laps.

22. **B** 105

1 First step: $\frac{2}{3} \times 420 = 280$. Remainder: $420 - 280 = 140$.

2 Second step: $\frac{1}{4}$ of the remainder: $\frac{1}{4} \times 140 = 35$.

3 Left: $140 - 35 = 105$.

WHY NOT (A) takes both fractions of the original amount instead of the remainder. (C) stops after the first step. (D) gives the first amount instead.

23. **C** 60

1 Subtract: $98 - 39 = 59$. This counts the steps from the first to the last.

2 Both ends are included, so add 1: $59 + 1 = 60$.

TIP Check with a small case: pages 3 through 5 are pages 3, 4, 5, which is 3 pages, not $5 - 3 = 2$.

WHY NOT (A) leaves out both ends. (B) subtracts but forgets to count the first one.

24. **D** 33

1 In 6 years the daughter will be $7 + 6 = 13$.

2 Then Brandon will be $3 \times 13 = 39$.

3 Today Brandon is 6 years younger than that: $39 - 6 = 33$.

WHY NOT (A) multiplies the daughter's age today instead of in 6 years.

25. **C** 36 minutes

1 Pieces: $7 (35 \div 5 = 7)$. Cuts = pieces $- 1 = 6$.

2 Time: $6 \times 6 = 36$ minutes.

WHY NOT (A) gives the number of cuts, not the time. (D) multiplies by the number of pieces instead of the number of cuts.

26. **B** Monday

1 Day 129 is $129 - 1 = 128$ days after day 1.

2 $128 \div 7 = 18$ weeks with 2 days left over, so move 2 days past Saturday: Monday.

WHY NOT (A) counts 129 days after day 1, but day 129 is only 128 days later. (C) counts backward instead of forward. (D) counts one day too few.

27. **A** 44

1 Let the daughter be a ; then Maria is $56 - a$.

2 In 4 years: $56 - a + 4 = 3(a + 4)$, so $60 - a = 3a + 12$.

3 Then $48 = 4a$, so $a = 12$, and Maria is $56 - 12 = 44$.

WHY NOT (B) gives Maria's age in 4 years.

28. **C** 160

1 After the first step, $1 - \frac{2}{5} = \frac{3}{5}$ of the whole is left.

2 The second step uses $\frac{1}{2}$ of that, leaving $1 - \frac{1}{2} = \frac{1}{2}$ of it: $\frac{3}{5} \times \frac{1}{2} = \frac{3}{10}$ of the whole.

3 So $\frac{3}{10}$ of the whole is 48: $48 \div \frac{3}{10} = 160$.

WHY NOT (A) undoes only the first step. (D) subtracts both fractions from the whole amount.

29. **D** 0310

1 Add the minutes: $55 + 15 = 70$ minutes = 1 hour 10 minutes.

2 Add the hours: $21 + 5 + 1 = 27$. Past midnight, subtract 24: $27 - 24 = 3$.

3 The answer is 0310 the next day.

WHY NOT (B) forgets to carry the extra hour from the minutes. (C) adds only the hours.

30. **C** \$1,800

1 After the first step, $1 - \frac{1}{6} = \frac{5}{6}$ of the whole is left.

2 The second step uses $\frac{2}{5}$ of that, leaving $1 - \frac{2}{5} = \frac{3}{5}$ of it: $\frac{5}{6} \times \frac{3}{5} = \frac{1}{2}$ of the whole.

3 So $\frac{1}{2}$ of the whole is \$900: $900 \div \frac{1}{2} = 1,800$.

WHY NOT (A) undoes only the first step. (B) increases the amount by each fraction instead of dividing.

31. **B** 11 hours 20 minutes

1 From 2025 to midnight (2400): 3 hours 35 minutes.

2 From midnight to 0745: 7 hours 45 minutes.

3 Add: 3 hours 35 minutes + 7 hours 45 minutes = 11 hours 20 minutes.

WHY NOT (A) counts only the time after midnight. (C) borrows 100 minutes instead of 60 when counting up to midnight. (D) subtracts 0745 from 2025 instead of going past midnight.

32. **B** 48

1 Let the daughter be a years old; then Yuki is $3a$.

2 8 years ago: $3a - 8 = 5(a - 8)$, so $3a - 8 = 5a - 40$.

3 Then $32 = 2a$, so $a = 16$ and Yuki is $3 \times 16 = 48$.

WHY NOT (A) gives the daughter's age. (D) uses the second ratio for today's ages.

33. **D** Thursday

1 $108 \div 7 = 15$ weeks with 3 days left over. 15 weeks ago was also a Sunday.

2 Go back 3 more days: Saturday, Friday, Thursday.

WHY NOT (A) counts forward instead of backward. (B) moves back 15 days (the number of weeks) instead of 3. (C) counts one day too few.

34. **D** 34

1 Spaces on one side: $240 \div 15 = 16$, so one side has $16 + 1 = 17$ trees.

2 Two sides: $2 \times 17 = 34$ trees.

WHY NOT (A) counts only one side of the road. (B) forgets the extra tree at the end of each side. (C) adds only one extra tree for both sides.

35. **C** \$126

1 Each gallon covers 350 square feet: $1,000 \div 350$ is 2 with a remainder of 300.

2 2 is not quite enough, so buy 3 gallons.

3 Cost: $3 \times 42 = 126$, so \$126.

WHY NOT (A) rounds the number of gallons down, which is not enough. (B) pays for part of a gallon, which is not sold.

PART V

Practice Tests

Four full-length math sections in the paper-and-pencil format:
30 Arithmetic Reasoning questions in 36 minutes and 25
Mathematics Knowledge questions in 24 minutes.

Practice Test 1

Practice Test 2

Practice Test 3

Practice Test 4

Practice Test 1

Instructions

This test has two parts, timed like the paper-and-pencil ASVAB. **Part 1, Arithmetic Reasoning:** 30 questions, 36 minutes. **Part 2, Mathematics Knowledge:** 25 questions, 24 minutes. No calculator. Use scratch paper. Answer every question—there is no penalty for guessing. Mark your answers on the answer sheet below.

Answer Sheet

Fill in one bubble per question. Photocopy this page to retake the test.

Part 1: Arithmetic Reasoning

- 1 ☐ A ☐ B ☐ C ☐ D
- 2 ☐ A ☐ B ☐ C ☐ D
- 3 ☐ A ☐ B ☐ C ☐ D
- 4 ☐ A ☐ B ☐ C ☐ D
- 5 ☐ A ☐ B ☐ C ☐ D
- 6 ☐ A ☐ B ☐ C ☐ D
- 7 ☐ A ☐ B ☐ C ☐ D
- 8 ☐ A ☐ B ☐ C ☐ D
- 9 ☐ A ☐ B ☐ C ☐ D
- 10 ☐ A ☐ B ☐ C ☐ D
- 11 ☐ A ☐ B ☐ C ☐ D
- 12 ☐ A ☐ B ☐ C ☐ D
- 13 ☐ A ☐ B ☐ C ☐ D
- 14 ☐ A ☐ B ☐ C ☐ D
- 15 ☐ A ☐ B ☐ C ☐ D
- 16 ☐ A ☐ B ☐ C ☐ D
- 17 ☐ A ☐ B ☐ C ☐ D
- 18 ☐ A ☐ B ☐ C ☐ D
- 19 ☐ A ☐ B ☐ C ☐ D
- 20 ☐ A ☐ B ☐ C ☐ D
- 21 ☐ A ☐ B ☐ C ☐ D
- 22 ☐ A ☐ B ☐ C ☐ D
- 23 ☐ A ☐ B ☐ C ☐ D
- 24 ☐ A ☐ B ☐ C ☐ D
- 25 ☐ A ☐ B ☐ C ☐ D
- 26 ☐ A ☐ B ☐ C ☐ D
- 27 ☐ A ☐ B ☐ C ☐ D
- 28 ☐ A ☐ B ☐ C ☐ D
- 29 ☐ A ☐ B ☐ C ☐ D
- 30 ☐ A ☐ B ☐ C ☐ D

Part 2: Mathematics Knowledge

- 1 ☐ A ☐ B ☐ C ☐ D
- 2 ☐ A ☐ B ☐ C ☐ D
- 3 ☐ A ☐ B ☐ C ☐ D
- 4 ☐ A ☐ B ☐ C ☐ D
- 5 ☐ A ☐ B ☐ C ☐ D
- 6 ☐ A ☐ B ☐ C ☐ D
- 7 ☐ A ☐ B ☐ C ☐ D
- 8 ☐ A ☐ B ☐ C ☐ D
- 9 ☐ A ☐ B ☐ C ☐ D
- 10 ☐ A ☐ B ☐ C ☐ D
- 11 ☐ A ☐ B ☐ C ☐ D
- 12 ☐ A ☐ B ☐ C ☐ D
- 13 ☐ A ☐ B ☐ C ☐ D
- 14 ☐ A ☐ B ☐ C ☐ D
- 15 ☐ A ☐ B ☐ C ☐ D
- 16 ☐ A ☐ B ☐ C ☐ D
- 17 ☐ A ☐ B ☐ C ☐ D
- 18 ☐ A ☐ B ☐ C ☐ D
- 19 ☐ A ☐ B ☐ C ☐ D
- 20 ☐ A ☐ B ☐ C ☐ D
- 21 ☐ A ☐ B ☐ C ☐ D
- 22 ☐ A ☐ B ☐ C ☐ D
- 23 ☐ A ☐ B ☐ C ☐ D
- 24 ☐ A ☐ B ☐ C ☐ D
- 25 ☐ A ☐ B ☐ C ☐ D

Part 1: Arithmetic Reasoning

Score: ____ / 30

Target time: 36 minutes

- At a recruiting station, one clerk can enter a stack of enlistment forms into the computer in 10 hours. A second clerk can enter the same stack in 6 hours. How long will it take the two clerks working together?
A 3 hours B 3 hours 45 minutes
C 8 hours D 16 hours
- A water trough on Jasmine's farm is 6 feet long, 3 feet wide, and 2 feet deep. How many gallons of water does it hold when it is full? (1 cubic foot holds 7.5 gallons.)
A 36 gal B 135 gal C 252 gal D 270 gal
- Leila works at a furniture store and earns a base salary of \$300 per week plus a 5% commission on her sales. Her sales this week total \$6,300. What is her total pay for the week?
A \$315 B \$330 C \$615 D \$3,450
- Keisha borrows \$8,000 from a credit union to buy a used car. The loan charges 8% simple interest per year. What is the total amount she must pay back at the end of 4 years?
A \$10,560 B \$11,560 C \$13,560 D \$33,600
- A spinner has 4 equal sections, and 1 of them is red. If the spinner is spun 60 times, about how many times should it land on red?
A 13 B 15 C 19 D 45
- A dining facility has 16 gallons of a drink that is 30% juice. How many gallons of water must be added so that the drink is 20% juice?
A 5 gallons B 8 gallons
C 10 gallons D 24 gallons
- Carlos's footlocker has a combination lock with a 3-digit code. Each digit can be any number from 0 to 9, and digits may repeat. How many different codes are possible?
A 720 B 1,000 C 1,100 D 1,300
- At a charity raffle, 150 tickets are sold and one winning ticket is drawn. Leila bought 3 tickets. What is the probability that she does *not* win?
A $\frac{1}{50}$ B $\frac{1}{49}$ C $\frac{49}{50}$ D $\frac{149}{150}$
- Petty Officer Jensen is furnishing an apartment off base. A gaming console that regularly costs \$510 is on sale for 30% off. A sales tax of 7% is charged on the sale price. How much does she pay in all?
A \$381.99 B \$392.70 C \$431.99 D \$545.70
- Two cars start 110 miles apart and travel toward each other on the same road. One goes 45 miles per hour and the other goes 65 miles per hour. In how many hours will they meet?
A 1 hour B 2 hours
C 3 hours D 5.5 hours
- An engineer platoon's chainsaws run on gasoline mixed with oil in the ratio 40 : 1 (gasoline to oil). How many fluid ounces of oil should be mixed with $2\frac{1}{2}$ gallons of gasoline? (1 gallon = 128 fluid ounces)
A 2 fluid ounces B 8 fluid ounces
C 16 fluid ounces D 100 fluid ounces
- A water can held 14 gal 1 qt of water. A squad used 1 gal 2 qt. How much water is left?
A 11 gal 3 qt B 12 gal 2 qt
C 12 gal 3 qt D 13 gal 1 qt
- Tyrone buys a mountain bike during a 15%-off sale. The regular price is \$300, and 7% sales tax is added to the sale price. How much does he pay, including tax?
A \$272.85 B \$276.00 C \$321.00 D \$422.85
- Sofia works at a car dealership and earns a base salary of \$2,200 per month plus a 5% commission on her sales. Her sales this month total \$23,000. What is her total pay for the month?
A \$1,150 B \$3,350 C \$13,700 D \$25,200

15. At a grocery store, bottled water costs \$0.89 per bottle, or \$18.24 for a case of 24. A camp director needs 64 bottles. The plan is to buy as many full cases as possible and the rest as single bottles. What is the total cost?
A \$36.48 B \$50.72 C \$54.72 D \$56.96
16. A submarine is at -900 feet (below sea level). It rises 30 feet per minute for 9 minutes and then dives 250 feet. What is its new position?
A $-1,420$ ft B $-1,120$ ft
C $-1,030$ ft D -880 ft
17. Notebooks cost \$3 each and pens cost \$1 each. Which expression gives the total cost, in dollars, of x notebooks and y pens?
A $4(x + y)$ B $3x + 1y$ C $4xy$ D $1x + 3y$
18. In a PT test, the soldiers of a squad did these numbers of push-ups: 56, 47, 25, 25, 62, 29. What is the median of the data?
A 25 B 29 C 37.5 D 38
19. A platoon starts a road march at 3 miles per hour. Two hours later, Private Chen follows the same route in a Humvee carrying water at 27 miles per hour. How many hours after the Humvee leaves will it catch up?
A 0.25 hours B 0.75 hours
C 1 hour D 2.25 hours
20. A platoon of 24 soldiers took a road march. $\frac{11}{12}$ of them carried the heavy rucksack, and $\frac{1}{2}$ of those soldiers finished in under three hours. How many soldiers carried the heavy rucksack and finished in under three hours?
A 8 B 11 C 12 D 22
21. A clock tower strikes 6 times at 6 o'clock, and it takes 20 seconds from the first strike to the last. At the same pace, how many seconds does it take to strike 12 times at 12 o'clock?
A 34 seconds B 40 seconds
C 44 seconds D 48 seconds
22. Brandon deposits \$2,600 in a savings account that pays 4% simple interest per year. How many years will it take for the account to earn \$208 in interest?
A 0.02 years B 1 year
C 2 years D 24 years
23. A ship's crew starts loading cargo at 2115, and the job takes 9 hours 55 minutes. At what military time will the loading be finished?
A 0710 B 0810 C 0840 D 1910
24. Priya leans a 15-foot ladder against a building, with its foot 9 feet from the wall. The foot slips until it is 12 feet from the wall. How far down the wall does the top of the ladder slide?
A 3 ft B 6 ft C 9 ft D 12 ft
25. Keisha is three times as old as her son. In 9 years, Keisha will be twice as old as the son. How old is Keisha now?
A 9 B 18 C 27 D 36
26. A charter bus carrying new recruits leaves the processing station at 9:15 a.m. and travels 60 miles to the training base at 60 miles per hour. After a 20-minute stop, it goes another 75 miles at 50 miles per hour. At what time does it arrive?
A 11:50 a.m. B 1:05 p.m.
C 11:45 a.m. D 12:05 p.m.
27. Derek buys 3.5 pounds of chicken at \$3.30 per pound and 4 pounds of ground beef at \$5.40 per pound. How much does he spend?
A \$16.20 B \$24.75 C \$32.15 D \$33.15
28. A Humvee patrol drives 120 miles to a checkpoint at 20 miles per hour and returns at 30 miles per hour. What is the average speed for the round trip?
A 12 mph B 18 mph C 24 mph D 25 mph

29. Two pipes feed a water tank on a farm. One pipe alone fills the empty tank in 5 hours, and the other alone in 3 hours. An outlet pipe used for irrigation drains a full tank in 2 hours. With all three pipes open, how long will it take to fill the empty tank?
- A 2 minutes B 4 hours
C 30 hours D 36 hours
30. Grace works as a hotel desk clerk and earns \$17 per hour, with time-and-a-half for every hour over 40 in a week. Last week she worked 8 hours a day Monday through Friday and 8 hours on Saturday. How much did she earn for the week?
- A \$816 B \$884 C \$952 D \$1,224

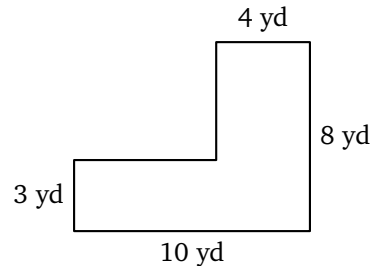
Part 2: Mathematics Knowledge

Score: ____ / 25

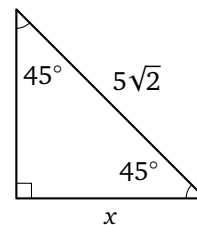
Target time: 24 minutes

- Evaluate $(-6)^3$.
A -256 B -236 C -216 D 216
- What is the area of a circle with a diameter of 38 feet?
A $361\pi \text{ ft}^2$ B $400\pi \text{ ft}^2$
C $722\pi \text{ ft}^2$ D $1,444\pi \text{ ft}^2$
- The volume of a cube-shaped water tank is 1,728 cubic meters. What is the total area of its six faces?
A 72 m^2 B 144 m^2 C 576 m^2 D 864 m^2
- Solve for x : $\frac{10}{x} = \frac{6}{15}$
A 4 B 10 C 19 D 25
- What is $\sqrt{14} \cdot \sqrt{12}$ in simplest form?
A $\sqrt{26}$ B $2\sqrt{42}$ C $4\sqrt{42}$ D 168
- A student has four sets of sticks. Which set can be put together, end to end, to form a triangle?
A 2, 6, 8 B 5, 6, 15
C 7, 11, 22 D 6, 12, 16
- What is the midpoint of the segment joining $(3, -3)$ and $(-5, -9)$?
A $(-6, -1)$ B $(-1, -6)$
C $(-4, -3)$ D $(-2, -12)$
- If $\frac{x}{6} - \frac{x}{10} = 3$, what is the value of x ?
A 12 B 45 C 46 D 180
- Which fraction is equal to 0.056?
A $\frac{7}{1250}$ B $\frac{7}{125}$ C $\frac{14}{25}$ D $\frac{125}{7}$
- What are the solutions of $2x^2 + 3x - 2 = 0$?
A $x = -4$ or $x = 1$ B $x = -\frac{1}{2}$ or $x = 2$
C $x = -2$ or $x = \frac{1}{2}$ D $x = \frac{1}{2}$
- What is the value of $38 - 4^2 \div 4 \times 5$?
A 10 B 16 C 18 D 28

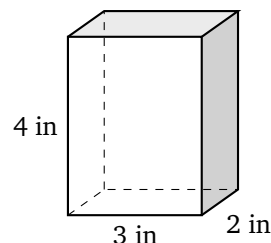
- The figure shows a garden. All corners are right angles. The sides shown measure 10 yards, 8 yards, 4 yards, and 3 yards. What is the perimeter?



- A 25 yd B 26 yd C 32 yd D 36 yd
- Simplify: $-6^2 + (-11)^2$
A -157 B -34 C 85 D 157
 - The figure shows a 45° - 45° - 90° triangle in which the hypotenuse is $5\sqrt{2}$. What is the value of x ?



- A $\frac{5\sqrt{2}}{2}$ B 5 C $5\sqrt{3}$ D 10
- What is 50% of 10% of 780?
A 29 B 34 C 39 D 468
 - If $0.2x - 0.5 = 1.7$, what is the value of x ?
A 11 B 12 C 13 D 110
 - The closed rectangular box shown is 3 inches long, 2 inches wide, and 4 inches high. What is its total surface area?



- A 24 in^2 B 26 in^2 C 40 in^2 D 52 in^2

18. What is the value of $4ab - b^2$ when $a = -1$ and $b = -1$?
A -3 B 3 C 5 D 6
19. Simplify $(3x^{-4})(4x^2)$ and write the answer with positive exponents.
A $12x^6$ B $\frac{12}{x^2}$ C $\frac{12}{x^8}$ D $\frac{7}{x^2}$
20. A rectangle that is 14 feet by 3 feet has a perimeter of 34 feet. If its length and width are both multiplied by 4, what is the perimeter of the new rectangle?
A 34 ft B 42 ft C 136 ft D 544 ft
21. What is the least common multiple of 14, 6, and 5?
A 210 B 224 C 290 D 420
22. Which of the following is the solution of $|x - 4| > 3$?
A $x < 1$ or $x > 7$ B $x < -7$ or $x > -1$
C $1 \leq x \leq 7$ D $x > 7$
23. What is $5\frac{1}{3} - 2\frac{5}{12}$?
A $2\frac{11}{12}$ B $3\frac{1}{12}$ C $3\frac{11}{12}$ D $7\frac{3}{4}$
24. What is the value of $x + y$ in the solution of this system?
$$3x - 2y = -32$$
$$4x - y = -26$$

A 6 B 7 C 8 D 10
25. Marcus runs laps around a path shaped like a regular polygon, turning 60° at each corner. How many sides does the path have?
A 2 B 3 C 4 D 6

Answer Key: Practice Test 1

The small number after each answer is the chapter that teaches the skill.

Part 1: Arithmetic Reasoning

1 B ch. 25	7 B ch. 26	13 A ch. 22	19 A ch. 24	25 C ch. 27
2 D ch. 20	8 C ch. 26	14 B ch. 23	20 B ch. 4	26 D ch. 24
3 C ch. 23	9 A ch. 22	15 B ch. 22	21 C ch. 27	27 D ch. 5
4 A ch. 23	10 A ch. 24	16 D ch. 2	22 C ch. 23	28 C ch. 24
5 B ch. 26	11 B ch. 25	17 B ch. 10	23 A ch. 27	29 C ch. 25
6 B ch. 25	12 C ch. 27	18 D ch. 26	24 A ch. 17	30 B ch. 22

Part 2: Mathematics Knowledge

1 C ch. 8	6 D ch. 17	11 C ch. 1	16 A ch. 11	21 A ch. 3
2 A ch. 19	7 B ch. 21	12 D ch. 18	17 D ch. 20	22 A ch. 12
3 D ch. 20	8 B ch. 11	13 C ch. 2	18 B ch. 10	23 A ch. 4
4 D ch. 7	9 B ch. 5	14 B ch. 17	19 B ch. 14	24 A ch. 13
5 B ch. 9	10 C ch. 15	15 C ch. 6	20 C ch. 18	25 D ch. 16

Scoring your practice test

Percent correct	AR (of 30)	MK (of 25)
90% or more — excellent	27–30	23–25
75–89% — good	23–26	19–22
60–74% — keep practicing	18–22	15–18
below 60% — review the chapters	0–17	0–14

These bands measure your progress in this book only. Your real AFQT percentile depends on all four AFQT subtests and is scaled by the Department of Defense.

Explanations: Arithmetic Reasoning

1. **B** 3 hours 45 minutes

1 Turn each time into a rate (the part of the job done in one hour): the first clerk does $\frac{1}{10}$ of the job per hour, and the second clerk does $\frac{1}{6}$.

2 Add the rates (common denominator 30): $\frac{1}{10} + \frac{1}{6} = \frac{3}{30} + \frac{5}{30} = \frac{8}{30} = \frac{4}{15}$ of the job per hour.

3 Time = 1 job ÷ rate: $1 \div \frac{4}{15} = \frac{15}{4} = 3\frac{3}{4}$ hours.

4 Change the fraction of an hour to minutes: $\frac{3}{4} \times 60 = 45$ minutes. So they need 3 hours 45 minutes.

TIP Product over sum: $\frac{6 \times 10}{6 + 10} = \frac{60}{16} = 3\frac{3}{4}$. The answer must be less than the faster time, 6 hours.

WHY NOT (A) halves the faster time, as if both worked as fast as the faster one. (C) averages the two times. (D) adds the two times instead of adding the two rates.

2. **D** 270 gal

1 Volume: $6 \times 3 \times 2 = 36$ cubic feet.

2 Each cubic foot holds 7.5 gallons: $36 \times 7.5 = 270$ gallons.

TIP Multiplying by 7.5 is the same as multiplying by 15 and halving: $36 \times 15 = 540$, and $540 \div 2 = 270$.

WHY NOT (A) finds the volume in cubic feet but forgets to change it to gallons. (C) uses 7 gallons per cubic foot instead of 7.5.

3. **C** \$615

1 Commission: $0.05 \times \$6,300 = \315 .

2 Add the base salary: $\$300 + \$315 = \$615$.

WHY NOT (A) is the commission only; it leaves out the base salary. (B) takes the commission on the base salary plus sales. (D) writes 5% as 0.5.

4. **A** \$10,560
 1 One year's interest: $\$8,000 \times 0.08 = \640 .
 2 Interest for the whole time: $\$640 \times 4 = \$2,560$.
 3 Add it to the principal: $\$8,000 + \$2,560 = \$10,560$.
WHY NOT (D) writes 8% as 0.8 instead of 0.08.
5. **B** 15
 1 $P(\text{red}) = \frac{1}{4}$.
 2 Expected count: $\frac{1}{4} \times 60 = 15$.
WHY NOT (D) is the expected number of spins that do *not* land on red.
6. **B** 8 gallons
 1 Pure juice now: $0.3 \times 16 = 4.8$ gallons. Adding water does not change this amount.
 2 In the new mixture these 4.8 gallons must be 20% of the total: $4.8 \div 0.2 = 24$ gallons in all.
 3 Water to add: $24 - 16 = 8$ gallons.
WHY NOT (D) is the final amount of the mixture, not the water added.
7. **B** 1,000
 1 Count the choices for each position: 10, 10, 10.
 2 Multiply (counting principle): $10 \times 10 \times 10 = 1,000$.
WHY NOT (A) does not let the digits repeat.
8. **C** $\frac{49}{50}$
 1 Tickets she did not buy: $150 - 3 = 147$.
 2 Probability = $\frac{147}{150} = \frac{49}{50}$.
WHY NOT (A) is the probability that she *does* win. (B) compares the tickets bought to the tickets not bought. (D) subtracts only one ticket.
9. **A** \$381.99
 1 Find the discount: $0.3 \times \$510 = \153 .
 2 Sale price: $\$510 - \$153 = \$357$.
 3 Tax on the *sale* price: $0.07 \times \$357 = \24.99 .
 4 Total: $\$357 + \$24.99 = \$381.99$.
TIP Or chain the multipliers: $\$510 \times 0.7 \times 1.07 = \381.99 .
WHY NOT (B) figures the tax on the regular price instead of the sale price. (D) adds the tax but forgets the discount.
10. **A** 1 hour
 1 Moving toward each other, the gap closes at the sum of the speeds: $45 + 65 = 110$ miles per hour.
 2 Time to close 110 miles: $110 \div 110 = 1$ hour.
TIP Check: in 1 hour they cover $45 + 65 = 110$ miles.
WHY NOT (B) divides by the average of the two speeds instead of their sum. (D) subtracts the speeds, but the two are closing the gap together.
11. **B** 8 fluid ounces
 1 Change gallons to fluid ounces: $2\frac{1}{2} \times 128 = 320$ fluid ounces of gasoline.
 2 The oil is $\frac{1}{40}$ of the gasoline amount: $320 \div 40 = 8$ fluid ounces.
WHY NOT (A) uses 32 ounces in a gallon (that is a quart). (D) multiplies by the ratio instead of dividing.
12. **C** 12 gal 3 qt
 1 You can't take 2 qt from 1 qt, so borrow 1 gal: 14 gal 1 qt becomes 13 gal 5 qt.
 2 Subtract: $5 - 2 = 3$ qt and $13 - 1 = 12$ gal.
 3 What is left: 12 gal 3 qt.
WHY NOT (D) subtracts the smaller number of quarts from the larger instead of borrowing.
13. **A** \$272.85
 1 Find the discount: $0.15 \times \$300 = \45 .
 2 Sale price: $\$300 - \$45 = \$255$.
 3 Tax on the *sale* price: $0.07 \times \$255 = \17.85 .
 4 Total: $\$255 + \$17.85 = \$272.85$.
TIP Or chain the multipliers: $\$300 \times 0.85 \times 1.07 = \272.85 .
WHY NOT (B) figures the tax on the regular price instead of the sale price. (C) adds the tax but forgets the discount.
14. **B** \$3,350
 1 Commission: $0.05 \times \$23,000 = \$1,150$.
 2 Add the base salary: $\$2,200 + \$1,150 = \$3,350$.
WHY NOT (A) is the commission only; it leaves out the base salary. (C) writes 5% as 0.5. (D) adds the sales instead of the commission.
15. **B** \$50.72
 1 Full cases: $64 \div 24$ is 2 with 16 left over, so buy 2 cases and 16 single bottles.
 2 Cases: $2 \times \$18.24 = \36.48 . Singles: $16 \times \$0.89 = \14.24 .
 3 Total: $\$36.48 + \$14.24 = \$50.72$.
WHY NOT (A) forgets the 16 single bottles. (C) buys one more full case instead of 16 single bottles. (D) buys every bottle singly.
16. **D** -880 ft
 1 The rise: $30 \times 9 = 270$ feet up.
 2 After rising: $-900 + 270 = -630$ feet.
 3 After diving: $-630 - 250 = -880$ feet.
WHY NOT (A) treats the rise as going down. (B) forgets to multiply the rate by the number of minutes.

17. **B** $3x + 1y$

- Each notebook costs \$3, so x of them cost $3x$ dollars.
- Each pen costs \$1, so y of them cost $1y$ dollars.
- Add the two costs: $3x + 1y$.

WHY NOT (A) charges both prices for every item. (C) combines unlike terms. (D) matches each price with the wrong item.

18. **D** 38

- Put the 6 values in order: 25, 25, 29, 47, 56, 62.
- With an even number of values, the median is the average of the two middle ones (the 3rd and 4th): $(29 + 47) \div 2 = 76 \div 2 = 38$.

WHY NOT (A) averages the two middle numbers without putting the list in order first. (B) uses only one of the two middle values.

19. **A** 0.25 hours

- Head start: in 2 hours the first one goes $3 \times 2 = 6$ miles.
- The gap closes at the difference of the speeds: $27 - 3 = 24$ miles per hour.
- Time to close it: $6 \div 24 = 0.25$ hours.

TIP Check: in 0.25 hours the chaser goes $27 \times 0.25 = 6.75$ miles, and the first one has gone $3 \times 2.25 = 6.75$ miles.

WHY NOT (D) counts the time from when the first one left.

20. **B** 11

- First group: $\frac{11}{12} \times 24 = 22$.
- Then take $\frac{1}{2}$ of that group: $\frac{1}{2} \times 22 = 11$.

WHY NOT (C) takes $\frac{1}{2}$ of the whole group instead of the smaller group. (D) is the size of the first group only.

21. **C** 44 seconds

- 6 strikes have $6 - 1 = 5$ gaps, so each gap is $20 \div 5 = 4$ seconds.
- 12 strikes have $12 - 1 = 11$ gaps: $11 \times 4 = 44$ seconds.

WHY NOT (B) sets up a simple proportion with the number of strikes. (D) counts the strikes instead of the gaps between them.

22. **C** 2 years

- One year's interest: $\$2,600 \times 0.04 = \104 .
- Number of years: $\$208 \div \$104 = 2$.

TIP Check: $\$104 \times 2 = \208 .

WHY NOT (A) forgets to change the percent to a decimal. (D) gives the number of months, not years.

23. **A** 0710

- Add the minutes: $15 + 55 = 70$ minutes = 1 hour 10 minutes.
- Add the hours: $21 + 9 + 1 = 31$. Past midnight, subtract 24: $31 - 24 = 7$.
- The answer is 0710 the next day.

WHY NOT (D) subtracts 12 hours instead of 24 after passing midnight.

24. **A** 3 ft

- Before: height = $\sqrt{15^2 - 9^2} = \sqrt{144} = 12$ feet.
- After: height = $\sqrt{15^2 - 12^2} = \sqrt{81} = 9$ feet.
- The top slides down $12 - 9 = 3$ feet.

WHY NOT (C) gives the final height of the ladder. (D) gives the starting height of the ladder.

25. **C** 27

- Let the son be a years old; then Keisha is $3a$.
- In 9 years: $3a + 9 = 2(a + 9)$, so $3a + 9 = 2a + 18$.
- Then $1a = 9$, so $a = 9$ and Keisha is $3 \times 9 = 27$.

WHY NOT (A) gives the son's age. (B) uses the second ratio for today's ages. (D) gives Keisha's age in 9 years.

26. **D** 12:05 p.m.

- First part: $60 \div 60 = 1$ hour. Second part: $75 \div 50 = 1.5$ hours (1 hour 30 minutes).
- Total time: 1 hour + 1 hour 30 minutes + 20 minutes = 2 hours 50 minutes.
- Add to the start time: 9:15 a.m. plus 2 hours 50 minutes is 12:05 p.m.

WHY NOT (A) uses the first speed for the whole trip. (C) forgets the 20-minute stop.

27. **D** \$33.15

- First item: $3.5 \times 3.30 = 11.55$.
- Second item: $4 \times 5.40 = 21.60$.
- Total: $11.55 + 21.60 = 33.15$, so \$33.15.

TIP For 3.5×3.30 , think 3×3.30 plus half of \$3.30.

WHY NOT (A) adds the weights and prices instead of multiplying. (B) uses the first price for both items.

28. **C** 24 mph

- Time going: $120 \div 20 = 6$ hours. Time returning: $120 \div 30 = 4$ hours.
- Total distance: $2 \times 120 = 240$ miles. Total time: $6 + 4 = 10$ hours.
- Average speed: $240 \div 10 = 24$ miles per hour.

TIP Shortcut for equal distances: $2ab \div (a + b) = 2 \times 20 \times 30 \div 50 = 24$.

WHY NOT (A) uses the one-way distance instead of the round-trip distance. (D) averages the two speeds, but more time is spent at the slower speed.

29. **C** 30 hours

- 1 Rates per hour: the feed pipes add $\frac{1}{5}$ and $\frac{1}{3}$ of the tank; the outlet pipe takes away $\frac{1}{2}$.
- 2 Net rate (common denominator 30): $\frac{1}{5} + \frac{1}{3} - \frac{1}{2} = \frac{6}{30} + \frac{10}{30} - \frac{15}{30} = \frac{1}{30}$ of the tank per hour.
- 3 Time = 1 tank $\div$ rate = $1 \div \frac{1}{30} = 30$ hours.

WHY NOT (A) stops at the net rate and forgets to flip it into a time. (B) averages the two fill times and ignores the outflow.

30. **B** \$884

- 1 Total hours: $8 \times 5 + 8 = 48$ hours.
- 2 Only hours over 40 are overtime: $48 - 40 = 8$ overtime hours.
- 3 Regular pay: $40 \times \$17 = \680 . Overtime rate: $1.5 \times \$17 = \25.50 per hour.
- 4 Overtime pay: $8 \times \$25.50 = \204 .
- 5 Total: $\$680 + \$204 = \$884$.

WHY NOT (A) pays every hour at the regular rate. (C) pays double time instead of time-and-a-half. (D) pays time-and-a-half for all the hours, not just the hours over 40.

Explanations: Mathematics Knowledge

1. **C** -216

- 1 The parentheses make -6 the base: $(-6)^3 = (-6) \times (-6) \times (-6)$.
- 2 Count the negative signs: there are 3, and an odd number of negative factors gives a negative product.
- 3 Since $6^3 = 216$, the answer is -216.

WHY NOT (D) gets the sign wrong: an odd number of negative factors gives a negative product.

2. **A** $361\pi \text{ ft}^2$

- 1 The formula $A = \pi r^2$ needs the radius, which is half the diameter: $r = 38 \div 2 = 19$ feet.
- 2 Square the radius: $19^2 = 361$.
- 3 So $A = \pi \times 361 = 361\pi$ square feet.

WHY NOT (C) squares the diameter and then halves it; halve first, then square. (D) uses the diameter in place of the radius.

3. **D** 864 m^2

- 1 Find the edge: which number cubed is 1,728? $12 \times 12 \times 12 = 1,728$, so $s = 12$ meters.
- 2 One face: $12^2 = 144$. Six faces: $6 \times 144 = 864$ square meters.

WHY NOT (A) forgets to square the edge. (B) finds the area of only one face. (C) counts only four of the six faces.

4. **D** 25

- 1 Cross-multiply: $6 \times x = 10 \times 15$, so $6x = 150$.
- 2 Divide both sides by 6: $x = 150 \div 6 = 25$.
- 3 Check: $\frac{10}{25}$ and $\frac{6}{15}$ both simplify to $\frac{2}{5}$. ✓

WHY NOT (A) solves as if x were on top, giving $10 \times 6 \div 15$. (C) adds the difference of the numerators instead of scaling.

5. **B** $2\sqrt{42}$

- 1 Multiply the numbers under the radicals: $\sqrt{14} \cdot \sqrt{12} = \sqrt{14 \times 12} = \sqrt{168}$.
- 2 Simplify: $\sqrt{168} = \sqrt{4 \cdot 42} = 2\sqrt{42}$.

WHY NOT (A) adds the numbers under the radicals instead of multiplying. (C) pulls out 4 instead of its square root, 2. (D) drops the radical sign.

6. **D** 6, 12, 16

- 1 For three lengths to form a triangle, the two shorter ones must add up to *more* than the longest one.
- 2 Test each set: 2, 6, 8: $2 + 6 = 8 = 8 \times$; 5, 6, 15: $5 + 6 = 11 < 15 \times$; 6, 12, 16: $6 + 12 = 18 > 16 \checkmark$; 7, 11, 22: $7 + 11 = 18 < 22 \times$.
- 3 Only 6, 12, 16 works.

WHY NOT (A) fails because $2 + 6 = 8$, which is not more than 8. (B) fails because $5 + 6 = 11$ is less than 15. (C) fails because $7 + 11 = 18$ is less than 22.

7. **B** $(-1, -6)$

- 1 Average the x-coordinates: $\frac{3+(-5)}{2} = \frac{-2}{2} = -1$.
- 2 Average the y-coordinates: $\frac{-3+(-9)}{2} = \frac{-12}{2} = -6$.
- 3 The midpoint is $(-1, -6)$.

WHY NOT (A) swaps the x- and y-coordinates. (C) subtracts the coordinates instead of adding them. (D) adds the coordinates but forgets to divide by 2.

8. **B** 45

- 1 Clear the fractions: multiply *every* term by the LCD of 6 and 10, which is 30.
- 2 $30 \cdot \frac{x}{6} = 5x$, $30 \cdot \frac{x}{10} = 3x$, and $30 \cdot 3 = 90$. So $5x - 3x = 90$.
- 3 Combine like terms: $2x = 90$.
- 4 Divide by 2: $x = \frac{90}{2} = 45$.
- 5 Check: $\frac{45}{6} - \frac{45}{10} = \frac{15}{2} - \frac{9}{2} = 3$. ✓

WHY NOT (A) subtracts the denominators, as if the result were $\frac{x}{4}$.

 9. **B** $\frac{7}{125}$

- 1 The last digit is in the thousandth place, so $0.056 = \frac{56}{1000}$.
- 2 Simplify by dividing the top and bottom by 8: $\frac{56}{1000} = \frac{7}{125}$.

WHY NOT (A) reads the last digit as ten-thousandths instead of thousandths. (C) reads the last digit as hundredths instead of thousandths. (D) flips the fraction.

 10. **C** $x = -2$ or $x = \frac{1}{2}$

- 1 Here $a = 2$, $b = 3$, $c = -2$.
- 2 Compute $b^2 - 4ac$: $3^2 - 4(2)(-2) = 9 + 16 = 25$, and $\sqrt{25} = 5$.
- 3 $x = \frac{-3 \pm 5}{4}$: either $x = \frac{-3+5}{4} = \frac{1}{2}$ or $x = \frac{-3-5}{4} = -2$.

TIP You can check by factoring: $2x^2 + 3x - 2 = (2x - 1)(x + 2)$.

WHY NOT (A) divides by 2 instead of by $2a = 4$. (B) uses 3 instead of -3 in the formula. (D) uses only the + sign of $\pm$.

 11. **C** 18

- 1 Exponent next: $4^2 = 4 \times 4 = 16$, which leaves $38 - 16 \div 4 \times 5$.
- 2 Divide (multiplication and division go left to right): $16 \div 4 = 4$, which leaves $38 - 4 \times 5$.
- 3 Multiply before you add or subtract: $4 \times 5 = 20$, which leaves $38 - 20$.
- 4 Subtract: $38 - 20 = 18$.

WHY NOT (D) doubles instead of squaring.

 12. **D** 36 yd

- 1 Find the missing corner: it is $10 - 4 = 6$ yd wide and $8 - 3 = 5$ yd tall.
 - 2 Go around the shape and add all six sides: $10 + 3 + 6 + 5 + 4 + 8 = 36$ yards.
- TIP** Shortcut: the two inside edges together make up for the missing pieces of the top and side, so the perimeter equals that of a 10×8 rectangle: $2 \times (10 + 8) = 36$.

WHY NOT (A) adds only the four labeled sides and leaves out the two edges of the cut-out corner.

 13. **C** 85

- 1 In -6^2 the exponent applies only to 6: $-6^2 = -(6 \times 6) = -36$.
- 2 In $(-11)^2$ the parentheses include the sign: $(-11)(-11) = 121$.
- 3 Add: $-36 + 121 = 85$.

WHY NOT (A) treats $(-11)^2$ as -11^2 . (B) multiplies by 2 instead of squaring. (D) treats -6^2 as $(-6)^2$.

 14. **B** 5

- 1 In a 45° - 45° - 90° triangle the sides are in the ratio $1 : 1 : \sqrt{2}$ (leg : leg : hypotenuse).
- 2 Leg = hypotenuse $\div \sqrt{2} = 5\sqrt{2} \div \sqrt{2} = 5$.

WHY NOT (A) halves the hypotenuse, the 30° - 60° - 90° rule. (C) uses $\sqrt{3}$ instead of $\sqrt{2}$. (D) multiplies by $\sqrt{2}$ instead of dividing.

 15. **C** 39

- 1 Work from the inside out. First, 10% of 780: $0.1 \times 780 = 78$.
- 2 Then 50% of that result: $0.5 \times 78 = 39$.

TIP Percents of percents multiply: $0.5 \times 0.1 = 0.05$, and $0.05 \times 780 = 39$.

WHY NOT (D) adds the two percents.

 16. **A** 11

- 1 Clear the decimals: multiply *every* term by 10. $2x - 5 = 17$.
- 2 Undo the subtraction: add 5 to both sides. $2x = 17 + 5 = 22$.
- 3 Divide both sides by 2: $x = \frac{22}{2} = 11$.

TIP You can also work with the decimals directly: $0.2x = 2.2$, and $2.2 \div 0.2 = 11$.

WHY NOT (D) slips the decimal point.

 17. **D** 52 in^2

- 1 The box has three pairs of matching faces: top and bottom ($3 \times 2 = 6$), front and back ($3 \times 4 = 12$), and the two ends ($2 \times 4 = 8$).
- 2 Add one face of each pair: $6 + 12 + 8 = 26$.
- 3 Double it for the matching faces: $2 \times 26 = 52$ square inches.

WHY NOT (A) finds the volume, not the surface area. (B) adds three faces but forgets that each face has a matching opposite face. (C) leaves out the top and bottom.

18. **B** 3

- 1 Substitute, putting each value in parentheses: $4(-1)(-1) - (-1)^2$.
- 2 Powers first: $(-1)^2 = 1$.
- 3 Multiply each term (watch the signs): $4 - 1$.
- 4 Add and subtract from left to right: $4 - 1 = 3$.

WHY NOT (C) treats $(-1)^2$ as -1 .19. **B** $\frac{12}{x^2}$

- 1 Multiply the coefficients: $3 \cdot 4 = 12$.
- 2 Add the exponents: $x^{-4} \cdot x^2 = x^{-4+2} = x^{-2}$, and $x^{-2} = \frac{1}{x^2}$.
- 3 So the product is $\frac{12}{x^2}$.

WHY NOT (A) ignores the negative sign on the exponent. (C) multiplies the exponents instead of adding them. (D) adds the coefficients instead of multiplying them.20. **C** 136 ft

- 1 Perimeter is a length (a sum of sides), so when every side is multiplied by 4, the perimeter is multiplied by 4 too.
- 2 $34 \times 4 = 136$, so the new perimeter is 136 feet.

WHY NOT (A) gives the original perimeter. (B) adds 4 to the length and width instead of multiplying. (D) multiplies by 16, which is how the area changes.21. **A** 210

- 1 Write each number as a product of primes: 5 (prime), $6 = 2 \times 3$, $14 = 2 \times 7$.
- 2 Take every prime that appears, with its highest power: $2 \times 3 \times 5 \times 7 = 210$.
- 3 Check: 210 divided by each of the numbers gives a whole number.

WHY NOT (D) is divisible by all of them, but it is not the smallest such number.22. **A** $x < 1$ or $x > 7$

- 1 $|x - 4| > 3$ means $x - 4$ is *more than* 3 away from zero, so there are two pieces: $x - 4 < -3$ or $x - 4 > 3$.
- 2 Solve each piece: $x < 1$ or $x > 7$.
- 3 Check with $x = 8$: $|8 - 4| = 4$, and $4 > 3$ is true. ✓

WHY NOT (B) uses the wrong sign: $x - 4$ is zero when $x = 4$, not $x = -4$. (C) gives the “between” set, which is for $<$. (D) keeps only the positive case; the inside can also be very negative.23. **A** $2\frac{11}{12}$

- 1 Rewrite the fractions with the common denominator 12: $5\frac{4}{12} - 2\frac{5}{12}$.
- 2 Since $\frac{4}{12}$ is less than $\frac{5}{12}$, regroup: take 1 from 5, so $5\frac{4}{12} = 4\frac{16}{12}$.
- 3 Subtract: whole numbers $4 - 2 = 2$, fractions $\frac{16}{12} - \frac{5}{12} = \frac{11}{12}$. The answer is $2\frac{11}{12}$.

WHY NOT (B) subtracts the smaller fraction from the larger instead of regrouping. (C) regroupes but forgets to take 1 from the whole number. (D) adds instead of subtracting.24. **A** 6

- 1 Multiply *every* term of the second equation by 2 so the y -terms will cancel: $8x - 2y = -52$.
- 2 Subtract the first equation from the new second (subtract *every* term). The y -terms cancel: $5x = -20$.
- 3 Divide both sides by 5: $x = \frac{-20}{5} = -4$.
- 4 Substitute $x = -4$ into $4x - y = -26$: $4(-4) - y = -26$, so $-y = -10$ and $y = 10$.
- 5 The question asks for $x + y$: $-4 + 10 = 6$.

WHY NOT (D) gives the value of y , not $x + y$.25. **D** 6

- 1 The exterior angles of any polygon add up to 360° ; the turn at each corner is an exterior angle.
- 2 All n exterior angles are equal, so $n = 360 \div 60 = 6$.

WHY NOT (B) divides 180° instead of 360° .

Practice Test 2

Instructions

This test has two parts, timed like the paper-and-pencil ASVAB. **Part 1, Arithmetic Reasoning:** 30 questions, 36 minutes. **Part 2, Mathematics Knowledge:** 25 questions, 24 minutes. No calculator. Use scratch paper. Answer every question—there is no penalty for guessing. Mark your answers on the answer sheet below.

Answer Sheet

Fill in one bubble per question. Photocopy this page to retake the test.

Part 1: Arithmetic Reasoning

- 1 ☐ A ☐ B ☐ C ☐ D
- 2 ☐ A ☐ B ☐ C ☐ D
- 3 ☐ A ☐ B ☐ C ☐ D
- 4 ☐ A ☐ B ☐ C ☐ D
- 5 ☐ A ☐ B ☐ C ☐ D
- 6 ☐ A ☐ B ☐ C ☐ D
- 7 ☐ A ☐ B ☐ C ☐ D
- 8 ☐ A ☐ B ☐ C ☐ D
- 9 ☐ A ☐ B ☐ C ☐ D
- 10 ☐ A ☐ B ☐ C ☐ D
- 11 ☐ A ☐ B ☐ C ☐ D
- 12 ☐ A ☐ B ☐ C ☐ D
- 13 ☐ A ☐ B ☐ C ☐ D
- 14 ☐ A ☐ B ☐ C ☐ D
- 15 ☐ A ☐ B ☐ C ☐ D
- 16 ☐ A ☐ B ☐ C ☐ D
- 17 ☐ A ☐ B ☐ C ☐ D
- 18 ☐ A ☐ B ☐ C ☐ D
- 19 ☐ A ☐ B ☐ C ☐ D
- 20 ☐ A ☐ B ☐ C ☐ D
- 21 ☐ A ☐ B ☐ C ☐ D
- 22 ☐ A ☐ B ☐ C ☐ D
- 23 ☐ A ☐ B ☐ C ☐ D
- 24 ☐ A ☐ B ☐ C ☐ D
- 25 ☐ A ☐ B ☐ C ☐ D
- 26 ☐ A ☐ B ☐ C ☐ D
- 27 ☐ A ☐ B ☐ C ☐ D
- 28 ☐ A ☐ B ☐ C ☐ D
- 29 ☐ A ☐ B ☐ C ☐ D
- 30 ☐ A ☐ B ☐ C ☐ D

Part 2: Mathematics Knowledge

- 1 ☐ A ☐ B ☐ C ☐ D
- 2 ☐ A ☐ B ☐ C ☐ D
- 3 ☐ A ☐ B ☐ C ☐ D
- 4 ☐ A ☐ B ☐ C ☐ D
- 5 ☐ A ☐ B ☐ C ☐ D
- 6 ☐ A ☐ B ☐ C ☐ D
- 7 ☐ A ☐ B ☐ C ☐ D
- 8 ☐ A ☐ B ☐ C ☐ D
- 9 ☐ A ☐ B ☐ C ☐ D
- 10 ☐ A ☐ B ☐ C ☐ D
- 11 ☐ A ☐ B ☐ C ☐ D
- 12 ☐ A ☐ B ☐ C ☐ D
- 13 ☐ A ☐ B ☐ C ☐ D
- 14 ☐ A ☐ B ☐ C ☐ D
- 15 ☐ A ☐ B ☐ C ☐ D
- 16 ☐ A ☐ B ☐ C ☐ D
- 17 ☐ A ☐ B ☐ C ☐ D
- 18 ☐ A ☐ B ☐ C ☐ D
- 19 ☐ A ☐ B ☐ C ☐ D
- 20 ☐ A ☐ B ☐ C ☐ D
- 21 ☐ A ☐ B ☐ C ☐ D
- 22 ☐ A ☐ B ☐ C ☐ D
- 23 ☐ A ☐ B ☐ C ☐ D
- 24 ☐ A ☐ B ☐ C ☐ D
- 25 ☐ A ☐ B ☐ C ☐ D

Part 1: Arithmetic Reasoning

Score: ____ / 30

Target time: 36 minutes

- Sofia drives at 60 miles per hour for 1 hour 15 minutes. How many miles does she drive?
A 69 miles B 75 miles
C 95 miles D 120 miles
- A unit store sells T-shirts for \$12 and pairs of socks for \$5. Which expression gives the total cost, in dollars, of x T-shirts and y pairs of socks?
A $17(x + y)$ B $12x + 5y$
C $17xy$ D $5x + 12y$
- For ruck-march snack packs, a unit mixes peanuts, raisins, and chocolate chips in the ratio 4 : 3 : 1 by weight. How many pounds of trail mix can be made if 32 pounds of peanuts are used?
A 64 pounds B 69 pounds
C 74 pounds D 256 pounds
- Two friends eat lunch together. The bill comes to \$116 before the tip. They leave an 18% tip and share the total equally. How much does each friend pay?
A \$10.44 B \$58.00 C \$58.44 D \$68.44
- On the Navy fitness test, Seaman Walker runs 1.5 miles in 15 minutes. What is his average speed in miles per hour?
A 0.1 mph B 3 mph C 5 mph D 6 mph
- A base dining facility mixes 60 pounds of 80%-lean ground beef costing \$4 per pound with 24 pounds of 93%-lean ground beef costing \$7.50 per pound. What is the cost per pound of the mixture?
A \$5.00 B \$5.75 C \$6.50 D \$420.00
- The speed limit on a road overseas is 48 kilometers per hour. About how many miles per hour is that? (1 mile $\approx$ 1.6 kilometers)
A 30 miles per hour B 32 miles per hour
C 76.8 miles per hour D 300 miles per hour
- Ana earns \$3,000 a month. She spends $\frac{1}{5}$ of it on rent and $\frac{1}{4}$ of the remainder on food. How much does she have left for other expenses?
A \$1,650 B \$1,800 C \$1,900 D \$2,400
- Airman Jensen puts an \$8,500 enlistment bonus into a savings account at the base credit union. The account pays 4% simple interest per year. How much interest will he earn in 18 months?
A \$510 B \$612 C \$6,120 D \$9,010
- The sum of three consecutive integers is 18. What is the largest of these integers?
A 5 B 7 C 10 D 11
- A medevac helicopter flies 30 miles to a field hospital at 120 miles per hour. How many minutes does the flight take?
A 15 minutes B 17 minutes
C 25 minutes D 240 minutes
- Tyrone is putting a fence around a circular garden that has a radius of 15 feet. How many feet of fencing does he need? Use 3.14 for π .
A 47.1 ft B 87.92 ft C 90 ft D 94.2 ft
- Tasha buys 5 granola bars at \$1.40 each and 6 notebooks at \$2.55 each. What is the total cost?
A \$22.30 B \$22.40 C \$23.30 D \$28.05
- In a technical training course, the unit tests count for 75% of the final grade and the final exam counts for 25%. Hannah has a test average of 80. What score does she need on the final exam to earn a final grade of 75?
A 15 B 54 C 60 D 75
- At a grocery store, Tyrone buys 4 jars of salsa at \$3.49 each and 3 loaves of bread at \$3.25 each. He pays with a \$50 bill. How much change should he receive?
A \$22.00 B \$23.71 C \$26.29 D \$27.29

16. Over 4 weeks, a recruiting office averaged 24 new enlistments per week. After one more week, the average for all 5 weeks is 22. How many people enlisted in that last week?
A 2 B 8 C 14 D 22
17. A water trough on Grace's farm is 10 feet long, 3 feet wide, and 2 feet deep. How many gallons of water does it hold when it is full? (1 cubic foot holds 7.5 gallons.)
A 8 gal B 60 gal C 420 gal D 450 gal
18. While training for a marathon, Andre ran 18, 9, 10, 8, and 14 miles in the first five weeks. If he runs the same distance in each of the next two weeks, how many miles must that be to average 17 miles per week over seven weeks?
A 17 B 26 C 30 D 60
19. A lab technician mixes 9 liters of a 50% acid solution with 3 liters of a 30% acid solution. What percent of the mixture is acid?
A 40% B 45% C 50% D 80%
20. Luis is saving for a new phone, which costs \$1,050. He has already saved \$300 and can save \$40 a week. How many weeks will it take him to have enough money?
A 19 weeks B 21 weeks
C 27 weeks D 34 weeks
21. Wire fencing comes in 25-foot rolls that cost \$45 each. How much will it cost to buy enough rolls to fence 133 feet?
A \$225 B \$245 C \$270 D \$315
22. A riding mower cuts grass four times as fast as a push mower. Using both mowers at the same time, a groundskeeper can mow a park in 4 hours. How long would it take using only the riding mower?
A 2 hours B 3 hours C 4 hours D 5 hours
23. Marcus deposits \$5,500 in an account that pays 8% yearly interest, compounded every six months (so 4% is added every six months). How much will be in the account after 1 year?
A \$448.80 B \$5,720.00
C \$5,940.00 D \$5,948.80
24. A rectangular garden is 132 feet by 120 feet. Stakes are placed every 12 feet around its edge, including the corners. How many stakes are used?
A 21 B 38 C 42 D 43
25. A bike shop buys bicycles for \$310 each and marks up the cost by 25%. How much does a customer pay for each bicycle?
A \$77.50 B \$387.50 C \$437.50 D \$487.50
26. A convoy covers 16 miles of highway in 20 minutes. What is the convoy's average speed in miles per hour?
A 0.8 mph B 33 mph C 48 mph D 80 mph
27. Emily deposits \$4,000 in an account that pays 8% interest per year, compounded once a year. How much interest will the account earn in 2 years?
A \$665.60 B \$765.60
C \$865.60 D \$4,665.60
28. Four friends, including Mei, line up for a photo. Mei must stand at the left end. How many different line-ups are possible?
A 3 B 4 C 5 D 6
29. Maria is buying cups and lids. Cups come in packs of 20, and lids come in packs of 15. She wants exactly the same number of cups and lids. What is the least number of packs of cups she can buy?
A 2 B 3 C 4 D 60
30. Specialist Patel is leaving the Army after four years and has two civilian job offers. Job A pays \$20.50 per hour, and Job B pays a salary of \$44,720 a year. At either job he would work 40 hours a week, 52 weeks a year. How much more per year does the better-paying job pay?
A \$40 B \$2,080 C \$5,360 D \$42,640

Part 2: Mathematics Knowledge

Score: ____ / 25

Target time: 24 minutes

1. Solve the system for
- y
- :

$$\begin{aligned}y &= -2x + 8 \\4x + y &= 14\end{aligned}$$

A -14 B 2 C 3 D 5

2. If
- y
- varies directly as
- x
- , and
- $y = 48$
- when
- $x = 8$
- , what is the value of
- y
- when
- $x = 4$
- ?

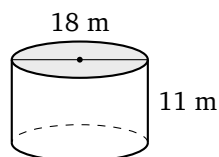
A 6 B 12 C 20 D 24

3. What is the value of
- $x + y$
- in the solution of this system?

$$\begin{aligned}4x + 3y &= -3 \\2x + 3y &= -9\end{aligned}$$

A -5 B -3 C -2 D 3

4. The cylinder shown has a diameter of 18 meters and a height of 11 meters. What is its volume?



A $198\pi \text{ m}^3$ B $891\pi \text{ m}^3$
C $1,782\pi \text{ m}^3$ D $3,564\pi \text{ m}^3$

5. Simplify
- $\sqrt{8} \cdot \sqrt{6}$
- .

A $2\sqrt{3}$ B $\sqrt{14}$ C $4\sqrt{3}$ D 48

6. A circle has a diameter of 20 feet. What is its area?

A $100\pi \text{ ft}^2$ B $121\pi \text{ ft}^2$ C $200\pi \text{ ft}^2$ D $400\pi \text{ ft}^2$

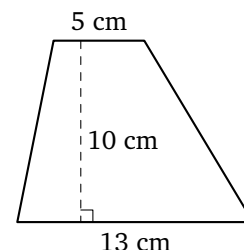
7. Given
- $f(x) = 3x^2 - x - 5$
- , find
- $f(4)$
- .

A 15 B 29 C 34 D 39

8. The six faces of a cube-shaped box have a total area of 24 square centimeters. What is its volume?

A 8 cm^3 B 12 cm^3 C 24 cm^3 D 27 cm^3

9. What is the area of a trapezoid with bases of 5 centimeters and 13 centimeters and a height of 10 centimeters?

A 50 cm^2 B 90 cm^2 C 130 cm^2 D 180 cm^2

10. Tyrone is building a tabletop in the shape of a regular decagon (10 sides). What is the measure of each interior angle of the tabletop?

A 36° B 72° C 124° D 144°

11. Which of the following is equal to
- $4(3x + 5) - 4(2x + 9)$
- ?

A $4x + 56$ B $20x + 56$
C $4x - 16$ D $4x - 4$

12. Which of the numbers 438, 675, 780, and 945 is divisible by both 5 and 6?

A 438 B 675 C 780 D 945

13. Divide
- $25x^3 + 20x^2 + 25x$
- by
- $5x$
- . (Assume
- $x \neq 0$
- .)

A $5x^2 + 4x$ B $5x^3 + 4x^2 + 5x$
C $5x^2 + 9x$ D $5x^2 + 4x + 5$

14. What is
- $\frac{2}{3} \div \frac{1}{2}$
- ?

A $\frac{1}{3}$ B $\frac{3}{4}$ C $1\frac{1}{3}$ D 3

15. If
- $x = -2$
- and
- $y = 11$
- , what is the value of
- $|5x - y| - |x|$
- ?

A -19 B 19 C 21 D 23

16. What is the slope of the line
- $4x + 2y = -10$
- ?

A 4 B -2 C 2 D $-\frac{1}{2}$

17. Multiply:
- $(x - 4)^2$

A $x^2 - 8x + 16$ B $x^2 - 8x - 16$
C $x^2 - 16$ D $x^2 + 16$

18. Solve the formula $y = mx + b$ for b .

A $b = y - mx$ B $b = y + mx$
 C $b = mx - y$ D $b = \frac{y}{mx}$

19. What is 20% of 75% of 860?

A 129 B 172 C 645 D 817

20. Which of the numbers 0.91, 0.091, 0.1, and 0.109 is the least?

A 0.091 B 0.91 C 0.1 D 0.109

21. Evaluate: $(10 - 6) \times 5 + 6^2 \div 3$

A 12 B 16 C 24 D 32

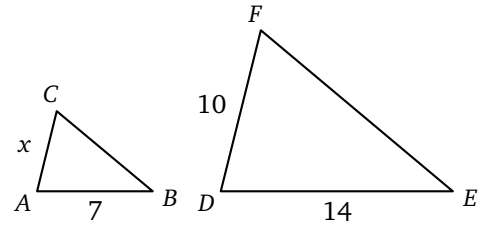
22. What is the value of $-x^2 - 4x + 9$ when $x = -5$?

A -36 B -4 C 2 D 4

23. Which of the following is the solution of $-2x - 11 \leq 3x - 36$?

A $x \leq 5$ B $x \geq -5$ C $x \geq 5$ D $x \geq -25$

24. In the figure, $\triangle ABC$ is similar to $\triangle DEF$, with $DE = 14$, $DF = 10$, and $AB = 7$. What is the value of x ?



A 5 B 8 C 9 D 20

25. What is $(3 \times 10^{-7})(4 \times 10^{12})$ written in scientific notation?

A 1.2×10^5 B 1.2×10^6
 C 2.2×10^6 D 1.2×10^{20}

Answer Key: Practice Test 2

The small number after each answer is the chapter that teaches the skill.

Part 1: Arithmetic Reasoning

1 B ch. 24	7 A ch. 27	13 A ch. 5	19 B ch. 25	25 B ch. 23
2 B ch. 10	8 B ch. 27	14 C ch. 26	20 A ch. 22	26 C ch. 24
3 A ch. 25	9 A ch. 23	15 C ch. 22	21 C ch. 27	27 A ch. 23
4 D ch. 22	10 B ch. 11	16 C ch. 26	22 D ch. 25	28 D ch. 26
5 D ch. 24	11 A ch. 24	17 D ch. 20	23 D ch. 23	29 B ch. 3
6 A ch. 25	12 D ch. 19	18 C ch. 26	24 C ch. 27	30 B ch. 22

Part 2: Mathematics Knowledge

1 B ch. 13	6 A ch. 19	11 C ch. 10	16 B ch. 21	21 D ch. 1
2 D ch. 7	7 D ch. 15	12 C ch. 3	17 A ch. 14	22 D ch. 10
3 C ch. 13	8 A ch. 20	13 D ch. 14	18 A ch. 11	23 C ch. 12
4 B ch. 20	9 B ch. 18	14 C ch. 4	19 A ch. 6	24 A ch. 17
5 C ch. 9	10 D ch. 16	15 B ch. 2	20 A ch. 5	25 B ch. 8

Scoring your practice test

Percent correct	AR (of 30)	MK (of 25)
90% or more — excellent	27–30	23–25
75–89% — good	23–26	19–22
60–74% — keep practicing	18–22	15–18
below 60% — review the chapters	0–17	0–14

These bands measure your progress in this book only. Your real AFQT percentile depends on all four AFQT subtests and is scaled by the Department of Defense.

Explanations: Arithmetic Reasoning

1. **B** 75 miles

- 1 Change the minutes to hours: $15 \text{ min} = \frac{15}{60} = \frac{1}{4}$ hour, so the time is $1\frac{1}{4}$ hours.
- 2 Distance: $60 \times 1\frac{1}{4} = 60 \times 1 + 60 \times \frac{1}{4} = 60 + 15 = 75$ miles.

WHY NOT (A) treats 1 hour 15 minutes as 1.15 hours. (D) rounds the time up to a whole hour.

2. **B** $12x + 5y$

- 1 Each T-shirt costs \$12, so x of them cost $12x$ dollars.
- 2 Each pair of socks costs \$5, so y of them cost $5y$ dollars.
- 3 Add the two costs: $12x + 5y$.

WHY NOT (A) charges both prices for every item. (C) combines unlike terms. (D) matches each price with the wrong item.

3. **A** 64 pounds

- 1 Peanuts are 4 parts, so one part is $32 \div 4 = 8$ pounds.
- 2 The whole batch is 8 parts: $8 \times 8 = 64$ pounds.

WHY NOT (D) multiplies by the number of parts without first finding the size of one part.

4. **D** \$68.44

- Find the tip: $0.18 \times \$116 = \20.88 .
- Total with tip: $\$116 + \$20.88 = \$136.88$.
- Split it 2 ways: $\$136.88 \div 2 = \68.44 .

WHY NOT (A) is one friend's share of the tip only. (B) forgets to add the tip.

 5. **D** 6 mph

- Change the time to hours: $15 \text{ min} = \frac{15}{60} = \frac{1}{4} \text{ hour}$.
- Speed: $1.5 \div \frac{1}{4} = 1.5 \times 4 = 6 \text{ miles per hour}$.

TIP Or: the pace is $15 \div 1.5 = 10$ minutes per mile, and $60 \div 10 = 6 \text{ mph}$.

WHY NOT (A) gives miles per minute, not miles per hour.

 6. **A** \$5.00

- Cost of each part: $60 \times 4.00 = 240.00$ and $24 \times 7.50 = 180.00$ dollars.
- Totals: $\$420$ for $60 + 24 = 84$ pounds.
- Price per pound: $420.00 \div 84 = 5.00$, or \$5 per pound.

TIP Check: \$5 is between \$4 and \$7.50, closer to the price of the item with more pounds.

WHY NOT (B) averages the two prices without weighting by the pounds. (C) weights each price by the wrong number of pounds. (D) is the total cost of the mixture, not the price per pound.

 7. **A** 30 miles per hour

- Fewer miles than kilometers, so divide by 1.6: $48 \div 1.6 = 480 \div 16 = 30 \text{ miles per hour}$.

WHY NOT (B) subtracts 16. (C) multiplies by 1.6 instead of dividing. (D) slips the decimal point.

 8. **B** \$1,800

- First step: $\frac{1}{5} \times 3,000 = 600$. Remainder: $3,000 - 600 = 2,400$.
- Second step: $\frac{1}{4}$ of the remainder: $\frac{1}{4} \times 2,400 = 600$.
- Left: $2,400 - 600 = 1,800$.

WHY NOT (A) takes both fractions of the original amount instead of the remainder. (D) stops after the first step.

 9. **A** \$510

- Time must be in years: $18 \text{ months} = \frac{18}{12} = 1\frac{1}{2}$ years.
- One year's interest: $\$8,500 \times 0.04 = \340 .
- For $1\frac{1}{2}$ years: $\$340 \times 1\frac{1}{2} = \510 .

WHY NOT (B) writes 18 months as 1.8 years. (C) uses 18 years instead of 18 months. (D) is the total with interest, not the interest alone.

 10. **B** 7

- Let n be the smallest. Consecutive integers go up by 1, so the numbers are $n, n + 1, n + 2$.
- Add them: $3n + 3 = 18$.
- Subtract 3: $3n = 15$. Divide by 3: $n = 5$.
- The numbers are 5, 6, 7. The largest is 7.

WHY NOT (A) is the smallest number, not the largest.

 11. **A** 15 minutes

- Time in hours: $t = d \div r = 30 \div 120 = \frac{30}{120} = \frac{1}{4}$ hour.

- Change to minutes: $\frac{1}{4} \times 60 = 15 \text{ minutes}$.

WHY NOT (C) reads 0.25 hour as 25 minutes. (D) divides the speed by the distance before multiplying by 60.

 12. **D** 94.2 ft

- The distance around a circle is its circumference, $C = \pi d$. The diameter is twice the radius: $d = 2 \times 15 = 30$ feet.

- $C \approx 3.14 \times 30 = 94.2$ feet.

TIP Multiply by 3 and by 0.14, then add: $3 \times 30 = 90$ and $0.14 \times 30 = 4.2$, so the total is 94.2.

WHY NOT (A) uses the radius in $C = \pi d$; the diameter is twice the radius. (C) uses 3 for π instead of 3.14.

 13. **A** \$22.30

- First item: $5 \times 1.40 = 7.00$.
- Second item: $6 \times 2.55 = 15.30$.
- Total: $7.00 + 15.30 = 22.30$, so \$22.30.

WHY NOT (D) charges every item at the higher price.

 14. **C** 60

- The tests contribute $0.75 \times 80 = 60$ points to the grade.
- The final must contribute $75 - 60 = 15$ points.
- The final counts 25%, so the exam score is $15 \div 0.25 = 60$.

TIP Check: $0.75(80) + 0.25(60) = 60 + 15 = 75$. ✓

WHY NOT (A) forgets to divide by the final exam's weight. (D) is the final grade wanted, not the exam score needed.

 15. **C** \$26.29

- Find the cost of each kind of item: $4 \times \$3.49 = \13.96 , and $3 \times \$3.25 = \9.75 .
- Add to get the total cost: $\$13.96 + \$9.75 = \$23.71$.
- Subtract the total from the amount paid: $\$50.00 - \$23.71 = \$26.29$.

WHY NOT (A) rounds each price up to a whole dollar instead of using the exact prices. (B) is the total cost, not the change. (D) forgets to borrow a dollar when subtracting the cents.

16. **C** 14

- 1 Total for the first 4: $4 \times 24 = 96$.
- 2 Total for all 5: $5 \times 22 = 110$.
- 3 The new value is the difference: $110 - 96 = 14$.

WHY NOT (A) gives the change in the average. (D) is the new average, not the new value.

17. **D** 450 gal

- 1 Volume: $10 \times 3 \times 2 = 60$ cubic feet.
- 2 Each cubic foot holds 7.5 gallons: $60 \times 7.5 = 450$ gallons.

TIP Multiplying by 7.5 is the same as multiplying by 15 and halving: $60 \times 15 = 900$, and $900 \div 2 = 450$.

WHY NOT (A) divides by 7.5 instead of multiplying. (B) finds the volume in cubic feet but forgets to change it to gallons. (C) uses 7 gallons per cubic foot instead of 7.5.

18. **C** 30

- 1 To average 17 over 7 weeks, the total must be $17 \times 7 = 119$.
- 2 The total so far is $18 + 9 + 10 + 8 + 14 = 59$.
- 3 The next two must add up to $119 - 59 = 60$, so each one is $60 \div 2 = 30$.

TIP Check: $18 + 9 + 10 + 8 + 14 + 30 + 30 = 119$, and $119 \div 7 = 17$. ✓

WHY NOT (A) is the target average, not the score needed. (D) is the total for the next two weeks, not the amount for each.

19. **B** 45%

- 1 Pure acid in each: $0.5 \times 9 = 4.5$ and $0.3 \times 3 = 0.9$ liters.
- 2 Totals: $4.5 + 0.9 = 5.4$ liters of acid in $9 + 3 = 12$ liters of mixture.
- 3 Percent: $\frac{5.4}{12} = 0.45 = 45\%$.

TIP Check: 45% is between 30% and 50%, closer to the percent of the larger amount.

WHY NOT (A) averages the two percents without weighting by the amounts. (D) adds the two percents.

20. **A** 19 weeks

- 1 Money still needed: $\$1,050 - \$300 = \$750$.
- 2 Divide by the weekly savings: $750 \div 40$ is 18 with 30 left over. After 18 weeks he has saved $18 \times \$40 = \720 , still \$30 short.
- 3 So he needs one more week: $18 + 1 = 19$ weeks.

WHY NOT (C) ignores the \$300 already saved. (D) adds the money already saved to the price instead of subtracting it.

21. **C** \$270

- 1 Each roll covers 25 feet: $133 \div 25$ is 5 with a remainder of 8.
- 2 5 is not quite enough, so buy 6 rolls.
- 3 Cost: $6 \times 45 = 270$, so \$270.

WHY NOT (A) rounds the number of rolls down, which is not enough.

22. **D** 5 hours

- 1 Let the push mower do r of the job per hour. Then the riding mower does $4r$, and together they do $r + 4r = 5r$ per hour.

- 2 Together they finish in 4 hours, so $5r = \frac{1}{4}$ and $r = \frac{1}{20}$. So the push mower alone needs 20 hours.

- 3 Working four times as fast, the riding mower needs one fourth of that time: $20 \div 4 = 5$ hours.

TIP Check: rates $\frac{1}{5} + \frac{1}{20} = \frac{1}{4}$ of the job per hour. ✓

WHY NOT (C) gives the time for both working together.

23. **D** \$5,948.80

- 1 Rate for each period: 4% (half of the yearly rate).
- 2 After the first six months: $\$5,500 + 0.04 \times \$5,500 = \$5,500 + \$220 = \$5,720$.
- 3 After the second: $\$5,720 + 0.04 \times \$5,720 = \$5,720 + \$228.80 = \$5,948.80$.

TIP The second period earns more (\$228.80 vs. \$220) because it earns interest on the interest.

WHY NOT (A) is the interest only, not the balance. (B) stops after the first period. (C) uses simple interest; it leaves out the interest earned on the first interest.

24. **C** 42

- 1 Perimeter: $2 \times (132 + 120) = 504$ feet.
- 2 Around a closed loop, posts = spaces: $504 \div 12 = 42$.

TIP A loop has no loose end, so you do not add 1.

WHY NOT (A) uses only two sides instead of the whole perimeter. (B) leaves out the corner posts. (D) adds 1 as if it were a straight fence; around a closed loop the first and last posts are the same.

25. **B** \$387.50

- 1 Markup: $0.25 \times \$310 = \77.50 .
- 2 Selling price: $\$310 + \$77.50 = \$387.50$.

TIP In one step: $1.25 \times \$310 = \387.50 .

WHY NOT (A) is the markup, not the selling price.

26. **C** 48 mph

- 1 Change the time to hours: $20 \text{ min} = \frac{20}{60} = \frac{1}{3} \text{ hour}$.
- 2 Speed: $16 \div \frac{1}{3} = 16 \times 3 = 48 \text{ miles per hour}$.

TIP Or scale up to an hour: 20 minutes goes into 60 minutes 3 times, and $16 \times 3 = 48$.

WHY NOT (A) gives miles per minute, not miles per hour. (D) treats 20 minutes as 0.2 hour.

27. **A** \$665.60

- 1 Rate for each period: 8%.
- 2 After the first year: $\$4,000 + 0.08 \times \$4,000 = \$4,000 + \$320 = \$4,320$.
- 3 After the second: $\$4,320 + 0.08 \times \$4,320 = \$4,320 + \$345.60 = \$4,665.60$.
- 4 Interest earned: $\$4,665.60 - \$4,000 = \$665.60$.

TIP The second year earns more (\$345.60 vs. \$320) because it earns interest on the interest.

WHY NOT (D) is the balance, not the interest.

28. **D** 6

- 1 Mei's place is fixed, so only the other 3 people need to be arranged.
- 2 $3 \times 2 \times 1 = 6$.

WHY NOT (A) counts only the choices for one position.

29. **B** 3

- 1 The number of each item must be a common multiple of 20 and 15; the least one is the LCM.
- 2 Multiples of 20: 20, 40, 60. The first that is also a multiple of 15 is 60, so she needs 60 of each.
- 3 Packs of cups: $60 \div 20 = 3$.

WHY NOT (C) is the number of packs of lids. (D) is the number of cups, not the number of packs.

30. **B** \$2,080

- 1 Hours in a year: $40 \times 52 = 2,080 \text{ hours}$.
- 2 Job A per year: $2,080 \times \$20.50 = \$42,640$. (Think $2,080 \times 20 = \$41,600$ plus $2,080 \times 0.50 = \$1,040$.)
- 3 Compare with Job B's \$44,720: $\$44,720 - \$42,640 = \$2,080$. Job B pays \$2,080 more.

WHY NOT (A) is the difference for one week, not for a year. (C) assumes 4 weeks in every month (48 weeks), but a year has 52 weeks. (D) is Job A's yearly pay, not the difference.

Explanations: Mathematics Knowledge

1. **B** 2

- 1 The first equation tells you what y equals. Replace y in the second equation with $(-2x + 8)$: $4x + (-2x + 8) = 14$.
- 2 Distribute: $4x - 2x + 8 = 14$, so $2x + 8 = 14$.
- 3 Solve: $2x = 6$, so $x = 3$.
- 4 Now use the first equation: $y = -2(3) + 8 = 2$.

WHY NOT (A) makes a sign error with the 8. (C) gives the value of x , not y . (D) gives $x + y$.

3. **C** -2

- 1 Subtract the second equation from the first (subtract every term). The y -terms cancel: $2x = 6$.
- 2 Divide both sides by 2: $x = \frac{6}{2} = 3$.
- 3 Substitute $x = 3$ into $4x + 3y = -3$: $4(3) + 3y = -3$, so $3y = -15$ and $y = -5$.
- 4 The question asks for $x + y$: $3 + (-5) = -2$.

WHY NOT (A) gives the value of y , not $x + y$. (D) gives the value of x , not $x + y$.

2. **D** 24

- 1 Direct variation means $y = kx$, where k is a constant.
- 2 Find k by dividing: $k = \frac{y}{x} = \frac{48}{8} = 6$.
- 3 Use it: $y = 6 \times 4 = 24$.

WHY NOT (A) stops after finding the constant k .

4. **B** $891\pi \text{ m}^3$

- 1 The radius is half the diameter: $r = 18 \div 2 = 9 \text{ meters}$.
- 2 Base: $\pi r^2 = \pi \times 9^2 = 81\pi \text{ square meters}$.
- 3 Volume: $V = \pi r^2 h = 81\pi \times 11 = 891\pi \text{ cubic meters}$.

WHY NOT (A) uses πd (the circumference) instead of the area of the base. (C) squares the diameter and then halves it; halve first. (D) uses the diameter in place of the radius.

5. **C** $4\sqrt{3}$

- 1 Multiply the numbers under the radicals: $\sqrt{8} \cdot \sqrt{6} = \sqrt{8 \times 6} = \sqrt{48}$.
- 2 Simplify: $\sqrt{48} = \sqrt{16 \cdot 3} = 4\sqrt{3}$.

WHY NOT (B) adds the numbers under the radicals instead of multiplying. (D) drops the radical sign.

6. **A** $100\pi \text{ ft}^2$

- 1 The formula $A = \pi r^2$ needs the radius, which is half the diameter: $r = 20 \div 2 = 10$ feet.
- 2 Square the radius: $10^2 = 100$.
- 3 So $A = \pi \times 100 = 100\pi$ square feet.

WHY NOT (C) squares the diameter and then halves it; halve first, then square. (D) uses the diameter in place of the radius.

7. **D** 39

- 1 Replace every x with (4): $f(4) = 3(4)^2 - (4) - 5$.
- 2 Square first (order of operations): $4^2 = 16$, then multiply: $3 \cdot 16 = 48$. So $f(4) = 48 - 4 - 5$.
- 3 Add: $f(4) = 39$.

WHY NOT (A) doubles 4 instead of squaring it.

8. **A** 8 cm^3

- 1 Each of the 6 faces has area $24 \div 6 = 4$ square centimeters.
- 2 A square with area 4 has side 2, so the edge is 2 centimeters.
- 3 Volume: $2^3 = 8$ cubic centimeters.

WHY NOT (C) gives the surface area, not the volume.

9. **B** 90 cm^2

- 1 Area of a trapezoid = $\frac{1}{2} \times (b_1 + b_2) \times h$: average the two parallel bases, then multiply by the height.
- 2 Add the bases: $13 + 5 = 18$.
- 3 $\frac{1}{2} \times 18 \times 10 = 9 \times 10 = 90$, so the area is 90 square centimeters.

WHY NOT (A) uses only the shorter base. (C) uses only the longer base. (D) forgets to multiply by $\frac{1}{2}$.

10. **D** 144°

- 1 A decagon has 10 sides. Sum of the interior angles: $(10 - 2) \times 180^\circ = 1,440^\circ$.
- 2 A regular polygon has 10 equal angles: $1,440^\circ \div 10 = 144^\circ$.

TIP Faster: each exterior angle is $360^\circ \div 10 = 36^\circ$, so each interior angle is $180^\circ - 36^\circ = 144^\circ$.

WHY NOT (A) is the exterior angle, not the interior angle.

11. **C** $4x - 16$

- 1 Distribute 4: $4(3x + 5) = 12x + 20$.
- 2 Distribute -4 (the minus sign comes with it): $-4(2x + 9) = -8x - 36$.
- 3 Combine like terms: $12x - 8x = 4x$ and $20 - 36 = -16$. The result is $4x - 16$.

TIP Check with $x = 1$: the original and $4x - 16$ both equal -12 .

WHY NOT (A) multiplies 9 by 4 instead of -4 ; the minus sign goes to every term. (B) adds the second product instead of subtracting it. (D) multiplies only the first term in each set of parentheses.

12. **C** 780

- 1 Divisible by 5: its last digit is 0 or 5. Divisible by 6: it is divisible by both 2 and 3.
- 2 780 passes both tests (ends in 0; even, digit sum 15): $780 \div 30 = 26$.
- 3 Each of the other choices fails one of the tests.

WHY NOT (A) is divisible by 6, but not by 5: it does not end in 0 or 5. (B) is divisible by 5, but not by 6: it is odd. (D) is divisible by 5, but not by 6: it is odd.

13. **D** $5x^2 + 4x + 5$

- 1 Divide *each* term of the top by $5x$.
- 2 $\frac{25x^3}{5x} = 5x^2$, $\frac{20x^2}{5x} = 4x$, and $\frac{25x}{5x} = 5$.
- 3 So the result is $5x^2 + 4x + 5$.

WHY NOT (A) drops the last term, but $\frac{25x}{5x}$ is 5, not 0. (B) divides the coefficients but forgets to subtract the exponents. (C) leaves an x on the last term.

14. **C** $1\frac{1}{3}$

- 1 Keep the first fraction, change $\div$ to $\times$, and flip the second fraction: $\frac{2}{3} \times \frac{2}{1}$.
- 2 Multiply top times top and bottom times bottom: $\frac{2}{3} \times 2 = \frac{4}{3} = 1\frac{1}{3}$.

WHY NOT (A) multiplies without flipping the second fraction. (B) divides in the wrong order (flips the answer). (D) flips both fractions.

15. **B** 19

- 1 Substitute: $|5(-2) - 11| - |-2|$.
- 2 Inside the first bars: $5(-2) - 11 = -10 - 11 = -21$, and $|-21| = 21$. Also $|-2| = 2$.
- 3 Subtract: $21 - 2 = 19$.

WHY NOT (A) keeps the signs instead of taking absolute values. (D) treats $-|x|$ as $+2$.

16. **B** -2

- 1 Solve for y . Move the x -term to the other side:
 $2y = -4x - 10$.
- 2 Divide every term by 2: $y = -2x - 5$.
- 3 The slope is the coefficient of x : -2.

WHY NOT (A) uses the coefficient of x without solving for y . (C) forgets to change the sign when moving the x -term. (D) flips the fraction (puts the x -coefficient on the bottom).

 17. **A** $x^2 - 8x + 16$

- 1 Use the pattern $(A - B)^2 = A^2 - 2AB + B^2$ with $A = x$ and $B = 4$.
- 2 $A^2 = (x)^2 = x^2$, $-2AB = -2(x)(4) = -8x$, and $B^2 = 4^2 = 16$.
- 3 So $(x - 4)^2 = x^2 - 8x + 16$.

TIP Check with $x = 1$: the original and your answer must give the same number.

WHY NOT (B) makes the last term negative, but $(-4)^2 = 16$. (C) squares each term, leaves out the middle term, and makes the last term negative. (D) squares each term but leaves out the middle term $-8x$.

 18. **A** $b = y - mx$

- 1 Treat every letter except b as if it were a number. Undo what is being done to b , in reverse order, doing the same thing to both sides.
- 2 b has mx added to it, so subtract mx from both sides: $y - mx = b$.
- 3 So $b = y - mx$.

TIP Check with easy numbers: pick values for the other letters, find b , and see that the original formula holds.

WHY NOT (B) adds mx instead of subtracting it. (C) subtracts in the wrong order. (D) divides by mx instead of subtracting it.

 19. **A** 129

- 1 Work from the inside out. First, 75% of 860:
 $0.75 \times 860 = 645$.
 - 2 Then 20% of that result: $0.2 \times 645 = 129$.
- TIP** Percents of percents multiply: $0.2 \times 0.75 = 0.15$, and $0.15 \times 860 = 129$.

WHY NOT (B) takes only 20% of 860. (C) stops after taking 75%. (D) adds the two percents.

 20. **A** 0.091

- 1 Give every number the same number of decimal places by adding zeros: 0.910, 0.091, 0.100, 0.109.
- 2 Now compare them like whole numbers. In order from least to greatest: $0.091 < 0.1 < 0.109 < 0.91$.
- 3 The least is 0.091.

WHY NOT (B) is greater than 0.091: in the tenths place it has 9, while 0.091 has 0. (C) is greater than 0.091: in the tenths place it has 1, while 0.091 has 0. (D) is greater than 0.091: in the tenths place it has 1, while 0.091 has 0.

 21. **D** 32

- 1 Parentheses first: $10 - 6 = 4$, which leaves $4 \times 5 + 6^2 \div 3$.
- 2 Exponent next: $6^2 = 6 \times 6 = 36$, which leaves $4 \times 5 + 36 \div 3$.
- 3 Multiply (multiplication and division go left to right): $4 \times 5 = 20$, which leaves $20 + 36 \div 3$.
- 4 Divide before you add or subtract: $36 \div 3 = 12$, which leaves $20 + 12$.
- 5 Add: $20 + 12 = 32$.

WHY NOT (C) doubles instead of squaring.

 22. **D** 4

- 1 Substitute, putting each value in parentheses:
 $-(-5)^2 - 4(-5) + 9$.
- 2 Powers first: $(-5)^2 = 25$.
- 3 Multiply each term (watch the signs): $-25 + 20 + 9$.
- 4 Add and subtract from left to right: $-25 + 20 + 9 = 4$.

WHY NOT (A) forgets that a negative times a negative is positive.

 23. **C** $x \geq 5$

- 1 Collect the x -terms on the left: subtract $3x$ from both sides. $-5x - 11 \leq -36$.
- 2 Add 11 to both sides (adding or subtracting never changes the sign): $-5x \leq -36 + 11$, so $-5x \leq -25$.
- 3 Divide both sides by -5 . Dividing by a *negative* number flips the sign: $\leq$ becomes $\geq$: $x \geq \frac{-25}{-5}$, so $x \geq 5$.

TIP To avoid dividing by a negative, collect the x -terms on the right instead: $25 \leq 5x$, so $5 \leq x$, which means the same as $x \geq 5$.

WHY NOT (A) forgets to flip the sign when dividing by -5 . (B) makes a sign error when dividing by -5 . (D) adds the x -terms instead of subtracting.

24. **A** 5

1 Corresponding sides are proportional. The scale factor from $\triangle ABC$ to $\triangle DEF$ is $14 \div 7 = 2$.

2 Divide the big triangle's side by the scale factor: $x = 10 \div 2 = 5$.

WHY NOT (D) multiplies by the scale factor instead of dividing.

25. **B** 1.2×10^6

1 Multiply the first factors: $3 \times 4 = 12$.

2 Add the exponents: $-7 + 12 = 5$, giving 12×10^5 .

3 12 is not less than 10, so write it as 1.2×10^1 and add 1 to the exponent: the answer is 1.2×10^6 .

WHY NOT (A) rewrites 12 as 1.2 but forgets to add 1 to the exponent. (D) ignores the negative sign on the first exponent.

Practice Test 3

Instructions

This test has two parts, timed like the paper-and-pencil ASVAB. **Part 1, Arithmetic Reasoning:** 30 questions, 36 minutes. **Part 2, Mathematics Knowledge:** 25 questions, 24 minutes. No calculator. Use scratch paper. Answer every question—there is no penalty for guessing. Mark your answers on the answer sheet below.

Answer Sheet

Fill in one bubble per question. Photocopy this page to retake the test.

Part 1: Arithmetic Reasoning

- 1 ☐ A ☐ B ☐ C ☐ D
- 2 ☐ A ☐ B ☐ C ☐ D
- 3 ☐ A ☐ B ☐ C ☐ D
- 4 ☐ A ☐ B ☐ C ☐ D
- 5 ☐ A ☐ B ☐ C ☐ D
- 6 ☐ A ☐ B ☐ C ☐ D
- 7 ☐ A ☐ B ☐ C ☐ D
- 8 ☐ A ☐ B ☐ C ☐ D
- 9 ☐ A ☐ B ☐ C ☐ D
- 10 ☐ A ☐ B ☐ C ☐ D
- 11 ☐ A ☐ B ☐ C ☐ D
- 12 ☐ A ☐ B ☐ C ☐ D
- 13 ☐ A ☐ B ☐ C ☐ D
- 14 ☐ A ☐ B ☐ C ☐ D
- 15 ☐ A ☐ B ☐ C ☐ D
- 16 ☐ A ☐ B ☐ C ☐ D
- 17 ☐ A ☐ B ☐ C ☐ D
- 18 ☐ A ☐ B ☐ C ☐ D
- 19 ☐ A ☐ B ☐ C ☐ D
- 20 ☐ A ☐ B ☐ C ☐ D
- 21 ☐ A ☐ B ☐ C ☐ D
- 22 ☐ A ☐ B ☐ C ☐ D
- 23 ☐ A ☐ B ☐ C ☐ D
- 24 ☐ A ☐ B ☐ C ☐ D
- 25 ☐ A ☐ B ☐ C ☐ D
- 26 ☐ A ☐ B ☐ C ☐ D
- 27 ☐ A ☐ B ☐ C ☐ D
- 28 ☐ A ☐ B ☐ C ☐ D
- 29 ☐ A ☐ B ☐ C ☐ D
- 30 ☐ A ☐ B ☐ C ☐ D

Part 2: Mathematics Knowledge

- 1 ☐ A ☐ B ☐ C ☐ D
- 2 ☐ A ☐ B ☐ C ☐ D
- 3 ☐ A ☐ B ☐ C ☐ D
- 4 ☐ A ☐ B ☐ C ☐ D
- 5 ☐ A ☐ B ☐ C ☐ D
- 6 ☐ A ☐ B ☐ C ☐ D
- 7 ☐ A ☐ B ☐ C ☐ D
- 8 ☐ A ☐ B ☐ C ☐ D
- 9 ☐ A ☐ B ☐ C ☐ D
- 10 ☐ A ☐ B ☐ C ☐ D
- 11 ☐ A ☐ B ☐ C ☐ D
- 12 ☐ A ☐ B ☐ C ☐ D
- 13 ☐ A ☐ B ☐ C ☐ D
- 14 ☐ A ☐ B ☐ C ☐ D
- 15 ☐ A ☐ B ☐ C ☐ D
- 16 ☐ A ☐ B ☐ C ☐ D
- 17 ☐ A ☐ B ☐ C ☐ D
- 18 ☐ A ☐ B ☐ C ☐ D
- 19 ☐ A ☐ B ☐ C ☐ D
- 20 ☐ A ☐ B ☐ C ☐ D
- 21 ☐ A ☐ B ☐ C ☐ D
- 22 ☐ A ☐ B ☐ C ☐ D
- 23 ☐ A ☐ B ☐ C ☐ D
- 24 ☐ A ☐ B ☐ C ☐ D
- 25 ☐ A ☐ B ☐ C ☐ D

Part 1: Arithmetic Reasoning

Score: ____ / 30

Target time: 36 minutes

- A gardener mixes 12 gallons of a 25% fertilizer solution with 3 gallons of a 10% fertilizer solution. What percent of the mixture is fertilizer?
A 17.5% B 22% C 26% D 35%
- A motor pool has 50 vehicles, and 8 of them are down for maintenance. If one vehicle is picked at random for a convoy, what is the probability that it is *not* down for maintenance?
A $\frac{4}{25}$ B $\frac{21}{25}$ C $\frac{22}{25}$ D $\frac{21}{4}$
- The average weight of the 9 soldiers in a squad is 182 pounds. Sergeant Brooks, who weighs 174 pounds, transfers out of the squad. What is the new average weight of the squad?
A 183 pounds B 188 pounds
C 190 pounds D 193 pounds
- A truck's trip odometer shows 6 miles. About how many kilometers is that? (1 mile $\approx$ 1.6 kilometers)
A 3.75 kilometers B 7.6 kilometers
C 9.5 kilometers D 9.6 kilometers
- A train covers 7 miles in 10 minutes. What is the train's average speed in miles per hour?
A 0.7 mph B 42 mph C 52 mph D 70 mph
- A box of 12 granola bars costs \$2.40. What is the price per bar?
A \$0.02 B \$0.20 C \$2.00 D \$5.00
- Luis is saving for a gaming computer, which costs \$900. He has already saved \$420 and can save \$65 a week. How many weeks will it take him to have enough money?
A 8 weeks B 9 weeks
C 14 weeks D 21 weeks
- Andre works at a furniture store and earns a base salary of \$500 per week plus a 6% commission on his sales. His sales this week total \$8,500. What is his total pay for the week?
A \$1,010 B \$1,210 C \$5,600 D \$9,000
- Nora has \$175 in a checking account. She pays \$108 for a utility bill and \$90 for groceries. Because the balance went below \$0, the bank also charges a \$25 overdraft fee. What is the account balance now?
A -\$398 B -\$68 C -\$53 D -\$48
- James's cookie recipe uses 150 grams of sugar per batch. How many batches can be made with a 2.4-kilogram bag of sugar?
A 1.6 B 14 C 16 D 2,400
- Jasmine runs every 6 days and swims every 9 days. Today is a run day and a swim day. In how many days will both happen on the same day again?
A 3 days B 15 days C 18 days D 54 days
- Aisha works as a delivery driver and earns \$22 per hour, with time-and-a-half for every hour over 40 in a week. Last week her pay was \$1,012. How many hours did she work last week?
A 4 hours B 44 hours
C 45 hours D 46 hours
- Staff Sergeant Nguyen hits a target $\frac{4}{5}$ of the time. If each shot is independent, what is the probability that Nguyen hits the target on both of the next 2 shots?
A $\frac{16}{25}$ B $\frac{17}{25}$ C $\frac{4}{5}$ D $\frac{24}{25}$
- Two pumps are filling an empty fuel bladder at a forward operating base. Alone, one pump would fill it in 15 hours and the other in 5 hours. A damaged valve leaks fuel and would empty a full bladder in 10 hours. With both pumps running and the valve leaking, how long will it take to fill the bladder?
A 6 hours B 7 hours
C 10 hours D 30 hours

15. Two pumps working together can drain a pond in 4 hours. The smaller pump alone would take 20 hours. How long would the larger pump take alone?
A 2 hours B 4 hours
C 5 hours D 16 hours
16. A convoy travels 125 miles on paved road at 50 miles per hour, then 10 miles on a dirt road at 20 miles per hour. What is the average speed for the whole trip?
A 35 mph B 45 mph C 54 mph D 70 mph
17. Victor deposits \$9,000 in an account that pays 6% yearly interest, compounded every six months (so 3% is added every six months). How much will be in the account after 1 year?
A \$548.10 B \$9,270.00
C \$9,540.00 D \$9,548.10
18. Eight years ago, Kevin was three times as old as his daughter was then. The daughter is 27 now. How old is Kevin now?
A 65 B 80 C 81 D 89
19. For a game, Ryan rolls a fair six-sided die 600 times. About how many times should he expect to roll an even number?
A 100 B 200 C 300 D 375
20. A vendor at the base's Fourth of July fair buys T-shirts for \$4.50 each and marks up the cost by 100%. How much does a customer pay for each T-shirt?
A \$4.50 B \$4.95 C \$8.00 D \$9.00
21. A box of supplies weighs 23 lb 11 oz. After Yuki removes 12 lb 13 oz of supplies, how much does the box weigh?
A 9 lb 14 oz B 10 lb 8 oz
C 10 lb 12 oz D 10 lb 14 oz
22. Lieutenant Santos flies a supply helicopter 300 miles to an outpost at 100 miles per hour and flies back at 150 miles per hour. What is the average speed for the round trip?
A 120 mph B 125 mph
C 150 mph D 250 mph
23. The wheel of Mei's wheelbarrow has a diameter of 14 inches. How many feet does it roll in 48 complete turns? Use $\frac{22}{7}$ for π .
A 88 ft B 176 ft C 352 ft D 2,112 ft
24. Corporal Chen's take-home pay is \$2,880 a month. She sends $\frac{1}{3}$ of the pay home to her family, and then puts $\frac{1}{2}$ of what is left into savings. How much money does she put into savings?
A \$760 B \$860 C \$960 D \$1,440
25. A grocery store has 20 pounds of peanuts priced at \$4 per pound. How many pounds of cashews priced at \$13 per pound must be mixed in to make a blend worth \$8 per pound?
A 16 pounds B 20 pounds
C 25 pounds D 36 pounds
26. A rental truck costs \$28 per day plus \$1.50 per mile. Which expression gives the cost, in dollars, of renting the truck for d days and driving it m miles?
A $28d + 1.50m$ B $28m + 1.50d$
C $(28 + 1.50)dm$ D $28 + 1.50m$
27. At the base commissary, protein bars come in three sizes: a box of 12 protein bars for \$20.40, a box of 15 protein bars for \$23.70, and a box of 20 protein bars for \$32. Which size is the best buy?
A All three cost the same per bar
B the box of 15
C the box of 12
D the box of 20
28. A shoe store marks a pair of running shoes down 40%. A week later it cuts the new price by another 25%. The two discounts together are the same as a single discount of what percent?
A 10% B 45% C 50% D 55%
29. Ryan rides a motorcycle at 88 feet per second. How many miles per hour is this? (1 mile = 5,280 feet and 1 hour = 3,600 seconds.)
A 1 mph B 50 mph
C 60 mph D 316,800 mph

30. Keisha is buying carpet for a living room that measures 24 feet by 15 feet. The carpet costs \$40 per square yard. What is the cost of the carpet?
- A \$40 B \$1,580 C \$1,600 D \$14,400

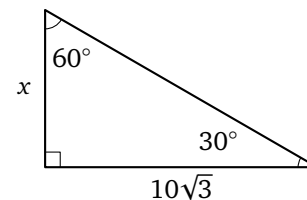
Part 2: Mathematics Knowledge

Score: ____ / 25

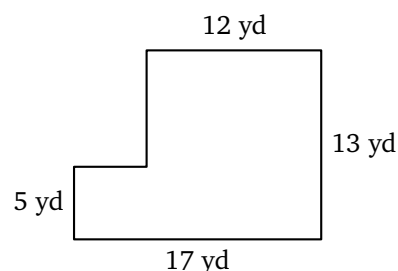
Target time: 24 minutes

1. Which expression is equal to $(x - 1)(x - 8)$?
A $x^2 - 9x + 8$ B $x^2 + 8$
C $x^2 - 9x - 8$ D $x^2 + 9x + 8$
2. A circular flower bed has a radius of 7 feet. A border of paving stones 4 feet wide is built all the way around it. What is the area of the border?
A $72\pi \text{ ft}^2$ B $121\pi \text{ ft}^2$ C $170\pi \text{ ft}^2$ D $176\pi \text{ ft}^2$
3. Find the product: 9×303
A 2,673 B 2,703 C 2,727 D 3,227
4. Evaluate: $|9 - 10| - |-8|$
A -12 B -9 C -8 D -7
5. What is $\sqrt{72}$ in simplest radical form?
A $6\sqrt{2}$ B $8\sqrt{2}$ C $9\sqrt{2}$ D $36\sqrt{2}$
6. What is the smallest positive number that is divisible by each of 4, 8, 3, and 7?
A 24 B 56 C 84 D 168
7. What is $\frac{5}{7}$ of 42?
A 30 B 34 C $58\frac{4}{5}$ D 210
8. If $a = 4$ and $b = -2$, what is the value of $a^2 - 2ab$?
A 0 B 32 C 47 D 64
9. Let n be a number. Which equation means "The difference between twice a number and 3 is 25"?
A $3 - 2n = 25$ B $2n + 3 = 25$
C $2(n - 3) = 25$ D $2n - 3 = 25$
10. The midpoint of segment AB is $M(-1, 1)$. If point A is $(8, 8)$, what are the coordinates of point B ?
A $(-9, -7)$ B $(17, 15)$
C $(7, 9)$ D $(-10, -6)$
11. What are the solutions of $x^2 = -x + 6$?
A $x = -1$ or $x = 6$ B $x = -3$ or $x = 2$
C $x = -2$ or $x = 3$ D $x = -3$

12. The volume of a sphere is $V = \frac{4}{3}\pi r^3$. What is the volume of a ball with diameter 18 inches?
A $108\pi \text{ in}^3$ B $324\pi \text{ in}^3$
C $922\pi \text{ in}^3$ D $972\pi \text{ in}^3$
13. Subtract: $30 - 28.35$
A 1 B 1.5 C 1.65 D 18.35
14. What is the distance between the points $(6, 3)$ and $(9, 7)$?
A 5 B 6 C 7 D 25
15. Which value of x satisfies $-17 < 5x + 3 < -2$?
A -5 B -4 C -2 D -1
16. In the 30° - 60° - 90° triangle shown, the longer leg measures $10\sqrt{3}$. What is the length x of the shorter leg?



- A $5\sqrt{3}$ B 10 C 20 D 30
17. If $\frac{24}{x} = \frac{12}{21}$, what is the value of x ?
A 33 B 42 C 57 D 504
18. Evaluate $10^{-6} \times 10^5$.
A $\frac{1}{10}$ B $\frac{1}{5}$ C $\frac{2}{5}$ D 10
19. The figure shows a garden. All corners are right angles. The sides shown measure 17 yards, 13 yards, 12 yards, and 5 yards. What is the perimeter?

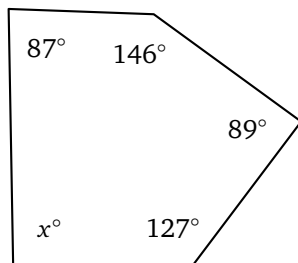


- A 47 yd B 60 yd C 86 yd D 181 yd

20. What is 80% of 50% of 60?

A 18 B 20 C 24 D 78

21. In the pentagon shown, four of the interior angles measure 127° , 89° , 146° , and 87° . What is the value of x ?

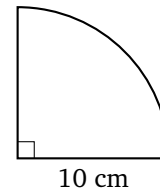


A 76 B 91 C 108 D 271

22. If $3x + 4y = 24$ and $4x + 3y = 39$, what is the value of $x + y$?

A -3 B 8 C 9 D 63

23. The figure shows one fourth of a circle. Each straight side is 10 centimeters long. What is the perimeter of the region?



A 5π cm B $(5\pi + 20)$ cm
C $(10\pi + 20)$ cm D $(20\pi + 20)$ cm

24. Carlos has a box with a volume of 50 cubic centimeters. He builds a second box whose length, width, and height are each 3 times as long as those of the first box. What is the volume of the second box?

A 150 cm^3 B 450 cm^3
C $1,350 \text{ cm}^3$ D $1,400 \text{ cm}^3$

25. If $\frac{x}{6} - \frac{x}{10} = 5$, what is the value of x ?

A $\frac{5}{2}$ B 20 C 75 D 300

Answer Key: Practice Test 3

The small number after each answer is the chapter that teaches the skill.

Part 1: Arithmetic Reasoning

1 B ch. 25	7 A ch. 22	13 A ch. 26	19 C ch. 26	25 A ch. 25
2 B ch. 26	8 A ch. 23	14 A ch. 25	20 D ch. 23	26 A ch. 10
3 A ch. 26	9 D ch. 2	15 C ch. 25	21 D ch. 27	27 B ch. 22
4 D ch. 27	10 C ch. 27	16 B ch. 24	22 A ch. 24	28 D ch. 23
5 B ch. 24	11 C ch. 3	17 D ch. 23	23 B ch. 19	29 C ch. 24
6 B ch. 5	12 B ch. 22	18 A ch. 27	24 C ch. 22	30 C ch. 18

Part 2: Mathematics Knowledge

1 A ch. 14	6 D ch. 3	11 B ch. 15	16 B ch. 17	21 B ch. 16
2 A ch. 19	7 A ch. 4	12 D ch. 20	17 B ch. 7	22 C ch. 13
3 C ch. 1	8 B ch. 10	13 C ch. 5	18 A ch. 8	23 B ch. 19
4 D ch. 2	9 D ch. 11	14 A ch. 21	19 B ch. 18	24 C ch. 20
5 A ch. 9	10 D ch. 21	15 C ch. 12	20 C ch. 6	25 C ch. 11

Scoring your practice test

Percent correct	AR (of 30)	MK (of 25)
90% or more — excellent	27–30	23–25
75–89% — good	23–26	19–22
60–74% — keep practicing	18–22	15–18
below 60% — review the chapters	0–17	0–14

These bands measure your progress in this book only. Your real AFQT percentile depends on all four AFQT subtests and is scaled by the Department of Defense.

Explanations: Arithmetic Reasoning

1. **B** 22%

1 Pure fertilizer in each: $0.25 \times 12 = 3$ and $0.1 \times 3 = 0.3$ gallons.

2 Totals: $3 + 0.3 = 3.3$ gallons of fertilizer in $12 + 3 = 15$ gallons of mixture.

3 Percent: $\frac{3.3}{15} = 0.22 = 22\%$.

TIP Check: 22% is between 10% and 25%, closer to the percent of the larger amount.

WHY NOT (A) averages the two percents without weighting by the amounts. (D) adds the two percents.

2. **B** $\frac{21}{25}$

1 Vehicles not in maintenance: $50 - 8 = 42$.

2 Probability = $\frac{42}{50} = \frac{21}{25}$.

WHY NOT (A) is the probability that the vehicle is down for maintenance. (D) compares good vehicles to broken ones instead of to the total.

3. **A** 183 pounds

1 Turn the average into a total: $9 \times 182 = 1,638$.

2 Subtract the value: $1,638 - 174 = 1,464$.

3 Divide by the new count, $9 - 1 = 8$: $1,464 \div 8 = 183$.

WHY NOT (C) subtracts the value from twice the average.

4. **D** 9.6 kilometers
 1 Each mile is about 1.6 kilometers, so multiply:
 $6 \times 1.6 = 9.6$ kilometers.
WHY NOT (A) divides by 1.6 instead of multiplying. (B) adds 1.6 instead of multiplying.
5. **B** 42 mph
 1 Change the time to hours: $10 \text{ min} = \frac{10}{60} = \frac{1}{6}$ hour.
 2 Speed: $7 \div \frac{1}{6} = 7 \times 6 = 42$ miles per hour.
TIP Or scale up to an hour: 10 minutes goes into 60 minutes 6 times, and $7 \times 6 = 42$.
WHY NOT (A) gives miles per minute, not miles per hour. (D) treats 10 minutes as 0.1 hour.
6. **B** \$0.20
 1 Price per bar = total price $\div$ number of bars.
 2 $2.40 \div 12 = 0.20$. Check: $12 \times 0.20 = 2.40$.
WHY NOT (A) puts the decimal point one place too far left. (C) puts the decimal point one place too far right. (D) divides the number of items by the price.
7. **A** 8 weeks
 1 Money still needed: $\$900 - \$420 = \$480$.
 2 Divide by the weekly savings: $480 \div 65$ is 7 with 25 left over. After 7 weeks he has saved $7 \times \$65 = \455 , still \$25 short.
 3 So he needs one more week: $7 + 1 = 8$ weeks.
WHY NOT (C) ignores the \$420 already saved. (D) adds the money already saved to the price instead of subtracting it.
8. **A** \$1,010
 1 Commission: $0.06 \times \$8,500 = \510 .
 2 Add the base salary: $\$500 + \$510 = \$1,010$.
WHY NOT (C) writes 6% as 0.6. (D) adds the sales instead of the commission.
9. **D** -\$48
 1 Start with \$175 and subtract both payments:
 $175 - 108 - 90 = -23$.
 2 Subtract the fee: $-23 - 25 = -48$.
 3 The balance is -\$48: the account is overdrawn by \$48.
WHY NOT (A) treats the starting balance as money owed instead of money in the account.
10. **C** 16
 1 Change kilograms to grams: $2.4 \times 1,000 = 2,400$ grams.
 2 Divide by one batch: $2,400 \div 150 = 16$ batches.
WHY NOT (A) uses 100 grams in a kilogram instead of 1,000. (D) gives the number of grams, not batches.
11. **C** 18 days
 1 The events line up again after a common multiple of 6 and 9; the first time is the least common multiple.
 2 Multiples of 9: 9, 18. The first one that is also a multiple of 6 is 18.
WHY NOT (A) is the greatest common factor, not the least common multiple. (B) adds the two numbers. (D) multiplies the two numbers; that is a common multiple, but not the first one.
12. **B** 44 hours
 1 Pay for the first 40 hours: $40 \times \$22 = \880 .
 2 The rest of the pay is overtime: $\$1,012 - \$880 = \$132$.
 3 Overtime rate: $1.5 \times \$22 = \33 per hour. Overtime hours: $\$132 \div \$33 = 4$.
 4 Total hours: $40 + 4 = 44$ hours.
WHY NOT (A) gives only the overtime hours, not the total. (D) divides the overtime pay by the regular rate instead of the overtime rate.
13. **A** $\frac{16}{25}$
 1 Each shot has probability $\frac{4}{5}$, and the events are independent, so multiply.
 2 $\frac{4}{5} \times \frac{4}{5} = \frac{16}{25}$.
WHY NOT (C) gives the probability for only one shot. (D) finds the probability of at least one, not all.
14. **A** 6 hours
 1 Rates per hour: the pumps add $\frac{1}{15}$ and $\frac{1}{5}$ of the bladder; the leak takes away $\frac{1}{10}$.
 2 Net rate (common denominator 30): $\frac{1}{15} + \frac{1}{5} - \frac{1}{10} = \frac{2}{30} + \frac{6}{30} - \frac{3}{30} = \frac{5}{30} = \frac{1}{6}$ of the bladder per hour.
 3 Time = 1 bladder $\div$ rate = $1 \div \frac{1}{6} = 6$ hours.
WHY NOT (C) averages the two fill times and ignores the outflow. (D) adds the three times.
15. **C** 5 hours
 1 Rates: together they do $\frac{1}{4}$ of the job per hour, and the smaller pump alone does $\frac{1}{20}$ of the job per hour.
 2 The other worker's rate is the difference: $\frac{1}{4} - \frac{1}{20} = \frac{5}{20} - \frac{1}{20} = \frac{4}{20} = \frac{1}{5}$ of the job per hour.
 3 Doing $\frac{1}{5}$ of the job each hour, the larger pump needs 5 hours alone.
TIP Check: $\frac{1}{20} + \frac{1}{5} = \frac{1}{4}$, the rate together. $\checkmark$
WHY NOT (D) subtracts the times instead of the rates.

16. **B** 45 mph

- 1 First part: $125 \div 50 = 2.5$ hours.
- 2 Second part: $10 \div 20 = 0.5$ hours.
- 3 Total distance: $125 + 10 = 135$ miles. Total time: $2.5 + 0.5 = 3$ hours.
- 4 Average speed: $135 \div 3 = 45$ miles per hour.

WHY NOT (A) averages the two speeds, but more time is spent at one of them. (C) divides the total distance by the first part's time only. (D) adds the two speeds.

17. **D** \$9,548.10

- 1 Rate for each period: 3% (half of the yearly rate).
- 2 After the first six months: $\$9,000 + 0.03 \times \$9,000 = \$9,000 + \$270 = \$9,270$.
- 3 After the second: $\$9,270 + 0.03 \times \$9,270 = \$9,270 + \$278.10 = \$9,548.10$.

TIP The second period earns more (\$278.10 vs. \$270) because it earns interest on the interest.

WHY NOT (A) is the interest only, not the balance. (B) stops after the first period. (C) uses simple interest; it leaves out the interest earned on the first interest.

18. **A** 65

- 1 8 years ago the daughter was $27 - 8 = 19$.
- 2 Then Kevin was $3 \times 19 = 57$.
- 3 Add the 8 years back: $57 + 8 = 65$.

WHY NOT (C) multiplies the daughter's age today instead of 8 years ago.

19. **C** 300

- 1 $P(\text{an even number}) = \frac{3}{6} = \frac{1}{2}$.
- 2 Expected count: $\frac{1}{2} \times 600 = 300$.

WHY NOT (A) counts only one favorable outcome out of six. (B) divides by the number of favorable outcomes.

20. **D** \$9.00

- 1 Markup: $1 \times \$4.50 = \4.50 .
- 2 Selling price: $\$4.50 + \$4.50 = \$9$.

TIP In one step: $2 \times \$4.50 = \9 .

WHY NOT (A) is the markup, not the selling price. (B) moves the decimal point one place too far when finding the markup.

21. **D** 10 lb 14 oz

- 1 You can't take 13 oz from 11 oz, so borrow 1 lb: 23 lb 11 oz becomes 22 lb 27 oz.
- 2 Subtract: $27 - 13 = 14$ oz and $22 - 12 = 10$ lb.
- 3 What is left: 10 lb 14 oz.

WHY NOT (B) borrows 10 ounces instead of 16.

22. **A** 120 mph

- 1 Time going: $300 \div 100 = 3$ hours. Time returning: $300 \div 150 = 2$ hours.
- 2 Total distance: $2 \times 300 = 600$ miles. Total time: $3 + 2 = 5$ hours.
- 3 Average speed: $600 \div 5 = 120$ miles per hour.

TIP Shortcut for equal distances: $2ab \div (a + b) = 2 \times 100 \times 150 \div 250 = 120$.

WHY NOT (B) averages the two speeds, but more time is spent at the slower speed. (D) adds the two speeds.

23. **B** 176 ft

- 1 In one turn the wheel rolls one circumference: $C = \pi d \approx \frac{22}{7} \times 14 = 22 \times 2 = 44$ inches.
- 2 In 48 turns: $44 \times 48 = 2,112$ inches.
- 3 Convert to feet (12 inches = 1 foot): $2,112 \div 12 = 176$ feet.

WHY NOT (A) uses the radius instead of the diameter. (C) uses the diameter in $2\pi r$. (D) forgets to convert inches to feet.

24. **C** \$960

- 1 First amount (sent home): $\frac{1}{3} \times \$2,880 = \960 .
- 2 What is left: $\$2,880 - \$960 = \$1,920$.
- 3 Second amount is $\frac{1}{2}$ of that: $\frac{1}{2} \times \$1,920 = \960 .

WHY NOT (D) takes $\frac{1}{2}$ of the whole pay instead of $\frac{1}{2}$ of what is left.

25. **A** 16 pounds

- 1 Let y be the pounds of cashews. The money must balance: $4(20) + 13y = 8(20 + y)$.
- 2 Multiply out: $80 + 13y = 160 + 8y$.
- 3 Collect terms: $13y - 8y = 160 - 80$, so $5y = 80$ and $y = 16$ pounds.

TIP Balance shortcut: each pound of peanuts is \$4 away from \$8, each pound of cashews is \$5 away; the totals must match: $20 \times 4 = 16 \times 5$.

WHY NOT (B) assumes equal amounts, which only works if the target is halfway between the prices. (C) swaps the two price differences. (D) gives the total weight of the blend.

26. **A** $28d + 1.50m$

- 1 Daily charges: $28d$ dollars.
- 2 Mileage charges: $1.50m$ dollars.
- 3 Total: $28d + 1.50m$.

WHY NOT (B) matches each rate with the wrong quantity. (C) multiplies everything together. (D) charges the daily rate only once instead of d times.

27. **B** the box of 15

1 Find each price per bar (price $\div$ number of bars):
 $\$20.40 \div 12 = \1.70 ; $\$23.70 \div 15 = \1.58 ; $\$32 \div 20 = \1.60 .

2 The lowest price per bar is \$1.58, so the box of 15 is the best buy.

TIP The biggest package is not always the best deal, so always compare unit prices.

WHY NOT (A) is not true: the prices per bar are \$1.70, \$1.58, and \$1.60. (C) has the lowest price tag, but it costs \$1.70 per bar, more than \$1.58. (D) is the largest size, but it costs \$1.60 per bar, more than \$1.58.

28. **D** 55%

1 After 40% off you pay 60% of the price, and after another 25% off you pay 75% of *that*.

2 Multiply: $0.6 \times 0.75 = 0.45$, so you pay 45% of the original price.

3 The single discount is $100\% - 45\% = 55\%$.

TIP Test it on \$100: 40% off gives \$60, and 25% off that gives \$45, which is \$55 off.

WHY NOT (A) multiplies the two percents. (B) is the percent of the price you still pay, not the discount.

29. **C** 60 mph

1 Feet per hour: $88 \times 3,600 = 316,800$.

2 Miles per hour: $316,800 \div 5,280 = 60$.

TIP Simplify first: $3,600 \div 5,280 = \frac{15}{22}$, so $88 \times \frac{15}{22} = 60$.

WHY NOT (A) changes seconds to minutes but not to hours. (D) changes seconds to hours but never changes feet to miles.

30. **C** \$1,600

1 Change each length to yards: $24 \div 3 = 8$ yd and $15 \div 3 = 5$ yd.

2 Area in square yards: $8 \times 5 = 40$ square yards.

3 Cost: $40 \times 40 = 1,600$, so the carpet costs \$1,600.

TIP Or find square feet first ($24 \times 15 = 360$) and divide by 9: $360 \div 9 = 40$ square yards.

WHY NOT (A) finds the number of square yards but forgets to multiply by the price. (D) forgets to change square feet to square yards.

Explanations: Mathematics Knowledge

1. **A** $x^2 - 9x + 8$

1 Use FOIL. First: $x \cdot x = x^2$. Outer: $x \cdot (-8) = -8x$. Inner: $(-1) \cdot x = -x$. Last: $(-1) \cdot (-8) = 8$.

2 Add them and combine the two middle terms:
 $x^2 - 8x - x + 8 = x^2 - 9x + 8$.

WHY NOT (B) multiplies only the First and Last terms and leaves out the middle. (C) gets the sign of the last term wrong. (D) gets the sign of the middle term wrong.

2. **A** 72π ft²

1 The border and the flower bed together form a large circle with radius $7 + 4 = 11$ feet.

2 Large circle: $\pi \times 11^2 = 121\pi$. The flower bed:
 $\pi \times 7^2 = 49\pi$.

3 The border is the difference: $121\pi - 49\pi = 72\pi$ square feet.

WHY NOT (B) includes the flower bed itself. (C) adds the two circle areas instead of subtracting. (D) adds the width twice when finding the outer radius.

3. **C** 2,727

1 Write 303 as $300 + 3$, an easy number plus a little.

2 Distribute the 9: $9 \times 300 + 9 \times 3 = 2,700 + 27$.

3 Add: $2,700 + 27 = 2,727$.

WHY NOT (A) subtracts 9×3 instead of adding it. (B) adds 3 instead of 9×3 .

4. **D** -7

1 Work inside the bars first: $9 - 10 = -1$, so $|9 - 10| = |-1| = 1$.

2 Also $|-8| = 8$.

3 Subtract: $1 - 8 = -7$.

WHY NOT (B) treats $|9 - 10|$ as -1 instead of 1.

5. **A** $6\sqrt{2}$

1 Find the largest perfect square that divides 72: it is 36, because $72 = 36 \times 2$.

2 Split the root: $\sqrt{72} = \sqrt{36} \times \sqrt{2} = 6\sqrt{2}$.

TIP Check: $(6\sqrt{2})^2 = 36 \times 2 = 72$. ✓

WHY NOT (D) pulls out 36 instead of its square root, 6.

6. **D** 168

- 1 Write each number as a product of primes: 3 (prime), $4 = 2^2$, 7 (prime), $8 = 2^3$.
- 2 Take every prime that appears, with its highest power: $2^3 \times 3 \times 7 = 168$.
- 3 Check: 168 divided by each of the numbers gives a whole number.

WHY NOT (A) is not divisible by 7. (B) is not divisible by 3. (C) is not divisible by 8.

 7. **A** 30

- 1 Divide by the bottom: $42 \div 7 = 6$.
- 2 Multiply by the top: $6 \times 5 = 30$.

WHY NOT (C) divides by the fraction instead of multiplying. (D) multiplies by 5 but forgets to divide by 7.

 8. **B** 32

- 1 Substitute, putting each value in parentheses: $(4)^2 - 2(4)(-2)$.
- 2 Powers first: $(4)^2 = 16$.
- 3 Multiply each term (watch the signs): $16 + 16$.
- 4 Add and subtract from left to right: $16 + 16 = 32$.

WHY NOT (A) forgets that a negative times a negative is positive.

 9. **D** $2n - 3 = 25$

- 1 Translate piece by piece. Let n be the number; "is," "gives," and "equals" all become "=".
- 2 "The difference between A and B " means $A - B$, in that order. Here $A = 2n$ and $B = 3$: $2n - 3$.
- 3 The equation is $2n - 3 = 25$.

WHY NOT (A) subtracts in the wrong order. (B) adds instead of subtracting. (C) subtracts 3 before multiplying by 2.

 10. **D** $(-10, -6)$

- 1 From A to M , x changes by $-1 - 8 = -9$ and y changes by $1 - 8 = -7$.
- 2 B is the same distance past M : $x = -1 + (-9) = -10$ and $y = 1 + (-7) = -6$.
- 3 Check: the midpoint of $(8, 8)$ and $(-10, -6)$ is $\left(\frac{-2}{2}, \frac{2}{2}\right) = (-1, 1)$. ✓

WHY NOT (A) finds only the change from A to M , not the endpoint. (B) goes from M through A instead of from A through M . (C) adds A and M without doubling M .

 11. **B** $x = -3$ or $x = 2$

- 1 Get 0 on one side: move every term to the left. $x^2 + x - 6 = 0$.
- 2 Factor: two numbers that multiply to -6 and add to 1 are 3 and -2 , so $(x + 3)(x - 2) = 0$.
- 3 Set each factor equal to 0: $x = -3$ or $x = 2$.

WHY NOT (A) factors with -6 and 1, which multiply to -6 but do not add to 1. (C) uses the numbers in the factors instead of their opposites. (D) finds only one of the two solutions.

 12. **D** $972\pi \text{ in}^3$

- 1 The radius is half the diameter: $18 \div 2 = 9$.
 - 2 Cube the radius: $9^3 = 9 \times 9 \times 9 = 729$.
 - 3 Substitute: $V = \frac{4}{3} \times \pi \times 729 = 972\pi$ cubic inches.
- TIP** Divide before you multiply: $729 \div 3 = 243$, then $4 \times 243 = 972$.

WHY NOT (A) squares the radius instead of cubing it. (B) uses $4\pi r^2$, which is the surface area.

 13. **C** 1.65

- 1 Line up the decimal points and fill in zeros so both numbers have 2 decimal places: $30.00 - 28.35$.
- 2 Subtract column by column, keeping the point in place: $30.00 - 28.35 = 1.65$.

WHY NOT (A) rounds 28.35 up to 29 and subtracts; that is only an estimate. (D) subtracts the smaller digit from the larger in each column instead of borrowing.

 14. **A** 5

- 1 Horizontal change: $|9 - 6| = 3$. Vertical change: $|7 - 3| = 4$.
- 2 These are the legs of a right triangle, and the distance is the hypotenuse: $d = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5$.

TIP The legs 3 and 4 belong to the Pythagorean triple 3-4-5.

WHY NOT (C) adds the horizontal and vertical distances instead of using the Pythagorean theorem. (D) forgets to take the square root.

 15. **C** -2

- 1 Solve first. Subtract 3 from all three parts: $-20 < 5x < -5$, then divide all three by 5: $-4 < x < -1$.
- 2 Only -2 is in this range. Check: $5(-2) + 3 = -7$, and $-17 < -7 < -2$. ✓

WHY NOT (A) gives $5(-5) + 3 = -22$, which is outside the range. (B) makes the middle equal to -17 , but that end is not included. (D) makes the middle equal to -2 , but that end is not included.

16. **B** 10

1 In a 30° - 60° - 90° triangle the sides are in the ratio $1 : \sqrt{3} : 2$ (short leg : long leg : hypotenuse). The short leg is across from the 30° angle.

2 Short leg = long leg $\div \sqrt{3} = 10\sqrt{3} \div \sqrt{3} = 10$.

WHY NOT (A) halves the long leg. (C) gives the hypotenuse, not the short leg. (D) multiplies by $\sqrt{3}$ instead of dividing.

17. **B** 42

1 Cross-multiply: $12 \times x = 24 \times 21$, so $12x = 504$.

2 Divide both sides by 12: $x = 504 \div 12 = 42$.

3 Check: $\frac{24}{42}$ and $\frac{12}{21}$ both simplify to $\frac{4}{7}$. ✓

WHY NOT (A) adds the difference of the numerators instead of scaling. (D) multiplies 24 by 21 but forgets to divide by 12.

18. **A** $\frac{1}{10}$

1 Same base, so add the exponents: $-6 + 5 = -1$, which gives 10^{-1} .

2 A negative exponent means a reciprocal: $10^{-1} = \frac{1}{10^1} = \frac{1}{10}$.

WHY NOT (D) subtracts the exponents in the wrong order.

19. **B** 60 yd

1 Find the missing corner: it is $17 - 12 = 5$ yd wide and $13 - 5 = 8$ yd tall.

2 Go around the shape and add all six sides: $17 + 5 + 5 + 8 + 12 + 13 = 60$ yards.

TIP Shortcut: the two inside edges together make up for the missing pieces of the top and side, so the perimeter equals that of a 17×13 rectangle: $2 \times (17 + 13) = 60$.

WHY NOT (A) adds only the four labeled sides and leaves out the two edges of the cut-out corner. (C) counts the two inside edges twice. (D) gives the area, not the perimeter.

20. **C** 24

1 Work from the inside out. First, 50% of 60: $0.5 \times 60 = 30$.

2 Then 80% of that result: $0.8 \times 30 = 24$.

TIP Percents of percents multiply: $0.8 \times 0.5 = 0.4$, and $0.4 \times 60 = 24$.

WHY NOT (A) subtracts the two percents. (D) adds the two percents.

21. **B** 91

1 A pentagon has 5 sides, so its interior angles add up to $(5 - 2) \times 180^\circ = 540^\circ$.

2 Add the known angles: $127 + 89 + 146 + 87 = 449$.

3 Subtract: $x = 540 - 449 = 91$.

WHY NOT (C) divides the angle sum by the number of angles, as if all the angles were equal. (D) uses $(n - 1) \times 180^\circ$ for the angle sum instead of $(n - 2) \times 180^\circ$.

22. **C** 9

1 Look at the coefficients: they are swapped. Add the two equations: $7x + 7y = 63$.

2 Every term has a factor of 7, so divide by 7: $x + y = \frac{63}{7} = 9$.

TIP No need to find x and y separately (they are 12 and -3).

WHY NOT (A) gives the value of y , not $x + y$. (D) forgets to divide by 7.

23. **B** $(5\pi + 20)$ cm

1 The distance around is the curved edge plus two straight edges, each a radius (10 centimeters).

2 The curved edge is one fourth of the circumference: $\frac{1}{4} \times 2\pi \times 10 = 5\pi$ centimeters.

3 Add the straight edges: $10 + 10 = 20$, so the total is $5\pi + 20$ centimeters.

WHY NOT (A) forgets the two straight edges (the radii). (C) uses half of the circumference instead of one fourth. (D) uses the whole circumference instead of one fourth.

24. **C** $1,350 \text{ cm}^3$

1 Each of the three dimensions is multiplied by 3, so the volume is multiplied by $3 \times 3 \times 3 = 27$.

2 New volume: $50 \times 27 = 1,350$ cubic centimeters.

WHY NOT (A) multiplies the volume by 3 instead of 3^3 . (B) multiplies the volume by 3^2 instead of 3^3 .

25. **C** 75

1 Clear the fractions: multiply every term by the LCD of 6 and 10, which is 30.

2 $30 \cdot \frac{x}{6} = 5x$, $30 \cdot \frac{x}{10} = 3x$, and $30 \cdot 5 = 150$. So $5x - 3x = 150$.

3 Combine like terms: $2x = 150$.

4 Divide by 2: $x = \frac{150}{2} = 75$.

5 Check: $\frac{75}{6} - \frac{75}{10} = \frac{25}{2} - \frac{15}{2} = 5$. ✓

WHY NOT (A) forgets to multiply the right side by 30. (B) subtracts the denominators, as if the result were $\frac{x}{4}$.

Practice Test 4

Instructions

This test has two parts, timed like the paper-and-pencil ASVAB. **Part 1, Arithmetic Reasoning:** 30 questions, 36 minutes. **Part 2, Mathematics Knowledge:** 25 questions, 24 minutes. No calculator. Use scratch paper. Answer every question—there is no penalty for guessing. Mark your answers on the answer sheet below.

Answer Sheet

Fill in one bubble per question. Photocopy this page to retake the test.

Part 1: Arithmetic Reasoning

- 1 ☐ A ☐ B ☐ C ☐ D
- 2 ☐ A ☐ B ☐ C ☐ D
- 3 ☐ A ☐ B ☐ C ☐ D
- 4 ☐ A ☐ B ☐ C ☐ D
- 5 ☐ A ☐ B ☐ C ☐ D
- 6 ☐ A ☐ B ☐ C ☐ D
- 7 ☐ A ☐ B ☐ C ☐ D
- 8 ☐ A ☐ B ☐ C ☐ D
- 9 ☐ A ☐ B ☐ C ☐ D
- 10 ☐ A ☐ B ☐ C ☐ D
- 11 ☐ A ☐ B ☐ C ☐ D
- 12 ☐ A ☐ B ☐ C ☐ D
- 13 ☐ A ☐ B ☐ C ☐ D
- 14 ☐ A ☐ B ☐ C ☐ D
- 15 ☐ A ☐ B ☐ C ☐ D
- 16 ☐ A ☐ B ☐ C ☐ D
- 17 ☐ A ☐ B ☐ C ☐ D
- 18 ☐ A ☐ B ☐ C ☐ D
- 19 ☐ A ☐ B ☐ C ☐ D
- 20 ☐ A ☐ B ☐ C ☐ D
- 21 ☐ A ☐ B ☐ C ☐ D
- 22 ☐ A ☐ B ☐ C ☐ D
- 23 ☐ A ☐ B ☐ C ☐ D
- 24 ☐ A ☐ B ☐ C ☐ D
- 25 ☐ A ☐ B ☐ C ☐ D
- 26 ☐ A ☐ B ☐ C ☐ D
- 27 ☐ A ☐ B ☐ C ☐ D
- 28 ☐ A ☐ B ☐ C ☐ D
- 29 ☐ A ☐ B ☐ C ☐ D
- 30 ☐ A ☐ B ☐ C ☐ D

Part 2: Mathematics Knowledge

- 1 ☐ A ☐ B ☐ C ☐ D
- 2 ☐ A ☐ B ☐ C ☐ D
- 3 ☐ A ☐ B ☐ C ☐ D
- 4 ☐ A ☐ B ☐ C ☐ D
- 5 ☐ A ☐ B ☐ C ☐ D
- 6 ☐ A ☐ B ☐ C ☐ D
- 7 ☐ A ☐ B ☐ C ☐ D
- 8 ☐ A ☐ B ☐ C ☐ D
- 9 ☐ A ☐ B ☐ C ☐ D
- 10 ☐ A ☐ B ☐ C ☐ D
- 11 ☐ A ☐ B ☐ C ☐ D
- 12 ☐ A ☐ B ☐ C ☐ D
- 13 ☐ A ☐ B ☐ C ☐ D
- 14 ☐ A ☐ B ☐ C ☐ D
- 15 ☐ A ☐ B ☐ C ☐ D
- 16 ☐ A ☐ B ☐ C ☐ D
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- 18 ☐ A ☐ B ☐ C ☐ D
- 19 ☐ A ☐ B ☐ C ☐ D
- 20 ☐ A ☐ B ☐ C ☐ D
- 21 ☐ A ☐ B ☐ C ☐ D
- 22 ☐ A ☐ B ☐ C ☐ D
- 23 ☐ A ☐ B ☐ C ☐ D
- 24 ☐ A ☐ B ☐ C ☐ D
- 25 ☐ A ☐ B ☐ C ☐ D

Part 1: Arithmetic Reasoning

Score: ____ / 30

Target time: 36 minutes

- Emily cuts an 11-foot board into pieces 7 inches long. How many full pieces can be cut?
A 1 B 6 C 16 D 18
- A recipe calls for $2\frac{3}{4}$ cups of milk. Marcus is making 3 batches. How many cups of milk does he need?
A $5\frac{3}{4}$ cups B $6\frac{3}{4}$ cups C $7\frac{1}{4}$ cups D $8\frac{1}{4}$ cups
- Specialist Patel is assigning ID codes to the soldiers in a training company. Each code is one letter from A to D followed by 3 digits (0 to 9, repeats allowed). How many different ID codes are possible?
A 34 B 40 C 2,880 D 4,000
- Aisha jogs at a steady 5 miles per hour. How many minutes does it take her to jog 4 miles?
A 48 minutes B 53 minutes
C 63 minutes D 80 minutes
- The average score of the 21 students who took a test is 88. A student who missed the test takes it later and scores 44. What is the new class average?
A 86 B 88 C 90 D 101
- A convoy must travel 240 miles. On the first day it covers $\frac{3}{8}$ of the distance, and on the second day it covers $\frac{3}{5}$ of the remaining distance. How many miles are left to travel?
A 60 B 70 C 90 D 150
- Mei's pickup truck gets 21 miles per gallon. Gas costs \$4.50 per gallon. How much will gas cost for a road trip of 210 miles?
A \$10.00 B \$35.00 C \$45.00 D \$94.50
- First squad can unload a supply truck in 90 minutes. Second squad can unload the same truck in 45 minutes. How long will it take if both squads work together?
A 15 minutes B 30 minutes
C 45 minutes D 135 minutes
- A road march covers 24 kilometers. About how many miles is that? (1 mile $\approx$ 1.6 kilometers)
A 14.4 miles B 15 miles
C 38.4 miles D 150 miles
- Derek is putting a fence around a rectangular yard that measures 40 feet by 30 feet. Fencing costs \$13 per foot. How much will it cost to fence the whole yard?
A \$910 B \$1,430 C \$1,820 D \$15,600
- Sofia drives 22 miles on the interstate in 20 minutes. What is her average speed in miles per hour?
A 1.1 mph B 46 mph C 56 mph D 66 mph
- Private Okafor borrowed \$1,500 at simple interest and paid \$675 in interest over 5 years. What was the yearly interest rate?
A 0.09% B 1.8% C 9% D 45%
- The angles of a triangle are in the ratio 1 : 6 : 2. What is the measure of the largest angle?
A 20° B 120° C 140° D 150°
- A 42-foot log is cut into 6-foot pieces. How many cuts are needed?
A 6 B 7 C 8 D 21
- A store sold a bicycle for \$664. It increased the price by 25%, then put the bicycle on sale at 40% off the new price. What is the sale price?
A \$348.00 B \$398.40 C \$498.00 D \$564.40
- A used-car dealer pays \$10,000 for a pickup truck and marks up the cost by 25%. What price is put on the truck?
A \$2,500 B \$7,500 C \$10,025 D \$12,500
- Combat engineers mix concrete for a bunker floor using cement, sand, and gravel in the ratio 1 : 2 : 3 by volume. How many cubic feet of sand are needed if 24 cubic feet of gravel are used?
A 10 cubic feet B 16 cubic feet
C 23 cubic feet D 48 cubic feet

18. Rosa is taking a photo of 6 soldiers standing in a line. In how many different orders can the 6 soldiers line up?
A 21 B 720 C 920 D 46,656
19. Priya's dinner bill is \$40 before tax. The sales tax is 7%, and she leaves an 18% tip on the \$40 (before tax). What is the total cost?
A \$42.80 B \$50.00 C \$60.00 D \$65.00
20. Before winter, a motor pool has 20 gallons of coolant that is 10% antifreeze. How many gallons of pure antifreeze must be added to make coolant that is 50% antifreeze?
A 8 gallons B 16 gallons
C 36 gallons D 80 gallons
21. A car is traveling at 88 feet per second on the highway. What is this speed in miles per hour? (1 mile = 5,280 feet and 1 hour = 3,600 seconds.)
A 1 mph B 55 mph
C 60 mph D 316,800 mph
22. A medic at an aid station has 650 milliliters of a 70% alcohol solution. How many milliliters of a 40% alcohol solution must be added to make a mixture that is 65% alcohol?
A 50 milliliters B 120 milliliters
C 130 milliliters D 3,250 milliliters
23. Keisha takes home \$3,780 a month. She spends $\frac{1}{3}$ of the pay on rent, and then puts $\frac{1}{4}$ of what is left into savings. How much money does she put into savings?
A \$630 B \$945 C \$1,890 D \$2,520
24. Yuki works as a delivery driver and earns \$24 per hour, with time-and-a-half for every hour over 40 in a week. Last week her pay was \$1,032. How many hours did she work last week?
A 42 hours B 43 hours
C 45 hours D 46 hours
25. A gym offers two options: pay \$10 per visit plus a \$15 monthly fee, or pay \$50 per month for unlimited visits. What is the greatest number of visits in a month for which paying per visit costs *less* than the unlimited plan?
A 1 visit B 2 visits C 3 visits D 4 visits
26. A dishwasher is priced at \$650. It is on sale for 20% off, and a 6% sales tax is charged on the sale price. What is the total cost?
A \$488.80 B \$501.20 C \$520.00 D \$551.20
27. A bowling league has 24 members. The 16 members of the morning group average 162, and the whole league averages 168. What is the average of the evening group?
A 165 B 168 C 174 D 180
28. Jamal leans a 25-foot ladder against a building, with its foot 7 feet from the wall. The foot slips until it is 15 feet from the wall. How far down the wall does the top of the ladder slide?
A 3 ft B 4 ft C 8 ft D 24 ft
29. Priya deposits \$9,000 in an account that pays 3% interest per year, compounded once a year. How much interest will the account earn in 2 years?
A \$548.10 B \$598.10
C \$698.10 D \$9,548.10
30. At the base commissary, sports drinks come in three sizes: a pack of 6 sports drinks for \$5.40, a pack of 8 sports drinks for \$7.76, and a pack of 24 sports drinks for \$17.76. Which size is the best buy?
A the pack of 24
B the pack of 6
C the pack of 8
D All three cost the same per bottle

Part 2: Mathematics Knowledge

Score: ____ / 25

Target time: 24 minutes

1. Which expression is equal to $(x - 5)^2$?
- A $x^2 + 25$ B $x^2 - 10x + 25$
 C $x^2 - 25$ D $x^2 - 10x - 25$

2. Solve the system below. What is the value of x ?

$$3x + y = 47$$

$$2x - y = 13$$

- A -12 B 1 C 11 D 12

3. If $f(x) = -2x^2 + x + 8$, what is $f(2)$?
- A -2 B 2 C 4 D 26

4. A circle has an area of 6π square meters. If the radius of the circle is multiplied by 4, what is the area of the new circle?

- A $24\pi \text{ m}^2$ B $48\pi \text{ m}^2$
 C $96\pi \text{ m}^2$ D $384\pi \text{ m}^2$

5. Which is the factored form of $x^2 - 81$?

- A $(x - 1)(x + 81)$ B $(x - 9)^2$
 C $(x - 9)(x + 9)$ D $(x + 9)^2$

6. Andre has \$70 saved and earns \$14 per hour. He wants to have no less than \$600. Which inequality shows the number of hours h he must work?

- A $70 + 14h < 600$ B $14(70 + h) \geq 600$
 C $70 + 14h > 600$ D $70 + 14h \geq 600$

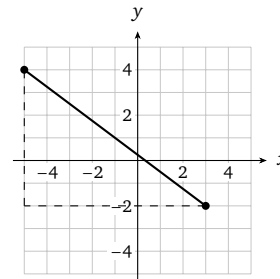
7. What is $7\frac{1}{3} + 2\frac{5}{6}$?

- A $9\frac{1}{6}$ B $9\frac{2}{3}$ C 10 D $10\frac{1}{6}$

8. 21 is what percent of 28?

- A 0.75% B 7.5% C 75% D 95%

9. In the figure, a segment joins the points $(3, -2)$ and $(-5, 4)$. How long is the segment?



- A 2 B 9 C 10 D 14

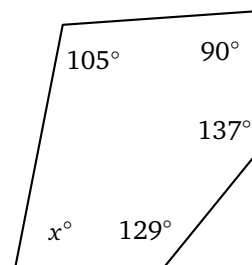
10. Solve for x : $\frac{x}{4} + \frac{x}{10} = 7$.

- A 1 B 18 C 19 D 20

11. What is $15 - 3.09$?

- A 11.91 B 12.09 C 12.91 D 18.09

12. In the pentagon shown, four of the interior angles measure 129° , 137° , 90° , and 105° . What is the value of x ?

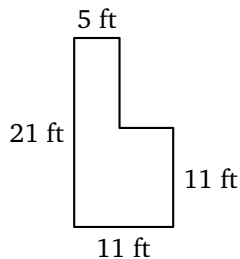


- A 59 B 79 C 108 D 259

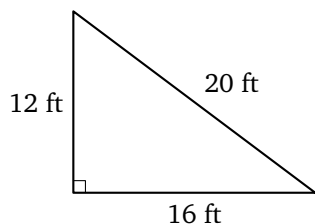
13. If $\frac{14}{x} = \frac{42}{24}$, what is the value of x ?

- A 4 B 6 C 8 D 336

14. The figure shows the floor of a barracks day room. All corners are right angles. The sides shown measure 11 feet, 21 feet, 5 feet, and 11 feet. What is the area?



- A 55 ft^2 B 171 ft^2 C 176 ft^2 D 231 ft^2
15. A cube-shaped box has a volume of 8 cubic centimeters. What is its total surface area?
- A 16 cm^2 B 24 cm^2 C 28 cm^2 D 32 cm^2
16. Which of the following is equal to $\frac{6^8 \times 6^6}{6^7}$?
- A 6^7 B 6^{21} C 6^2 D 6^{41}
17. If n stands for a number, which expression represents “half of a number, decreased by 12”?
- A $2n - 12$ B $\frac{n-12}{2}$ C $\frac{n}{2} - 12$ D $12 - \frac{n}{2}$
18. The right triangle shown has sides of 12 feet, 16 feet, and 20 feet. What is its area?



19. The radius of a cylinder is doubled, and its height stays the same. The volume of the new cylinder is how many times the volume of the original?

A 4 B 2 C 3 D 8

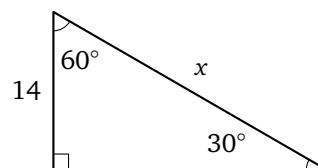
20. What is the least common multiple of 6, 5, 4, and 9?

A 90 B 180 C 360 D 1,080

21. Evaluate: $12^2 - 2 \times (6 + 2) \div 4$

A 140 B 160 C 220 D 284

22. In a 30° - 60° - 90° triangle, the shorter leg is 14 long. How long is the hypotenuse?



A 7 B $14\sqrt{2}$ C $14\sqrt{3}$ D 28

23. Simplify: $(-4)^3 - (-6)^2$

A -100 B -28 C -24 D 28

24. Simplify $\frac{27}{\sqrt{6}}$ so that there is no radical in the denominator.

A $\frac{9}{2}$ B $\frac{9\sqrt{6}}{2}$ C 27 D $27\sqrt{6}$

25. Simplify $\frac{x^2y^{-4}}{x^{-3}y}$ and write the answer with positive exponents.

A $\frac{x^5}{y^5}$ B $\frac{y^3}{x}$ C $\frac{y^5}{x^5}$ D x^5y^5

A 96 ft^2 B 160 ft^2
C 192 ft^2 D $1,920 \text{ ft}^2$

Answer Key: Practice Test 4

The small number after each answer is the chapter that teaches the skill.

Part 1: Arithmetic Reasoning

1 D ch. 27	7 C ch. 24	13 B ch. 7	19 B ch. 23	25 C ch. 12
2 D ch. 4	8 B ch. 25	14 A ch. 27	20 B ch. 25	26 D ch. 22
3 D ch. 26	9 B ch. 27	15 C ch. 6	21 C ch. 24	27 D ch. 26
4 A ch. 24	10 C ch. 18	16 D ch. 23	22 C ch. 25	28 B ch. 17
5 A ch. 26	11 D ch. 24	17 B ch. 25	23 A ch. 22	29 A ch. 23
6 A ch. 27	12 C ch. 23	18 B ch. 26	24 A ch. 22	30 A ch. 22

Part 2: Mathematics Knowledge

1 B ch. 14	6 D ch. 12	11 A ch. 5	16 A ch. 8	21 A ch. 1
2 D ch. 13	7 D ch. 4	12 B ch. 16	17 C ch. 10	22 D ch. 17
3 B ch. 15	8 C ch. 6	13 C ch. 7	18 A ch. 18	23 A ch. 2
4 C ch. 19	9 C ch. 21	14 B ch. 18	19 A ch. 20	24 B ch. 9
5 C ch. 15	10 D ch. 11	15 B ch. 20	20 B ch. 3	25 A ch. 14

Scoring your practice test

Percent correct	AR (of 30)	MK (of 25)
90% or more — excellent	27–30	23–25
75–89% — good	23–26	19–22
60–74% — keep practicing	18–22	15–18
below 60% — review the chapters	0–17	0–14

These bands measure your progress in this book only. Your real AFQT percentile depends on all four AFQT subtests and is scaled by the Department of Defense.

Explanations: Arithmetic Reasoning

1. **D** 18

- 1 Change feet to inches: $11 \times 12 = 132$ inches.
- 2 Divide: $132 \div 7$ is 18 with a remainder of 6.
- 3 Only full ones count, so round down to 18.

WHY NOT (A) forgets to change feet to inches. (B) gives the leftover amount, not the number asked for.

2. **D** $8\frac{1}{4}$ cups

- 1 Multiply the whole number and the fraction by 3: $2 \times 3 = 6$ and $\frac{3}{4} \times 3 = \frac{9}{4} = 2\frac{1}{4}$.
- 2 Add: $6 + 2\frac{1}{4} = 8\frac{1}{4}$.

WHY NOT (A) adds 3 instead of multiplying by 3. (B) multiplies only the whole number. (C) drops the extra whole number from $\frac{9}{4}$.

3. **D** 4,000

- 1 Count the choices for each position: 4, 10, 10, 10.
- 2 Multiply (counting principle): $4 \times 10 \times 10 \times 10 = 4,000$.

WHY NOT (A) adds the choices instead of multiplying them. (B) counts only one digit. (C) does not let the digits repeat.

4. **A** 48 minutes

- 1 Time in hours: $4 \div 5 = \frac{4}{5}$ hour.
- 2 Change to minutes: $\frac{4}{5} \times 60 = 48$ minutes.

TIP Or: at 5 mph one mile takes $60 \div 5 = 12$ minutes, and $4 \times 12 = 48$.

WHY NOT (D) reads 0.8 hour as 80 minutes.

5. **A** 86

- 1 Turn the average into a total: $21 \times 88 = 1,848$.
- 2 Add the value: $1,848 + 44 = 1,892$.
- 3 Divide by the new count, $21 + 1 = 22$: $1,892 \div 22 = 86$.

WHY NOT (B) assumes the average does not change. (C) adds the new value divided by the new count to the old average.

6. **A** 60

- 1 First step: $\frac{3}{8} \times 240 = 90$. Remainder: $240 - 90 = 150$.
- 2 Second step: $\frac{3}{5}$ of the remainder: $\frac{3}{5} \times 150 = 90$.
- 3 Left: $150 - 90 = 60$.

WHY NOT (C) answers a different part of the question. (D) stops after the first step.

7. **C** \$45.00

- 1 Gallons needed: $210 \div 21 = 10$ gallons.
- 2 Cost: $10 \times \$4.50 = \45 .

WHY NOT (A) gives the number of gallons, not the cost. (D) multiplies the miles per gallon by the price.

8. **B** 30 minutes

- 1 Turn each time into a rate (the part of the job done in one minute): first squad does $\frac{1}{90}$ of the job per minute, and second squad does $\frac{1}{45}$.
- 2 Add the rates (common denominator 90): $\frac{1}{90} + \frac{1}{45} = \frac{1}{90} + \frac{2}{90} = \frac{3}{90} = \frac{1}{30}$ of the job per minute.
- 3 Time = 1 job $\div$ rate: $1 \div \frac{1}{30} = 30$ minutes.

TIP The answer must be less than the faster time, 45 minutes.

WHY NOT (C) subtracts the two times. (D) adds the two times instead of adding the two rates.

9. **B** 15 miles

- 1 A mile is longer than a kilometer, so there are fewer miles. Divide by 1.6: $24 \div 1.6$.
- 2 Clear the decimal: $240 \div 16 = 15$ miles.

WHY NOT (A) multiplies by 0.6. (C) multiplies by 1.6 instead of dividing. (D) slips the decimal point.

10. **C** \$1,820

- 1 A fence goes around the edge, so find the perimeter: $2 \times 40 + 2 \times 30 = 80 + 60 = 140$ feet.
- 2 Multiply by the price per foot: $140 \times 13 = 1,820$, so the fence costs \$1,820.

WHY NOT (A) fences only one length and one width (half the perimeter). (B) leaves out one side. (D) uses the area instead of the perimeter.

11. **D** 66 mph

- 1 Change the time to hours: $20 \text{ min} = \frac{20}{60} = \frac{1}{3} \text{ hour}$.
- 2 Speed: $22 \div \frac{1}{3} = 22 \times 3 = 66$ miles per hour.

TIP Or scale up to an hour: 20 minutes goes into 60 minutes 3 times, and $22 \times 3 = 66$.

WHY NOT (A) gives miles per minute, not miles per hour.

12. **C** 9%

- 1 Interest for one year: $\$675 \div 5 = \135 .
- 2 Divide by the principal: $\$135 \div \$1,500 = 0.09$.
- 3 Change to a percent: $0.09 = 9\%$.

TIP Check: $\$1,500 \times 0.09 \times 5 = \675 .

WHY NOT (A) forgets to change the decimal to a percent. (B) divides by 5 twice. (D) forgets to divide by the 5 years.

13. **B** 120°

- 1 The three angles of a triangle add up to 180°. Add the terms of the ratio: $1 + 6 + 2 = 9$ equal parts.
- 2 One part is $180 \div 9 = 20$ degrees.
- 3 The largest angle is 6 parts: $6 \times 20 = 120^\circ$.

TIP Check: $20 + 120 + 40 = 180$.

WHY NOT (A) is the smallest angle.

14. **A** 6

- 1 Number of pieces: $42 \div 6 = 7$.
- 2 The last piece needs no cut of its own, so cuts = pieces -1 : $7 - 1 = 6$.

TIP One cut makes 2 pieces, two cuts make 3 pieces: always one fewer cut than pieces.

WHY NOT (B) gives the number of pieces, not the number of cuts.

15. **C** \$498.00

- 1 Raise the price by 25%: $664 \times 1.25 = 830$.
- 2 Take 40% off the new price: $830 \times 0.6 = 498$.
- 3 The final price is \$498.

WHY NOT (B) applies only the decrease. (D) combines the changes into a single 15% decrease.

16. **D** \$12,500

- 1 Markup: $0.25 \times \$10,000 = \$2,500$.
- 2 Selling price: $\$10,000 + \$2,500 = \$12,500$.

TIP In one step: $1.25 \times \$10,000 = \$12,500$.

WHY NOT (A) is the markup, not the selling price. (B) subtracts the markup instead of adding it. (C) adds the percent as if it were dollars.

17. **B** 16 cubic feet

- Gravel is 3 parts, so one part is $24 \div 3 = 8$ cubic feet.
- Sand is 2 parts: $2 \times 8 = 16$ cubic feet.

WHY NOT (C) adds the difference in parts instead of using a multiple. (D) multiplies by the number of parts without first finding the size of one part.

18. **B** 720

- First place: 6 choices; second place: 5 (one is used); and so on down to 1.
- Multiply: $6 \times 5 \times 4 \times 3 \times 2 \times 1 = 720$.

WHY NOT (A) adds the choices instead of multiplying them. (D) lets the same item fill more than one place.

19. **B** \$50.00

- Tax: $0.07 \times \$40 = \2.80 .
- Tip: $0.18 \times \$40 = \7.20 .
- Total: $\$40 + \$2.80 + \$7.20 = \50 .

TIP Both percents are of the same amount, so add them first: $7\% + 18\% = 25\%$, and $1.25 \times \$40 = \50 .

WHY NOT (A) forgets the tip.

20. **B** 16 gallons

- Let y be the gallons of pure antifreeze added (pure means 100%). Pure antifreeze before and after: $0.1(20) + y = 0.5(20 + y)$.
- Multiply out: $2 + y = 10 + 0.5y$.
- Collect terms: $0.5y = 8$, so $y = 16$ gallons.

WHY NOT (A) takes the rise in percent of the original amount only. (C) gives the total amount of the new mixture.

21. **C** 60 mph

- Feet per hour: $88 \times 3,600 = 316,800$.
- Miles per hour: $316,800 \div 5,280 = 60$.

TIP Simplify first: $3,600 \div 5,280 = \frac{15}{22}$, so $88 \times \frac{15}{22} = 60$.

WHY NOT (A) changes seconds to minutes but not to hours. (D) changes seconds to hours but never changes feet to miles.

22. **C** 130 milliliters

- Let y be the milliliters added. The pure alcohol going in must equal the pure alcohol in the mixture: $0.7(650) + 0.4y = 0.65(650 + y)$.
- Multiply out: $455 + 0.4y = 422.5 + 0.65y$.
- Collect the y terms: $32.5 = 0.25y$, so $y = 130$ milliliters.

TIP Balance shortcut: a 70% alcohol solution is 5 points from 65% and a 40% alcohol solution is 25 points away, so $650 \times 5 = y \times 25$, which gives $y = 130$.

WHY NOT (A) treats the added solution as if it were plain water. (D) swaps the two differences.

23. **A** \$630

- First amount (spent on rent): $\frac{1}{3} \times \$3,780 = \$1,260$.
- What is left: $\$3,780 - \$1,260 = \$2,520$.
- Second amount is $\frac{1}{4}$ of that: $\frac{1}{4} \times \$2,520 = \630 .

WHY NOT (B) takes $\frac{1}{4}$ of the whole pay instead of $\frac{1}{4}$ of what is left. (C) is the amount remaining, not the amount asked for. (D) is what is left after the amount spent on rent, before the second step.

24. **A** 42 hours

- Pay for the first 40 hours: $40 \times \$24 = \960 .
- The rest of the pay is overtime: $\$1,032 - \$960 = \$72$.
- Overtime rate: $1.5 \times \$24 = \36 per hour. Overtime hours: $\$72 \div \$36 = 2$.
- Total hours: $40 + 2 = 42$ hours.

WHY NOT (B) divides the overtime pay by the regular rate instead of the overtime rate.

25. **C** 3 visits

- Let v be the number of visits. Paying per visit costs less when $15 + 10v < 50$.
- Subtract 15: $10v < 35$.
- Divide by 10: $v < 35 \div 10 = 3\frac{1}{2}$.
- The greatest whole number below that is 3.

WHY NOT (B) adds the \$15 fee to the price of each visit. (D) rounds up, which makes paying per visit cost more.

26. **D** \$551.20

- Find the discount: $0.2 \times \$650 = \130 .
- Sale price: $\$650 - \$130 = \$520$.
- Tax on the sale price: $0.06 \times \$520 = \31.20 .
- Total: $\$520 + \$31.20 = \$551.20$.

TIP Or chain the multipliers: $\$650 \times 0.8 \times 1.06 = \551.20 .

WHY NOT (A) subtracts the tax instead of adding it. (C) forgets to add the sales tax.

27. **D** 180

- Total for everyone: $24 \times 168 = 4,032$.
- Total for the first group: $16 \times 162 = 2,592$.
- The evening group has $4,032 - 2,592 = 1,440$ points in all, so their average is $1,440 \div 8 = 180$.

WHY NOT (B) gives the average of the whole group. (C) assumes the two groups are the same size.

28. **B** 4 ft

- Before: height = $\sqrt{25^2 - 7^2} = \sqrt{576} = 24$ feet.
- After: height = $\sqrt{25^2 - 15^2} = \sqrt{400} = 20$ feet.
- The top slides down $24 - 20 = 4$ feet.

WHY NOT (C) assumes the top slides down as far as the foot slides out. (D) gives the starting height of the ladder.

29. **A** \$548.10

- Rate for each period: 3%.
- After the first year: $\$9,000 + 0.03 \times \$9,000 = \$9,000 + \$270 = \$9,270$.
- After the second: $\$9,270 + 0.03 \times \$9,270 = \$9,270 + \$278.10 = \$9,548.10$.
- Interest earned: $\$9,548.10 - \$9,000 = \$548.10$.

TIP The second year earns more (\$278.10 vs. \$270) because it earns interest on the interest.

WHY NOT (D) is the balance, not the interest.

30. **A** the pack of 24

- Find each price per bottle (price $\div$ number of bottles): $\$5.40 \div 6 = \0.90 ; $\$7.76 \div 8 = \0.97 ; $\$17.76 \div 24 = \0.74 .
- The lowest price per bottle is \$0.74, so the pack of 24 is the best buy.

WHY NOT (B) has the lowest price tag, but it costs \$0.90 per bottle, more than \$0.74. (C) costs \$0.97 per bottle, more than \$0.74. (D) is not true: the prices per bottle are \$0.90, \$0.97, and \$0.74.

Explanations: Mathematics Knowledge

1. **B** $x^2 - 10x + 25$

- Use the pattern $(A - B)^2 = A^2 - 2AB + B^2$ with $A = x$ and $B = 5$.
- $A^2 = (x)^2 = x^2$, $-2AB = -2(x)(5) = -10x$, and $B^2 = 5^2 = 25$.
- So $(x - 5)^2 = x^2 - 10x + 25$.

TIP Check with $x = 1$: the original and your answer must give the same number.

WHY NOT (A) squares each term but leaves out the middle term $-10x$. (C) squares each term, leaves out the middle term, and makes the last term negative. (D) makes the last term negative, but $(-5)^2 = 25$.

2. **D** 12

- Add the equations. The y -terms cancel: $5x = 60$.
- Divide both sides by 5: $x = \frac{60}{5} = 12$.
- Substitute $x = 12$ into $3x + y = 47$: $3(12) + y = 47$, so $y = 11$.

WHY NOT (A) makes a sign error. (B) gives the value of $x - y$, not x . (C) gives the value of y , not x .

3. **B** 2

- Replace every x with (2): $f(2) = -2(2)^2 + (2) + 8$.
- Square first (order of operations): $2^2 = 4$, then multiply: $-2 \cdot 4 = -8$. So $f(2) = -8 + 2 + 8$.
- Add: $f(2) = 2$.

WHY NOT (A) makes a sign error with x . (D) squares $-2(2)$ instead of squaring only 2.

4. **C** $96\pi \text{ m}^2$

- Area depends on the radius squared, so multiplying the radius by 4 multiplies the area by $4^2 = 16$.
- New area: $16 \times 6\pi = 96\pi$ square meters.

WHY NOT (A) multiplies the area by 4 instead of by 16. (B) multiplies the area by 2×4 instead of 4^2 . (D) multiplies the area by $4^3 = 64$ instead of 4^2 .

5. **C** $(x - 9)(x + 9)$

- $x^2 - 81$ is a difference of squares: $(x)^2 - 9^2$.
- Use $A^2 - B^2 = (A + B)(A - B)$: $x^2 - 81 = (x - 9)(x + 9)$.

WHY NOT (A) uses 81 and -1 , which multiply to -81 but do not add to 0. (B) is a perfect square, not a difference of squares. (D) is a perfect square, not a difference of squares.

6. **D** $70 + 14h \geq 600$

- "No less than 600" means the quantity is greater than or equal to 600: it includes 600 itself, so the symbol is $\geq$.
- The quantity is $70 + 14h$, so the inequality is $70 + 14h \geq 600$.

WHY NOT (A) points the sign the wrong way and gets the boundary wrong. (B) multiplies the savings by 14 too. (C) leaves out the boundary number, which is allowed.

7. **D** $10\frac{1}{6}$
- 1 Add the whole numbers: $7 + 2 = 9$.
 - 2 Add the fractions with the common denominator 6: $\frac{2}{6} + \frac{5}{6} = \frac{7}{6} = 1\frac{1}{6}$.
 - 3 Combine: $9 + 1\frac{1}{6} = 10\frac{1}{6}$.
- WHY NOT** (A) drops the extra whole number from $\frac{7}{6}$. (B) adds the numerators and adds the denominators of the fractions.
8. **C** 75%
- 1 Write the part over the whole: $\frac{21}{28}$.
 - 2 Simplify and convert: $\frac{21}{28} = 0.75 = 75\%$.
- WHY NOT** (A) forgets to multiply by 100. (B) moves the decimal point only one place.
9. **C** 10
- 1 Horizontal change: $|-5 - 3| = 8$. Vertical change: $|4 - (-2)| = 6$.
 - 2 These are the legs of a right triangle, and the distance is the hypotenuse: $d = \sqrt{8^2 + 6^2} = \sqrt{64 + 36} = \sqrt{100} = 10$.
- TIP** The legs 6 and 8 belong to the Pythagorean triple 6-8-10.
- WHY NOT** (A) subtracts the two distances. (D) adds the horizontal and vertical distances instead of using the Pythagorean theorem.
10. **D** 20
- 1 Clear the fractions: multiply *every* term by the LCD of 4 and 10, which is 20.
 - 2 $20 \cdot \frac{x}{4} = 5x$, $20 \cdot \frac{x}{10} = 2x$, and $20 \cdot 7 = 140$. So $5x + 2x = 140$.
 - 3 Combine like terms: $7x = 140$.
 - 4 Divide by 7: $x = \frac{140}{7} = 20$.
 - 5 Check: $\frac{20}{4} + \frac{20}{10} = 5 + 2 = 7$. ✓
- WHY NOT** (A) forgets to multiply the right side by 20.
11. **A** 11.91
- 1 Line up the decimal points and fill in zeros so both numbers have 2 decimal places: $15.00 - 3.09$.
 - 2 Subtract column by column, keeping the point in place: $15.00 - 3.09 = 11.91$.
- WHY NOT** (B) subtracts the smaller digit from the larger in each column instead of borrowing. (C) borrows but forgets to reduce the ones digit it borrowed from. (D) adds instead of subtracting.
12. **B** 79
- 1 A pentagon has 5 sides, so its interior angles add up to $(5 - 2) \times 180^\circ = 540^\circ$.
 - 2 Add the known angles: $129 + 137 + 90 + 105 = 461$.
 - 3 Subtract: $x = 540 - 461 = 79$.
- WHY NOT** (C) divides the angle sum by the number of angles, as if all the angles were equal. (D) uses $(n - 1) \times 180^\circ$ for the angle sum instead of $(n - 2) \times 180^\circ$.
13. **C** 8
- 1 Cross-multiply: $42 \times x = 14 \times 24$, so $42x = 336$.
 - 2 Divide both sides by 42: $x = 336 \div 42 = 8$.
 - 3 Check: $\frac{14}{8}$ and $\frac{42}{24}$ both simplify to $\frac{7}{4}$. ✓
- WHY NOT** (D) multiplies 14 by 24 but forgets to divide by 42.
14. **B** 171 ft^2
- 1 Find the missing corner: it is $11 - 5 = 6$ ft wide and $21 - 11 = 10$ ft tall.
 - 2 Area of the whole 11×21 rectangle: $11 \times 21 = 231$.
 - 3 Subtract the missing corner: $231 - 6 \times 10 = 231 - 60 = 171$ square feet.
- TIP** Or split the shape into two rectangles: $5 \times 21 = 105$ and $6 \times 11 = 66$; $105 + 66 = 171$.
- WHY NOT** (A) multiplies the two shorter sides only. (C) subtracts the wrong rectangle. (D) forgets to subtract the missing corner.
15. **B** 24 cm^2
- 1 Find the edge: which number cubed is 8? $2 \times 2 \times 2 = 8$, so $s = 2$ centimeters.
 - 2 One face: $2^2 = 4$. Six faces: $6 \times 4 = 24$ square centimeters.
- WHY NOT** (A) counts only four of the six faces.
16. **A** 6^7
- 1 Numerator: same base, so add the exponents. $6^8 \times 6^6 = 6^{14}$.
 - 2 Divide: subtract the exponents. $\frac{6^{14}}{6^7} = 6^{14-7} = 6^7$.
- WHY NOT** (B) adds the exponent in the denominator instead of subtracting it. (C) divides the exponents instead of subtracting. (D) multiplies the exponents in the numerator instead of adding them.
17. **C** $\frac{n}{2} - 12$
- 1 “Half of a number” is $\frac{n}{2}$; “decreased by 12” subtracts 12 from that.
 - 2 With n for the number, the expression is $\frac{n}{2} - 12$.
- WHY NOT** (A) doubles instead of halving. (B) subtracts before taking half. (D) reverses the order of the subtraction.

18. **A** 96 ft²

1 In a right triangle the two legs (the sides that form the right angle) are the base and the height. The longest side, 20 feet, is the hypotenuse and is not used.

2 Area of a triangle = $\frac{1}{2} \times \text{base} \times \text{height}$.

3 $\frac{1}{2} \times 16 \times 12 = \frac{1}{2} \times 192 = 96$, so the area is 96 square feet.

WHY NOT (B) uses the hypotenuse as the height. (C) forgets to multiply by $\frac{1}{2}$. (D) multiplies all three sides.

 19. **A** 4

1 In $V = \pi r^2 h$, the radius is squared.

2 Replace r with $2r$: $\pi(2r)^2 h = 4\pi r^2 h$, so the volume is multiplied by 4.

WHY NOT (B) assumes the volume grows by the same factor as the radius. (D) cubes the factor, but only the radius changed, not the height.

 20. **B** 180

1 Write each number as a product of primes: $4 = 2^2$, 5 (prime), $6 = 2 \times 3$, $9 = 3^2$.

2 Take every prime that appears, with its highest power: $2^2 \times 3^2 \times 5 = 180$.

3 Check: 180 divided by each of the numbers gives a whole number.

WHY NOT (A) is not divisible by 4. (C) is divisible by all of them, but it is not the smallest such number. (D) multiplies all the numbers; that works, but it is not the smallest.

 21. **A** 140

1 Parentheses first: $6 + 2 = 8$, which leaves $12^2 - 2 \times 8 \div 4$.

2 Exponent next: $12^2 = 12 \times 12 = 144$, which leaves $144 - 2 \times 8 \div 4$.

3 Multiply (multiplication and division go left to right): $2 \times 8 = 16$, which leaves $144 - 16 \div 4$.

4 Divide before you add or subtract: $16 \div 4 = 4$, which leaves $144 - 4$.

5 Subtract: $144 - 4 = 140$.

WHY NOT (D) adds or subtracts before multiplying or dividing.

 22. **D** 28

1 In a 30° - 60° - 90° triangle the sides are in the ratio $1 : \sqrt{3} : 2$ (short leg : long leg : hypotenuse). The short leg is across from the 30° angle.

2 Hypotenuse = $2 \times \text{short leg} = 2 \times 14 = 28$.

WHY NOT (A) halves the short leg instead of doubling it. (B) uses the 45° - 45° - 90° rule. (C) gives the long leg, not the hypotenuse.

 23. **A** -100

1 $(-4)^3 = (-4)(-4)(-4) = -64$: three negative factors give a negative.

2 $(-6)^2 = (-6)(-6) = 36$: two negative factors give a positive.

3 Subtract: $-64 - 36 = -100$.

WHY NOT (B) treats $(-6)^2$ as -36 . (C) multiplies by the exponents instead of using them as powers. (D) thinks $(-4)^3$ is positive; an odd power of a negative is negative.

 24. **B** $\frac{9\sqrt{6}}{2}$

1 Multiply the top and bottom by $\sqrt{6}$: $\frac{27}{\sqrt{6}} \times \frac{\sqrt{6}}{\sqrt{6}} = \frac{27\sqrt{6}}{6}$, because $\sqrt{6} \cdot \sqrt{6} = 6$.

2 Simplify $\frac{27}{6}$: divide top and bottom by 3, giving $\frac{9\sqrt{6}}{2}$.

WHY NOT (A) multiplies only the bottom by $\sqrt{6}$. (C) cancels the 6 on the bottom against the $\sqrt{6}$ on top. (D) multiplies the top by $\sqrt{6}$ but forgets that the bottom becomes 6.

 25. **A** $\frac{x^5}{y^5}$

1 Subtract exponents of like bases (top minus bottom). For x : $2 - (-3) = 5$, so x^5 .

2 For y : $-4 - 1 = -5$, so $y^{-5} = \frac{1}{y^5}$.

3 Together: $\frac{x^5}{y^5}$.

WHY NOT (B) ignores the negative signs and just subtracts the exponents. (C) puts the variables on the wrong sides of the fraction bar. (D) drops the negative sign on the y -exponent.

Formula Sheet

The ASVAB does not give you formulas. Everything on this page is fair game—learn it until you can write it from memory.

Percents & money

- part = percent $\times$ whole
- percent change = $\frac{\text{new} - \text{original}}{\text{original}} \times 100\%$
- sale price = price $\times$ (1 – discount rate)
- total with tax = price $\times$ (1 + tax rate)
- simple interest: $I = Prt$ (t in years)

Rates

- distance = rate $\times$ time, $d = rt$
- average speed = $\frac{\text{total distance}}{\text{total time}}$
- work together: $\frac{1}{a} + \frac{1}{b} = \frac{1}{t}$
- unit rate = $\frac{\text{amount}}{\text{one unit}}$

Statistics & probability

- mean = $\frac{\text{sum of values}}{\text{number of values}}$
- median: middle value of the *sorted* list
- mode: most frequent value; range = max – min
- $P(\text{event}) = \frac{\text{favorable outcomes}}{\text{total outcomes}}$
- $P(\text{not } A) = 1 - P(A)$; independent: $P(A \text{ and } B) = P(A)P(B)$

Exponents & roots

- $a^m \cdot a^n = a^{m+n}$, $\frac{a^m}{a^n} = a^{m-n}$, $(a^m)^n = a^{mn}$
- $a^0 = 1$, $a^{-n} = \frac{1}{a^n}$, $(ab)^n = a^n b^n$
- $\sqrt{ab} = \sqrt{a} \sqrt{b}$, $\sqrt{\frac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}}$

Algebra

- $(a + b)^2 = a^2 + 2ab + b^2$
- $(a - b)^2 = a^2 - 2ab + b^2$
- $(a + b)(a - b) = a^2 - b^2$
- quadratic formula: $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

- slope $m = \frac{y_2 - y_1}{x_2 - x_1}$; line: $y = mx + b$
- distance = $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
- midpoint = $\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$

Angles & triangles

- complementary: sum 90° ; supplementary: sum 180°
- angles of a triangle add to 180°
- polygon with n sides: angle sum $(n - 2) \times 180^\circ$
- Pythagorean theorem: $a^2 + b^2 = c^2$
- common triples: 3-4-5, 5-12-13, 8-15-17, 7-24-25
- 45-45-90: sides $x, x, x\sqrt{2}$; 30-60-90: $x, x\sqrt{3}, 2x$

Perimeter & area

- rectangle: $P = 2l + 2w$, $A = lw$
- square: $P = 4s$, $A = s^2$
- triangle: $A = \frac{1}{2}bh$
- parallelogram: $A = bh$ (height, not slanted side)
- trapezoid: $A = \frac{1}{2}(b_1 + b_2)h$

Circles & solids

- $d = 2r$, circumference $C = 2\pi r = \pi d$
- area $A = \pi r^2$; $\pi \approx 3.14 \approx \frac{22}{7}$
- box: $V = lwh$, $SA = 2lw + 2lh + 2wh$
- cube: $V = s^3$, $SA = 6s^2$
- cylinder: $V = \pi r^2 h$
- cone: $V = \frac{1}{3}\pi r^2 h$; sphere: $V = \frac{4}{3}\pi r^3$

Quick Reference Tables

Squares and cubes

$1^2 = 1$	$14^2 = 196$	$1^3 = 1$
$2^2 = 4$	$15^2 = 225$	$2^3 = 8$
$3^2 = 9$	$16^2 = 256$	$3^3 = 27$
$4^2 = 16$	$17^2 = 289$	$4^3 = 64$
$5^2 = 25$	$18^2 = 324$	$5^3 = 125$
$6^2 = 36$	$19^2 = 361$	$6^3 = 216$
$7^2 = 49$	$20^2 = 400$	$7^3 = 343$
$8^2 = 64$	$21^2 = 441$	$8^3 = 512$
$9^2 = 81$	$22^2 = 484$	$9^3 = 729$
$10^2 = 100$	$23^2 = 529$	$10^3 = 1,000$
$11^2 = 121$	$24^2 = 576$	
$12^2 = 144$	$25^2 = 625$	
$13^2 = 169$		

Fractions, decimals, percents

Fraction	Decimal	Percent
$\frac{1}{2}$	0.5	50%
$\frac{1}{3}$	0.333...	$33\frac{1}{3}\%$
$\frac{2}{3}$	0.666...	$66\frac{2}{3}\%$
$\frac{1}{4}$	0.25	25%
$\frac{3}{4}$	0.75	75%
$\frac{1}{5}$	0.2	20%
$\frac{2}{5}$	0.4	40%
$\frac{3}{5}$	0.6	60%
$\frac{4}{5}$	0.8	80%
$\frac{1}{6}$	0.1666...	$16\frac{2}{3}\%$
$\frac{1}{8}$	0.125	12.5%
$\frac{3}{8}$	0.375	37.5%
$\frac{5}{8}$	0.625	62.5%
$\frac{7}{8}$	0.875	87.5%
$\frac{1}{10}$	0.1	10%
$\frac{1}{20}$	0.05	5%
$\frac{1}{25}$	0.04	4%
$\frac{1}{50}$	0.02	2%
$\frac{1}{100}$	0.01	1%

Prime numbers less than 100

2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97

Divisibility rules

- 2 last digit is even
- 3 sum of digits divisible by 3
- 4 last two digits divisible by 4
- 5 last digit 0 or 5
- 6 divisible by 2 and by 3
- 9 sum of digits divisible by 9
- 10 last digit 0

Units of measure

12 inches = 1 foot; 3 feet = 1 yard
 5,280 feet = 1 mile; 1,760 yards = 1 mile
 8 fluid ounces = 1 cup; 2 cups = 1 pint
 2 pints = 1 quart; 4 quarts = 1 gallon
 16 ounces = 1 pound; 2,000 pounds = 1 ton
 60 seconds = 1 minute; 60 minutes = 1 hour
 $1 \text{ yard}^2 = 9 \text{ feet}^2$; $1 \text{ foot}^2 = 144 \text{ inches}^2$
 $1 \text{ yard}^3 = 27 \text{ feet}^3$

Metric: kilo = 1,000; centi = $\frac{1}{100}$; milli = $\frac{1}{1000}$
 1 km = 1,000 m; 1 m = 100 cm = 1,000 mm
 1 kg = 1,000 g; 1 L = 1,000 mL

24-hour (military) time

Hours after noon: add 12. 1:00 p.m. = 1300,
 6:45 p.m. = 1845, midnight = 0000, 12:30 a.m.
 = 0030.

Glossary of Math Terms

Mathematics Knowledge questions often test vocabulary directly (“Which of the following is a prime number?”). Make sure every term below is familiar.

Absolute value. The distance of a number from 0 on the number line; never negative. $|-7| = 7$.

Acute angle. An angle greater than 0° and less than 90° .

Area. The amount of surface a flat figure covers, measured in square units.

Average (mean). The sum of a set of values divided by the number of values.

Binomial. A polynomial with two terms, such as $x + 5$.

Circumference. The distance around a circle: $C = 2\pi r = \pi d$.

Coefficient. The number multiplying a variable. In $7x^2$, the coefficient is 7.

Complementary angles. Two angles whose measures add up to 90° .

Composite number. A whole number greater than 1 that has more than two factors, such as 9 or 12.

Congruent. Having exactly the same size and shape (equal sides, equal angles).

Consecutive integers. Integers that follow one another in order, such as 7, 8, 9.

Constant. A term with no variable, such as the 5 in $3x + 5$.

Denominator. The bottom number of a fraction; it tells how many equal parts make a whole.

Diameter. A segment through the center of a circle from side to side; twice the radius.

Difference. The result of a subtraction.

Digit. Any of the symbols 0–9 used to write numbers.

Divisor. The number you divide by. In $20 \div 4$, the divisor is 4.

Equation. A statement that two expressions are equal, such as $2x + 3 = 11$.

Equilateral triangle. A triangle with three equal sides and three 60° angles.

Even number. An integer divisible by 2.

Exponent. The small raised number that tells how many times the base is used as a factor: $2^3 = 2 \cdot 2 \cdot 2$.

Factor. A number that divides another number evenly. The factors of 12 are 1, 2, 3, 4, 6, 12.

Greatest common factor (GCF). The largest number that divides each of two or more numbers.

Hypotenuse. The longest side of a right triangle, opposite the right angle.

Improper fraction. A fraction whose numerator is greater than or equal to its denominator, such as $\frac{7}{4}$.

Inequality. A statement using $<$, $>$, $\leq$, or $\geq$ to compare two expressions.

Integer. A whole number or its opposite: $\dots, -2, -1, 0, 1, 2, \dots$

Irrational number. A number that cannot be written as a fraction of integers, such as $\sqrt{2}$ or π .

Isosceles triangle. A triangle with at least two equal sides; the angles opposite them are equal.

Least common multiple (LCM). The smallest positive number that is a multiple of each of two or more numbers.

Like terms. Terms with the same variables raised to the same powers, such as $3x^2$ and $-5x^2$.

Linear equation. An equation whose graph is a straight line; the variable has no exponent above 1.

Median. The middle value of a sorted list (or the average of the two middle values).

Mixed number. A whole number and a fraction together, such as $2\frac{1}{3}$.

Mode. The value that occurs most often in a data set.

Monomial. A polynomial with one term, such as $-4x^3y$.

Multiple. The product of a number and a whole number. Multiples of 6: 6, 12, 18, $\dots$

Numerator. The top number of a fraction.

Obtuse angle. An angle greater than 90° and less than 180° .

Odd number. An integer that is not divisible by 2.

Parallel lines. Lines in the same plane that never meet; they have equal slopes.

Percent. Per hundred: $35\% = \frac{35}{100} = 0.35$.

Perimeter. The distance around a figure.

Perpendicular lines. Lines that meet at a right angle; their slopes are negative reciprocals.

Polygon. A closed figure made of straight sides (triangle, quadrilateral, pentagon, $\dots$).

Polynomial. A sum of terms of the form ax^n , such as $3x^2 - 5x + 2$.

Prime number. A whole number greater than 1 whose only factors are 1 and itself: 2, 3, 5, 7, 11, ...

Probability. The chance that an event occurs, from 0 (impossible) to 1 (certain).

Product. The result of a multiplication.

Proportion. An equation stating that two ratios are equal, such as $\frac{3}{4} = \frac{9}{12}$.

Quadrant. One of the four regions of the coordinate plane, numbered I–IV counterclockwise from the upper right.

Quadratic equation. An equation of the form $ax^2 + bx + c = 0$ with $a \neq 0$.

Quotient. The result of a division.

Radius. The distance from the center of a circle to any point on it.

Range. The difference between the largest and smallest values in a data set.

Ratio. A comparison of two quantities by division, written $3 : 4$ or $\frac{3}{4}$.

Rational number. Any number that can be written as a fraction of two integers.

Reciprocal. The number that multiplies with a given number to give 1. The reciprocal of $\frac{2}{3}$ is $\frac{3}{2}$.

Right angle. An angle of exactly 90° .

Scientific notation. A number written as $a \times 10^n$ with $1 \leq a < 10$, such as 4.2×10^5 .

Slope. The steepness of a line: rise over run, $\frac{y_2 - y_1}{x_2 - x_1}$.

Square root. A number that, multiplied by itself, gives the original number: $\sqrt{49} = 7$.

Sum. The result of an addition.

Supplementary angles. Two angles whose measures add up to 180° .

Trinomial. A polynomial with three terms, such as $x^2 + 5x + 6$.

Variable. A letter that stands for an unknown or changing number.

Vertex. A corner point of an angle, polygon, or solid (plural: vertices).

Vertical angles. The opposite angles formed by two intersecting lines; they are equal.

Volume. The amount of space a solid occupies, measured in cubic units.

Whole numbers. The numbers 0, 1, 2, 3, ...

Progress Tracker

Record your score the first time you complete each practice set, then redo the problems you missed about a week later and record that score too.

Ch.	Practice set	Score	Date	Redo
1	Whole Numbers & Order of Operations	/ 25		/ 25
2	Negative Numbers & Absolute Value	/ 25		/ 25
3	Factors, Multiples, Primes, GCF & LCM	/ 25		/ 25
4	Fractions & Mixed Numbers	/ 25		/ 25
5	Decimals	/ 25		/ 25
6	Percents	/ 25		/ 25
7	Ratios & Proportions	/ 25		/ 25
8	Exponents & Scientific Notation	/ 25		/ 25
9	Square Roots & Radicals	/ 25		/ 25
10	Algebraic Expressions	/ 25		/ 25
11	Linear Equations	/ 25		/ 25
12	Inequalities	/ 25		/ 25
13	Systems of Equations	/ 25		/ 25
14	Polynomials & Exponent Rules	/ 25		/ 25
15	Factoring & Quadratic Equations	/ 25		/ 25
16	Lines & Angles	/ 25		/ 25
17	Triangles & the Pythagorean Theorem	/ 25		/ 25
18	Perimeter & Area	/ 25		/ 25
19	Circles	/ 25		/ 25
20	Volume & Surface Area	/ 25		/ 25
21	Coordinate Geometry	/ 25		/ 25
22	Money: Shopping, Discounts, Tax & Wages	/ 40		/ 40
23	Interest, Commission & Percent Change	/ 35		/ 35
24	Rate, Distance & Time	/ 40		/ 40
25	Work Rates & Mixtures	/ 35		/ 35
26	Averages, Statistics & Probability	/ 40		/ 40
27	Measurement, Units & Multi-Step Problems	/ 35		/ 35
–	Diagnostic Test	/ 30		
–	Practice Test 1	/ 55		
–	Practice Test 2	/ 55		
–	Practice Test 3	/ 55		
–	Practice Test 4	/ 55		

Thank you, and good luck!

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